JOSE RIZAL UNIVERSITY

COLLEGE OF COMPUTER STUDIES AND ENGINEERING

COMPUTER ENGINEERING DEPARTMENT

CPE C313

ARTIFICIAL INTELLIGENCE

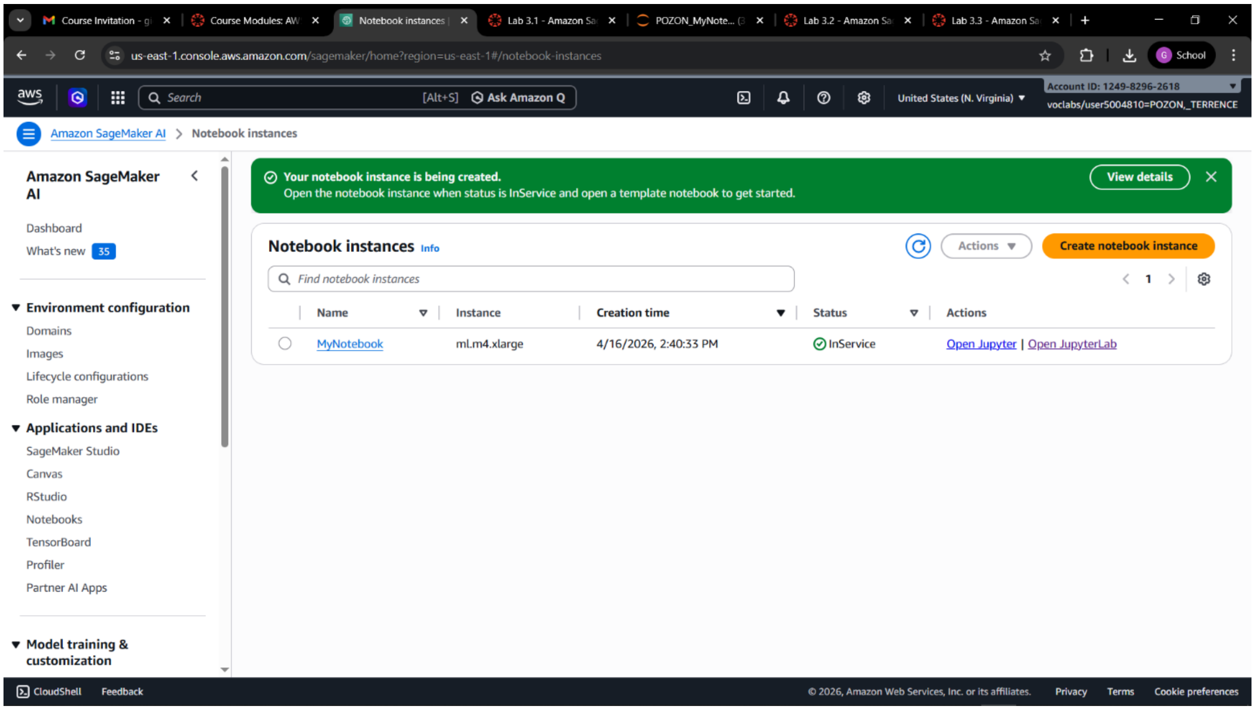
<u>LABORATORY</u>

1&2

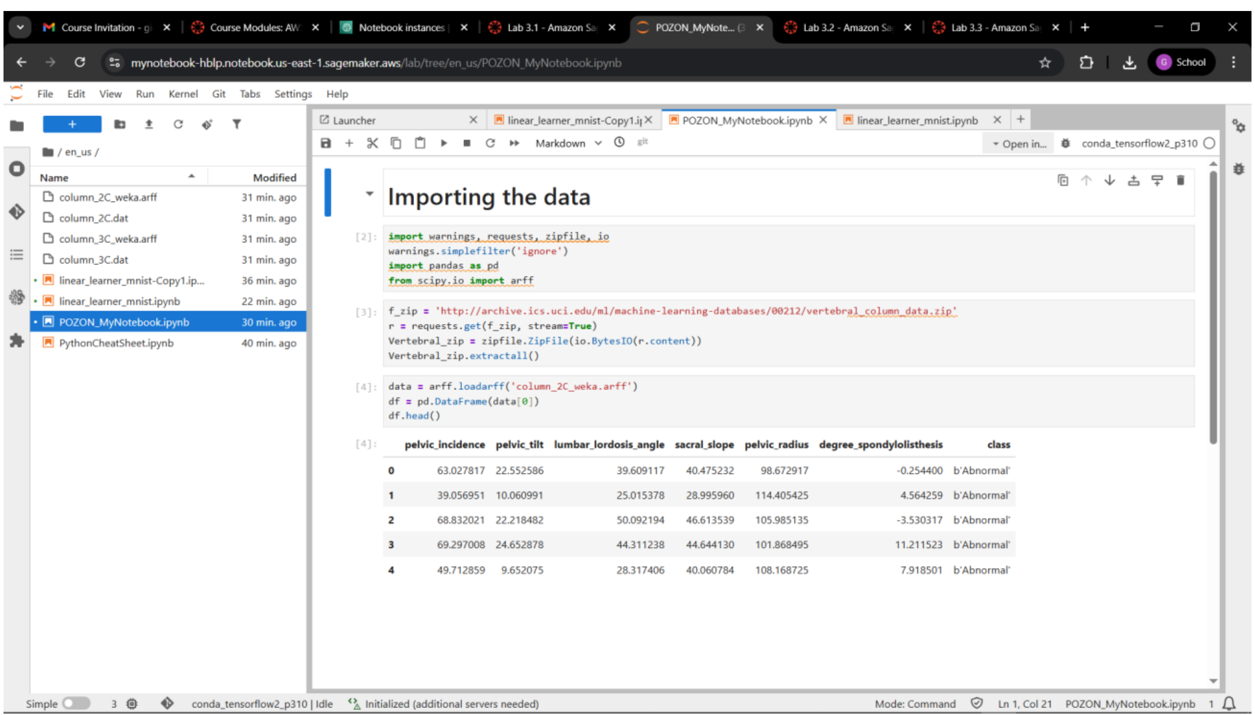
Pozon, Gion Terrence B.
301G
2026-04-16

LABORATORY 1

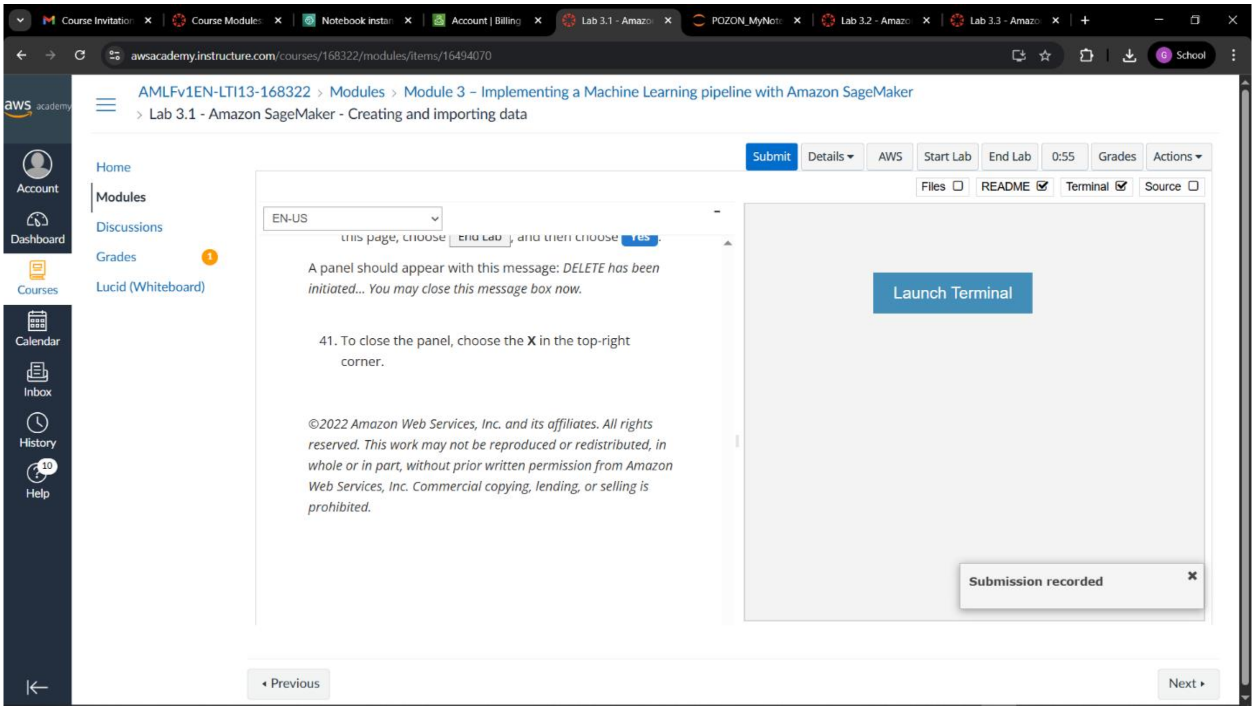
1. AWS Console account
2. MyNotebook instance configuration



3. Output after the code below is executed



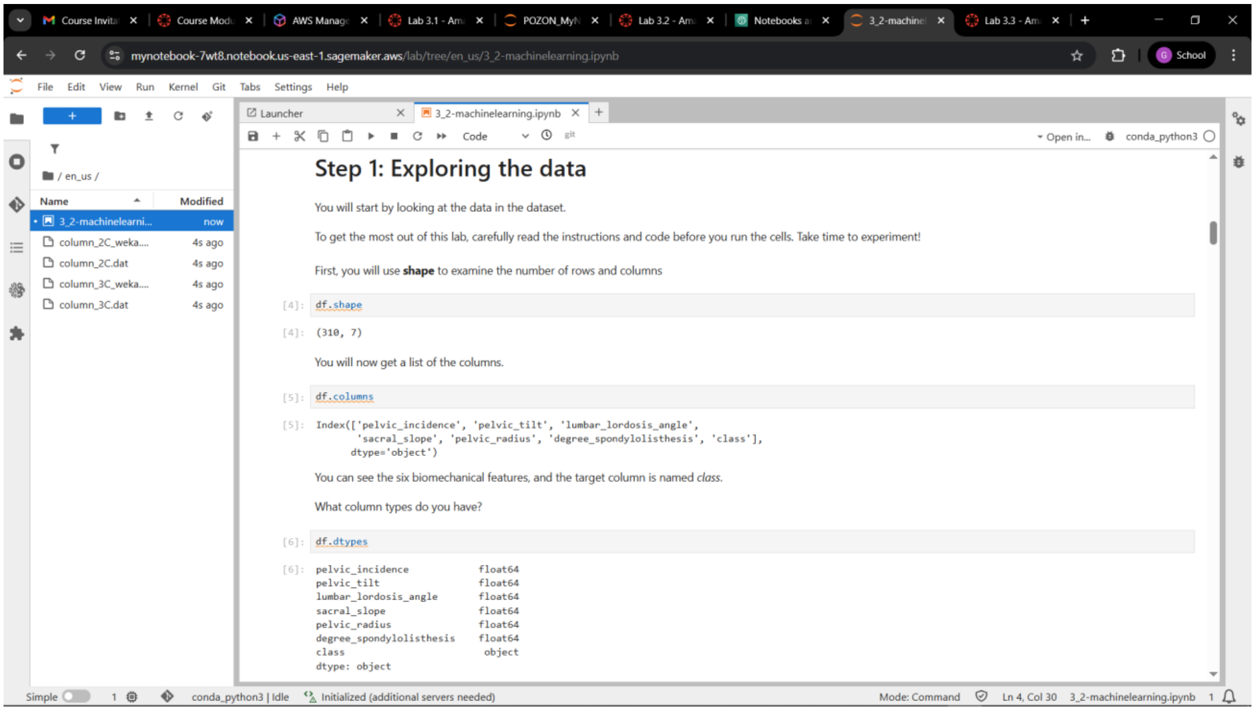
4. Screenshot of the Submitted activity.



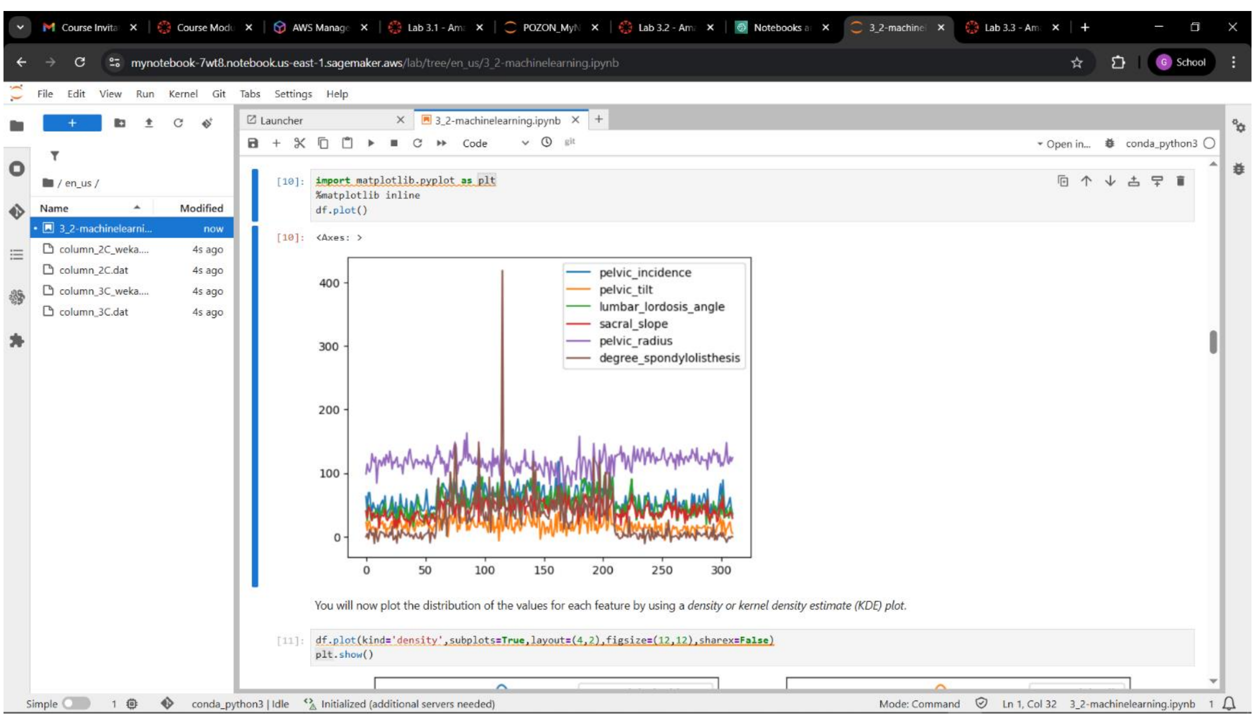
LABORATORY 2

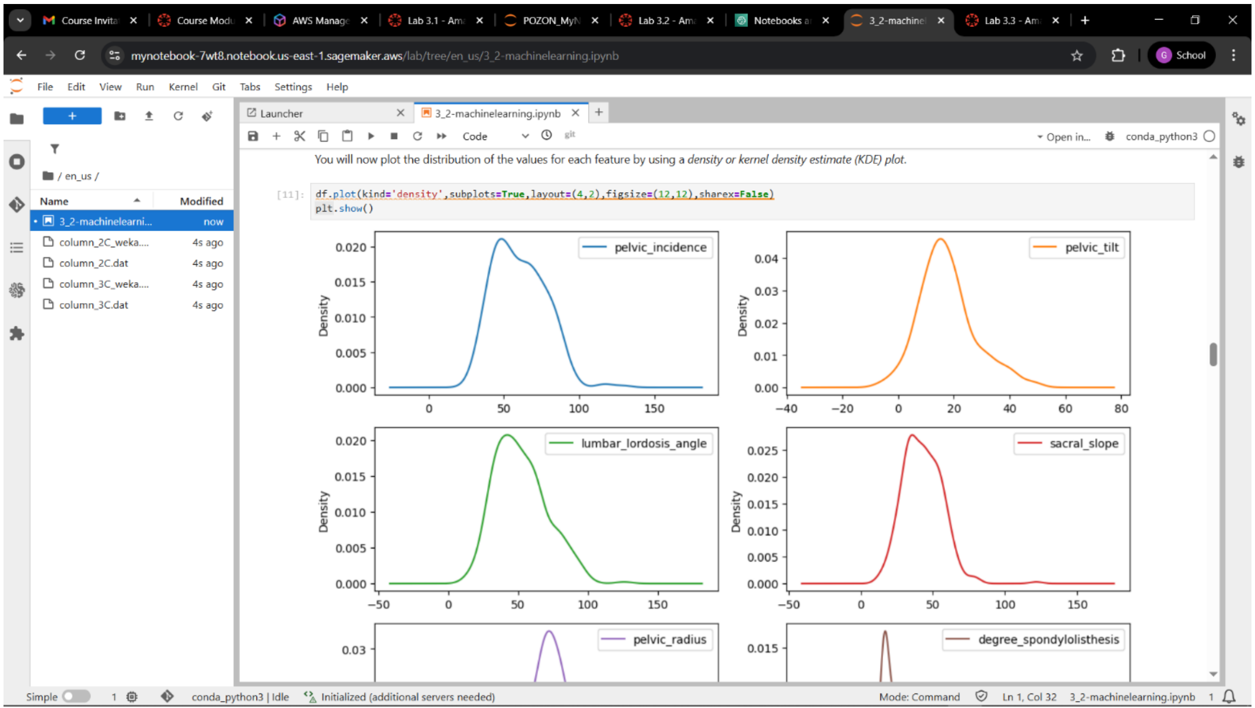
1. Output of this last task:

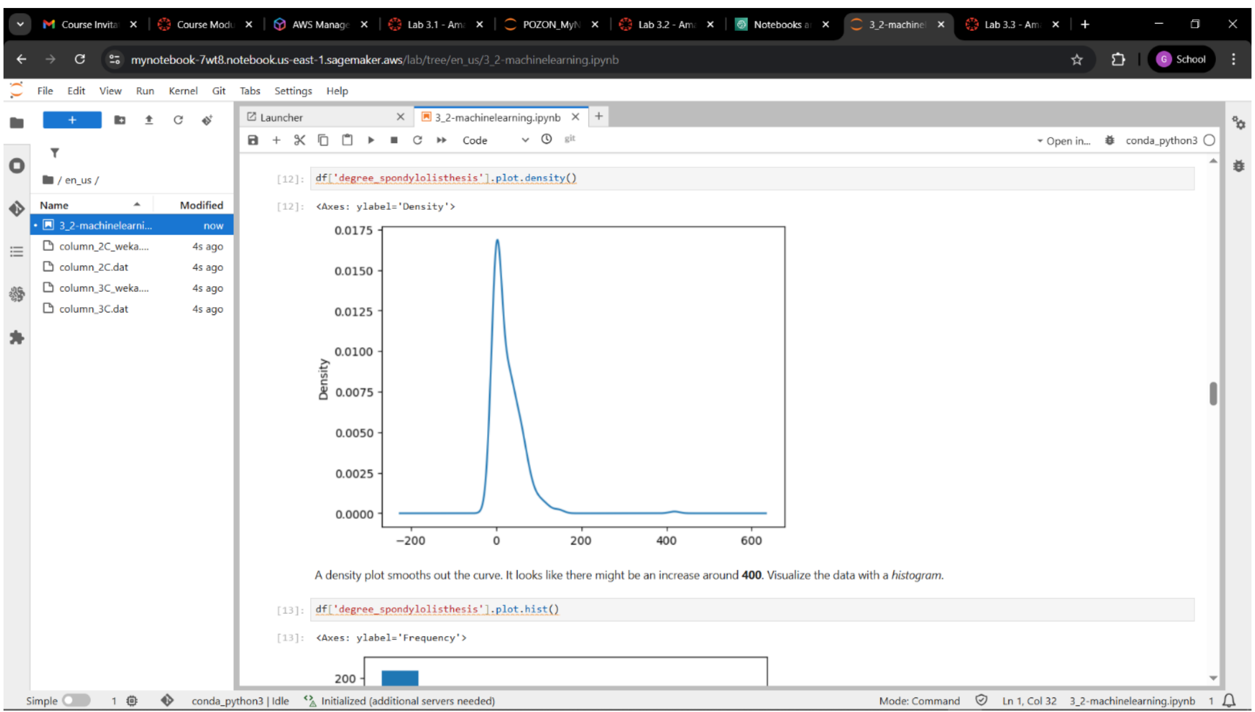
- Examined the structure of a dataset by using pandas

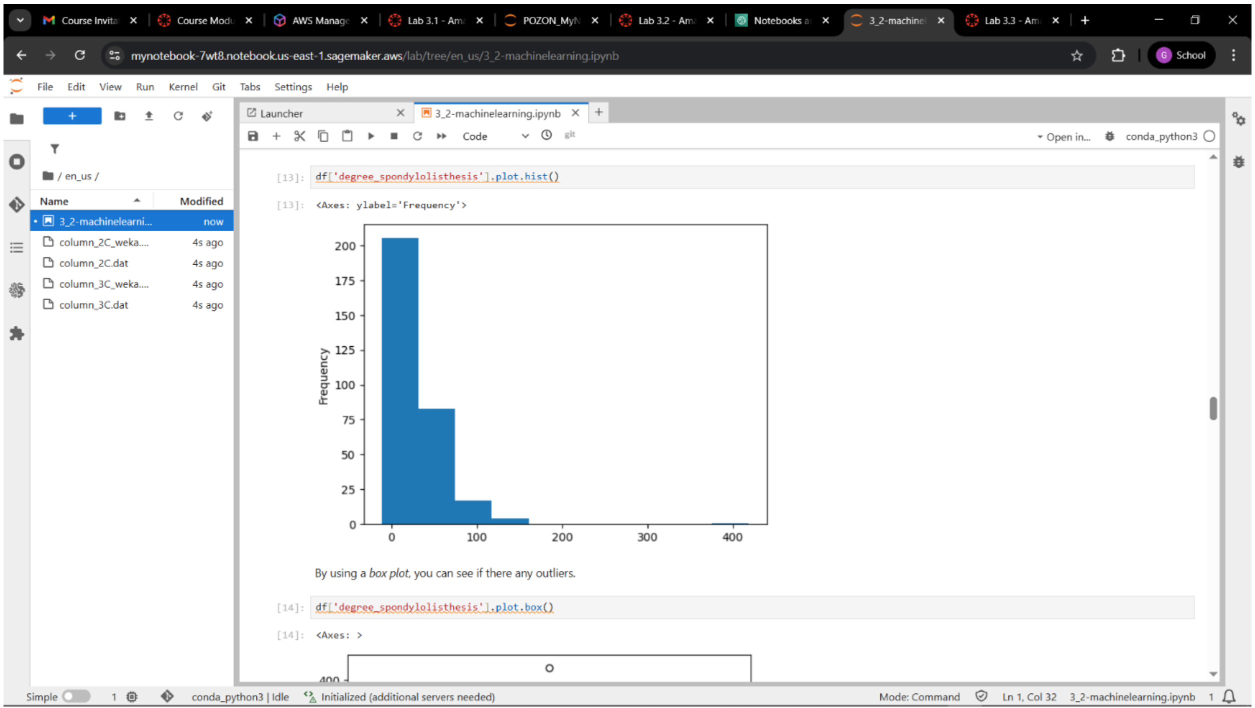


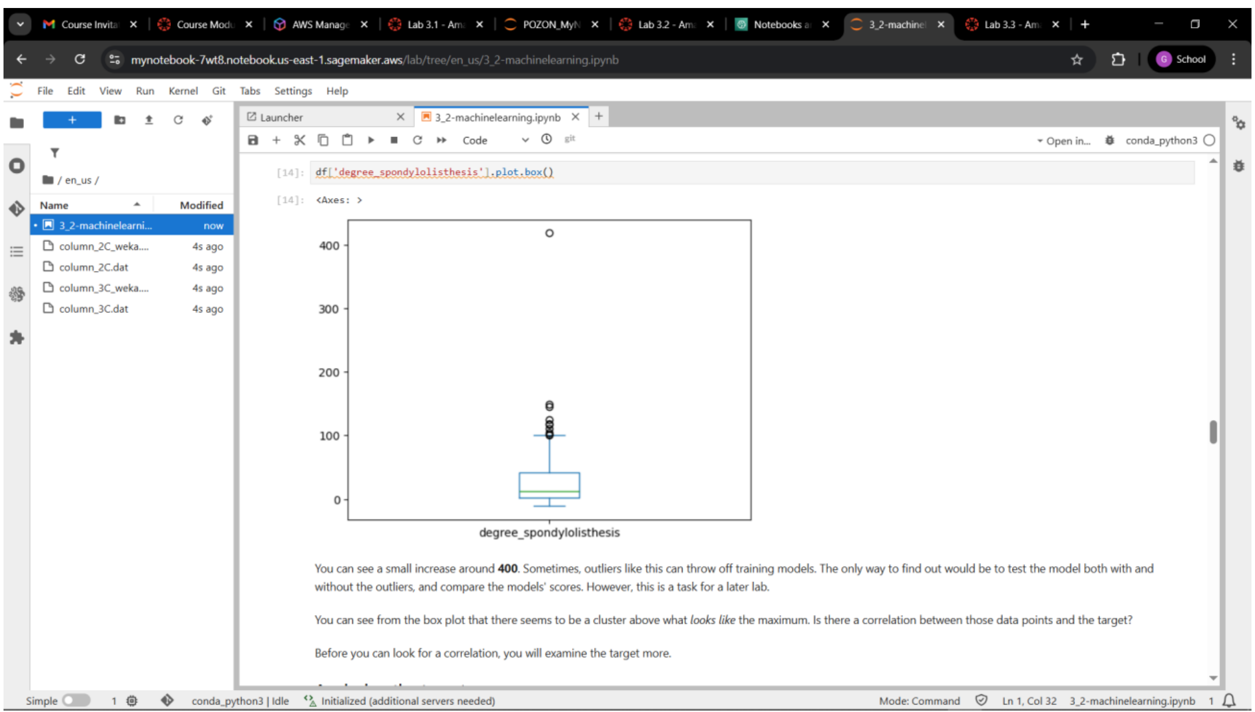
- Viewed statistics with pandas and Matplotlib



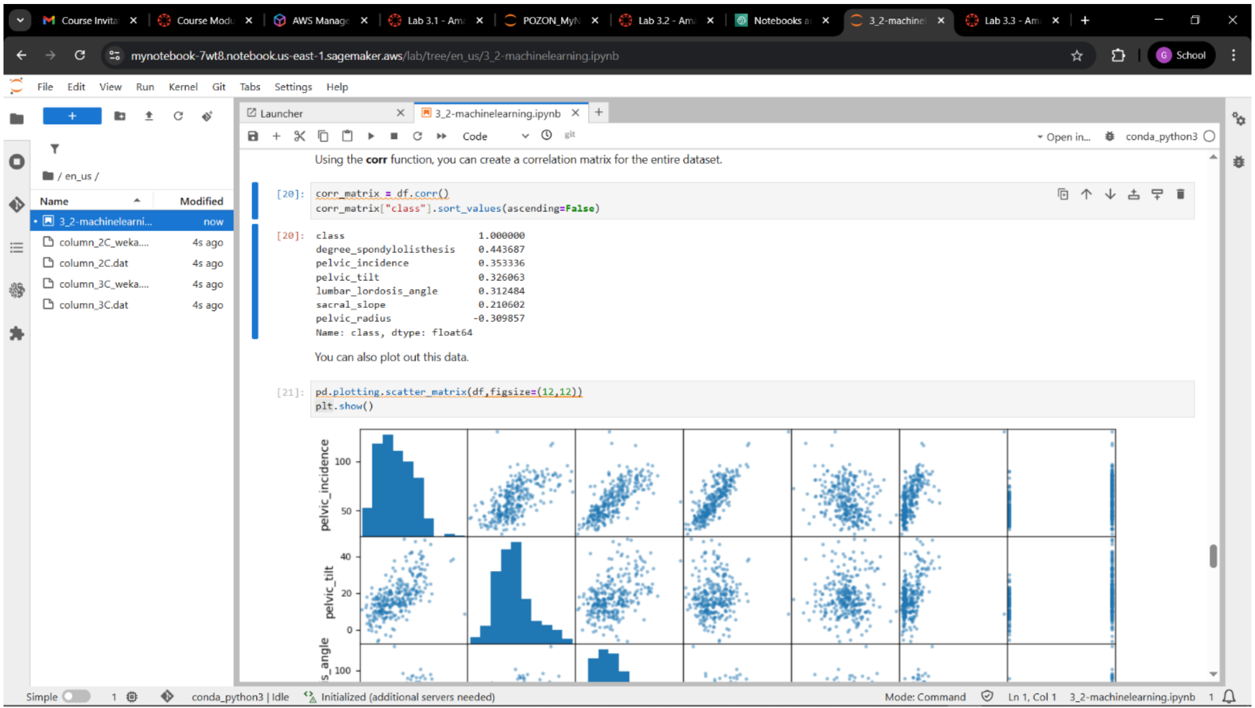

mynotebook-7wt8.notebook.us-east-1.sagemaker.aws/lab/tree/en_us/3_2-machinelearning.ipynb
File Edit View Run Kernel Git Tabs Settings Help
Launcher
3_2-machinelearning.ipynb
Name
Modified
3_2-machinelearni...
now
column_2C_weka....
4s ago
column_2C.dat
4s ago
column_3C_weka....
4s ago
column_3C.dat
4s ago
You will now plot the distribution of the values for each feature by using a density or kernel density estimate (KDE) plot.
[11]: df.plot(kind='density',subplots=True,layout=(4,2),figsize=(12,12),sharex=False)
plt.show()
pelvic_incidence
pelvic_tilt
lumbar_lordosis_angle
sacral_slope
pelvic_radius
degree_spondylolisthesis
Density
Simple
conda_python3 | Idle
Initialized (additional servers needed)
Mode: Command
Ln 1, Col 32
3_2-machinelearning.ipynb

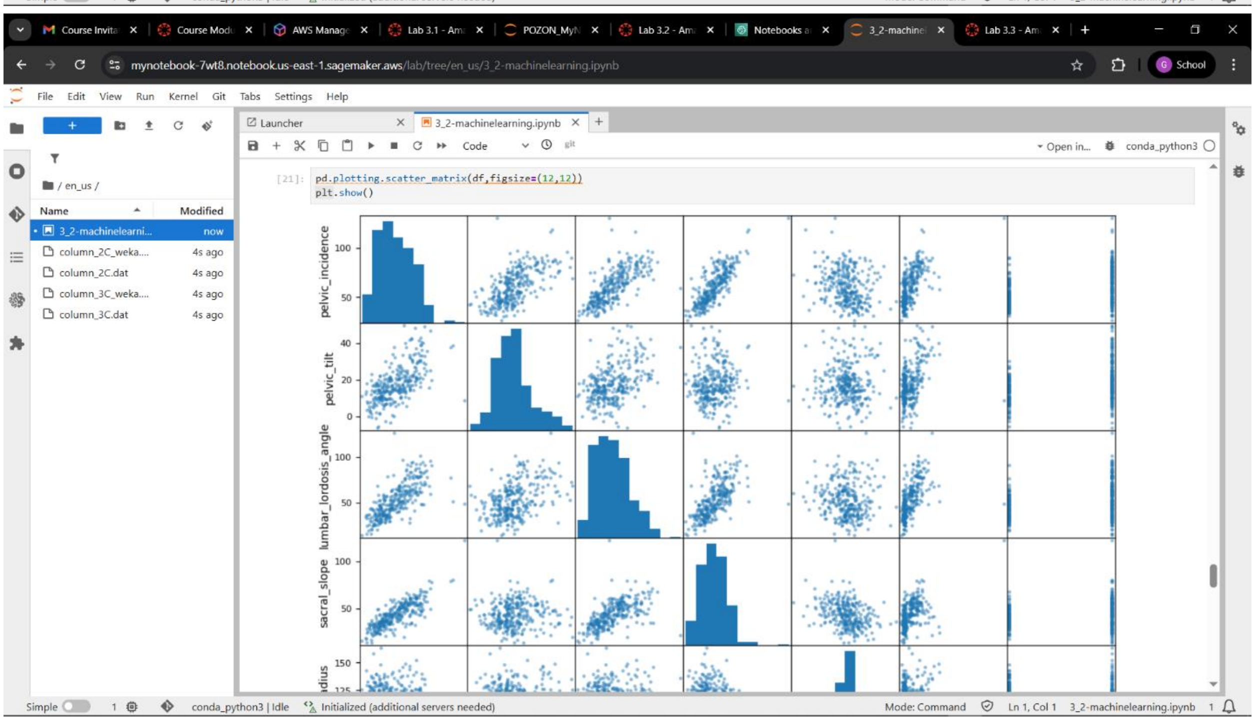

mynotebook-7wt8.notebook.us-east-1.sagemaker.aws/lab/tree/en_us/3_2-machinelearning.ipynb
File Edit View Run Kernel Git Tabs Settings Help
Launcher
3_2-machinelearning.ipynb
[12]: df['degree_spondylolisthesis'].plot.density()
[12]: <Axes: ylabel='Density'>
Density
A density plot smooths out the curve. It looks like there might be an increase around 400. Visualize the data with a histogram.
[13]: df['degree_spondylolisthesis'].plot.hist()
[13]: <Axes: ylabel='Frequency'>
Simple
conda_python3 | Idle
Initialized (additional servers needed)
Mode: Command
Ln 1, Col 32
3_2-machinelearning.ipynb

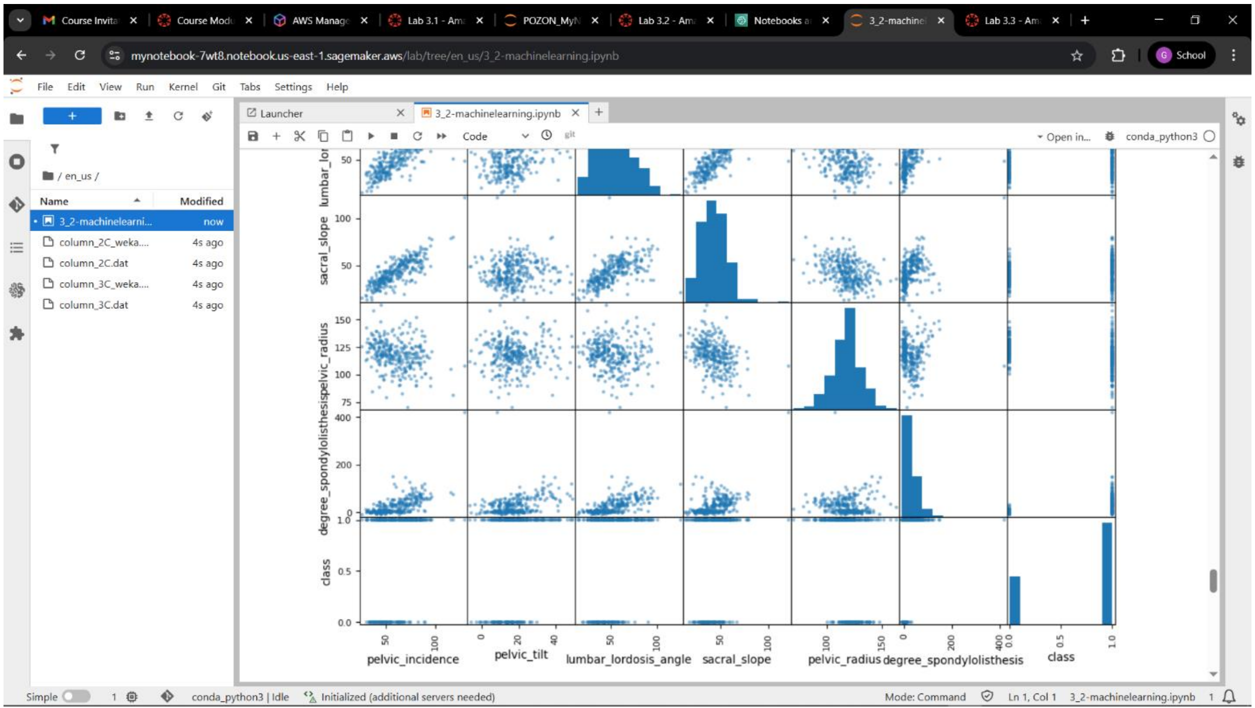

mynotebook-7wt8.notebook.us-east-1.sagemaker.aws/lab/tree/en_us/3_2-machinelearning.ipynb
File Edit View Run Kernel Git Tabs Settings Help
Launcher
3_2-machinelearning.ipynb
/ en_us /
Name
Modified
3_2-machinelearni...
now
column_2C_weka....
4s ago
column_2C.dat
4s ago
column_3C_weka....
4s ago
column_3C.dat
4s ago
[13]: df['degree_spondylolisthesis'].plot.hist()
[13]: <Axes: ylabel='Frequency'>
Frequency
By using a box plot, you can see if there any outliers.
[14]: df['degree_spondylolisthesis'].plot.box()
[14]: <Axes: >
Mode: Command
Ln 1, Col 32
conda_python3 | Idle
Initialized (additional servers needed)


mynotebook-7wt8.notebook.us-east-1.sagemaker.aws/lab/tree/en_us/3_2-machinelearning.ipynb
[14]: df['degree_spondylolisthesis'].plot.box()
[14]: <Axes: >
degree_spondylolisthesis
You can see a small increase around 400. Sometimes, outliers like this can throw off training models. The only way to find out would be to test the model both with and without the outliers, and compare the models' scores. However, this is a task for a later lab.
You can see from the box plot that there seems to be a cluster above what looks like the maximum. Is there a correlation between those data points and the target?
Before you can look for a correlation, you will examine the target more.
Mode: Command
Ln 1, Col 32
conda_python3 | Idle
Initialized (additional servers needed)

- Looked for correlation between features in a dataset







2. Explain the correlation of the different variables:

1. **Question:** Are there any features that aren't well-distributed? Are there any features with outliers that you want to look at? Does it look like there might be any correlations between features?

    - degree_spondylolisthesis has the most outliers since its max is almost 10 times the value of 75% percentile, meaning there are 25% of people that have the values between 41 and 418, lumbar_lordosis_angle, pelvic_tilt, pelvic_incidence have similar values with each other which can indicate correlation

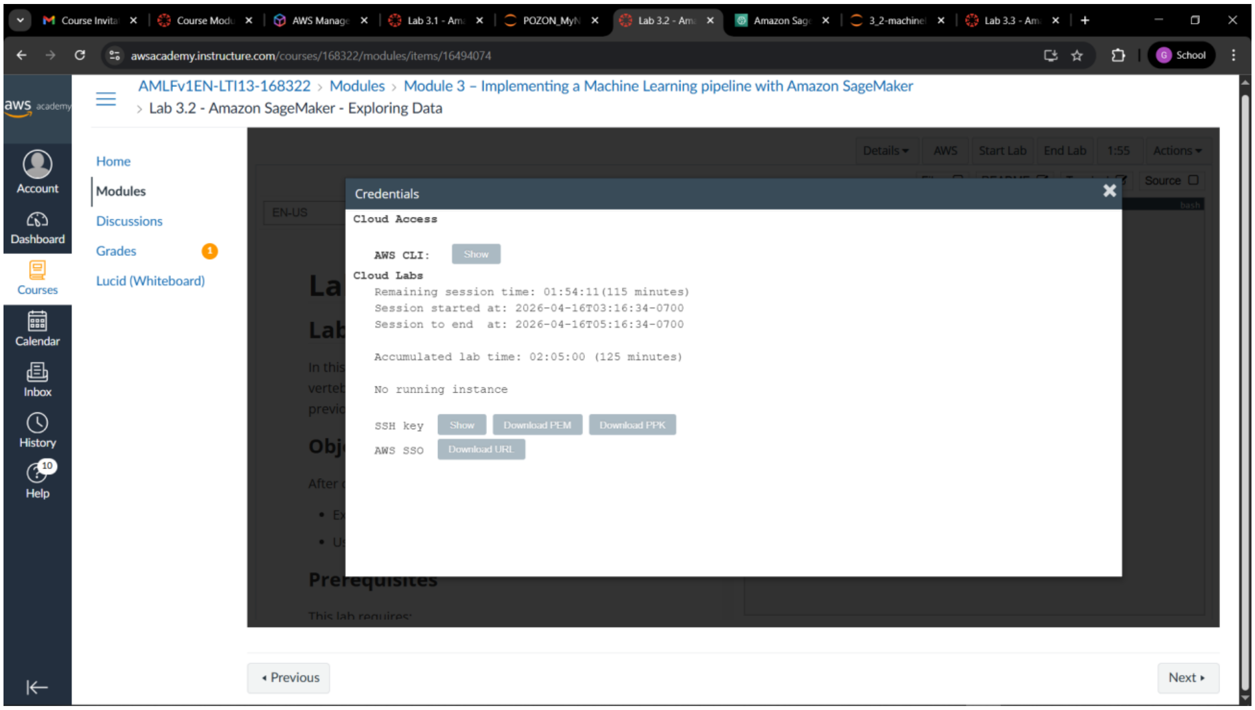
2. Which variables are highly correlated?

    - lumbar_lordosis_angle, pelvic_tilt, pelvic_incidence since they are closely together with the values of 0.31, 0.32, and 0.35 respectively

3. Which variables may have little or no effect on the target variables.

    - pelvic_radius since it has negative value which is -0.30

Canvas Details



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CPE C313

ARTIFICIAL INTELLIGENCE

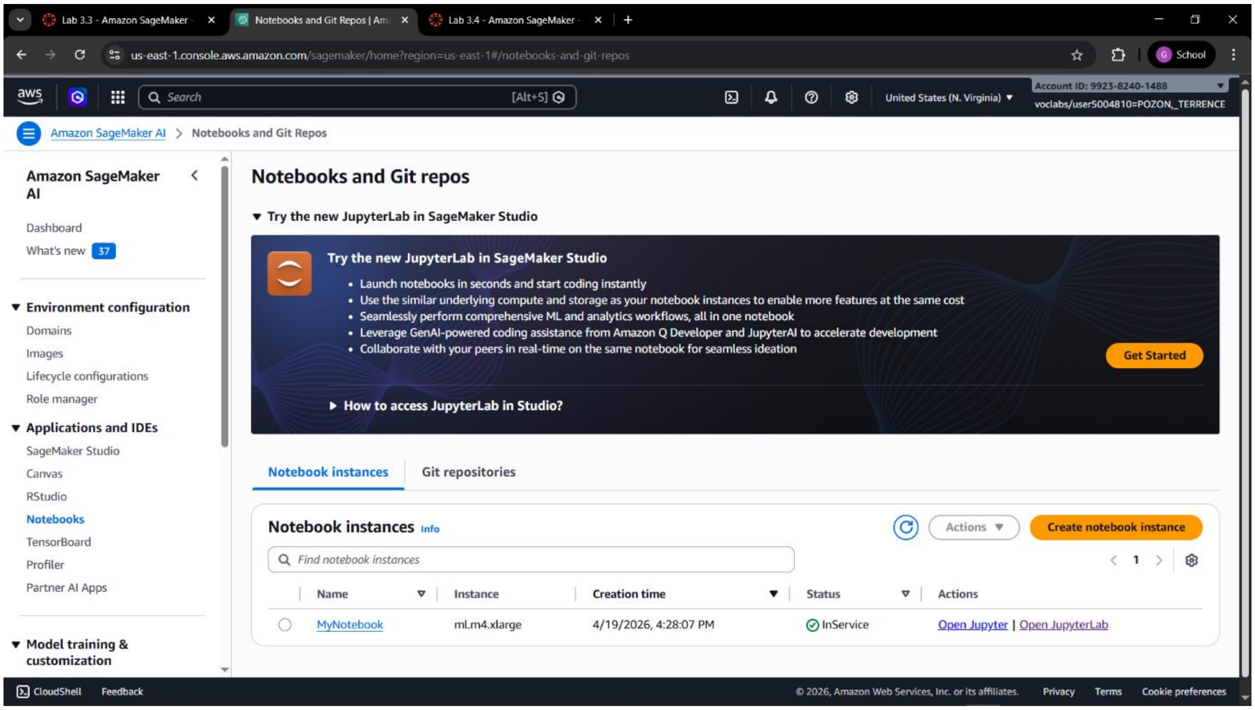
<u>LABORATORY</u>

3&4

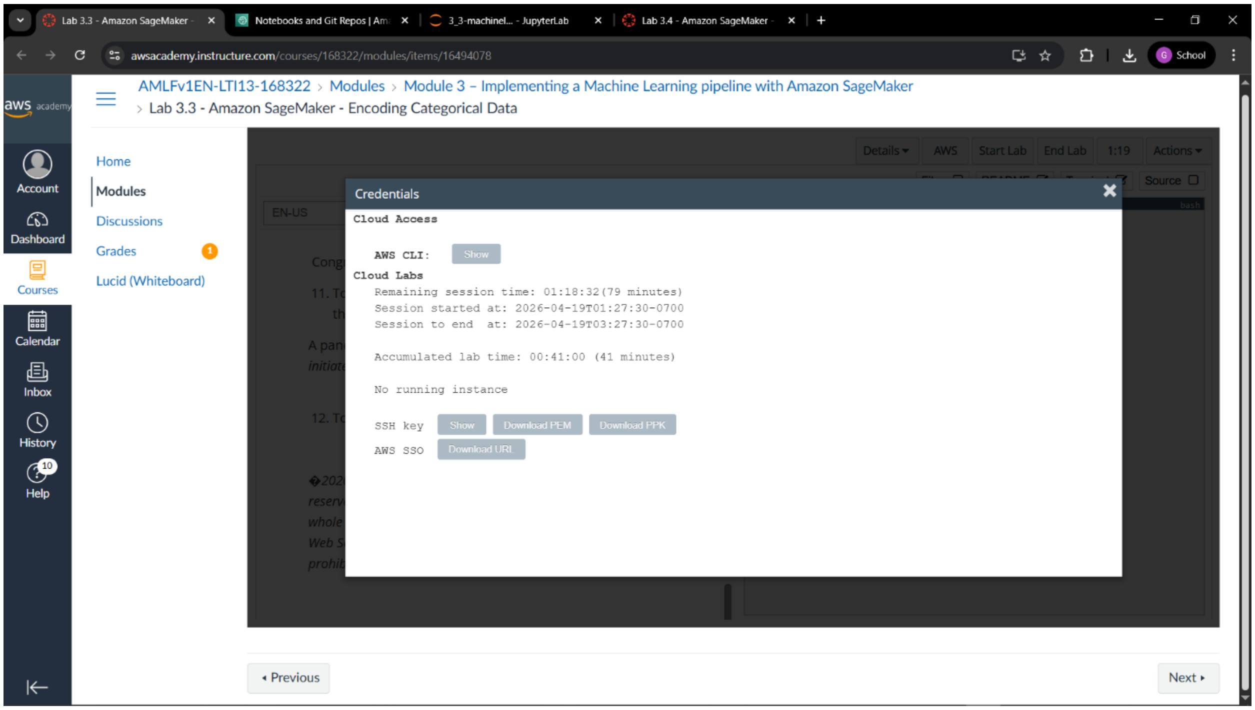
Pozon, Gion Terrence B.
301G
2026-04-19

Laboratory Activity 3

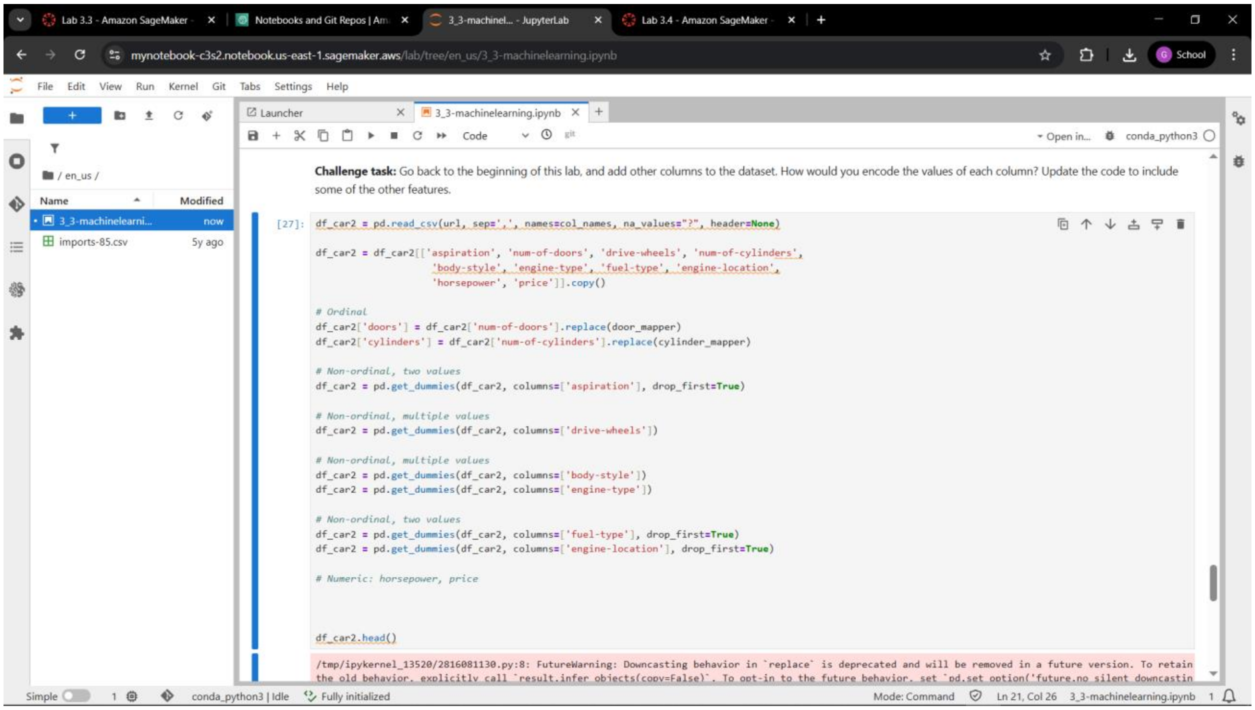
AWS Console account & MyNotebook instance configuration

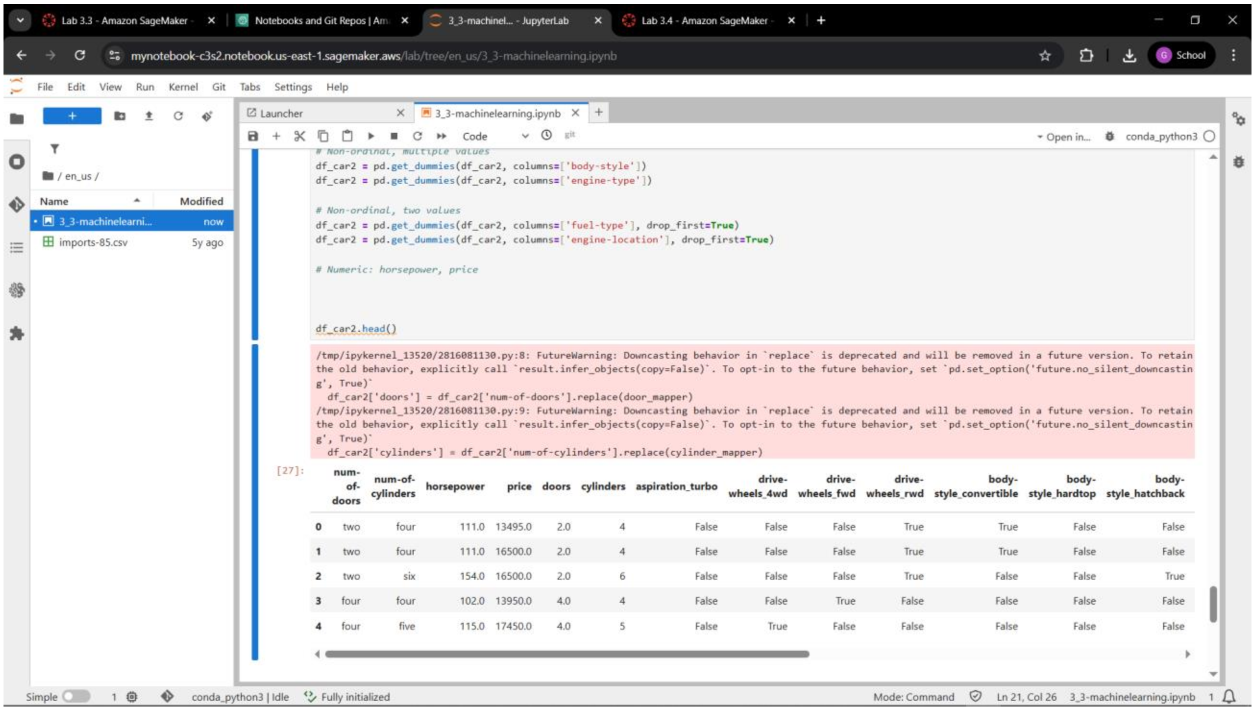


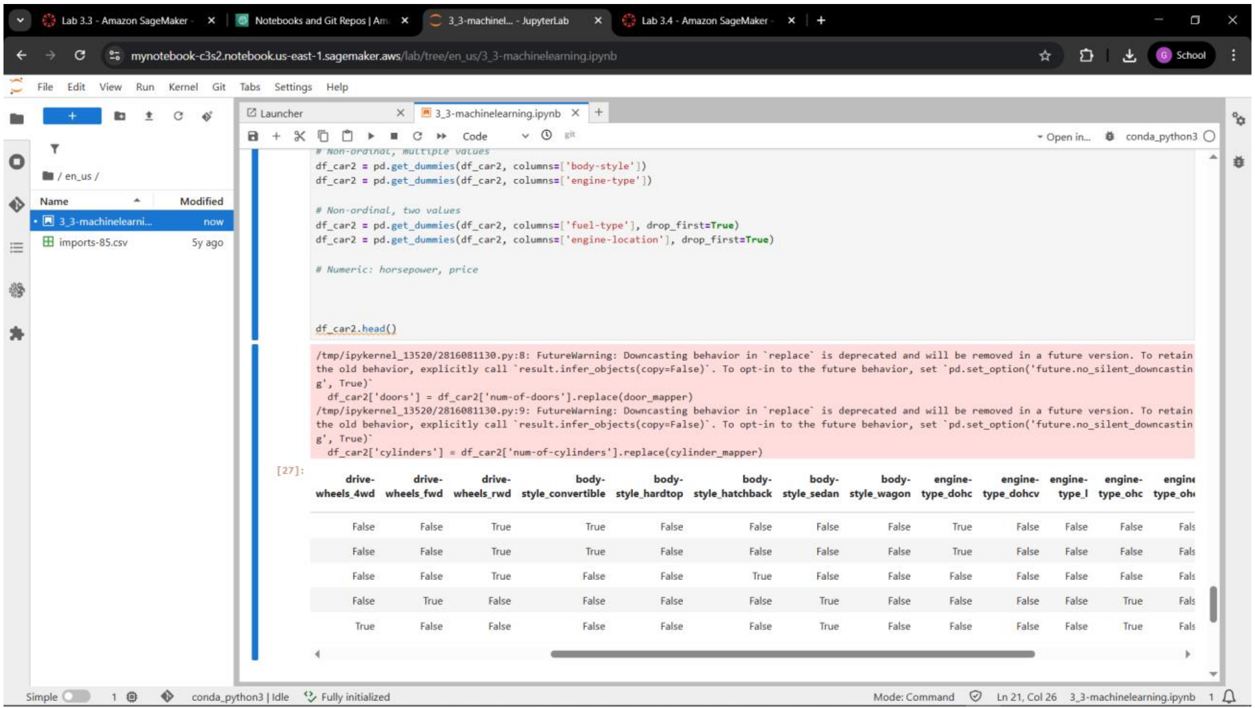
Canvas Screenshot



Challenge Task





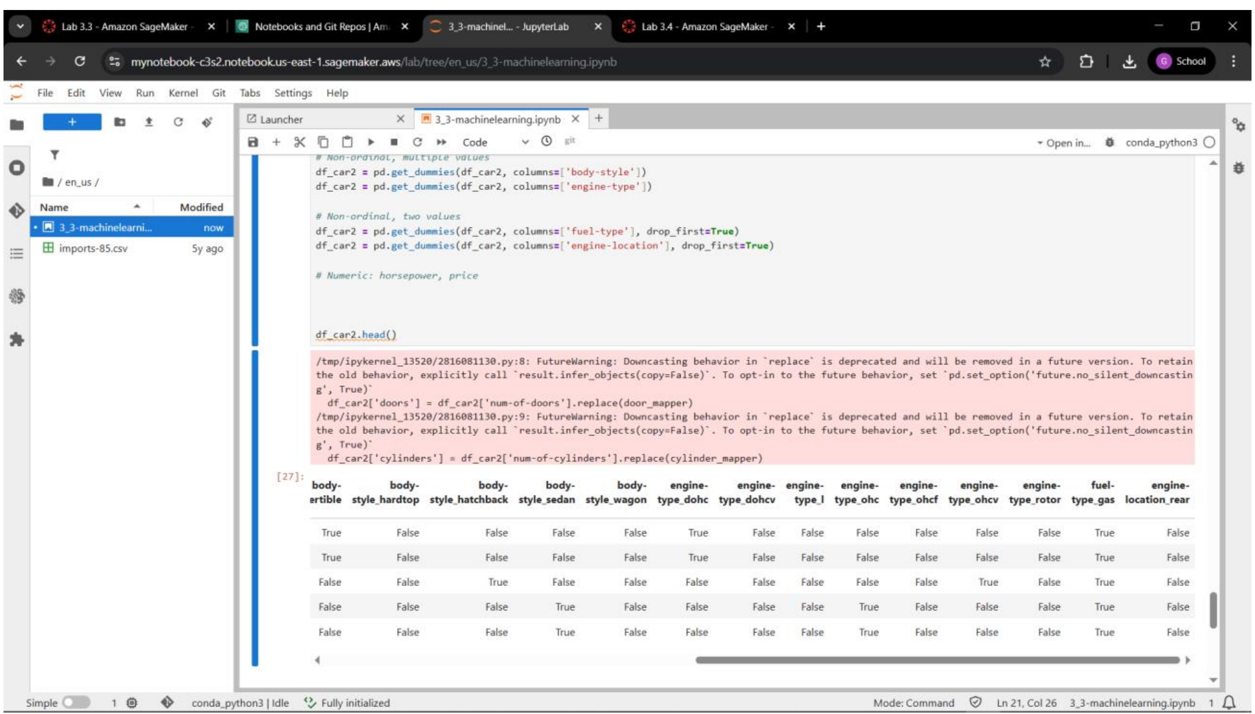


```
# Non-ordinal, multiple values
df_car2 = pd.get_dummies(df_car2, columns=['body-style'])
df_car2 = pd.get_dummies(df_car2, columns=['engine-type'])

# Non-ordinal, two values
df_car2 = pd.get_dummies(df_car2, columns=['fuel-type'], drop_first=True)
df_car2 = pd.get_dummies(df_car2, columns=['engine-location'], drop_first=True)

# Numeric: horsepower, price



df_car2.head()
```

```
/tmp/ipykernel_13520/2816081130.py:8: FutureWarning: Downcasting behavior in `replace` is deprecated and will be removed in a future version. To retain the old behavior, explicitly call `result.infer_objects(copy=False)`. To opt-in to the future behavior, set `pd.set_option('future.no_silent_downcasting', True)`
  df_car2['doors'] = df_car2['num-of-doors'].replace(door_mapper)
/tmp/ipykernel_13520/2816081130.py:9: FutureWarning: Downcasting behavior in `replace` is deprecated and will be removed in a future version. To retain the old behavior, explicitly call `result.infer_objects(copy=False)`. To opt-in to the future behavior, set `pd.set_option('future.no_silent_downcasting', True)`
  df_car2['cylinders'] = df_car2['num-of-cylinders'].replace(cylinder_mapper)
```

[27]:

drive-wheels_4wd	drive-wheels_fwd	drive-wheels_rwd	body-style_convertible	body-style_hardtop	body-style_hatchback	body-style_sedan	body-style_wagon	engine-type_dohc	engine-type_dohcv	engine-type_l	engine-type_ohc	engine-type_oh
False	False	True	True	False	False	False	False	True	False	False	False	Fals
False	False	True	True	False	False	False	False	True	False	False	False	Fals
False	False	True	False	False	True	False	False	False	False	False	False	Fals
False	True	False	False	False	False	True	False	False	False	False	True	Fals
True	False	False	False	False	False	True	False	False	False	False	True	Fals



```
# Non-ordinal, multiple values
df_car2 = pd.get_dummies(df_car2, columns=['body-style'])
df_car2 = pd.get_dummies(df_car2, columns=['engine-type'])

# Non-ordinal, two values
df_car2 = pd.get_dummies(df_car2, columns=['fuel-type'], drop_first=True)
df_car2 = pd.get_dummies(df_car2, columns=['engine-location'], drop_first=True)

# Numeric: horsepower, price



df_car2.head()
```

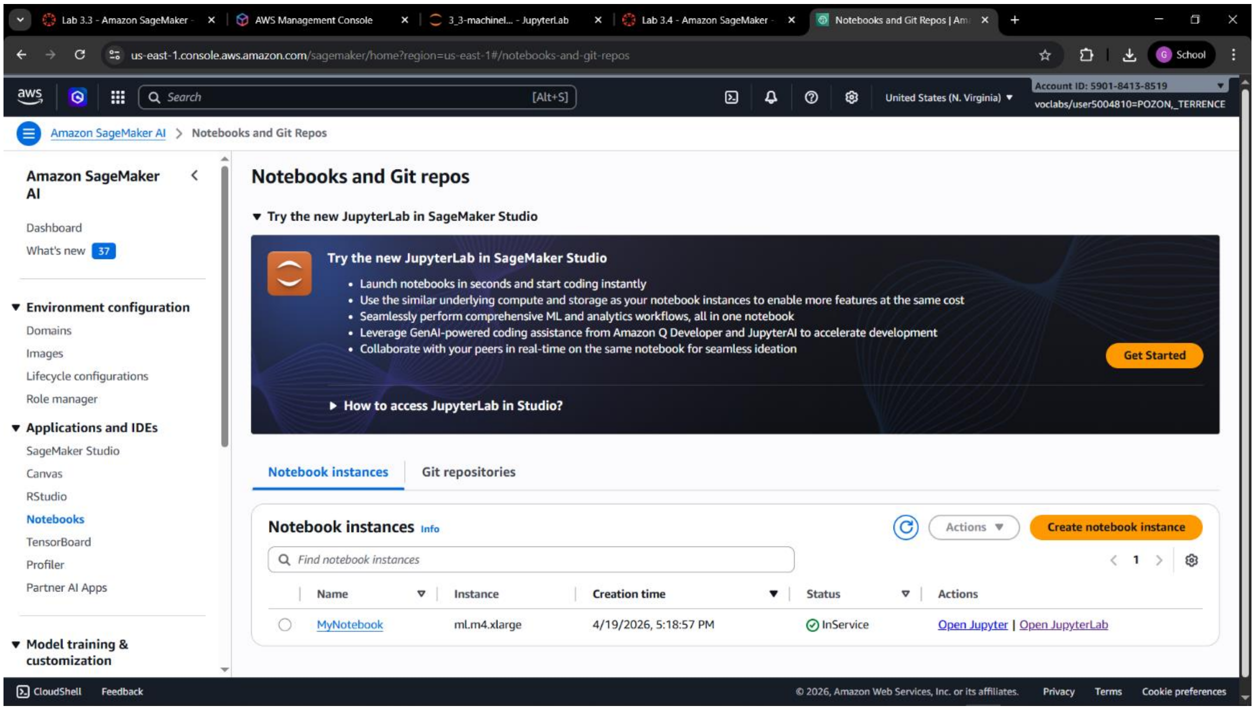
```
/tmp/ipykernel_13520/2816081130.py:8: FutureWarning: Downcasting behavior in `replace` is deprecated and will be removed in a future version. To retain the old behavior, explicitly call `result.infer_objects(copy=False)`. To opt-in to the future behavior, set `pd.set_option('future.no_silent_downcasting', True)`
  df_car2['doors'] = df_car2['num-of-doors'].replace(door_mapper)
/tmp/ipykernel_13520/2816081130.py:9: FutureWarning: Downcasting behavior in `replace` is deprecated and will be removed in a future version. To retain the old behavior, explicitly call `result.infer_objects(copy=False)`. To opt-in to the future behavior, set `pd.set_option('future.no_silent_downcasting', True)`
  df_car2['cylinders'] = df_car2['num-of-cylinders'].replace(cylinder_mapper)
```

[27]:

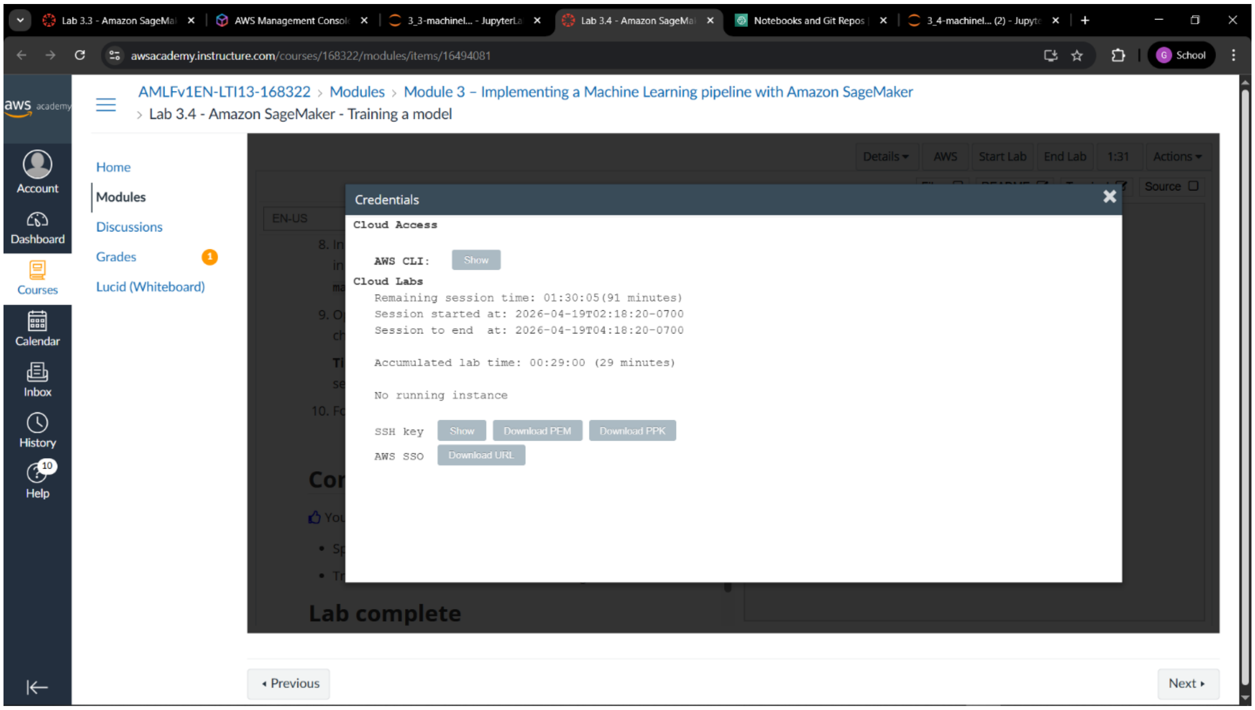
body-ertible	body-style_hardtop	body-style_hatchback	body-style_sedan	body-style_wagon	engine-type_dohc	engine-type_dohcv	engine-type_l	engine-type_ohc	engine-type_ohcf	engine-type_ohcv	engine-type_rotor	fuel-type_gas	engine-location_rear
True	False	False	False	False	True	False	False	False	False	False	False	True	False
True	False	False	False	False	True	False	False	False	False	False	False	True	False
False	False	True	False	False	False	False	False	False	False	True	False	True	False
False	False	False	True	False	False	False	False	True	False	False	False	True	False
False	False	False	True	False	False	False	False	True	False	False	False	True	False

Laboratory Activity 4

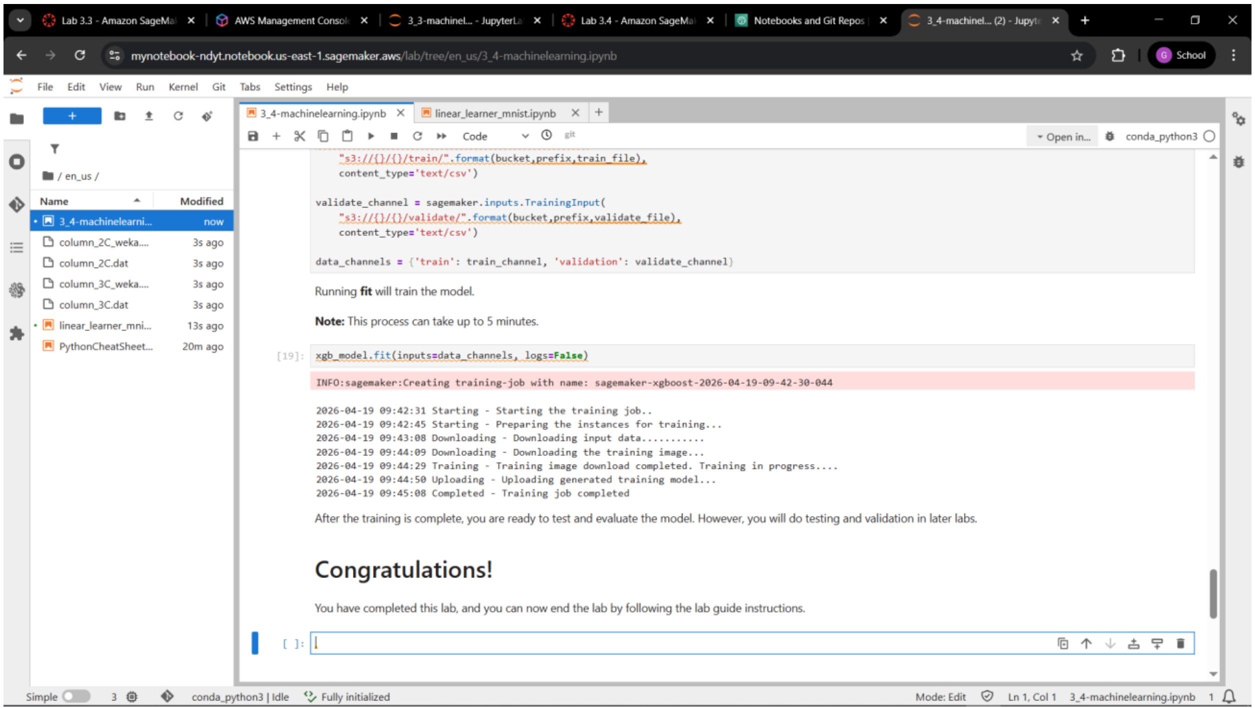
AWS Console account & MyNotebook instance configuration



Canvas Screenshot



Model Trained



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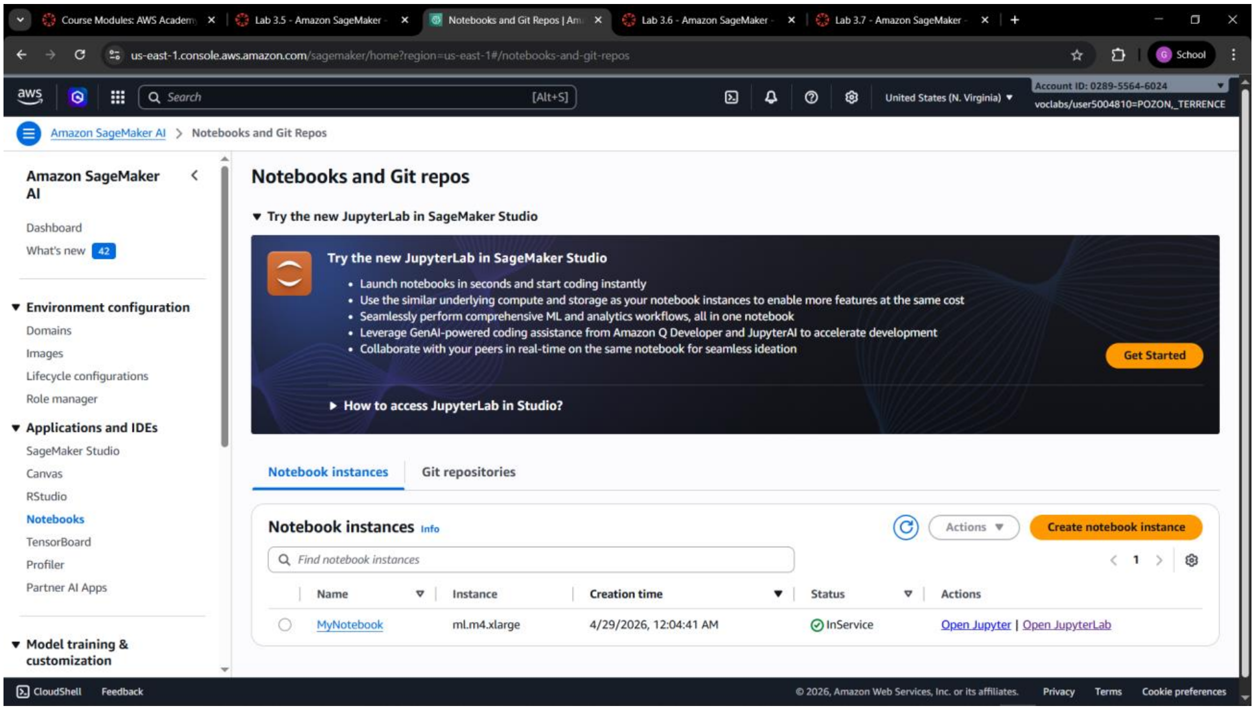
LABORATORY

3.5 - 3.7

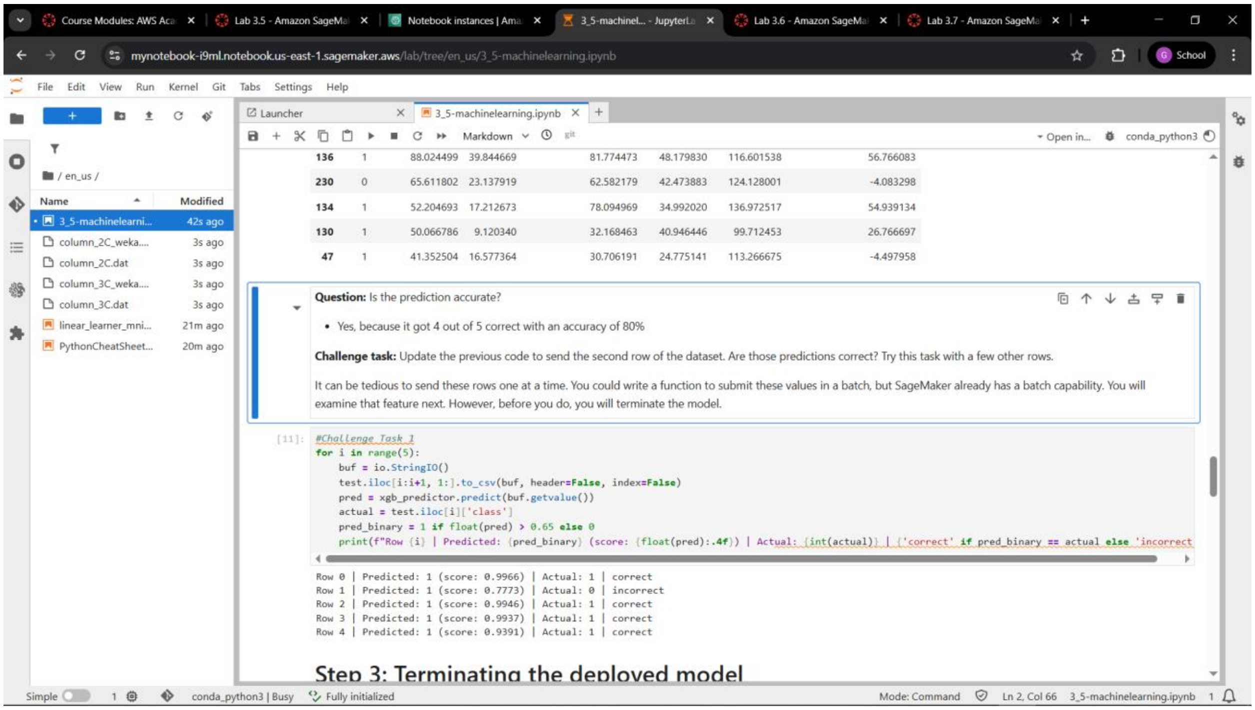
Pozon, Gion Terrence B.
301G
2026-04-29

Activity Lab 3.5

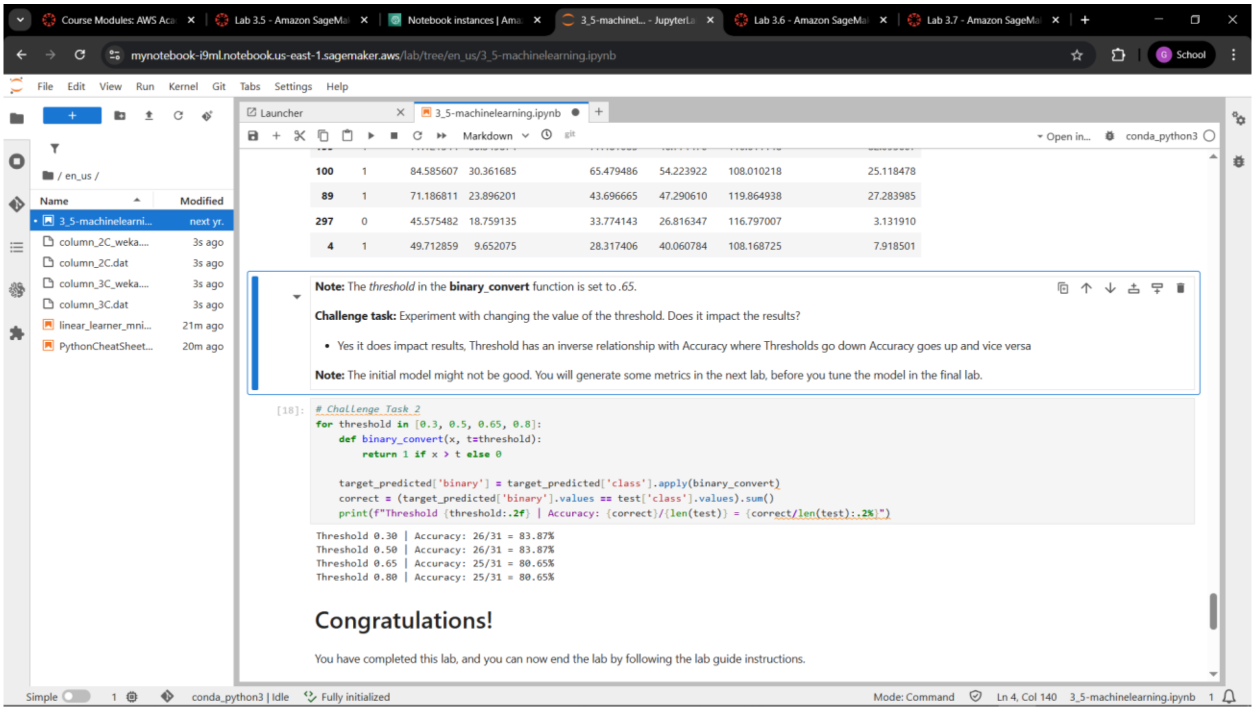
AWS Console account & MyNotebook



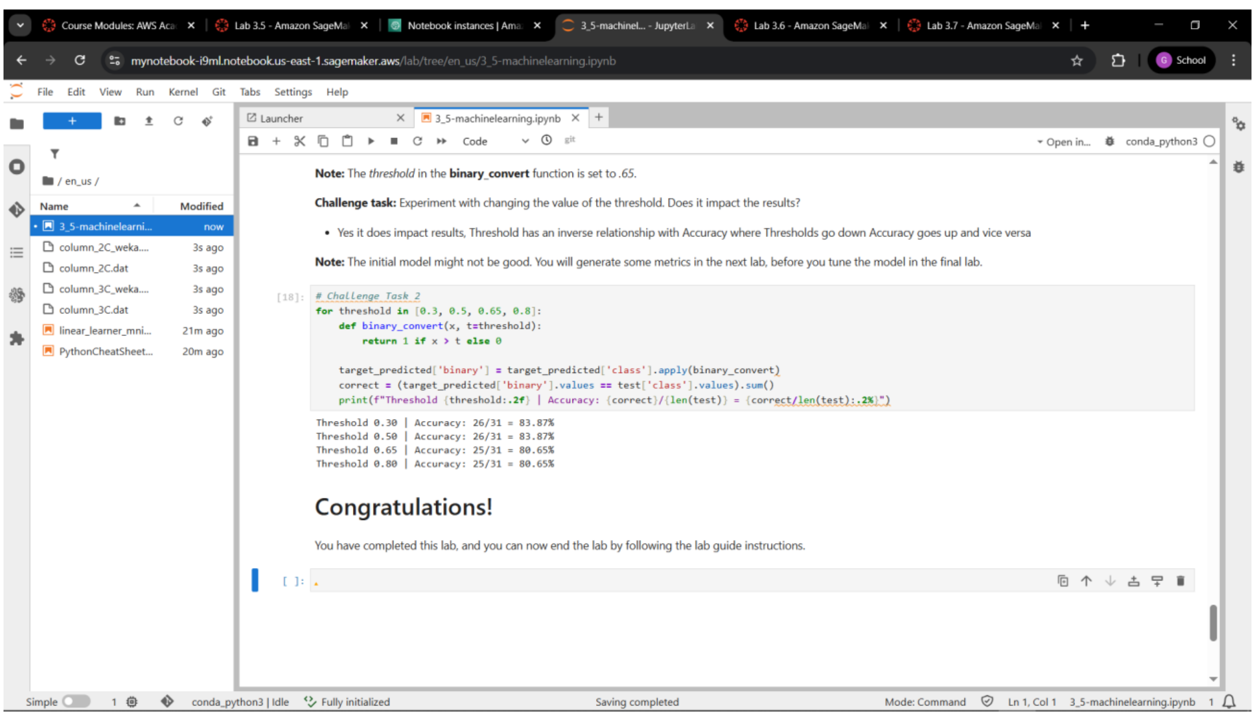
Challenge Task 1



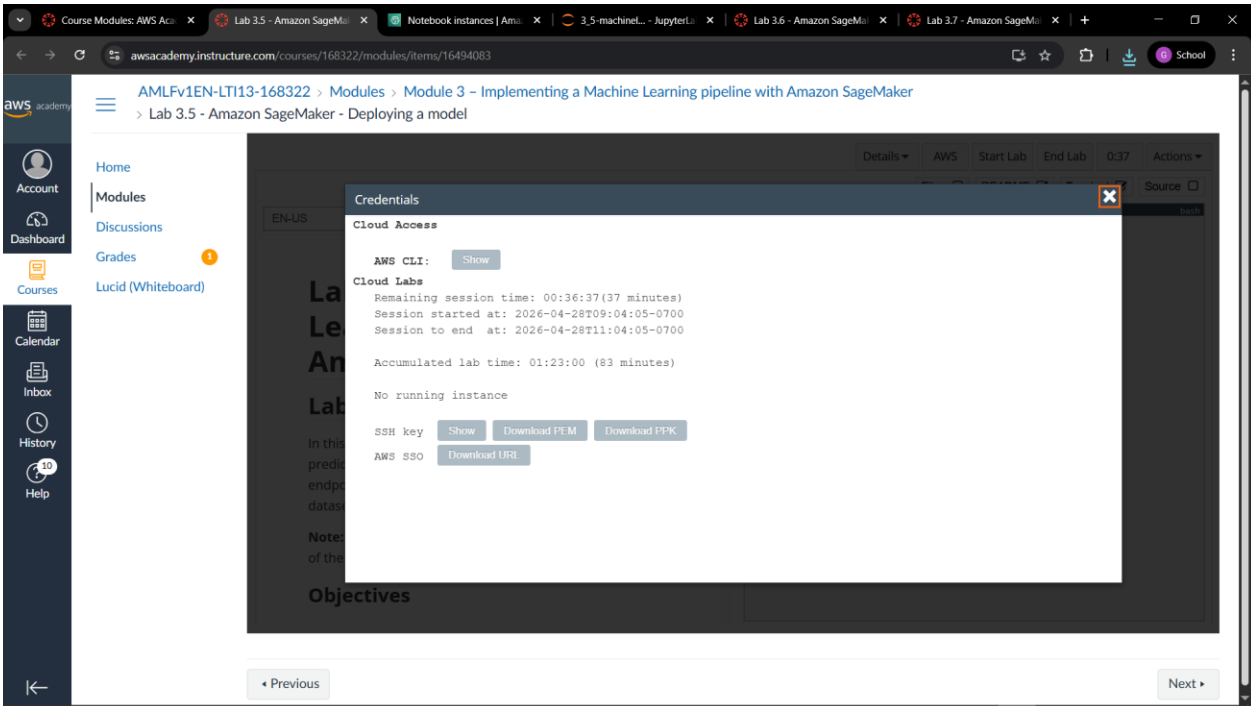
Challenge Task 2



Last Codes

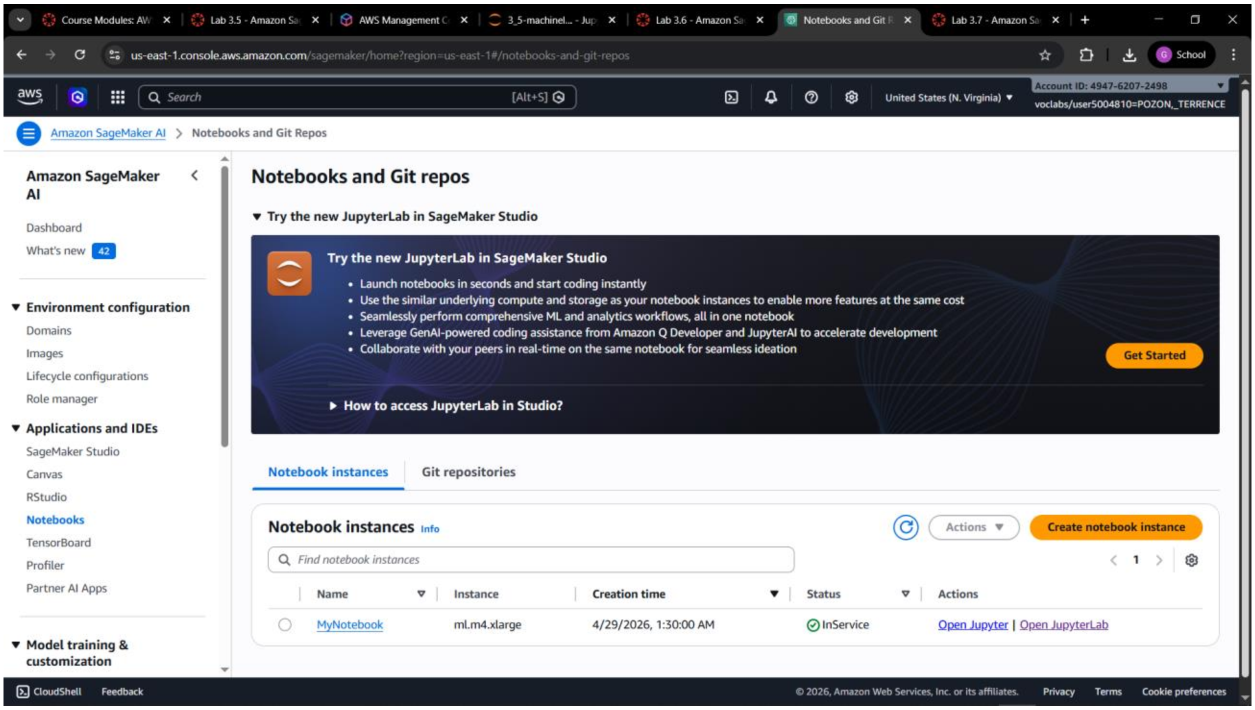


Canvas Details

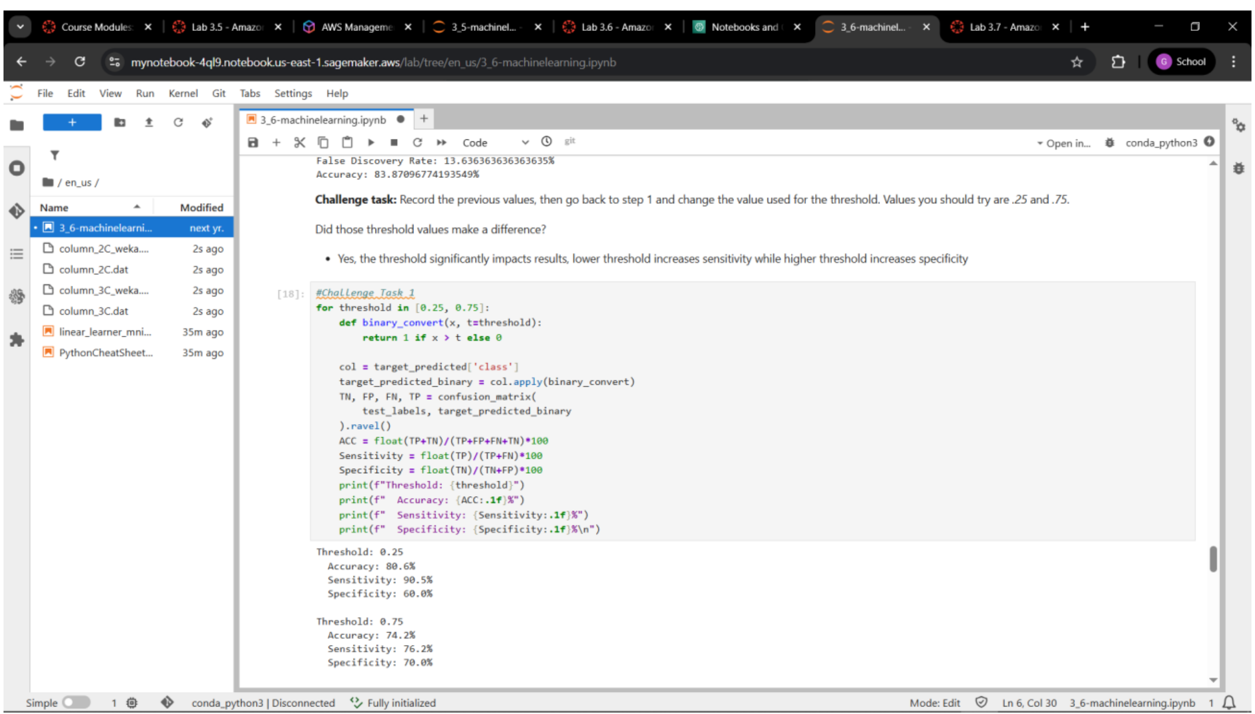


Activity Lab 3.6

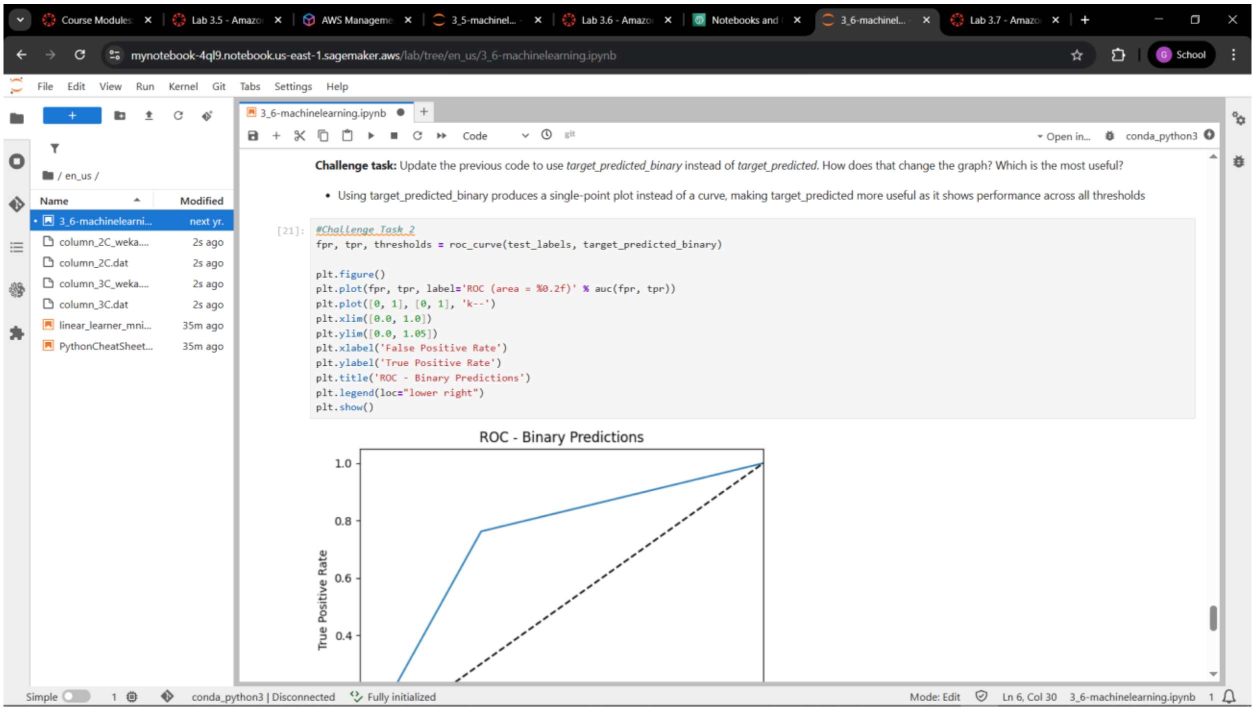
AWS Console account & MyNotebook

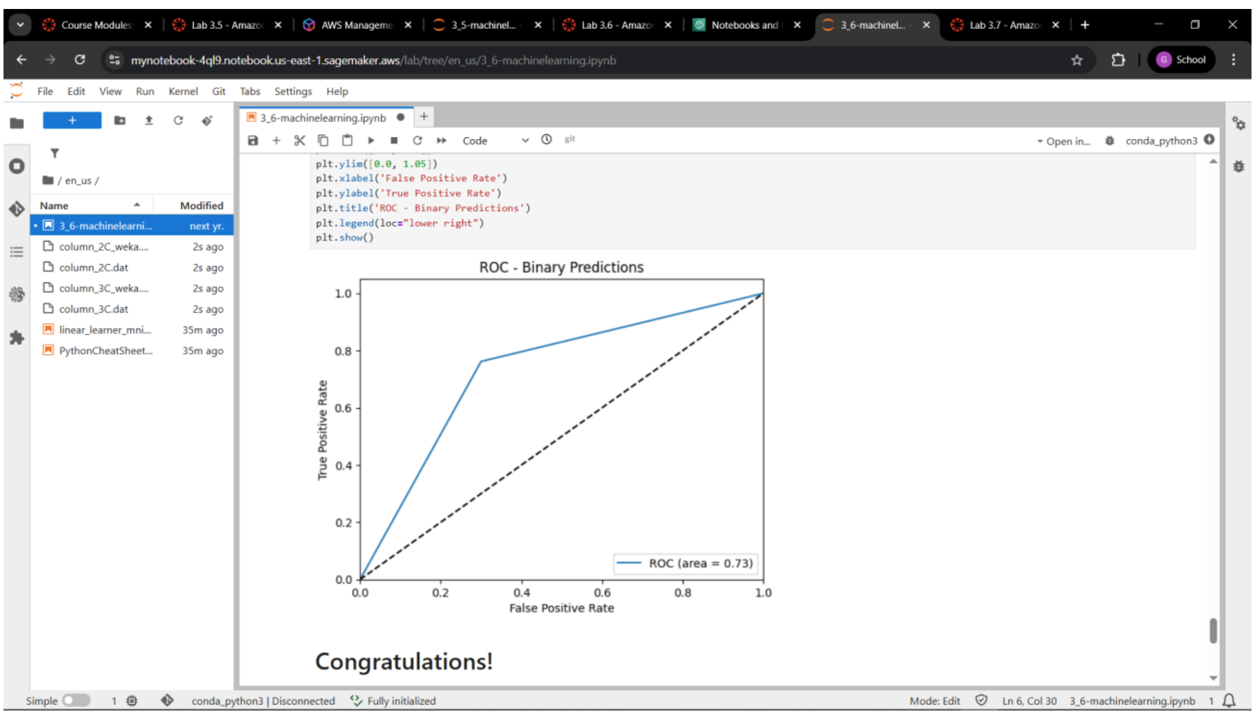


Challenge Task 1

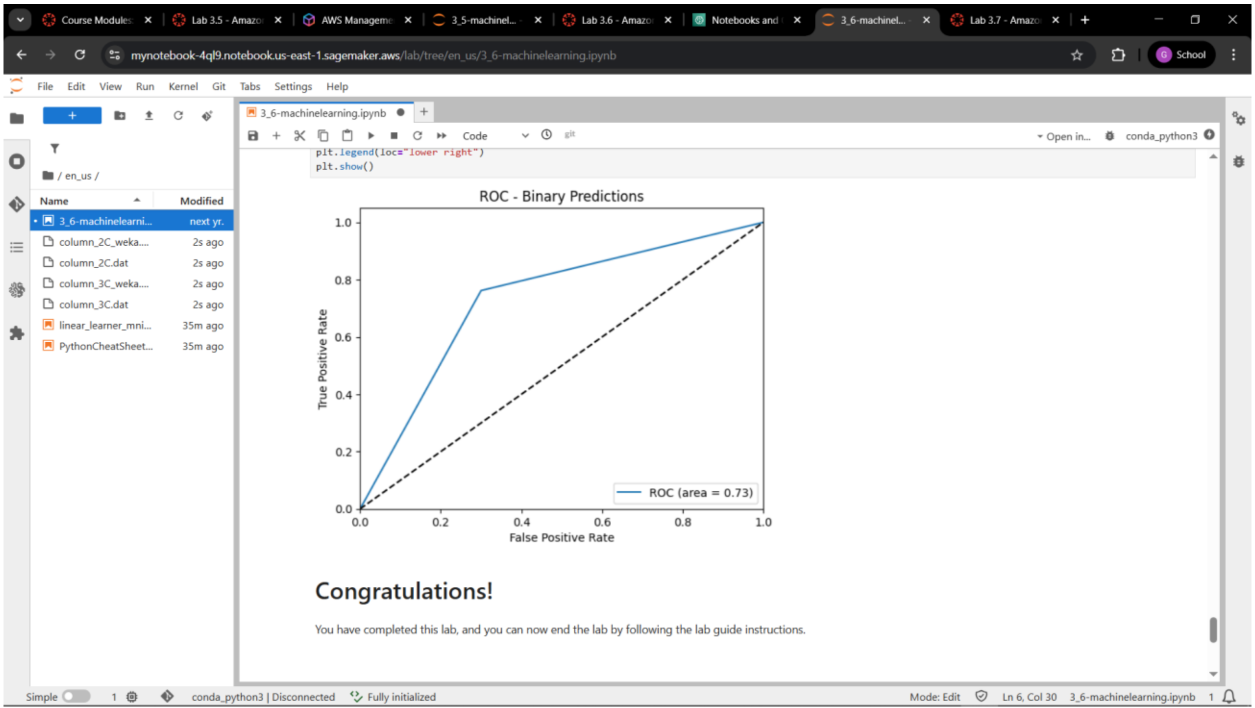


Challenge Task 2

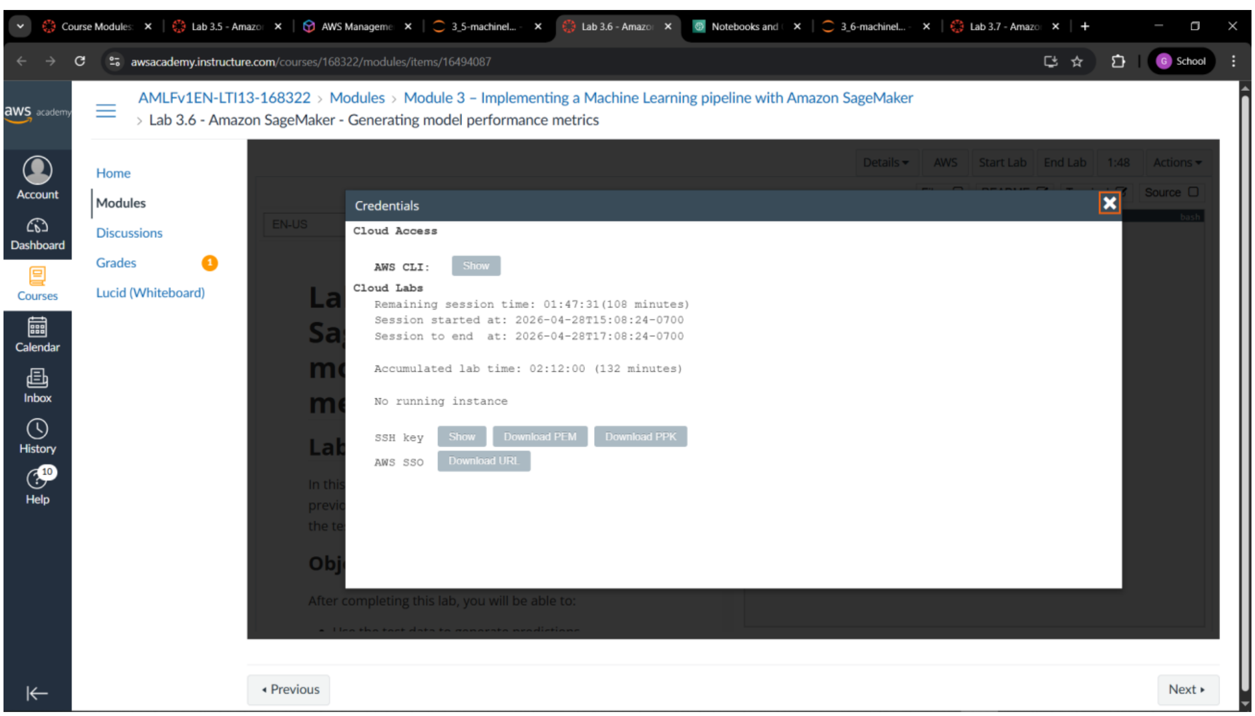




Last Codes

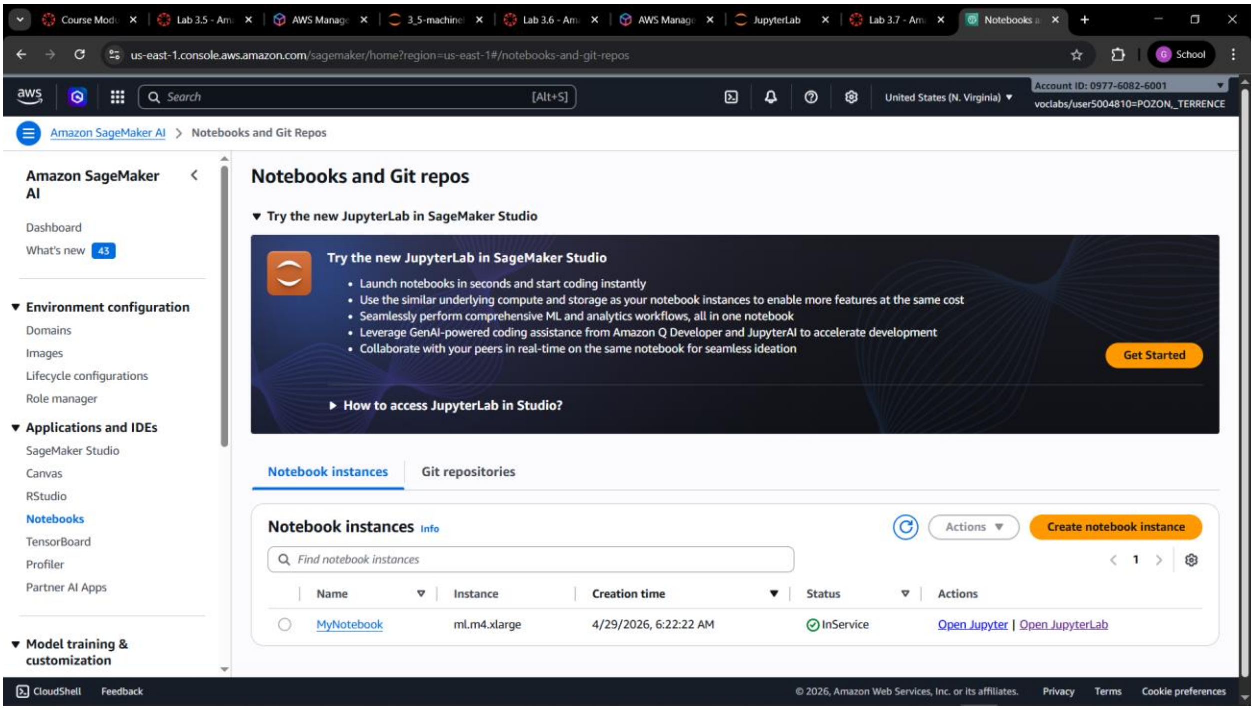


Canvas Details

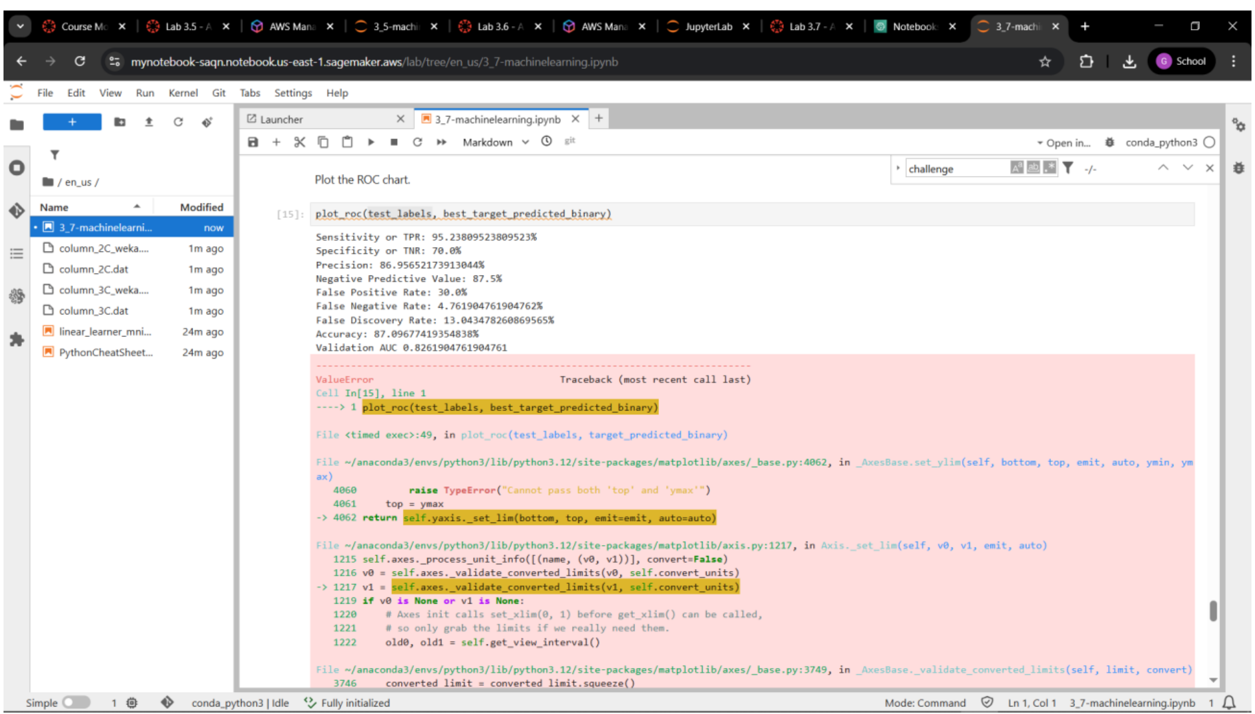


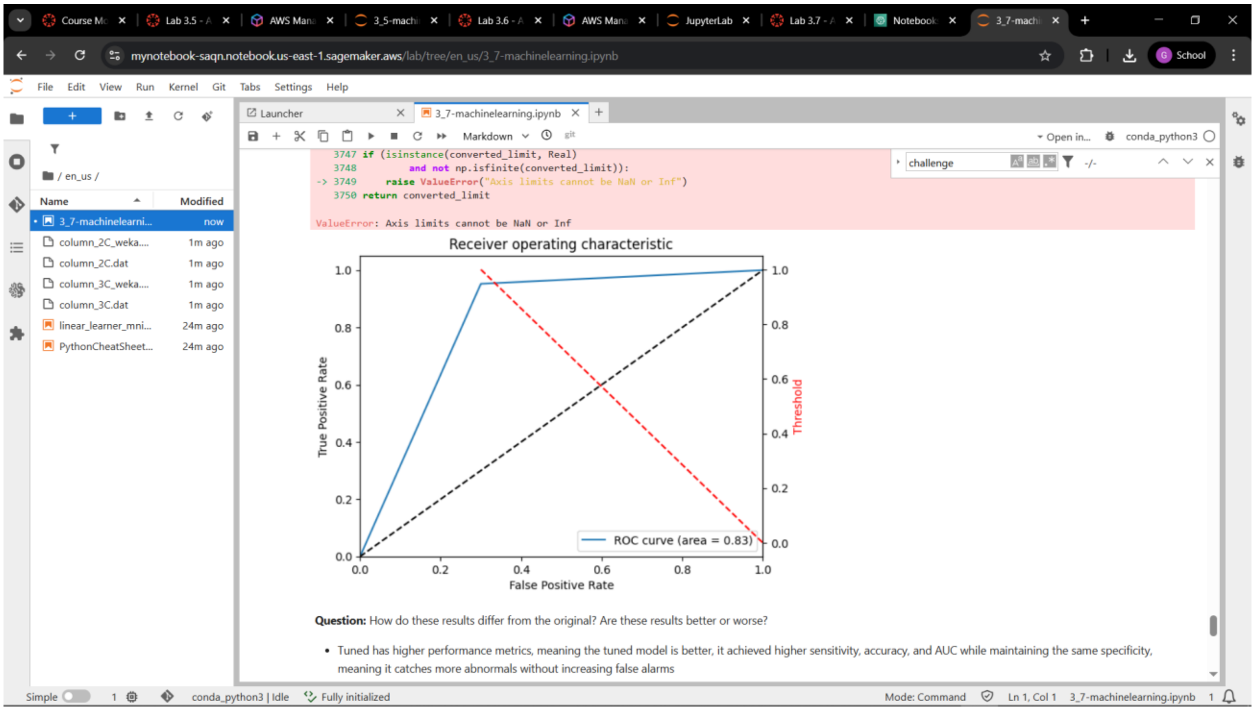
Activity Lab 3.7

AWS Console account & MyNotebook

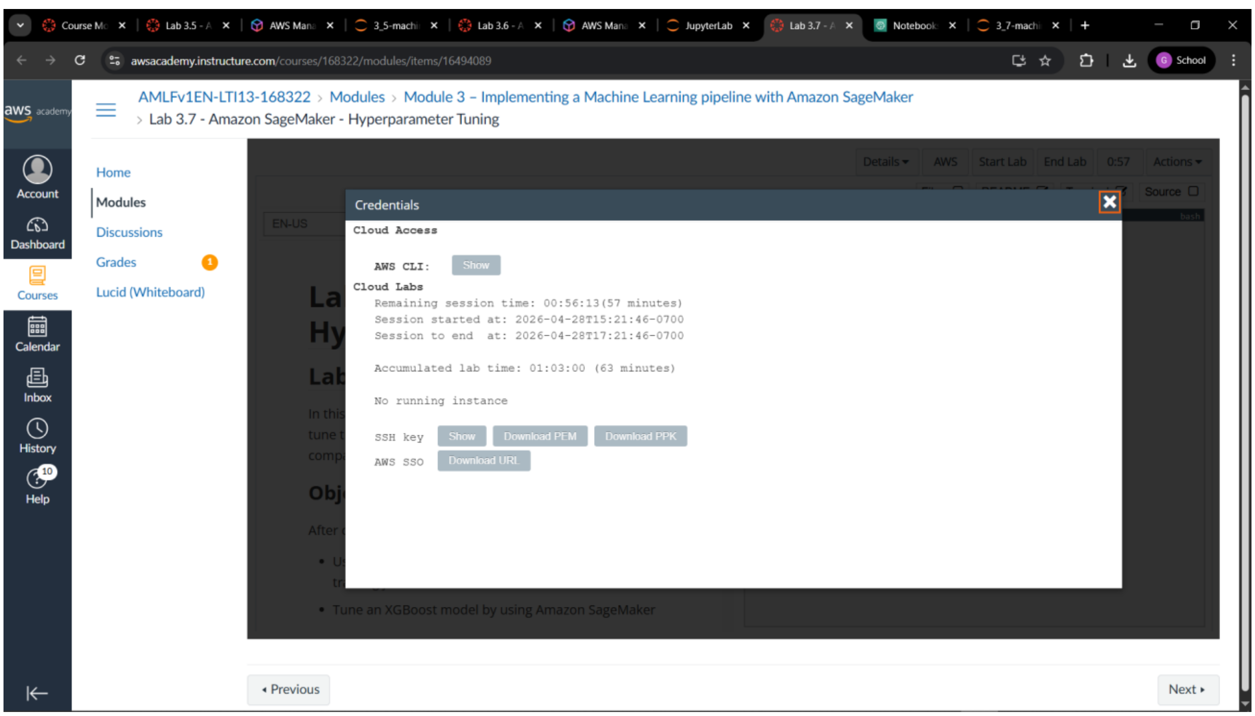


Last Codes





Canvas Details



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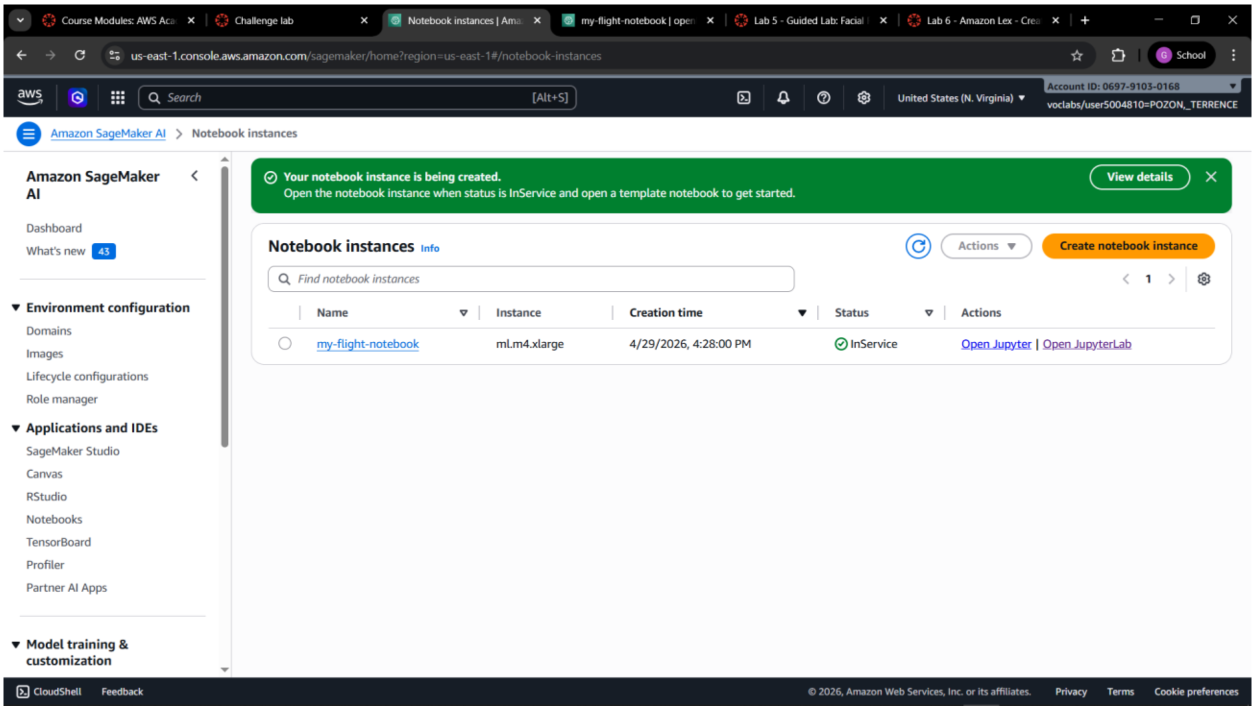
<u>LABORATORY</u>

Challenge

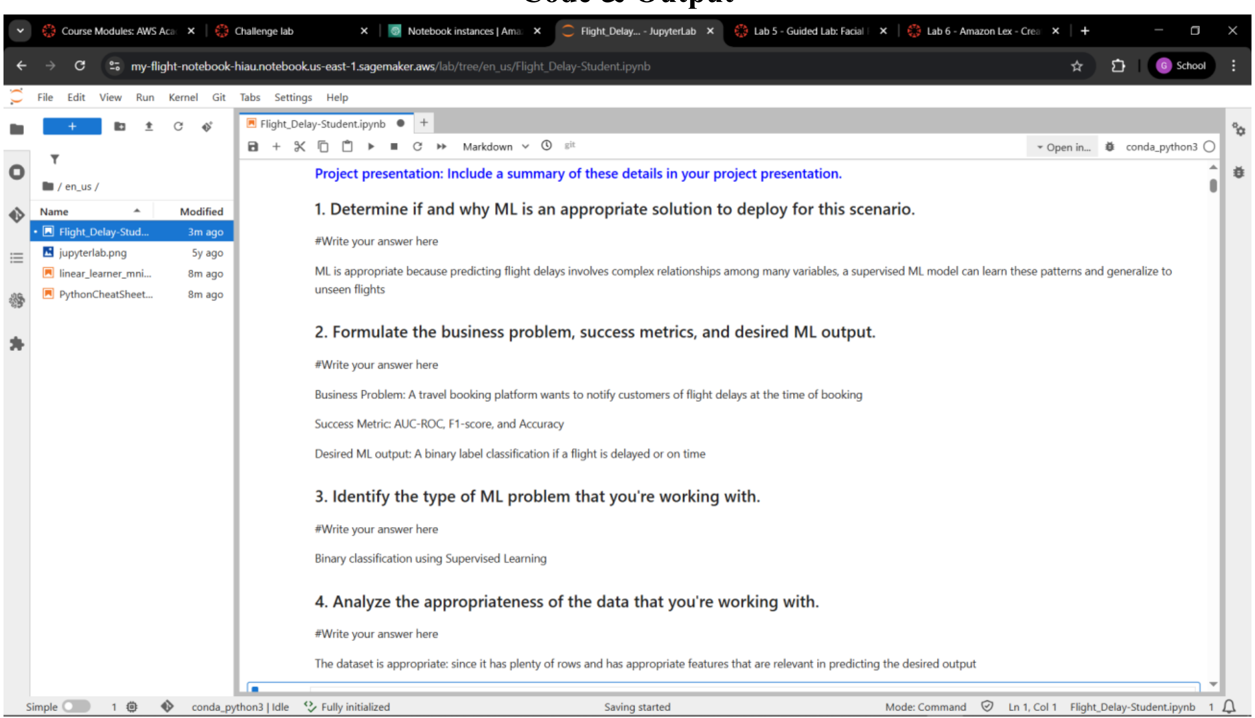
Pozon, Gion Terrence B.
301G
2026-04-29

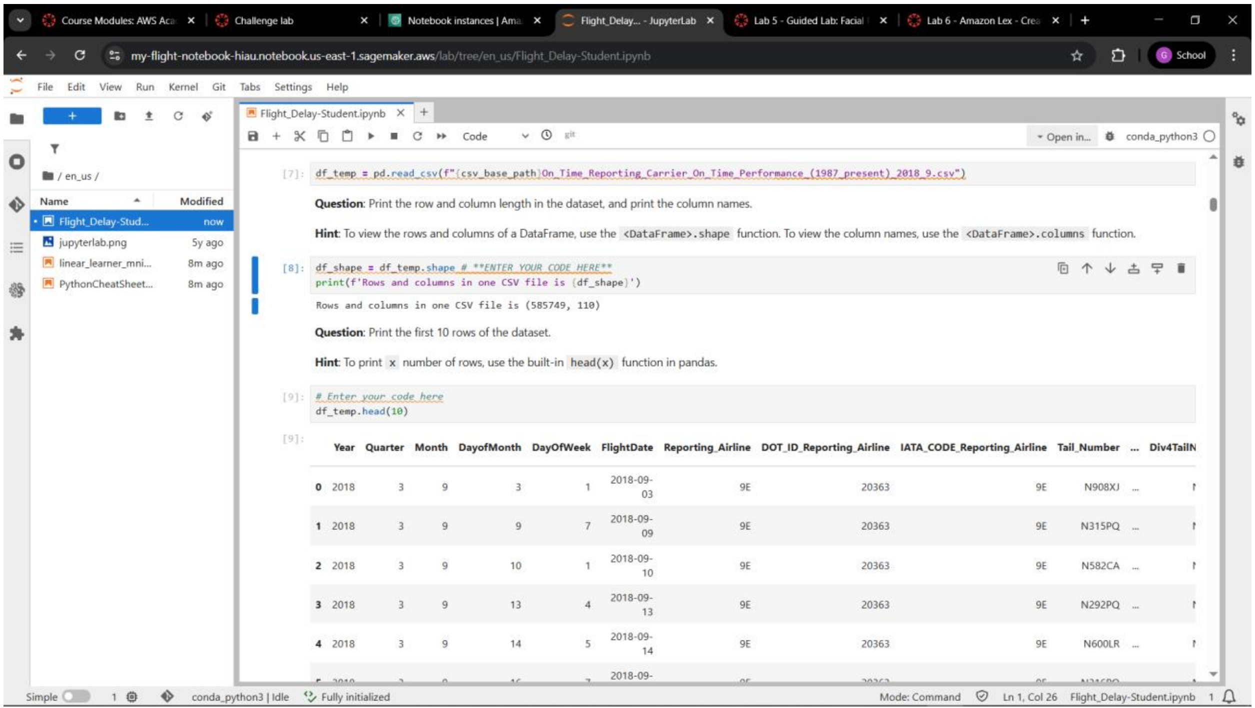
Challenge Lab

AWS Console account & MyNotebook



Code & Output





```
[7]: df_temp = pd.read_csv(f"{csv_base_path}On_Time_Reporting_Carrier_On_Time_Performance_(1987_present)_2018_9.csv")
```

Question: Print the row and column length in the dataset, and print the column names.

Hint: To view the rows and columns of a DataFrame, use the `<DataFrame>.shape` function. To view the column names, use the `<DataFrame>.columns` function.

```
[8]: df_shape = df_temp.shape # **ENTER YOUR CODE HERE**
print(f'Rows and columns in one CSV file is {df_shape}')
```

```
Rows and columns in one CSV file is (585749, 110)
```

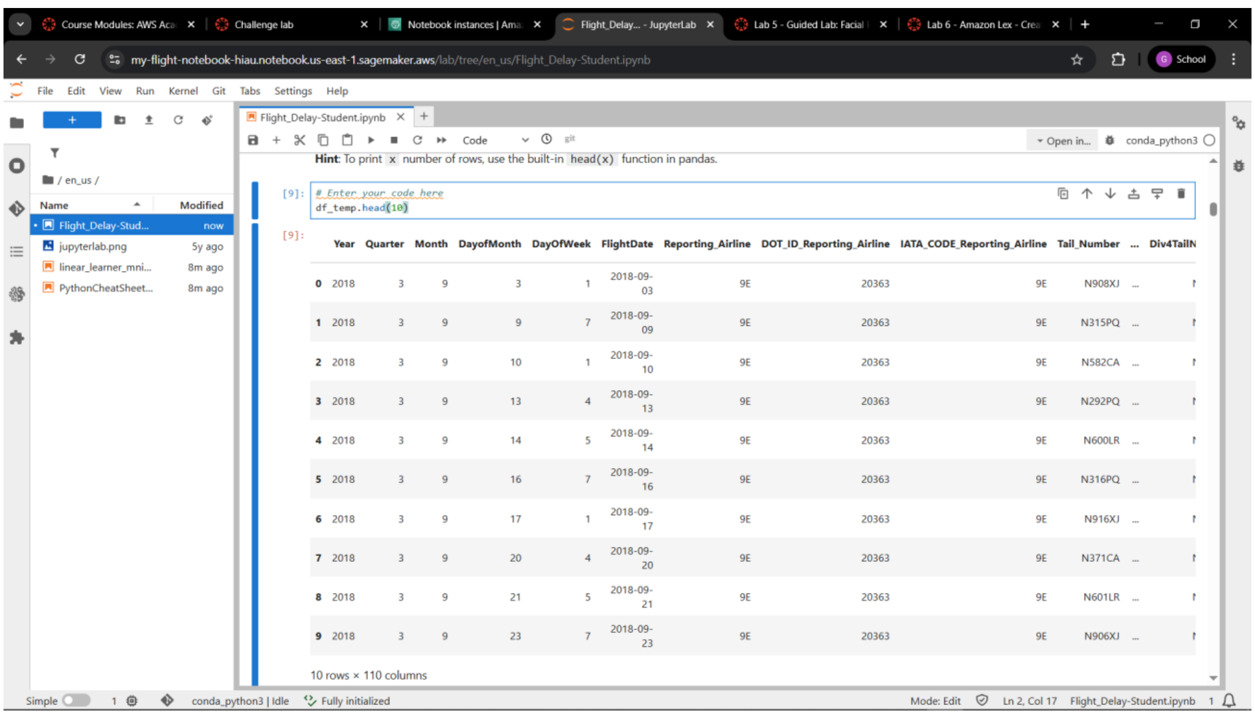
Question: Print the first 10 rows of the dataset.

Hint: To print `x` number of rows, use the built-in `head(x)` function in pandas.

```
[9]: # Enter your code here
df_temp.head(10)
```

[9]:

	Year	Quarter	Month	DayofMonth	DayOfWeek	FlightDate	Reporting_Airline	DOT_ID_Reporting_Airline	IATA_CODE_Reporting_Airline	Tail_Number	...	Div4TailN
0	2018	3	9	3	1	2018-09-03	9E	20363	9E	N908XJ	...	[illegible]
1	2018	3	9	9	7	2018-09-09	9E	20363	9E	N315PQ	...	[illegible]
2	2018	3	9	10	1	2018-09-10	9E	20363	9E	N582CA	...	[illegible]
3	2018	3	9	13	4	2018-09-13	9E	20363	9E	N292PQ	...	[illegible]
4	2018	3	9	14	5	2018-09-14	9E	20363	9E	N600LR	...	[illegible]



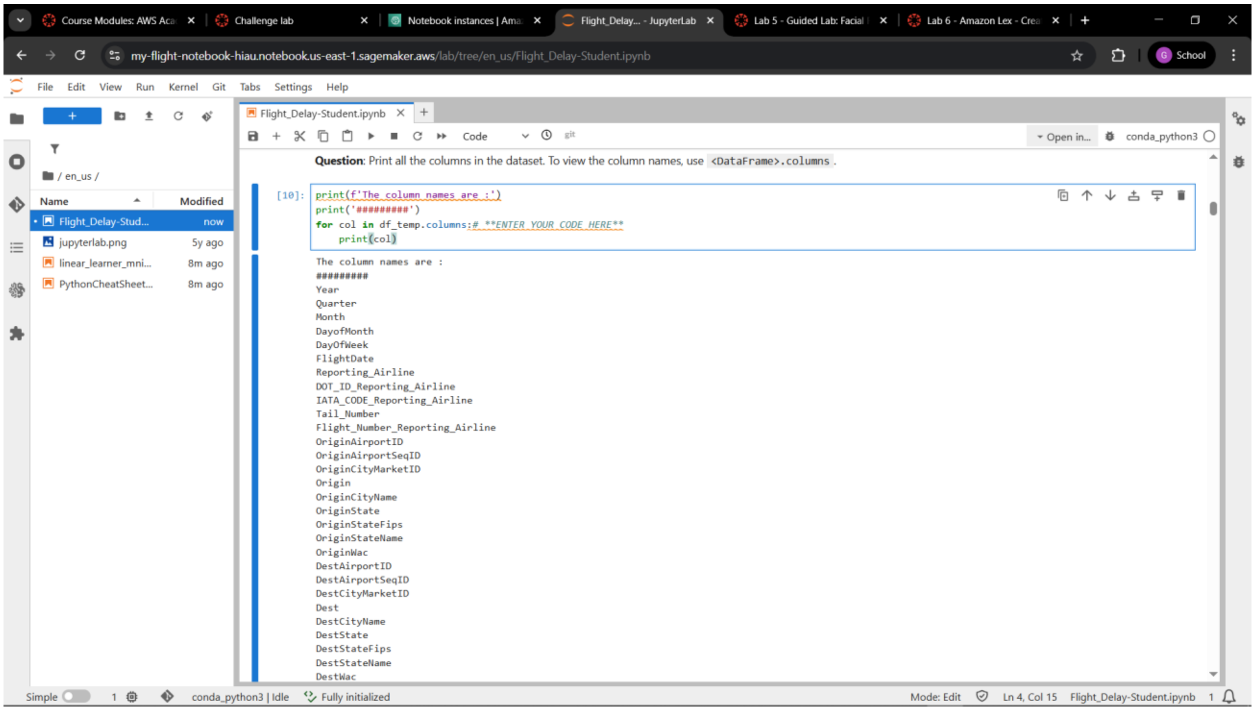
Hint: To print `x` number of rows, use the built-in `head(x)` function in pandas.

```
[9]: # Enter your code here
df_temp.head(10)
```

[9]:

	Year	Quarter	Month	DayofMonth	DayOfWeek	FlightDate	Reporting_Airline	DOT_ID_Reporting_Airline	IATA_CODE_Reporting_Airline	Tail_Number	...	Div4TailN
0	2018	3	9	3	1	2018-09-03	9E	20363	9E	N908XJ	...	[illegible]
1	2018	3	9	9	7	2018-09-09	9E	20363	9E	N315PQ	...	[illegible]
2	2018	3	9	10	1	2018-09-10	9E	20363	9E	N582CA	...	[illegible]
3	2018	3	9	13	4	2018-09-13	9E	20363	9E	N292PQ	...	[illegible]
4	2018	3	9	14	5	2018-09-14	9E	20363	9E	N600LR	...	[illegible]
5	2018	3	9	16	7	2018-09-16	9E	20363	9E	N316PQ	...	[illegible]
6	2018	3	9	17	1	2018-09-17	9E	20363	9E	N916XJ	...	[illegible]
7	2018	3	9	20	4	2018-09-20	9E	20363	9E	N371CA	...	[illegible]
8	2018	3	9	21	5	2018-09-21	9E	20363	9E	N601LR	...	[illegible]
9	2018	3	9	23	7	2018-09-23	9E	20363	9E	N906XJ	...	[illegible]

10 rows × 110 columns


Course Modules: AWS Aca
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Lab 5 - Guided Lab: Facial
Lab 6 - Amazon Lex - Crea
my-flight-notebook-hiau.notebook.us-east-1.sagemaker.aws/lab/tree/en_us/Flight_Delay-Student.ipynb
School
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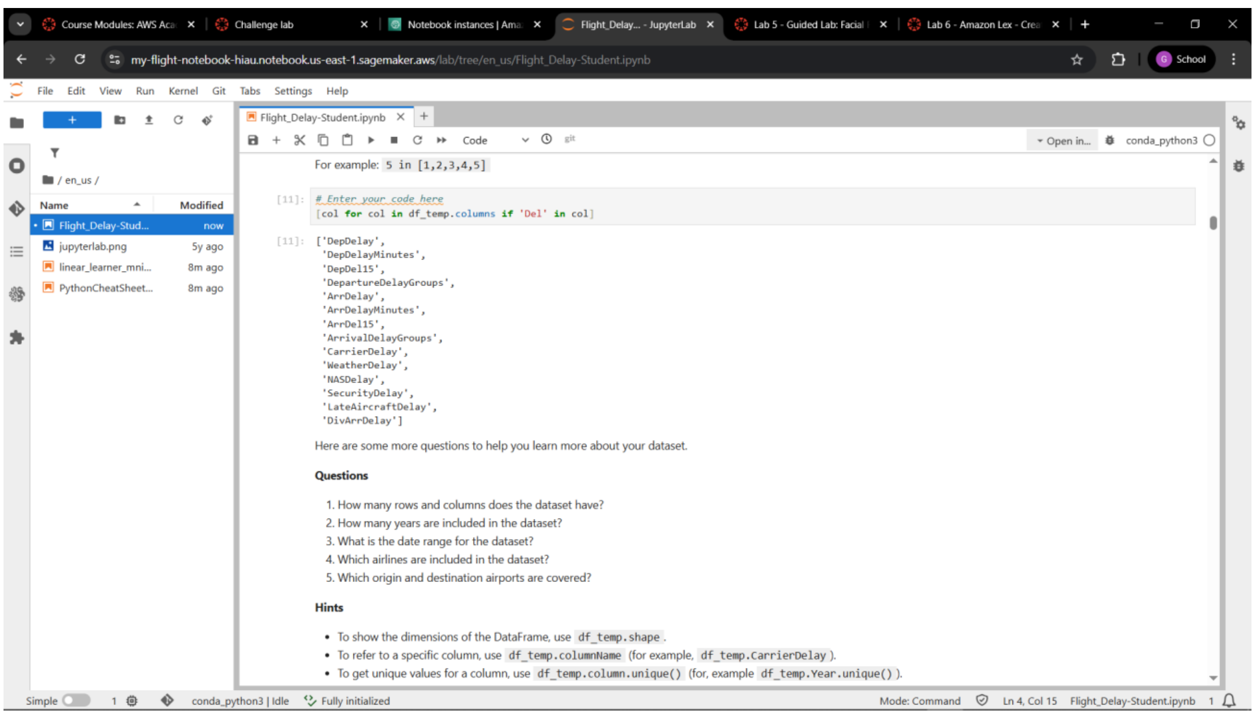
5y ago
linear_learner_mni...
8m ago
PythonCheatSheet...
8m ago
Flight_Delay-Student.ipynb
Code
Open in...
conda_python3
Question: Print all the columns in the dataset. To view the column names, use <DataFrame>.columns .
[10]:
print(f'The column names are :')
print('#########')
for col in df_temp.columns:# **ENTER YOUR CODE HERE**
print(col)
The column names are :
#########
Year
Quarter
Month
DayofMonth
DayOfWeek
FlightDate
Reporting_Airline
DOT_ID_Reporting_Airline
IATA_CODE_Reporting_Airline
Tail_Number
Flight_Number_Reporting_Airline
OriginAirportID
OriginAirportSeqID
OriginCityMarketID
Origin
OriginCityName
OriginState
OriginStateFips
OriginStateName
OriginWac
DestAirportID
DestAirportSeqID
DestCityMarketID
Dest
DestCityName
DestState
DestStateFips
DestStateName
DestWac
Simple
conda_python3 | Idle
Fully initialized
Mode: Edit
Ln 4, Col 15
Flight_Delay-Student.ipynb


Course Modules: AWS Aca
Challenge lab
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Flight_Delay... - JupyterLab
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my-flight-notebook-hiau.notebook.us-east-1.sagemaker.aws/lab/tree/en_us/Flight_Delay-Student.ipynb
School
File
Edit
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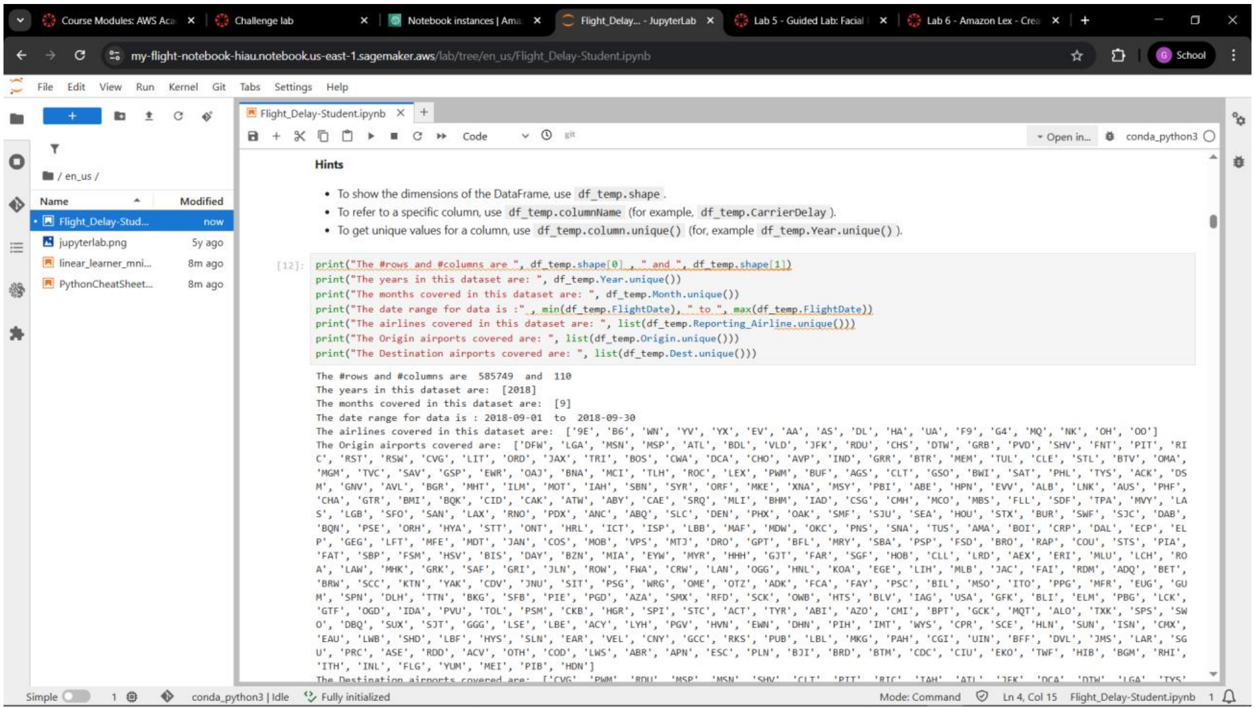
Settings
Help
/ en_us /
Name
Modified
Flight_Delay-Stud...
now
jupyterlab.png
5y ago
linear_learner_mni...
8m ago
PythonCheatSheet...
8m ago
Flight_Delay-Student.ipynb
Code
Open in...
conda_python3
For example: 5 in [1,2,3,4,5]
[11]:
Enter your code here
[col for col in df_temp.columns if 'Del' in col]
[11]:
['DepDelay',
'DepDelayMinutes',
'DepDel15',
'DepartureDelayGroups',
'ArrDelay',
'ArrDelayMinutes',
'ArrDel15',
'ArrivalDelayGroups',
'CarrierDelay',
'WeatherDelay',
'NASDelay',
'SecurityDelay',
'LateAircraftDelay',
'DivArrDelay']
Here are some more questions to help you learn more about your dataset.
Questions
1. How many rows and columns does the dataset have?
2. How many years are included in the dataset?
3. What is the date range for the dataset?
4. Which airlines are included in the dataset?
5. Which origin and destination airports are covered?
Hints
To show the dimensions of the DataFrame, use df_temp.shape .
To refer to a specific column, use df_temp.columnName (for example, df_temp.CarrierDelay).
To get unique values for a column, use df_temp.column.unique() (for, example df_temp.Year.unique()).
Simple
conda_python3 | Idle
Fully initialized
Mode: Command
Ln 4, Col 15
Flight_Delay-Student.ipynb

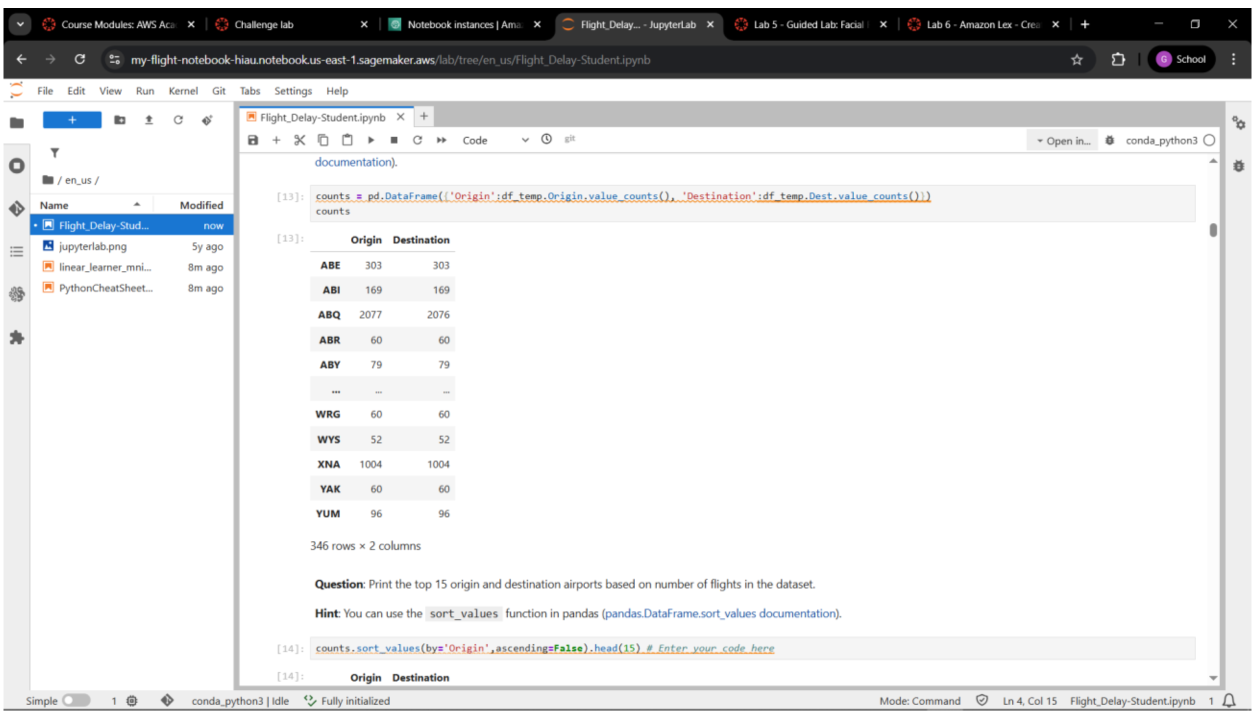


Hints

- To show the dimensions of the DataFrame, use `df_temp.shape` .
- To refer to a specific column, use `df_temp.columnName` (for example, `df_temp.CarrierDelay`).
- To get unique values for a column, use `df_temp.column.unique()` (for, example `df_temp.Year.unique()`).

```
[12]: print("The #rows and #columns are ", df_temp.shape[0] , " and ", df_temp.shape[1])
print("The years in this dataset are: ", df_temp.Year.unique())
print("The months covered in this dataset are: ", df_temp.Month.unique())
print("The date range for data is :" , min(df_temp.FlightDate), " to ", max(df_temp.FlightDate))
print("The airlines covered in this dataset are: ", list(df_temp.Reporting_Airline.unique()))
print("The Origin airports covered are: ", list(df_temp.Origin.unique()))
print("The Destination airports covered are: ", list(df_temp.Dest.unique()))
```

```
The #rows and #columns are  585749  and  110
The years in this dataset are:  [2018]
The months covered in this dataset are:  [9]
The date range for data is : 2018-09-01  to  2018-09-30
The airlines covered in this dataset are:  ['9E', 'B6', 'WN', 'YV', 'YX', 'EV', 'AA', 'AS', 'DL', 'HA', 'UA', 'F9', 'G4', 'MQ', 'NK', 'OH', 'OO']
The Origin airports covered are:  ['DFW', 'LGA', 'MSN', 'MSP', 'ATL', 'BDL', 'VLD', 'JFK', 'RDU', 'CHS', 'DTW', 'GRB', 'PVD', 'SHV', 'FNT', 'PIT', 'RI
C', 'RST', 'RSW', 'CVG', 'LIT', 'ORD', 'JAX', 'TRI', 'BOS', 'CWA', 'DCA', 'CHO', 'AVP', 'IND', 'GRR', 'BTR', 'MEM', 'TUL', 'CLE', 'STL', 'BTV', 'OMA',
'MGM', 'TVC', 'SAV', 'GSP', 'EWR', 'OAJ', 'BNA', 'MCI', 'TLH', 'ROC', 'LEX', 'PWM', 'BUF', 'AGS', 'CLT', 'GSO', 'BWI', 'SAT', 'PHL', 'TYS', 'ACK', 'DS
M', 'GNV', 'AVL', 'BGR', 'MHT', 'ILM', 'MOT', 'IAH', 'SBN', 'SYR', 'ORF', 'MKE', 'XNA', 'MSY', 'PBI', 'ABE', 'HPN', 'EVV', 'ALB', 'LNK', 'AUS', 'PHF',
'CHA', 'GTR', 'BMI', 'BQK', 'CID', 'CAK', 'ATW', 'ABY', 'CAE', 'SRQ', 'MLI', 'BHM', 'IAD', 'CSG', 'CMH', 'MCO', 'MBS', 'FLL', 'SDF', 'TPA', 'MVY', 'LA
S', 'LGB', 'SFO', 'SAN', 'LAX', 'RNO', 'PDX', 'ANC', 'ABQ', 'SLC', 'DEN', 'PHX', 'OAK', 'SMF', 'SJU', 'SEA', 'HOU', 'STX', 'BUR', 'SWF', 'SJC', 'DAB',
'BQN', 'PSE', 'ORH', 'HYA', 'STT', 'ONT', 'HRL', 'ICT', 'ISP', 'LBB', 'MAF', 'MDW', 'OKC', 'PNS', 'SNA', 'TUS', 'AMA', 'BOI', 'CRP', 'DAL', 'ECP', 'EL
P', 'GEG', 'LFT', 'MFE', 'MDT', 'JAN', 'COS', 'MOB', 'VPS', 'MTJ', 'DRO', 'GPT', 'BFL', 'MRY', 'SBA', 'PSP', 'FSD', 'BRO', 'RAP', 'COU', 'STS', 'PIA',
'FAT', 'SBP', 'FSM', 'HSV', 'BIS', 'DAY', 'BZN', 'MIA', 'EYW', 'MYR', 'HHH', 'GJT', 'FAR', 'SGF', 'HOB', 'CLL', 'LRD', 'AEX', 'ERI', 'MLU', 'LCH', 'RO
A', 'LAW', 'MHK', 'GRK', 'SAF', 'GRI', 'JLN', 'ROW', 'FWA', 'CRW', 'LAN', 'OGG', 'HNL', 'KOA', 'EGE', 'LIH', 'MLB', 'JAC', 'FAI', 'RDM', 'ADQ', 'BET',
'BRW', 'SCC', 'KTN', 'YAK', 'CDV', 'JNU', 'SIT', 'PSG', 'WRG', 'OME', 'OTZ', 'ADK', 'FCA', 'FAY', 'PSC', 'BIL', 'MSO', 'ITO', 'PPG', 'MFR', 'EUG', 'GU
M', 'SPN', 'DLH', 'TTN', 'BKG', 'SFB', 'PIE', 'PGD', 'AZA', 'SMX', 'RFD', 'SCK', 'OWB', 'HTS', 'BLV', 'IAG', 'USA', 'GFK', 'BLI', 'ELM', 'PBG', 'LCK',
'GTF', 'OGD', 'IDA', 'PVU', 'TOL', 'PSM', 'CKB', 'HGR', 'SPI', 'STC', 'ACT', 'TYR', 'ABI', 'AZO', 'CMI', 'BPT', 'GCK', 'MQT', 'ALO', 'TXK', 'SPS', 'SW
O', 'DBQ', 'SUX', 'SJT', 'GGG', 'LSE', 'LBE', 'ACY', 'LYH', 'PGV', 'HVN', 'EWN', 'DHN', 'PIH', 'IMT', 'WYS', 'CPR', 'SCE', 'HLN', 'SUN', 'ISN', 'CMX',
'EAU', 'LWB', 'SHD', 'LBF', 'HYS', 'SLN', 'EAR', 'VEL', 'CNY', 'GCC', 'RKS', 'PUB', 'LBL', 'MKG', 'PAH', 'CGI', 'UIN', 'BFF', 'DVL', 'JMS', 'LAR', 'SG
U', 'PRC', 'ASE', 'RDD', 'ACV', 'OTH', 'COD', 'LWS', 'ABR', 'APN', 'ESC', 'PLN', 'BJI', 'BRD', 'BTM', 'CDC', 'CIU', 'EKO', 'TWF', 'HIB', 'BGM', 'RHI',
'ITH', 'INL', 'FLG', 'YUM', 'MEI', 'PIB', 'HDN']
```



documentation).

```
[13]: counts = pd.DataFrame({'Origin':df_temp.Origin.value_counts(), 'Destination':df_temp.Dest.value_counts()})
counts
```

[13]:

	Origin	Destination
ABE	303	303
ABI	169	169
ABQ	2077	2076
ABR	60	60
ABY	79	79
...	...	...
WRG	60	60
WYS	52	52
XNA	1004	1004
YAK	60	60
YUM	96	96

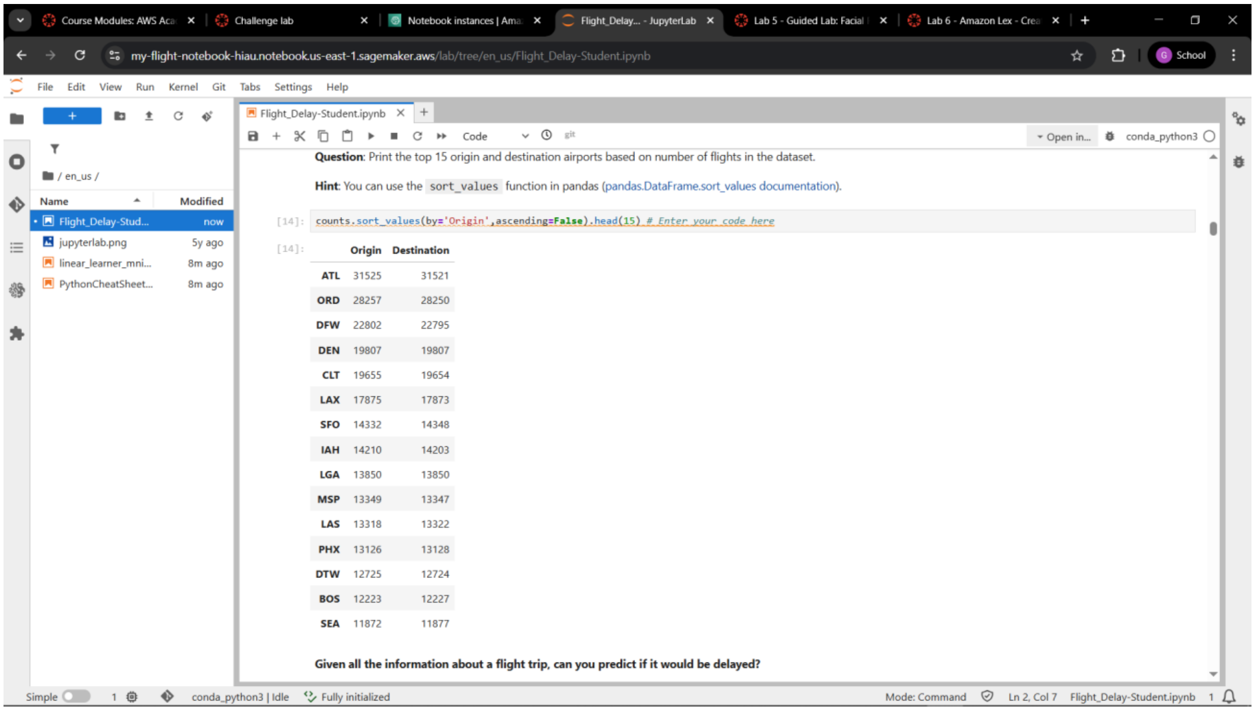
346 rows × 2 columns

Question: Print the top 15 origin and destination airports based on number of flights in the dataset.

Hint: You can use the `sort_values` function in pandas (pandas.DataFrame.sort_values documentation).

```
[14]: counts.sort_values(by='Origin',ascending=False).head(15) # Enter your code here
```

[14]: Origin Destination



Question: Print the top 15 origin and destination airports based on number of flights in the dataset.

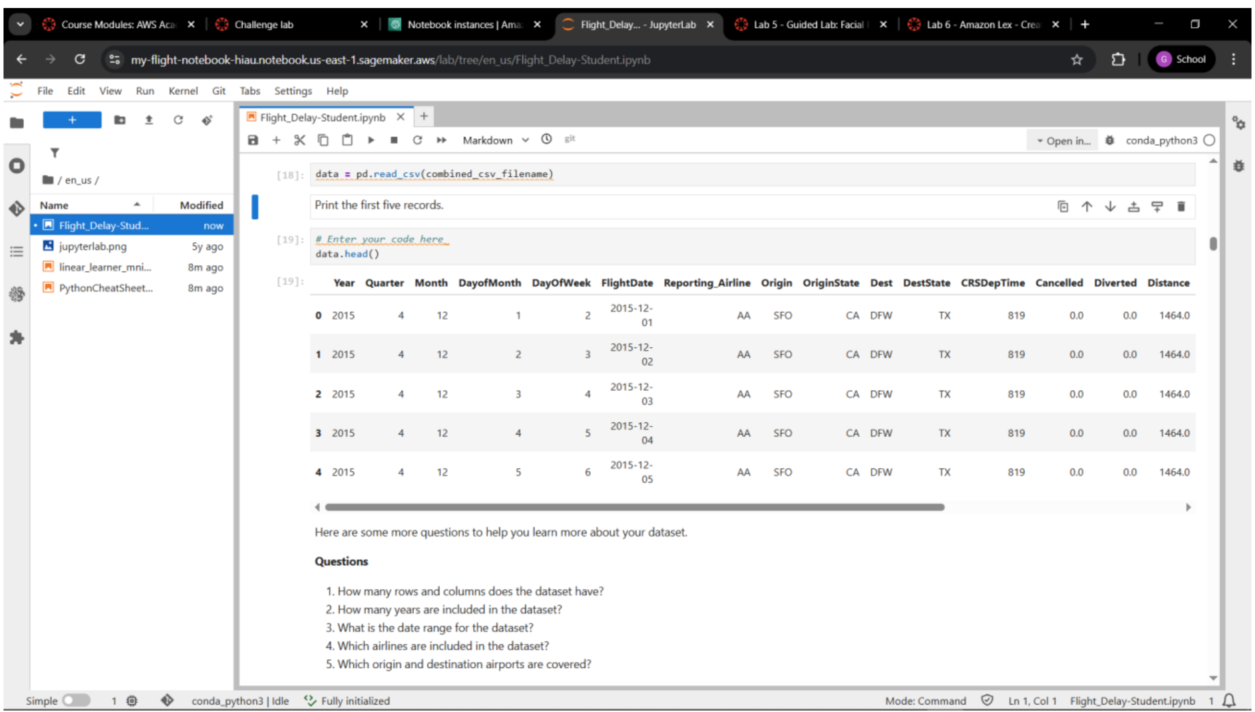
Hint: You can use the `sort_values` function in pandas (pandas.DataFrame.sort_values documentation).

```
[14]: counts.sort_values(by='Origin',ascending=False).head(15) # Enter your code here
```

[14]:

	Origin	Destination
ATL	31525	31521
ORD	28257	28250
DFW	22802	22795
DEN	19807	19807
CLT	19655	19654
LAX	17875	17873
SFO	14332	14348
IAH	14210	14203
LGA	13850	13850
MSP	13349	13347
LAS	13318	13322
PHX	13126	13128
DTW	12725	12724
BOS	12223	12227
SEA	11872	11877

Given all the information about a flight trip, can you predict if it would be delayed?



```
[18]: data = pd.read_csv(combined_csv_filename)
```

Print the first five records.

```
[19]: # Enter your code here
data.head()
```

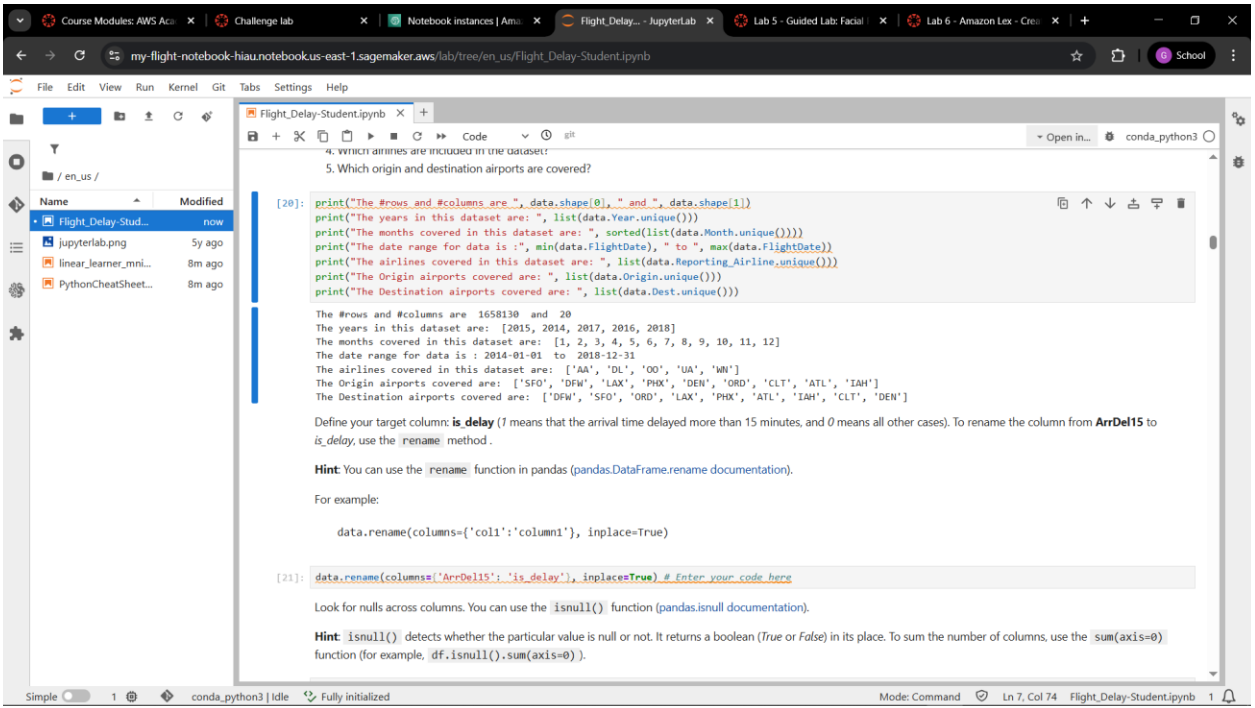
[19]:

	Year	Quarter	Month	DayofMonth	DayOfWeek	FlightDate	Reporting_Airline	Origin	OriginState	Dest	DestState	CRSDepTime	Cancelled	Diverted	Distance
0	2015	4	12	1	2	2015-12-01	AA	SFO	CA	DFW	TX	819	0.0	0.0	1464.0
1	2015	4	12	2	3	2015-12-02	AA	SFO	CA	DFW	TX	819	0.0	0.0	1464.0
2	2015	4	12	3	4	2015-12-03	AA	SFO	CA	DFW	TX	819	0.0	0.0	1464.0
3	2015	4	12	4	5	2015-12-04	AA	SFO	CA	DFW	TX	819	0.0	0.0	1464.0
4	2015	4	12	5	6	2015-12-05	AA	SFO	CA	DFW	TX	819	0.0	0.0	1464.0

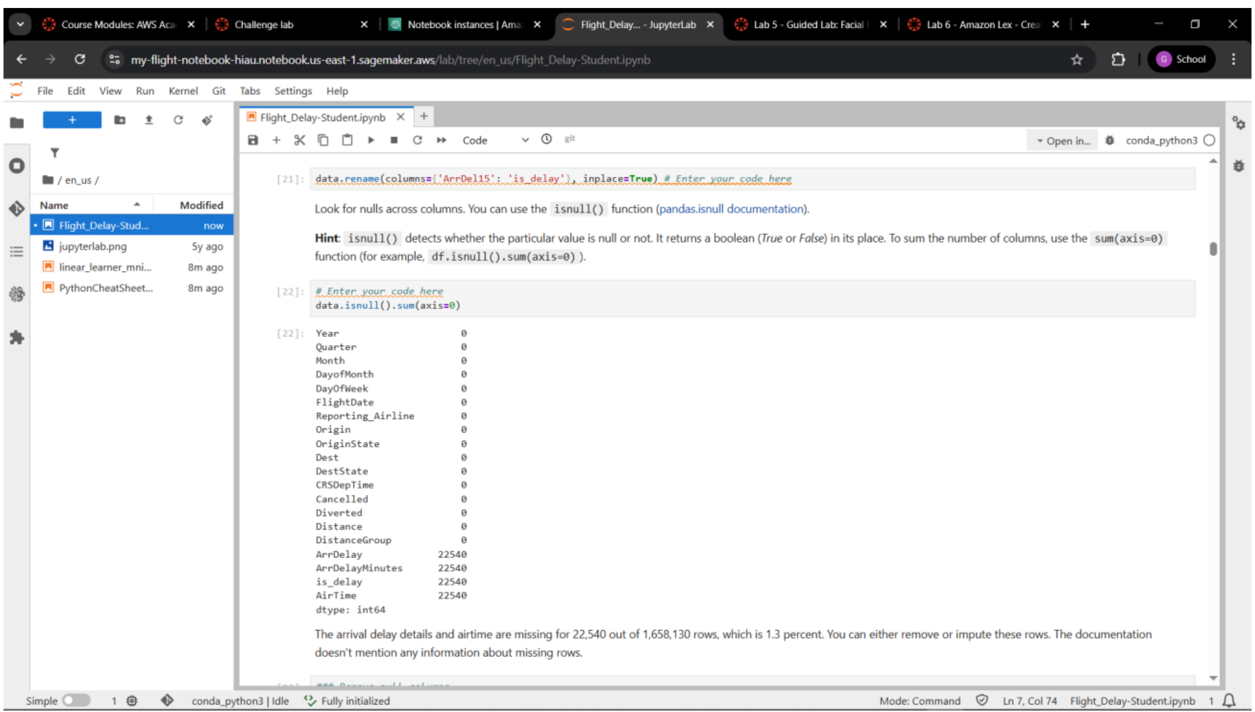
Here are some more questions to help you learn more about your dataset.

Questions

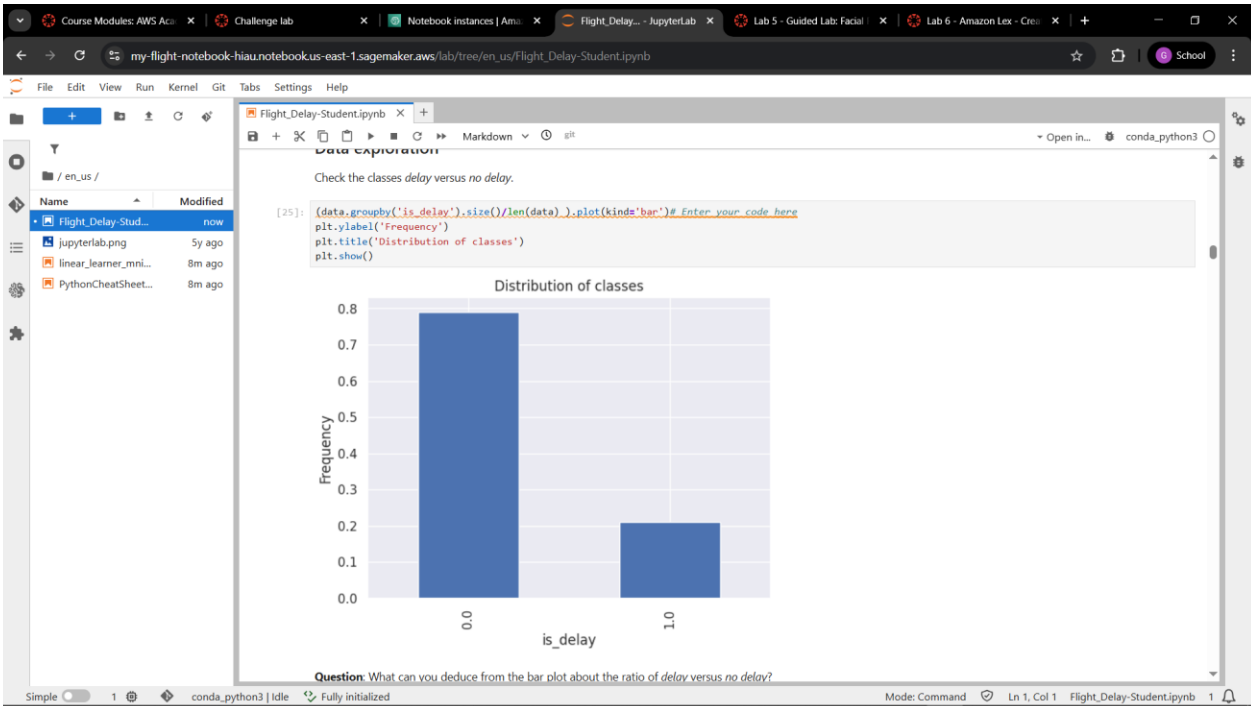
1. How many rows and columns does the dataset have?
2. How many years are included in the dataset?
3. What is the date range for the dataset?
4. Which airlines are included in the dataset?
5. Which origin and destination airports are covered?



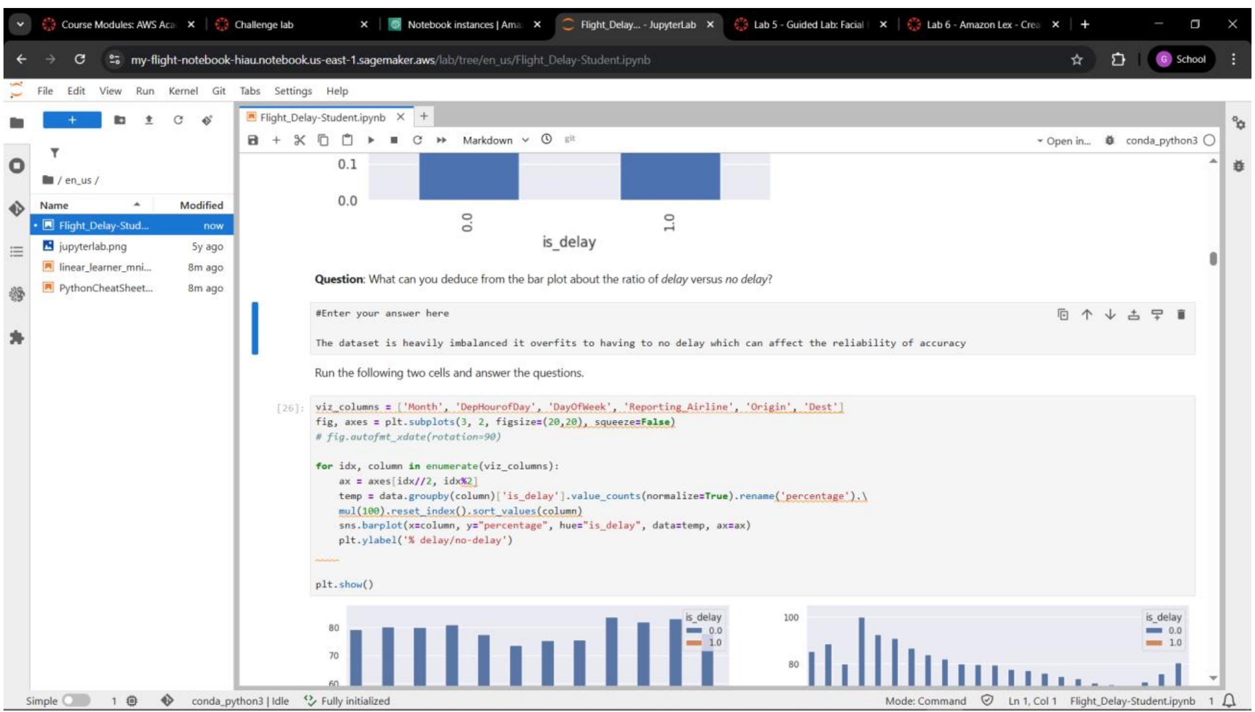
5. Which origin and destination airports are covered?
[20]:
print("The #rows and #columns are ", data.shape[0], " and ", data.shape[1])
print("The years in this dataset are: ", list(data.Year.unique()))
print("The months covered in this dataset are: ", sorted(list(data.Month.unique())))
print("The date range for data is :", min(data.FlightDate), " to ", max(data.FlightDate))
print("The airlines covered in this dataset are: ", list(data.Reporting_Airline.unique()))
print("The Origin airports covered are: ", list(data.Origin.unique()))
print("The Destination airports covered are: ", list(data.Dest.unique()))
The #rows and #columns are 1658130 and 20
The years in this dataset are: [2015, 2014, 2017, 2016, 2018]
The months covered in this dataset are: [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12]
The date range for data is : 2014-01-01 to 2018-12-31
The airlines covered in this dataset are: ['AA', 'DL', 'OO', 'UA', 'WN']
The Origin airports covered are: ['SFO', 'DFW', 'LAX', 'PHX', 'DEN', 'ORD', 'CLT', 'ATL', 'IAH']
The Destination airports covered are: ['DFW', 'SFO', 'ORD', 'LAX', 'PHX', 'ATL', 'IAH', 'CLT', 'DEN']
Define your target column: is_delay (1 means that the arrival time delayed more than 15 minutes, and 0 means all other cases). To rename the column from ArrDel15 to is_delay, use the rename method.
Hint: You can use the rename function in pandas (pandas.DataFrame.rename documentation).
For example:
data.rename(columns={'col1':'column1'}, inplace=True)
[21]: data.rename(columns={'ArrDel15': 'is_delay'}, inplace=True) # Enter your code here
Look for nulls across columns. You can use the isnull() function (pandas.isnull documentation).
Hint: isnull() detects whether the particular value is null or not. It returns a boolean (True or False) in its place. To sum the number of columns, use the sum(axis=0) function (for example, df.isnull().sum(axis=0)).



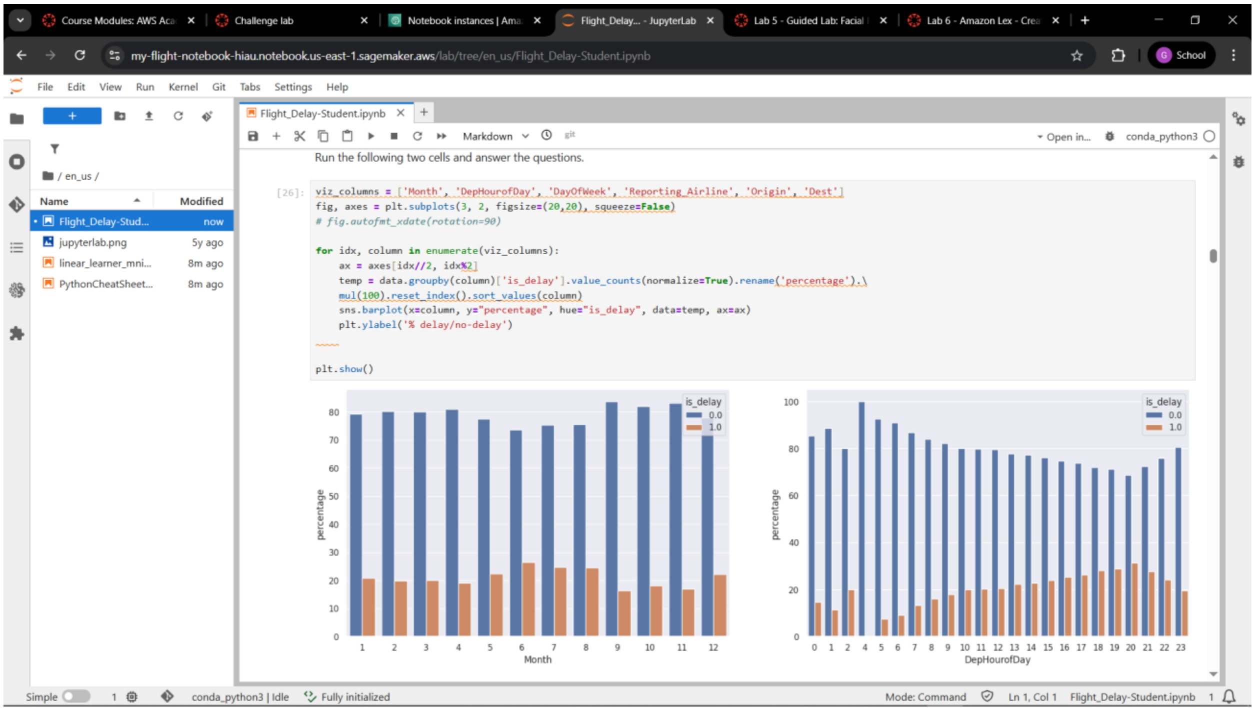
[21]: data.rename(columns={'ArrDel15': 'is_delay'}, inplace=True) # Enter your code here
Look for nulls across columns. You can use the isnull() function (pandas.isnull documentation).
Hint: isnull() detects whether the particular value is null or not. It returns a boolean (True or False) in its place. To sum the number of columns, use the sum(axis=0) function (for example, df.isnull().sum(axis=0)).
[22]: # Enter your code here
data.isnull().sum(axis=0)
[22]:
Year 0
Quarter 0
Month 0
DayofMonth 0
DayOfWeek 0
FlightDate 0
Reporting_Airline 0
Origin 0
OriginState 0
Dest 0
DestState 0
CRSDepTime 0
Cancelled 0
Diverted 0
Distance 0
DistanceGroup 0
ArrDelay 22540
ArrDelayMinutes 22540
is_delay 22540
AirTime 22540
dtype: int64
The arrival delay details and airtime are missing for 22,540 out of 1,658,130 rows, which is 1.3 percent. You can either remove or impute these rows. The documentation doesn't mention any information about missing rows.

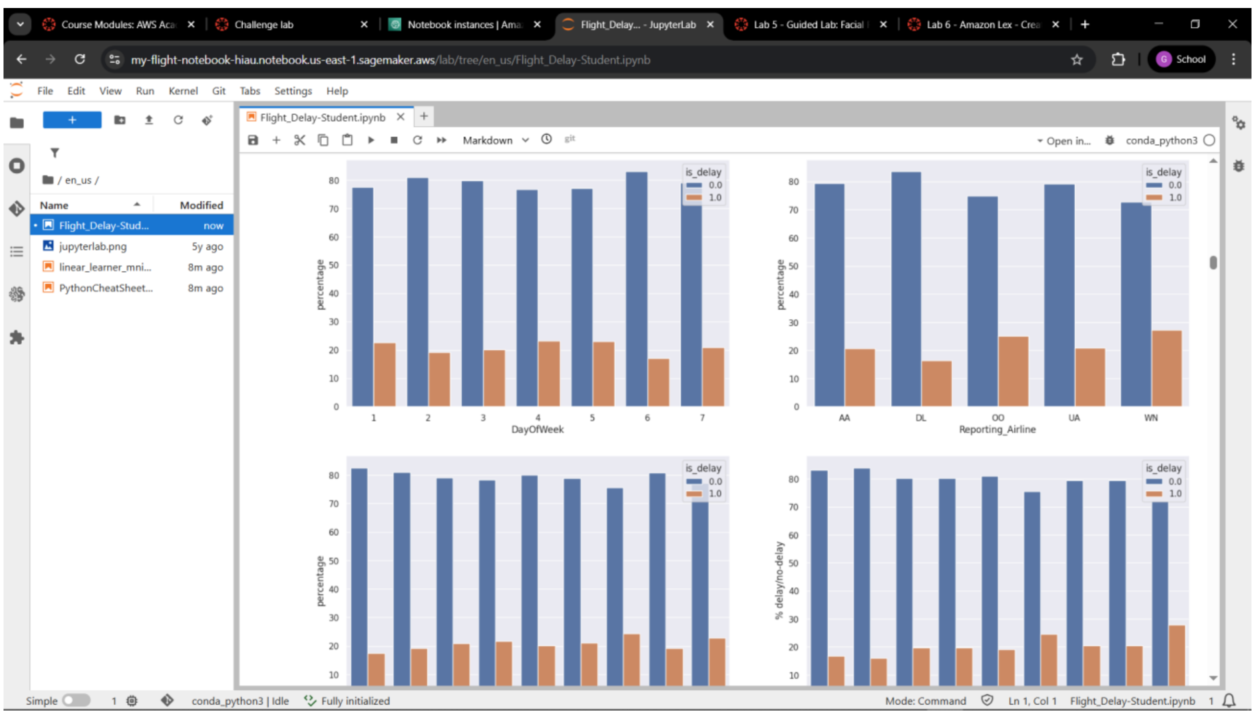


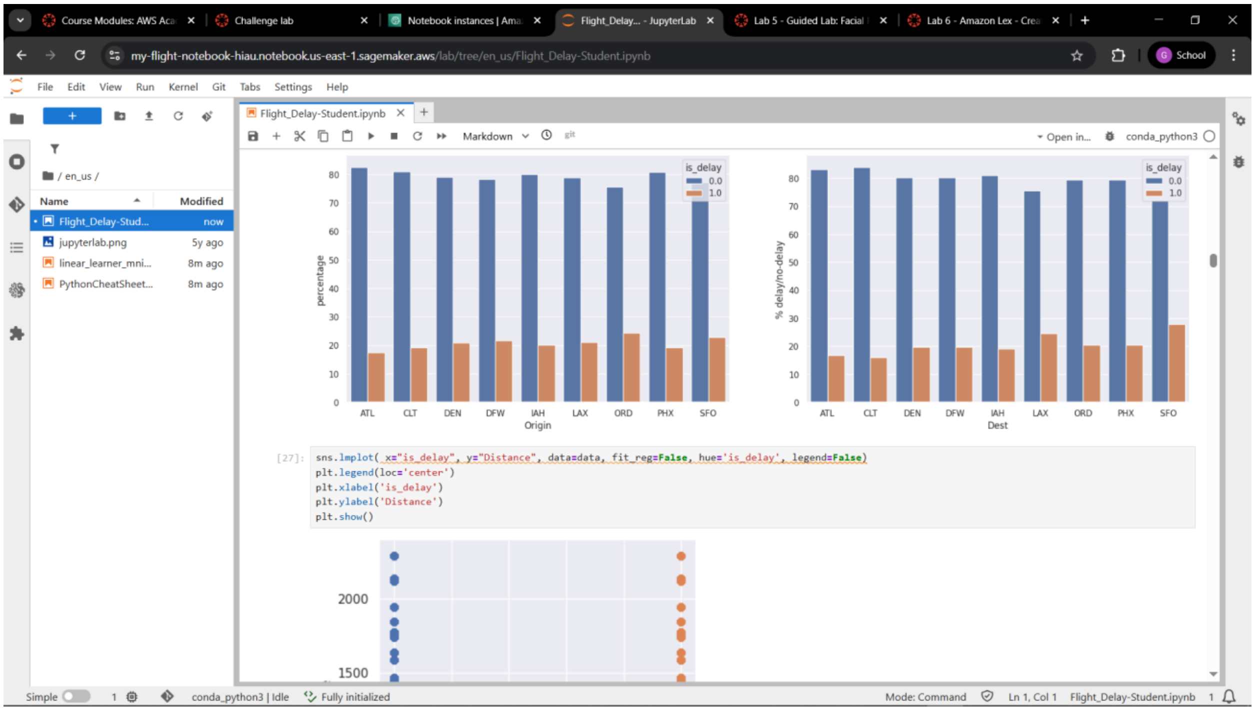
my-flight-notebook-hiau.notebook.us-east-1.sagemaker.aws/lab/tree/en_us/Flight_Delay-Student.ipynb
Flight_Delay-Student.ipynb
Check the classes *delay* versus *no delay*.
[25]:
(data.groupby('is_delay').size()/len(data)).plot(kind='bar')# Enter your code here
plt.ylabel('Frequency')
plt.title('Distribution of classes')
plt.show()
Distribution of classes
Frequency
is_delay
Question: What can you deduce from the bar plot about the ratio of *delay* versus *no delay*?

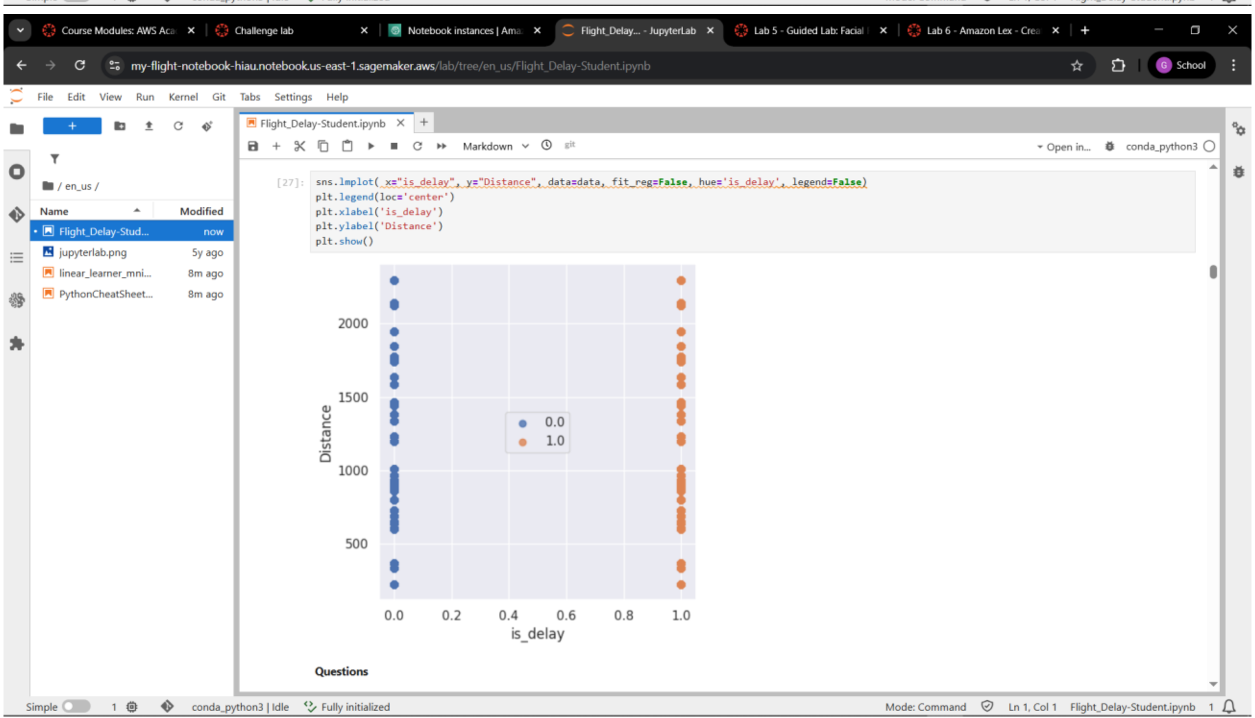


my-flight-notebook-hiau.notebook.us-east-1.sagemaker.aws/lab/tree/en_us/Flight_Delay-Student.ipynb
is_delay
Question: What can you deduce from the bar plot about the ratio of *delay* versus *no delay*?
#Enter your answer here
The dataset is heavily imbalanced it overfits to having to no delay which can affect the reliability of accuracy
Run the following two cells and answer the questions.
[26]:
viz_columns = ['Month', 'DepHourofDay', 'DayOfWeek', 'Reporting_Airline', 'Origin', 'Dest']
fig, axes = plt.subplots(3, 2, figsize=(20,20), squeeze=False)
# fig.autofmt_xdate(rotation=90)
for idx, column in enumerate(viz_columns):
    ax = axes[idx//2, idx%2]
    temp = data.groupby(column)['is_delay'].value_counts(normalize=True).rename('percentage').\
    mul(100).reset_index().sort_values(column)
    sns.barplot(x=column, y="percentage", hue="is_delay", data=temp, ax=ax)
    plt.ylabel('% delay/no-delay')
plt.show()
is_delay


my-flight-notebook-hiau.notebook.us-east-1.sagemaker.aws/lab/tree/en_us/Flight_Delay-Student.ipynb
Flight_Delay-Student.ipynb
Run the following two cells and answer the questions.
[26]:
viz_columns = ['Month', 'DepHourofDay', 'DayOfWeek', 'Reporting_Airline', 'Origin', 'Dest']
fig, axes = plt.subplots(3, 2, figsize=(20,20), squeeze=False)
# fig.autofmt_xdate(rotation=90)
for idx, column in enumerate(viz_columns):
ax = axes[idx//2, idx%2]
temp = data.groupby(column)['is_delay'].value_counts(normalize=True).rename('percentage').\
mul(100).reset_index().sort_values(column)
sns.barplot(x=column, y="percentage", hue="is_delay", data=temp, ax=ax)
plt.ylabel('% delay/no-delay')
plt.show()
is_delay
0.0
1.0
percentage
Month
DepHourofDay
Mode: Command
Ln 1, Col 1
conda_python3 | Idle
Fully initialized


my-flight-notebook-hiau.notebook.us-east-1.sagemaker.aws/lab/tree/en_us/Flight_Delay-Student.ipynb
Flight_Delay-Student.ipynb
is_delay
0.0
1.0
percentage
DayOfWeek
AA
DL
OO
UA
WN
Reporting_Airline
% delay/no-delay
Mode: Command
Ln 1, Col 1
conda_python3 | Idle
Fully initialized


Course Modules: AWS Aca
Challenge lab
Notebook instances | Ama
Flight_Delay... - JupyterLab
Lab 5 - Guided Lab: Facial
Lab 6 - Amazon Lex - Crea
my-flight-notebook-hiau.notebook.us-east-1.sagemaker.aws/lab/tree/en_us/Flight_Delay-Student.ipynb
School
File Edit View Run Kernel Git Tabs Settings Help
/ en_us /
Name
Modified
Flight_Delay-Stud... now
jupyterlab.png 5y ago
linear_learner_mni... 8m ago
PythonCheatSheet... 8m ago
Flight_Delay-Student.ipynb
Markdown
Open in...
conda_python3
is_delay
0.0
1.0
percentage
ATL CLT DEN DFW IAH LAX ORD PHX SFO
Origin
% delay/no-delay
Dest
[27]:
sns.lmplot(x="is_delay", y="Distance", data=data, fit_reg=False, hue='is_delay', legend=False)
plt.legend(loc='center')
plt.xlabel('is_delay')
plt.ylabel('Distance')
plt.show()
2000
1500
Simple
conda_python3 | Idle
Fully initialized
Mode: Command
Ln 1, Col 1
Flight_Delay-Student.ipynb


Course Modules: AWS Aca
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Lab 5 - Guided Lab: Facial
Lab 6 - Amazon Lex - Crea
my-flight-notebook-hiau.notebook.us-east-1.sagemaker.aws/lab/tree/en_us/Flight_Delay-Student.ipynb
School
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/ en_us /
Name
Modified
Flight_Delay-Stud... now
jupyterlab.png 5y ago
linear_learner_mni... 8m ago
PythonCheatSheet... 8m ago
Flight_Delay-Student.ipynb
Markdown
Open in...
conda_python3
[27]:
sns.lmplot(x="is_delay", y="Distance", data=data, fit_reg=False, hue='is_delay', legend=False)
plt.legend(loc='center')
plt.xlabel('is_delay')
plt.ylabel('Distance')
plt.show()
2000
1500
1000
500
Distance
0.0
1.0
0.0 0.2 0.4 0.6 0.8 1.0
is_delay
Questions
Simple
conda_python3 | Idle
Fully initialized
Mode: Command
Ln 1, Col 1
Flight_Delay-Student.ipynb

Questions

Using the data from the previous charts, answer these questions:

- Which months have the most delays?
- What time of the day has the most delays?
- What day of the week has the most delays?
- Which airline has the most delays?
- Which origin and destination airports have the most delays?
- Is flight distance a factor in the delays?

#Enter your answers here

- Most delayed months: May, June, July, December
- Most delayed time of day: Evening at 8 PM
- Most delayed day of week: Monday, Thursday, Friday
- Most delayed airline: OO and WN
- Most delayed airports: ORD
- Distance: no since the values are similar with each other indicating little to no impact as a factor in predicting delays

Features

Look at all the columns and what their specific types are.

[28]:
```
data.columns
```

[28]:
```
Index(['Year', 'Quarter', 'Month', 'DayofMonth', 'DayOfWeek', 'FlightDate',
       'Reporting_Airline', 'Origin', 'OriginState', 'Dest', 'DestState',
       'CRSDepTime', 'Cancelled', 'Diverted', 'Distance', 'DistanceGroup',
       'ArrDelay', 'ArrDelayMinutes', 'is_delay', 'AirTime', 'DepHourofDay'],
      dtype='object')
```

[31]:
```
data_dummies = pd.get_dummies(data[categorical_columns], drop_first=True) # Enter your code here
data_dummies = data_dummies.replace({True: 1, False: 0})
data = pd.concat([data, data_dummies], axis = 1)
data.drop(categorical_columns,axis=1, inplace=True)
```

Check the length of the dataset and the new columns.

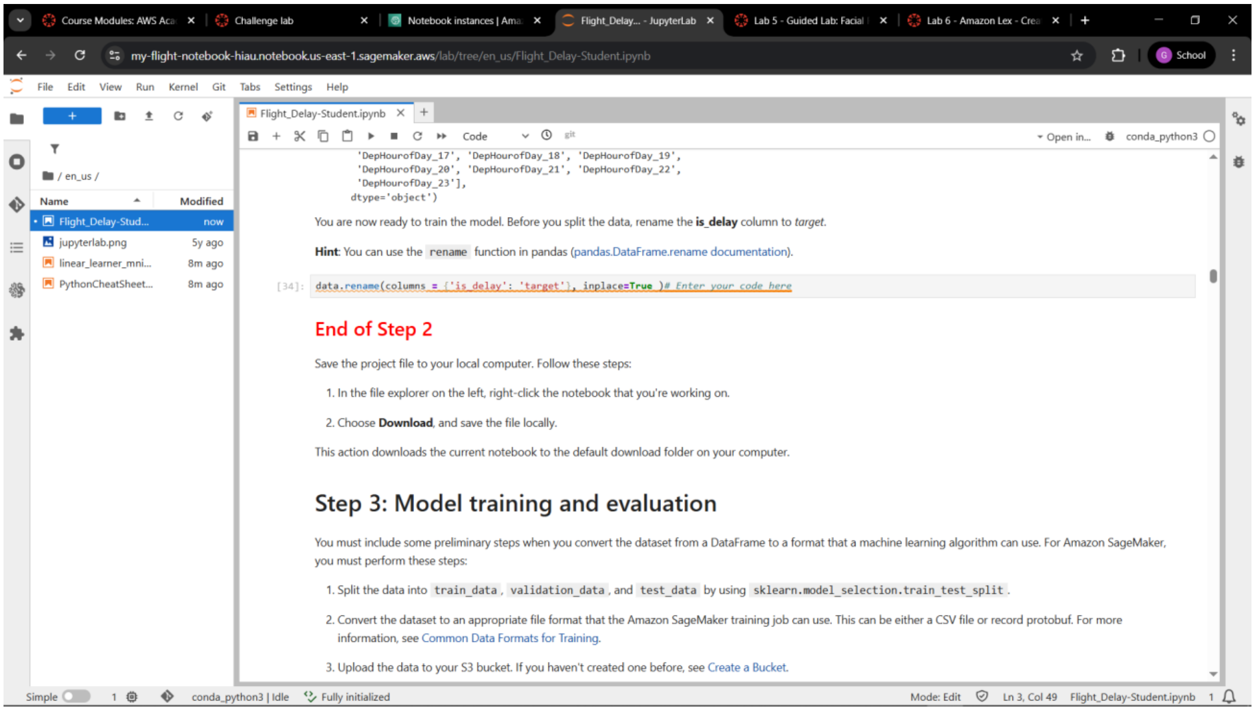
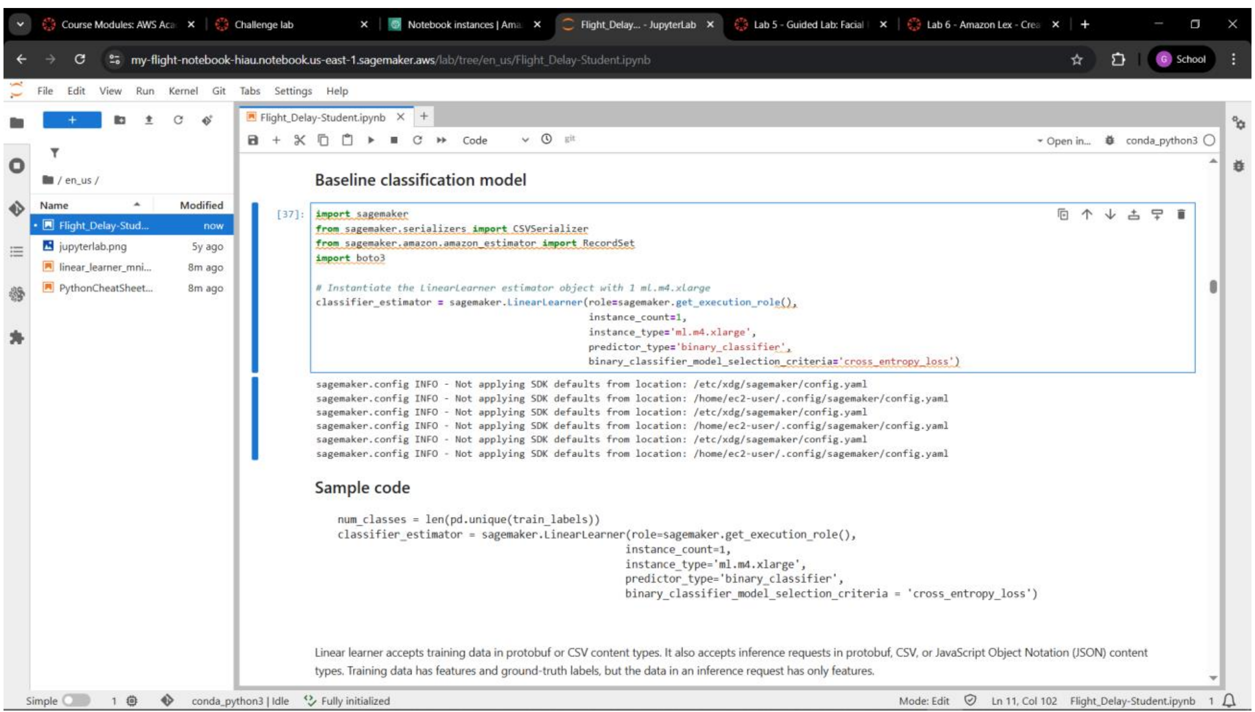
Hint: Use the `shape` and `columns` properties.

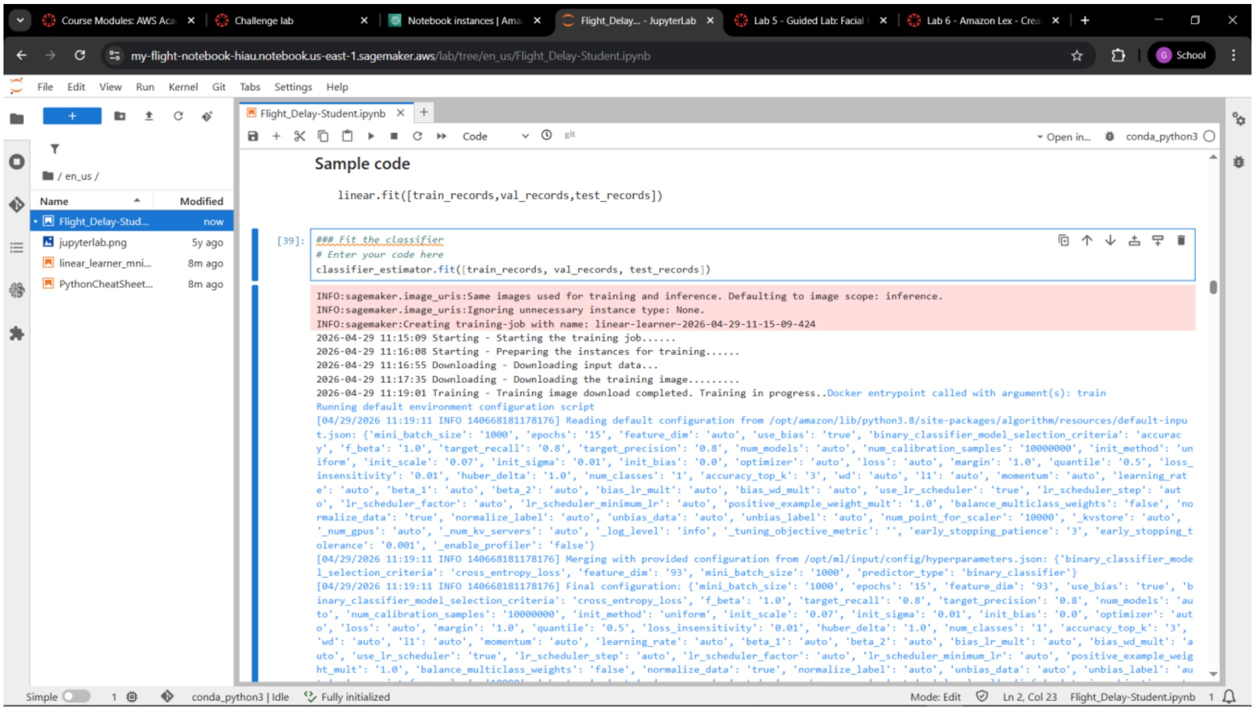
[32]:
```
# Enter your code here
data.shape
```

[32]:
```
(1635590, 94)
```

[33]:
```
# Enter your code here
data.columns
```

[33]:
```
Index(['is_delay', 'Distance', 'Quarter_2', 'Quarter_3', 'Quarter_4',
       'Month_2', 'Month_3', 'Month_4', 'Month_5', 'Month_6', 'Month_7',
       'Month_8', 'Month_9', 'Month_10', 'Month_11', 'Month_12',
       'DayofMonth_2', 'DayofMonth_3', 'DayofMonth_4', 'DayofMonth_5',
       'DayofMonth_6', 'DayofMonth_7', 'DayofMonth_8', 'DayofMonth_9',
       'DayofMonth_10', 'DayofMonth_11', 'DayofMonth_12', 'DayofMonth_13',
       'DayofMonth_14', 'DayofMonth_15', 'DayofMonth_16', 'DayofMonth_17',
       'DayofMonth_18', 'DayofMonth_19', 'DayofMonth_20', 'DayofMonth_21',
       'DayofMonth_22', 'DayofMonth_23', 'DayofMonth_24', 'DayofMonth_25',
       'DayofMonth_26', 'DayofMonth_27', 'DayofMonth_28', 'DayofMonth_29',
       'DayofMonth_30', 'DayofMonth_31', 'DayOfWeek_2', 'DayOfWeek_3',
       'DayOfWeek_4', 'DayOfWeek_5', 'DayOfWeek_6', 'DayOfWeek_7',
       'Reporting_Airline_DL', 'Reporting_Airline_OO', 'Reporting_Airline_UA',
       'Reporting_Airline_WN', 'Origin_CLT', 'Origin_DEN', 'Origin_DFW',
       'Origin_IAH', 'Origin_LAX', 'Origin_ORD', 'Origin_PHX', 'Origin_SFO',
       'Dest_CLT', 'Dest_DEN', 'Dest_DFW', 'Dest_IAH', 'Dest_LAX', 'Dest_ORD',
       'Dest_PHX', 'Dest_SFO', 'DepHourofDay_1', 'DepHourofDay_2',
       'DepHourofDay_4', 'DepHourofDay_5', 'DepHourofDay_6', 'DepHourofDay_7',
       'DepHourofDay_8', 'DepHourofDay_9', 'DepHourofDay_10',
```


Flight_Delay-Student.ipynb
'DepHourofDay_17', 'DepHourofDay_18', 'DepHourofDay_19',
'DepHourofDay_20', 'DepHourofDay_21', 'DepHourofDay_22',
'DepHourofDay_23'],
dtype='object')
You are now ready to train the model. Before you split the data, rename the is_delay column to target.
Hint: You can use the rename function in pandas (pandas.DataFrame.rename documentation).
[34]: data.rename(columns = {'is_delay': 'target'}, inplace=True)# Enter your code here
End of Step 2
Save the project file to your local computer. Follow these steps:
1. In the file explorer on the left, right-click the notebook that you're working on.
2. Choose Download, and save the file locally.
This action downloads the current notebook to the default download folder on your computer.
Step 3: Model training and evaluation
You must include some preliminary steps when you convert the dataset from a DataFrame to a format that a machine learning algorithm can use. For Amazon SageMaker, you must perform these steps:
1. Split the data into train_data, validation_data, and test_data by using sklearn.model_selection.train_test_split.
2. Convert the dataset to an appropriate file format that the Amazon SageMaker training job can use. This can be either a CSV file or record protobuf. For more information, see Common Data Formats for Training.
3. Upload the data to your S3 bucket. If you haven't created one before, see Create a Bucket.

Flight_Delay-Student.ipynb
Baseline classification model
[37]: import sagemaker
from sagemaker.serializers import CSVSerializer
from sagemaker.amazon.amazon_estimator import RecordSet
import boto3
# Instantiate the LinearLearner estimator object with 1 ml.m4.xlarge
classifier_estimator = sagemaker.LinearLearner(role=sagemaker.get_execution_role(),
instance_count=1,
instance_type='ml.m4.xlarge',
predictor_type='binary_classifier',
binary_classifier_model_selection_criteria='cross_entropy_loss')
sagemaker.config INFO - Not applying SDK defaults from location: /etc/xdg/sagemaker/config.yaml
sagemaker.config INFO - Not applying SDK defaults from location: /home/ec2-user/.config/sagemaker/config.yaml
Sample code
num_classes = len(pd.unique(train_labels))
classifier_estimator = sagemaker.LinearLearner(role=sagemaker.get_execution_role(),
instance_count=1,
instance_type='ml.m4.xlarge',
predictor_type='binary_classifier',
binary_classifier_model_selection_criteria = 'cross_entropy_loss')
Linear learner accepts training data in protobuf or CSV content types. It also accepts inference requests in protobuf, CSV, or JavaScript Object Notation (JSON) content types. Training data has features and ground-truth labels, but the data in an inference request has only features.



Course Modules: AWS Aca
Challenge lab
Notebook instances | Ama
Flight_Delay... - JupyterLab
Lab 5 - Guided Lab: Facial
Lab 6 - Amazon Lex - Crea
my-flight-notebook-hiau.notebook.us-east-1.sagemaker.aws/lab/tree/en_us/Flight_Delay-Student.ipynb
School
File Edit View Run Kernel Git Tabs Settings Help
/ en_us /
Name Modified
Flight_Delay-Stud... now
jupyterlab.png 5y ago
linear_learner_mni... 8m ago
PythonCheatSheet... 8m ago
Flight_Delay-Student.ipynb
Code
Open in...
conda_python3
Sample code
linear.fit([train_records,val_records,test_records])
[39]:
Fit the classifier
# Enter your code here
classifier_estimator.fit([train_records, val_records, test_records])
INFO:sagemaker.image_uris:Same images used for training and inference. Defaulting to image scope: inference.
INFO:sagemaker.image_uris:Ignoring unnecessary instance type: None.
INFO:sagemaker:Creating training-job with name: linear-learner-2026-04-29-11-15-09-424
2026-04-29 11:15:09 Starting - Starting the training job......
2026-04-29 11:16:08 Starting - Preparing the instances for training......
2026-04-29 11:16:55 Downloading - Downloading input data...
2026-04-29 11:17:35 Downloading - Downloading the training image.........
2026-04-29 11:19:01 Training - Training image download completed. Training in progress..Docker entrypoint called with argument(s): train
Running default environment configuration script
[04/29/2026 11:19:11 INFO 140668181178176] Reading default configuration from /opt/amazon/lib/python3.8/site-packages/algorithm/resources/default-input.json: {'mini_batch_size': '1000', 'epochs': '15', 'feature_dim': 'auto', 'use_bias': 'true', 'binary_classifier_model_selection_criteria': 'accuracy', 'f_beta': '1.0', 'target_recall': '0.8', 'target_precision': '0.8', 'num_models': 'auto', 'num_calibration_samples': '10000000', 'init_method': 'uniform', 'init_scale': '0.07', 'init_sigma': '0.01', 'init_bias': '0.0', 'optimizer': 'auto', 'loss': 'auto', 'margin': '1.0', 'quantile': '0.5', 'loss_insensitivity': '0.01', 'huber_delta': '1.0', 'num_classes': '1', 'accuracy_top_k': '3', 'wd': 'auto', 'l1': 'auto', 'momentum': 'auto', 'learning_rate': 'auto', 'beta_1': 'auto', 'beta_2': 'auto', 'bias_lr_mult': 'auto', 'bias_wd_mult': 'auto', 'use_lr_scheduler': 'true', 'lr_scheduler_step': 'auto', 'lr_scheduler_factor': 'auto', 'lr_scheduler_minimum_lr': 'auto', 'positive_example_weight_mult': '1.0', 'balance_multiclass_weights': 'false', 'normalize_data': 'true', 'normalize_label': 'auto', 'unbias_data': 'auto', 'unbias_label': 'auto', 'num_point_for_scaler': '10000', '_kvstore': 'auto', '_num_gpus': 'auto', '_num_kv_servers': 'auto', '_log_level': 'info', '_tuning_objective_metric': '', 'early_stopping_patience': '3', 'early_stopping_tolerance': '0.001', '_enable_profiler': 'false'}
[04/29/2026 11:19:11 INFO 140668181178176] Merging with provided configuration from /opt/ml/input/config/hyperparameters.json: {'binary_classifier_model_selection_criteria': 'cross_entropy_loss', 'feature_dim': '93', 'mini_batch_size': '1000', 'predictor_type': 'binary_classifier'}
[04/29/2026 11:19:11 INFO 140668181178176] Final configuration: {'mini_batch_size': '1000', 'epochs': '15', 'feature_dim': '93', 'use_bias': 'true', 'binary_classifier_model_selection_criteria': 'cross_entropy_loss', 'f_beta': '1.0', 'target_recall': '0.8', 'target_precision': '0.8', 'num_models': 'auto', 'num_calibration_samples': '10000000', 'init_method': 'uniform', 'init_scale': '0.07', 'init_sigma': '0.01', 'init_bias': '0.0', 'optimizer': 'auto', 'loss': 'auto', 'margin': '1.0', 'quantile': '0.5', 'loss_insensitivity': '0.01', 'huber_delta': '1.0', 'num_classes': '1', 'accuracy_top_k': '3', 'wd': 'auto', 'l1': 'auto', 'momentum': 'auto', 'learning_rate': 'auto', 'beta_1': 'auto', 'beta_2': 'auto', 'bias_lr_mult': 'auto', 'bias_wd_mult': 'auto', 'use_lr_scheduler': 'true', 'lr_scheduler_step': 'auto', 'lr_scheduler_factor': 'auto', 'lr_scheduler_minimum_lr': 'auto', 'positive_example_weight_mult': '1.0', 'balance_multiclass_weights': 'false', 'normalize_data': 'true', 'normalize_label': 'auto', 'unbias_data': 'auto', 'unbias_label': 'au
Simple
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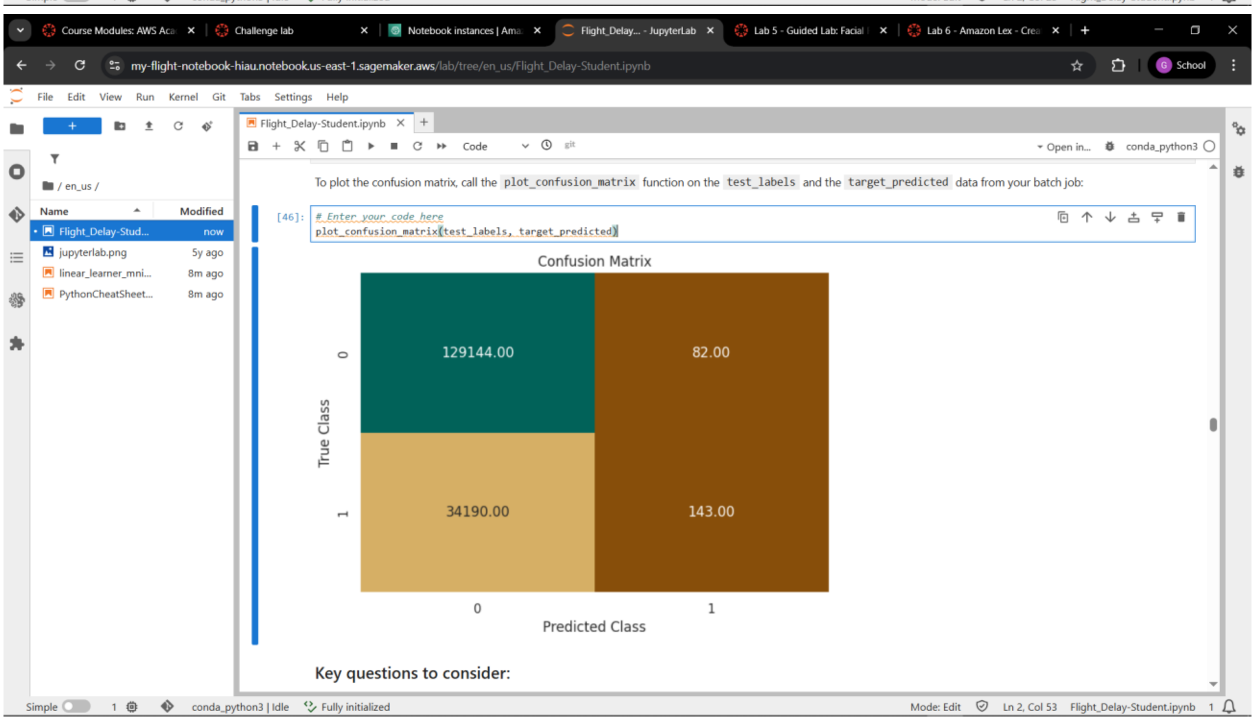
Ln 2, Col 23
Flight_Delay-Student.ipynb



Course Modules: AWS Aca
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my-flight-notebook-hiau.notebook.us-east-1.sagemaker.aws/lab/tree/en_us/Flight_Delay-Student.ipynb
School
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Name Modified
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linear_learner_mni... 8m ago
PythonCheatSheet... 8m ago
Flight_Delay-Student.ipynb
Code
Open in...
conda_python3
To plot the confusion matrix, call the plot_confusion_matrix function on the test_labels and the target_predicted data from your batch job:
[46]:
# Enter your code here
plot_confusion_matrix(test_labels, target_predicted)
Confusion Matrix
True Class
0
1
129144.00
82.00
34190.00
143.00
0
1
Predicted Class
Key questions to consider:
Simple
conda_python3 | Idle
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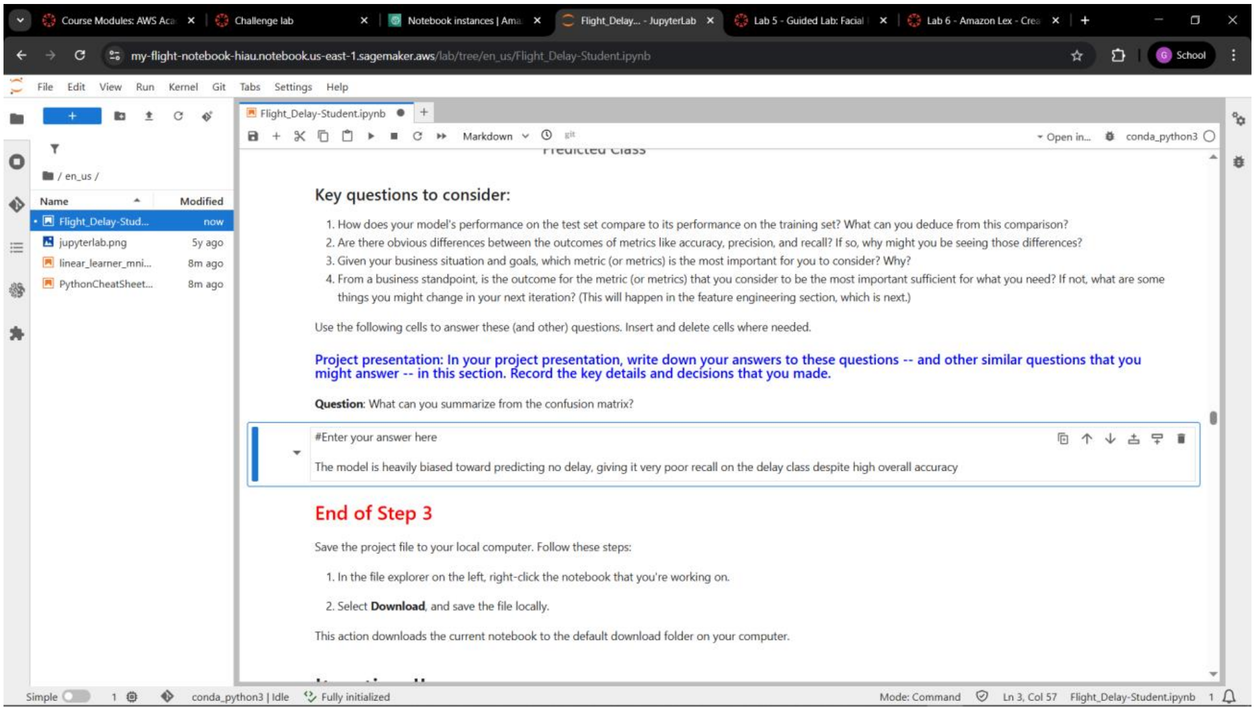
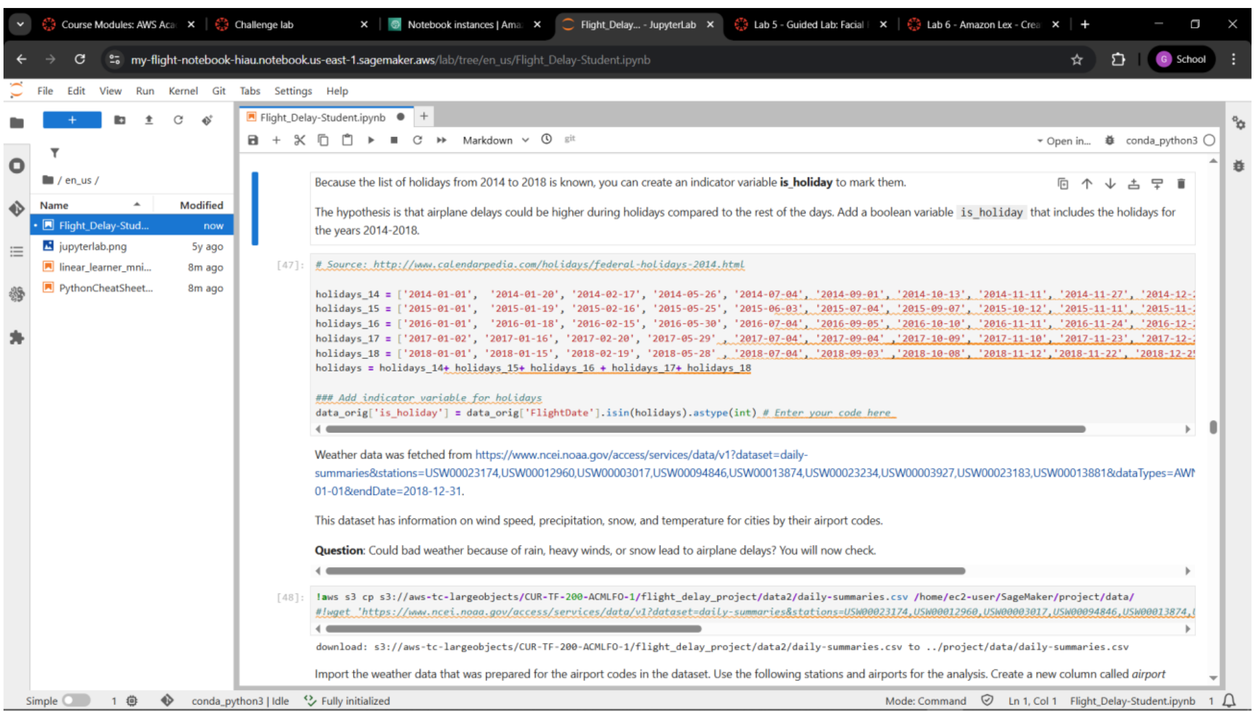
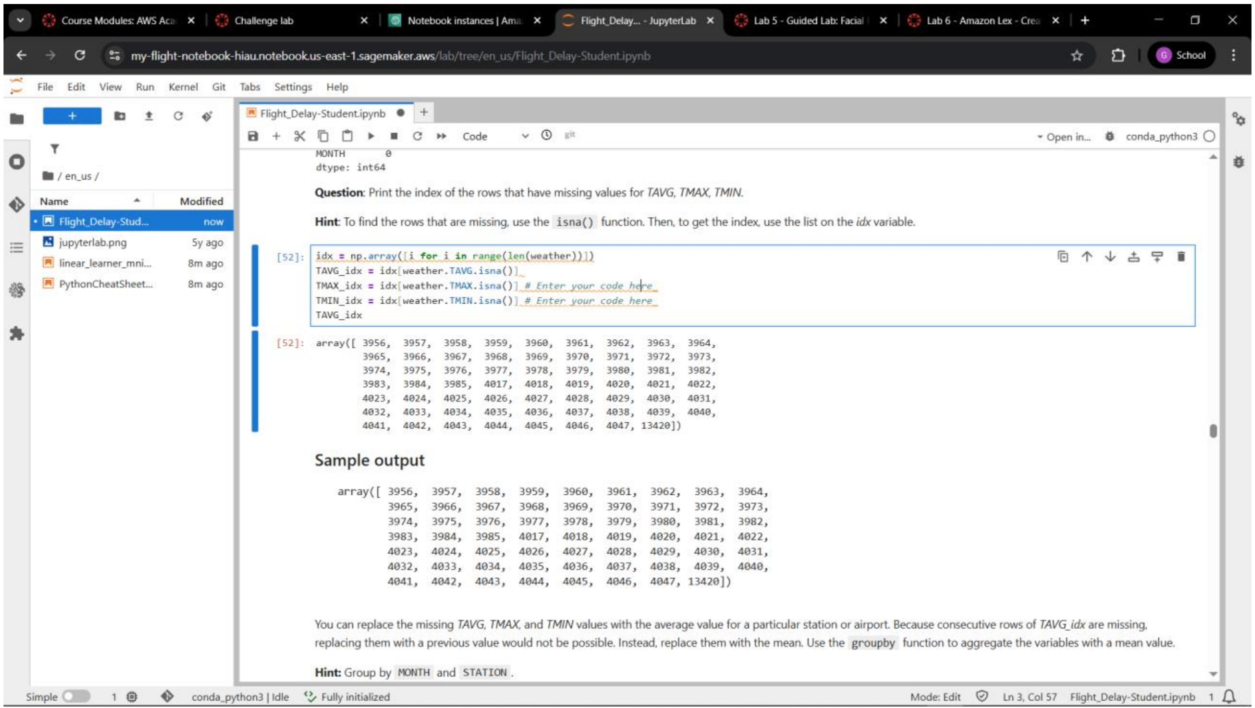
Mode: Edit
Ln 2, Col 53
Flight_Delay-Student.ipynb



Key questions to consider:
1. How does your model's performance on the test set compare to its performance on the training set? What can you deduce from this comparison?
2. Are there obvious differences between the outcomes of metrics like accuracy, precision, and recall? If so, why might you be seeing those differences?
3. Given your business situation and goals, which metric (or metrics) is the most important for you to consider? Why?
4. From a business standpoint, is the outcome for the metric (or metrics) that you consider to be the most important sufficient for what you need? If not, what are some things you might change in your next iteration? (This will happen in the feature engineering section, which is next.)
Use the following cells to answer these (and other) questions. Insert and delete cells where needed.
Project presentation: In your project presentation, write down your answers to these questions -- and other similar questions that you might answer -- in this section. Record the key details and decisions that you made.
Question: What can you summarize from the confusion matrix?
#Enter your answer here
The model is heavily biased toward predicting no delay, giving it very poor recall on the delay class despite high overall accuracy
End of Step 3
Save the project file to your local computer. Follow these steps:
1. In the file explorer on the left, right-click the notebook that you're working on.
2. Select Download, and save the file locally.
This action downloads the current notebook to the default download folder on your computer.



Because the list of holidays from 2014 to 2018 is known, you can create an indicator variable is_holiday to mark them.
The hypothesis is that airplane delays could be higher during holidays compared to the rest of the days. Add a boolean variable is_holiday that includes the holidays for the years 2014-2018.
[47]:
# Source: http://www.calendarpedia.com/holidays/federal-holidays-2014.html
holidays_14 = ['2014-01-01', '2014-01-20', '2014-02-17', '2014-05-26', '2014-07-04', '2014-09-01', '2014-10-13', '2014-11-11', '2014-11-27', '2014-12-
holidays_15 = ['2015-01-01', '2015-01-19', '2015-02-16', '2015-05-25', '2015-06-03', '2015-07-04', '2015-09-07', '2015-10-12', '2015-11-11', '2015-11-
holidays_16 = ['2016-01-01', '2016-01-18', '2016-02-15', '2016-05-30', '2016-07-04', '2016-09-05', '2016-10-10', '2016-11-11', '2016-11-24', '2016-12-
holidays_17 = ['2017-01-02', '2017-01-16', '2017-02-20', '2017-05-29' , '2017-07-04', '2017-09-04' ,'2017-10-09', '2017-11-10', '2017-11-23', '2017-12-
holidays_18 = ['2018-01-01', '2018-01-15', '2018-02-19', '2018-05-28' , '2018-07-04', '2018-09-03' ,'2018-10-08', '2018-11-12','2018-11-22', '2018-12-2
holidays = holidays_14+ holidays_15+ holidays_16 + holidays_17+ holidays_18
Add indicator variable for holidays
data_orig['is_holiday'] = data_orig['FlightDate'].isin(holidays).astype(int) # Enter your code here
Weather data was fetched from https://www.ncei.noaa.gov/access/services/data/v1?dataset=daily-summaries&stations=USW00023174,USW00012960,USW00003017,USW00094846,USW00013874,USW00023234,USW00003927,USW00023183,USW00013881&dataTypes=AWN
01-01&endDate=2018-12-31.
This dataset has information on wind speed, precipitation, snow, and temperature for cities by their airport codes.
Question: Could bad weather because of rain, heavy winds, or snow lead to airplane delays? You will now check.
[48]:
!aws s3 cp s3://aws-tc-largeobjects/CUR-TF-200-ACMLFO-1/flight_delay_project/data2/daily-summaries.csv /home/ec2-user/SageMaker/project/data/
#!wget 'https://www.ncei.noaa.gov/access/services/data/v1?dataset=daily-summaries&stations=USW00023174,USW00012960,USW00003017,USW00094846,USW00013874,
download: s3://aws-tc-largeobjects/CUR-TF-200-ACMLFO-1/flight_delay_project/data2/daily-summaries.csv to ../project/data/daily-summaries.csv
Import the weather data that was prepared for the airport codes in the dataset. Use the following stations and airports for the analysis. Create a new column called airport



my-flight-notebook-hiau.notebook.us-east-1.sagemaker.aws/lab/tree/en_us/Flight_Delay-Student.ipynb

Flight_Delay-Student.ipynb

```
MONTH        0
dtype: int64
```

Question: Print the index of the rows that have missing values for *TAVG, TMAX, TMIN*.

Hint: To find the rows that are missing, use the `isna()` function. Then, to get the index, use the list on the *idx* variable.

[52]:
```
idx = np.array([i for i in range(len(weather))])
TAVG_idx = idx[weather.TAVG.isna()]
TMAX_idx = idx[weather.TMAX.isna()] # Enter your code here
TMIN_idx = idx[weather.TMIN.isna()] # Enter your code here
TAVG_idx
```

[52]:
```
array([ 3956,  3957,  3958,  3959,  3960,  3961,  3962,  3963,  3964,
        3965,  3966,  3967,  3968,  3969,  3970,  3971,  3972,  3973,
        3974,  3975,  3976,  3977,  3978,  3979,  3980,  3981,  3982,
        3983,  3984,  3985,  4017,  4018,  4019,  4020,  4021,  4022,
        4023,  4024,  4025,  4026,  4027,  4028,  4029,  4030,  4031,
        4032,  4033,  4034,  4035,  4036,  4037,  4038,  4039,  4040,
        4041,  4042,  4043,  4044,  4045,  4046,  4047, 13420])
```

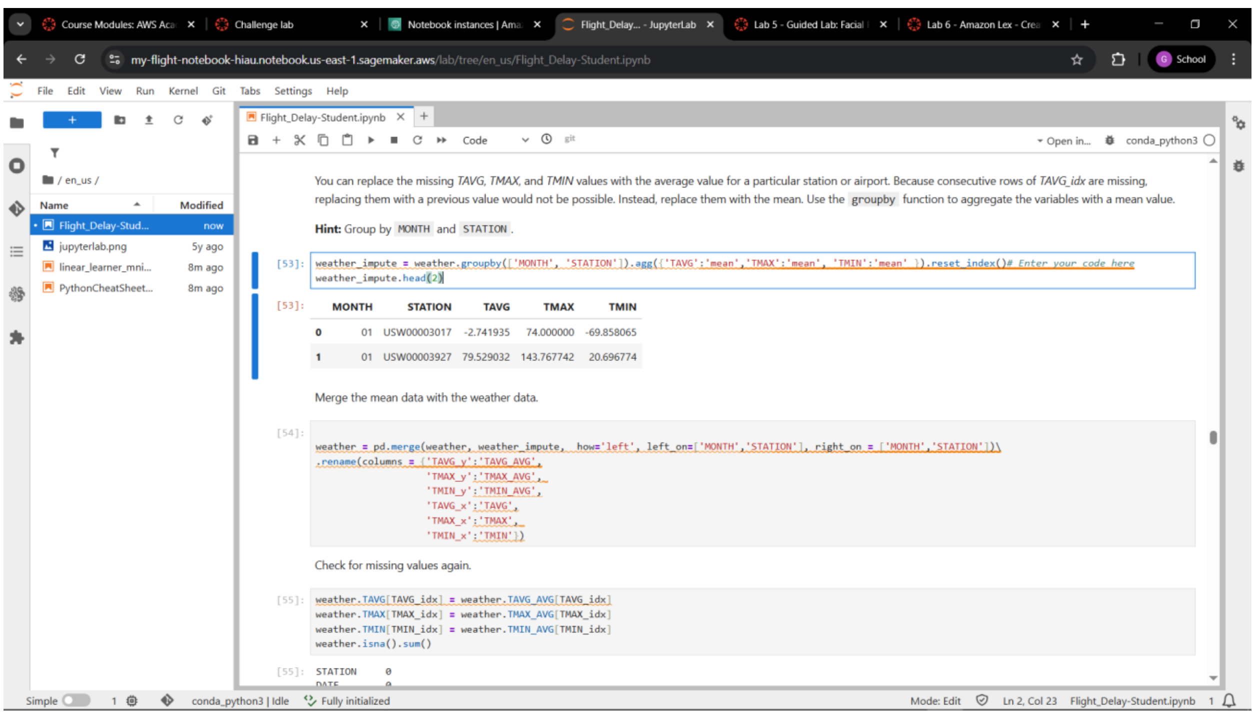
Sample output

```
array([ 3956,  3957,  3958,  3959,  3960,  3961,  3962,  3963,  3964,
        3965,  3966,  3967,  3968,  3969,  3970,  3971,  3972,  3973,
        3974,  3975,  3976,  3977,  3978,  3979,  3980,  3981,  3982,
        3983,  3984,  3985,  4017,  4018,  4019,  4020,  4021,  4022,
        4023,  4024,  4025,  4026,  4027,  4028,  4029,  4030,  4031,
        4032,  4033,  4034,  4035,  4036,  4037,  4038,  4039,  4040,
        4041,  4042,  4043,  4044,  4045,  4046,  4047, 13420])
```

You can replace the missing *TAVG, TMAX,* and *TMIN* values with the average value for a particular station or airport. Because consecutive rows of *TAVG_idx* are missing, replacing them with a previous value would not be possible. Instead, replace them with the mean. Use the `groupby` function to aggregate the variables with a mean value.

Hint: Group by `MONTH` and `STATION`.

Simple | conda_python3 | Idle | Fully initialized | Mode: Edit | Ln 3, Col 57 | Flight_Delay-Student.ipynb



my-flight-notebook-hiau.notebook.us-east-1.sagemaker.aws/lab/tree/en_us/Flight_Delay-Student.ipynb

Flight_Delay-Student.ipynb

You can replace the missing *TAVG, TMAX,* and *TMIN* values with the average value for a particular station or airport. Because consecutive rows of *TAVG_idx* are missing, replacing them with a previous value would not be possible. Instead, replace them with the mean. Use the `groupby` function to aggregate the variables with a mean value.

Hint: Group by `MONTH` and `STATION`.

[53]:
```
weather_impute = weather.groupby(['MONTH', 'STATION']).agg({'TAVG':'mean','TMAX':'mean', 'TMIN':'mean' }).reset_index()# Enter your code here
weather_impute.head(2)
```

[53]:

	MONTH	STATION	TAVG	TMAX	TMIN
0	01	USW00003017	-2.741935	74.000000	-69.858065
1	01	USW00003927	79.529032	143.767742	20.696774

Merge the mean data with the weather data.

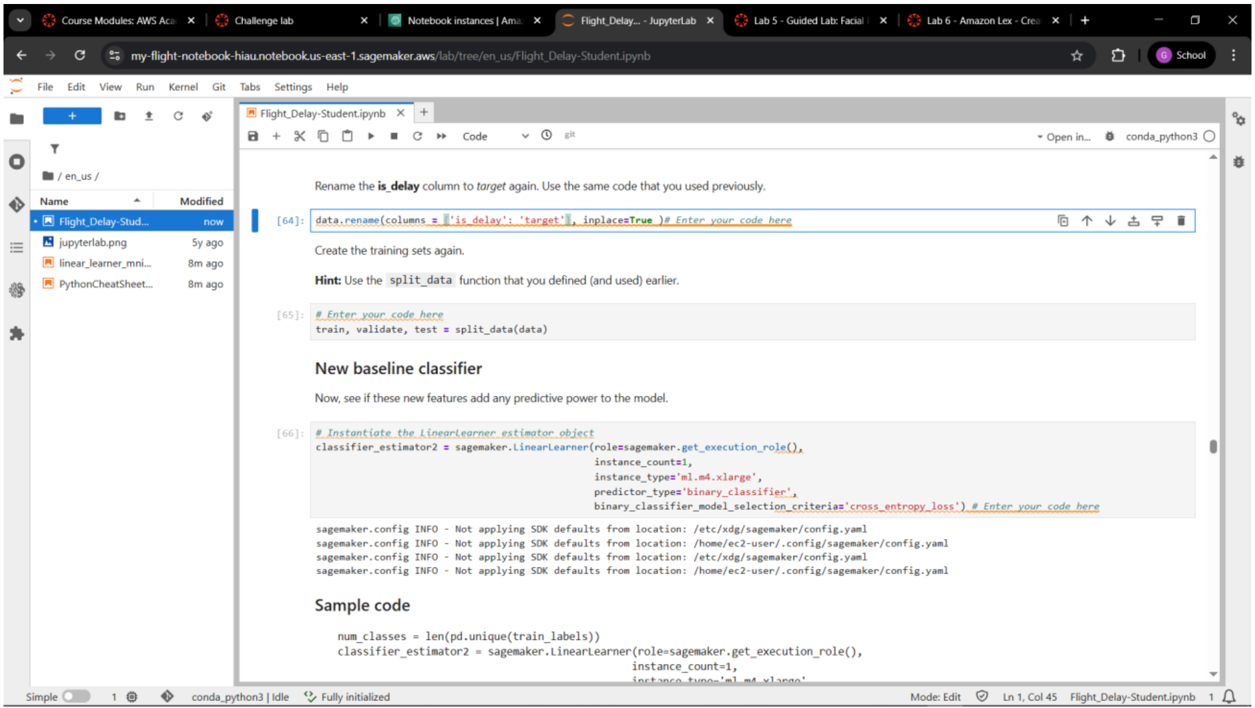
[54]:
```
weather = pd.merge(weather, weather_impute,  how='left', left_on=['MONTH','STATION'], right_on = ['MONTH','STATION'])\
.rename(columns = {'TAVG_y':'TAVG_AVG',
                   'TMAX_y':'TMAX_AVG',
                   'TMIN_y':'TMIN_AVG',
                   'TAVG_x':'TAVG',
                   'TMAX_x':'TMAX',
                   'TMIN_x':'TMIN'})
```

Check for missing values again.

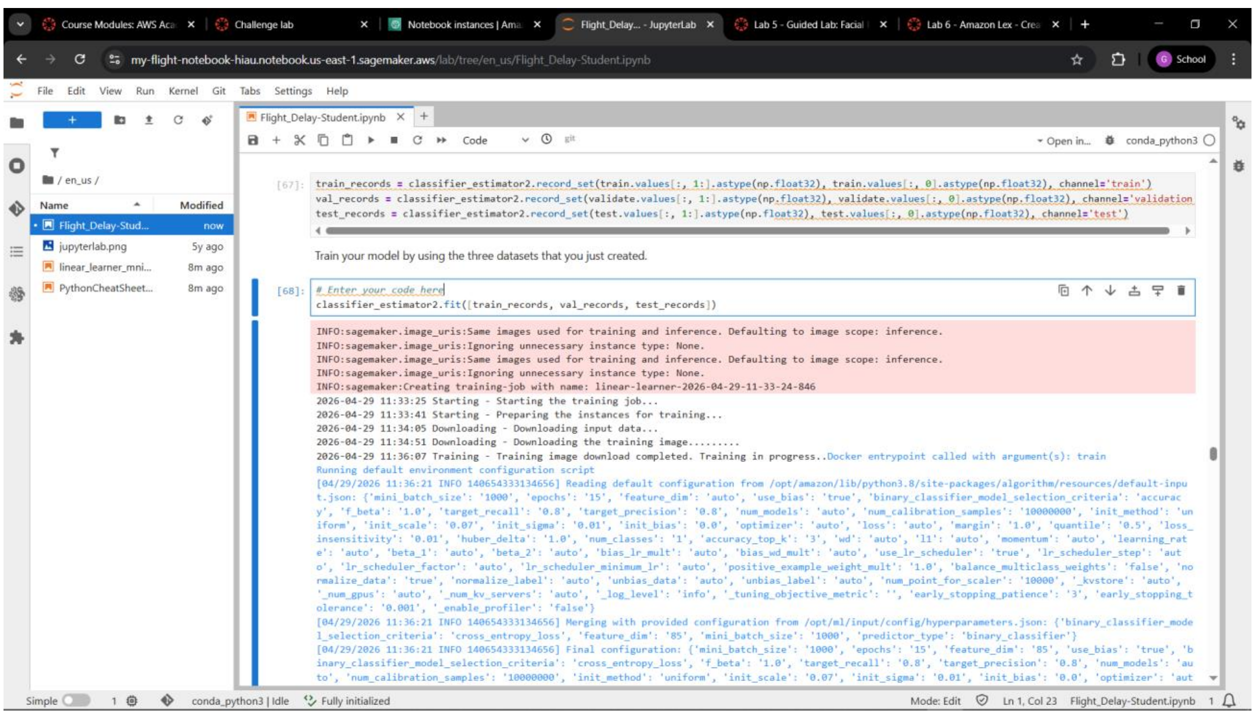
[55]:
```
weather.TAVG[TAVG_idx] = weather.TAVG_AVG[TAVG_idx]
weather.TMAX[TMAX_idx] = weather.TMAX_AVG[TMAX_idx]
weather.TMIN[TMIN_idx] = weather.TMIN_AVG[TMIN_idx]
weather.isna().sum()
```

[55]:
```
STATION     0
DATE        0
```

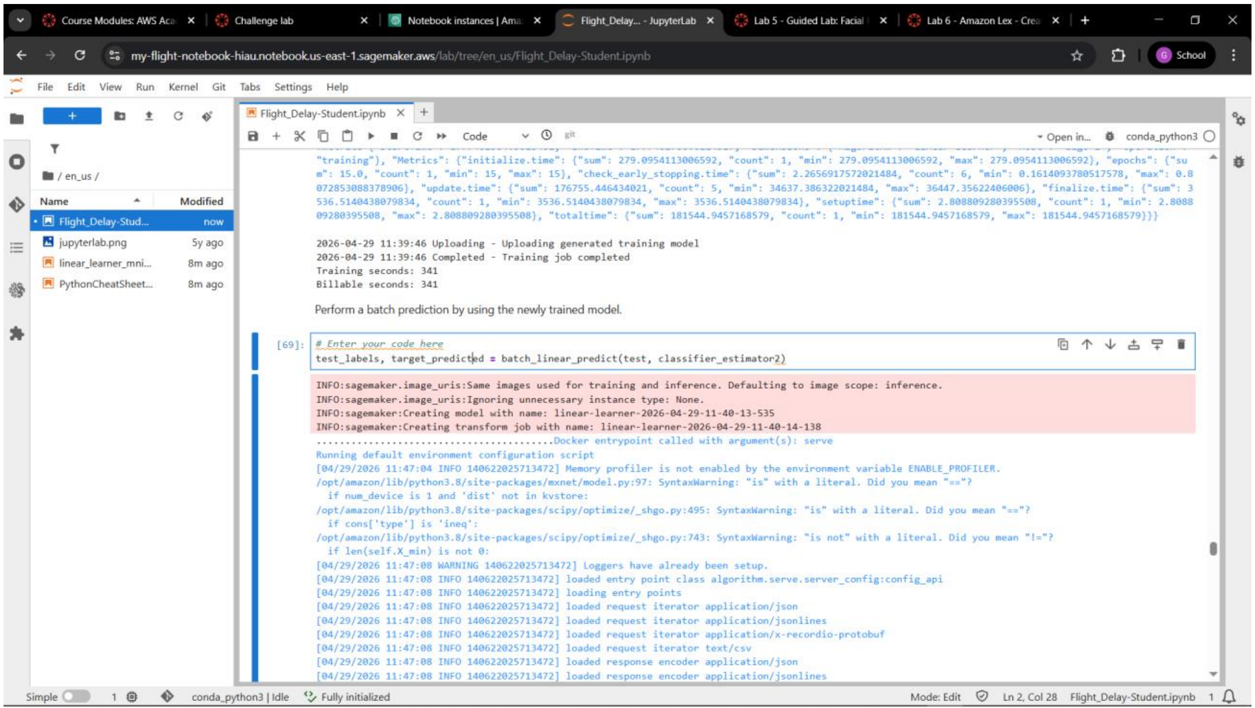
Simple | conda_python3 | Idle | Fully initialized | Mode: Edit | Ln 2, Col 23 | Flight_Delay-Student.ipynb

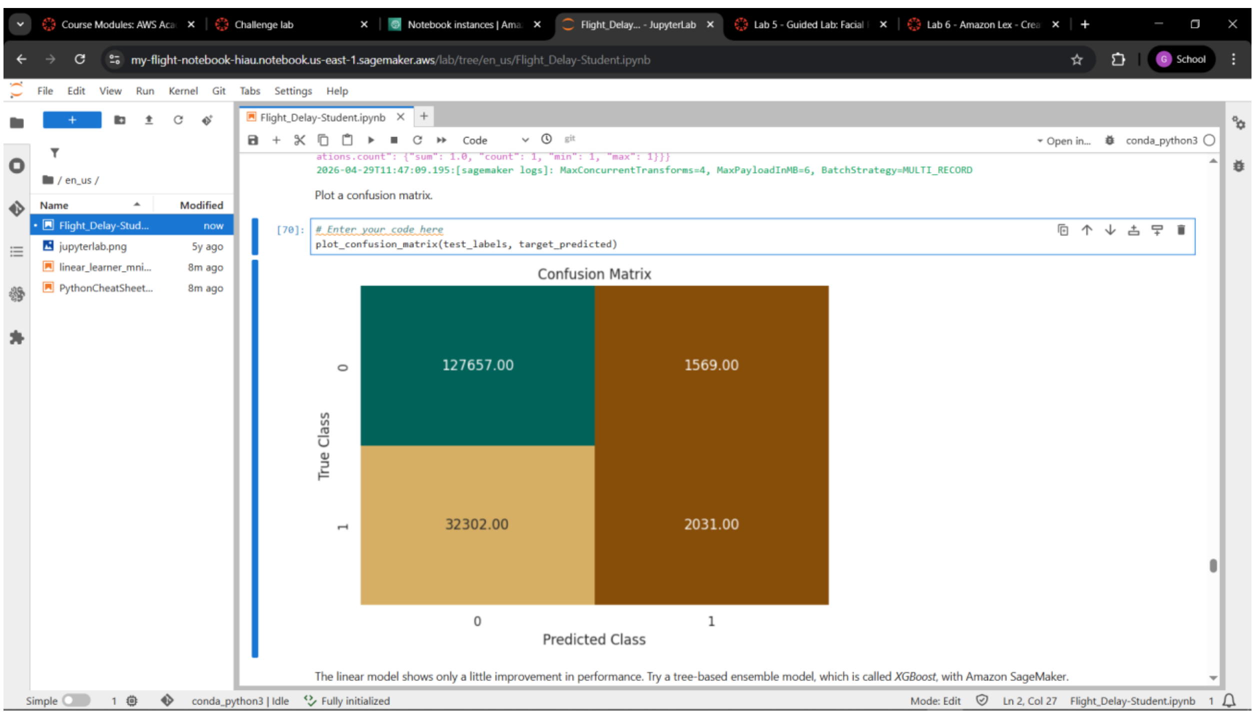


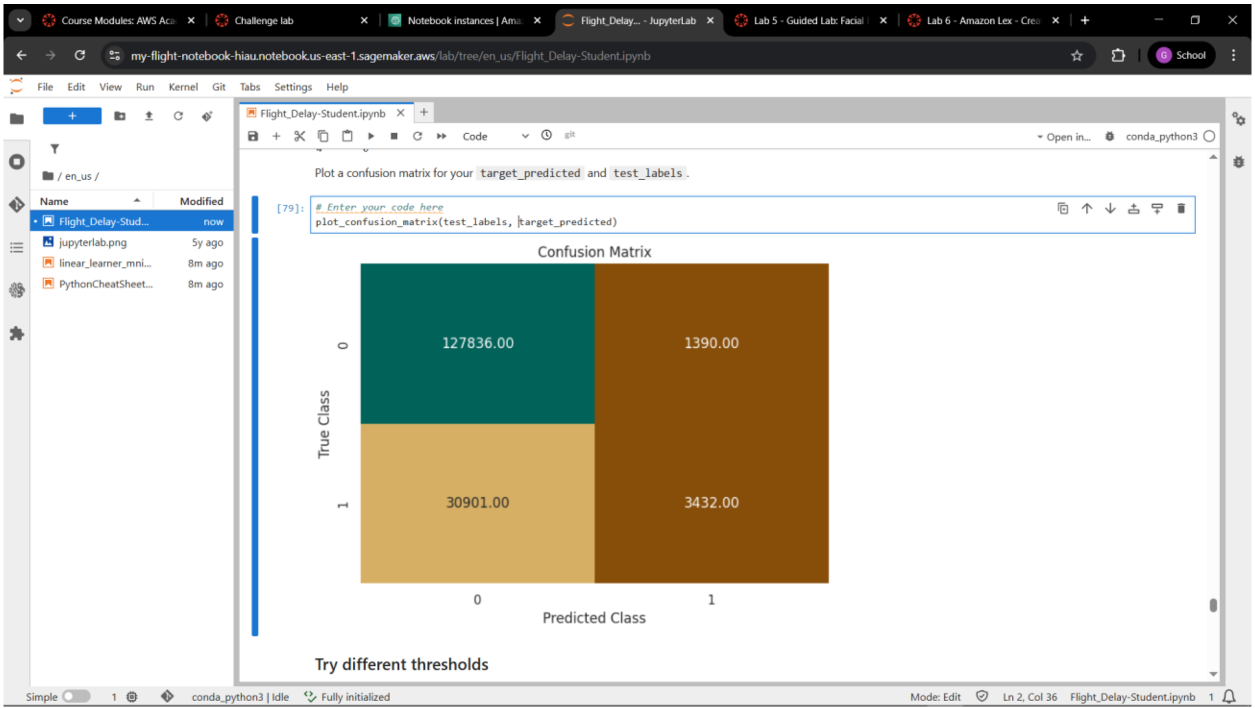
Flight_Delay-Student.ipynb
Rename the is_delay column to target again. Use the same code that you used previously.
[64]: data.rename(columns = {'is_delay': 'target'}, inplace=True)# Enter your code here
Create the training sets again.
Hint: Use the split_data function that you defined (and used) earlier.
[65]: # Enter your code here
train, validate, test = split_data(data)
New baseline classifier
Now, see if these new features add any predictive power to the model.
[66]: # Instantiate the LinearLearner estimator object
classifier_estimator2 = sagemaker.LinearLearner(role=sagemaker.get_execution_role(),
instance_count=1,
instance_type='ml.m4.xlarge',
predictor_type='binary_classifier',
binary_classifier_model_selection_criteria='cross_entropy_loss') # Enter your code here
sagemaker.config INFO - Not applying SDK defaults from location: /etc/xdg/sagemaker/config.yaml
sagemaker.config INFO - Not applying SDK defaults from location: /home/ec2-user/.config/sagemaker/config.yaml
sagemaker.config INFO - Not applying SDK defaults from location: /etc/xdg/sagemaker/config.yaml
sagemaker.config INFO - Not applying SDK defaults from location: /home/ec2-user/.config/sagemaker/config.yaml
Sample code
num_classes = len(pd.unique(train_labels))
classifier_estimator2 = sagemaker.LinearLearner(role=sagemaker.get_execution_role(),
instance_count=1,

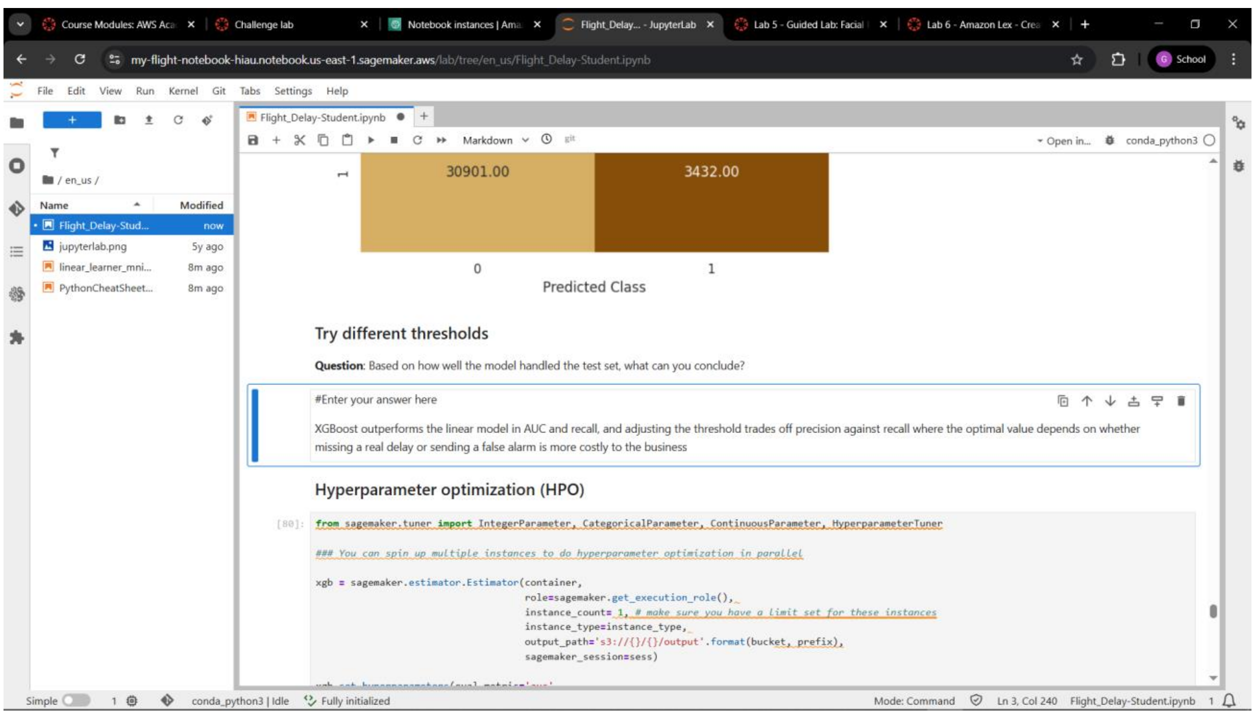


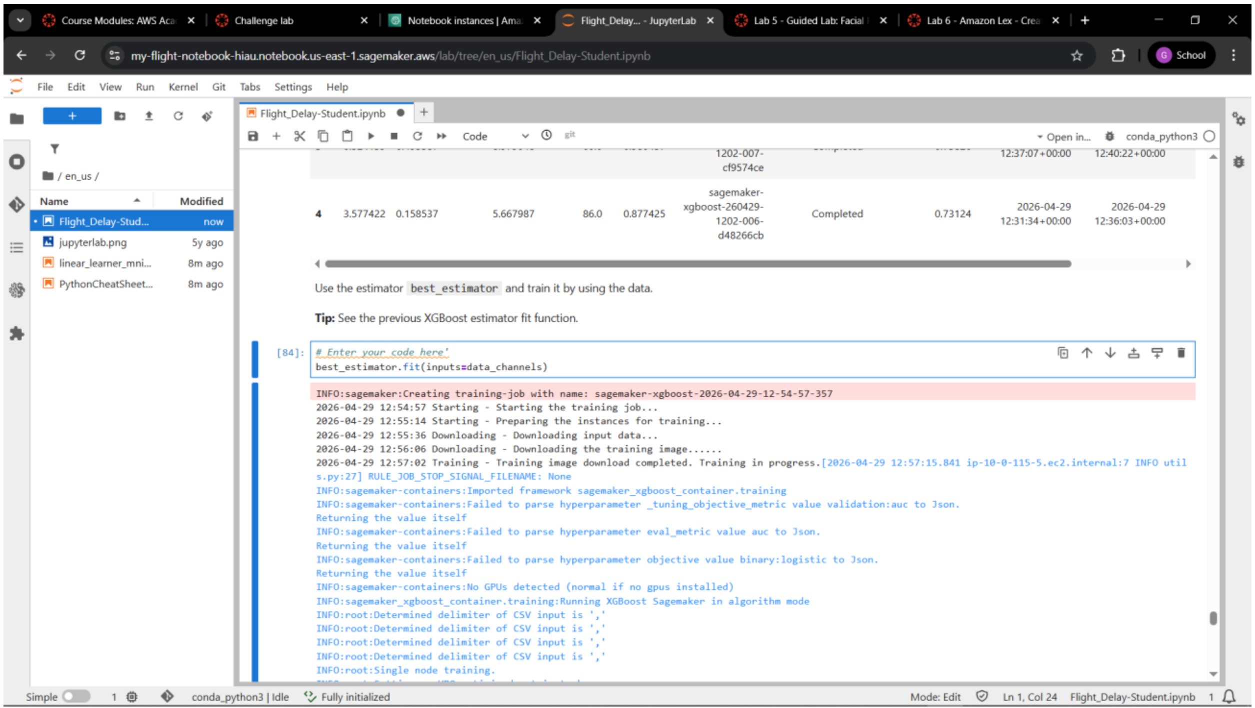
Flight_Delay-Student.ipynb
[67]: train_records = classifier_estimator2.record_set(train.values[:, 1:].astype(np.float32), train.values[:, 0].astype(np.float32), channel='train')
val_records = classifier_estimator2.record_set(validate.values[:, 1:].astype(np.float32), validate.values[:, 0].astype(np.float32), channel='validation
test_records = classifier_estimator2.record_set(test.values[:, 1:].astype(np.float32), test.values[:, 0].astype(np.float32), channel='test')
Train your model by using the three datasets that you just created.
[68]: # Enter your code here
classifier_estimator2.fit([train_records, val_records, test_records])
INFO:sagemaker.image_uris:Same images used for training and inference. Defaulting to image scope: inference.
INFO:sagemaker.image_uris:Ignoring unnecessary instance type: None.
INFO:sagemaker.image_uris:Same images used for training and inference. Defaulting to image scope: inference.
INFO:sagemaker.image_uris:Ignoring unnecessary instance type: None.
INFO:sagemaker:Creating training-job with name: linear-learner-2026-04-29-11-33-24-846
2026-04-29 11:33:25 Starting - Starting the training job...
2026-04-29 11:33:41 Starting - Preparing the instances for training...
2026-04-29 11:34:05 Downloading - Downloading input data...
2026-04-29 11:34:51 Downloading - Downloading the training image.........
2026-04-29 11:36:07 Training - Training image download completed. Training in progress..Docker entrypoint called with argument(s): train
Running default environment configuration script
[04/29/2026 11:36:21 INFO 140654333134656] Reading default configuration from /opt/amazon/lib/python3.8/site-packages/algorithm/resources/default-input.json: {'mini_batch_size': '1000', 'epochs': '15', 'feature_dim': 'auto', 'use_bias': 'true', 'binary_classifier_model_selection_criteria': 'accuracy', 'f_beta': '1.0', 'target_recall': '0.8', 'target_precision': '0.8', 'num_models': 'auto', 'num_calibration_samples': '10000000', 'init_method': 'uniform', 'init_scale': '0.07', 'init_sigma': '0.01', 'init_bias': '0.0', 'optimizer': 'auto', 'loss': 'auto', 'margin': '1.0', 'quantile': '0.5', 'loss_insensitivity': '0.01', 'huber_delta': '1.0', 'num_classes': '1', 'accuracy_top_k': '3', 'wd': 'auto', 'l1': 'auto', 'momentum': 'auto', 'learning_rate': 'auto', 'beta_1': 'auto', 'beta_2': 'auto', 'bias_lr_mult': 'auto', 'bias_wd_mult': 'auto', 'use_lr_scheduler': 'true', 'lr_scheduler_step': 'auto', 'lr_scheduler_factor': 'auto', 'lr_scheduler_minimum_lr': 'auto', 'positive_example_weight_mult': '1.0', 'balance_multiclass_weights': 'false', 'normalize_data': 'true', 'normalize_label': 'auto', 'unbias_data': 'auto', 'unbias_label': 'auto', 'num_point_for_scaler': '10000', '_kvstore': 'auto', '_num_gpus': 'auto', '_num_kv_servers': 'auto', '_log_level': 'info', '_tuning_objective_metric': '', 'early_stopping_patience': '3', 'early_stopping_tolerance': '0.001', '_enable_profiler': 'false'}
[04/29/2026 11:36:21 INFO 140654333134656] Merging with provided configuration from /opt/ml/input/config/hyperparameters.json: {'binary_classifier_model_selection_criteria': 'cross_entropy_loss', 'feature_dim': '85', 'mini_batch_size': '1000', 'predictor_type': 'binary_classifier'}
[04/29/2026 11:36:21 INFO 140654333134656] Final configuration: {'mini_batch_size': '1000', 'epochs': '15', 'feature_dim': '85', 'use_bias': 'true', 'binary_classifier_model_selection_criteria': 'cross_entropy_loss', 'f_beta': '1.0', 'target_recall': '0.8', 'target_precision': '0.8', 'num_models': 'auto', 'num_calibration_samples': '10000000', 'init_method': 'uniform', 'init_scale': '0.07', 'init_sigma': '0.01', 'init_bias': '0.0', 'optimizer': 'aut

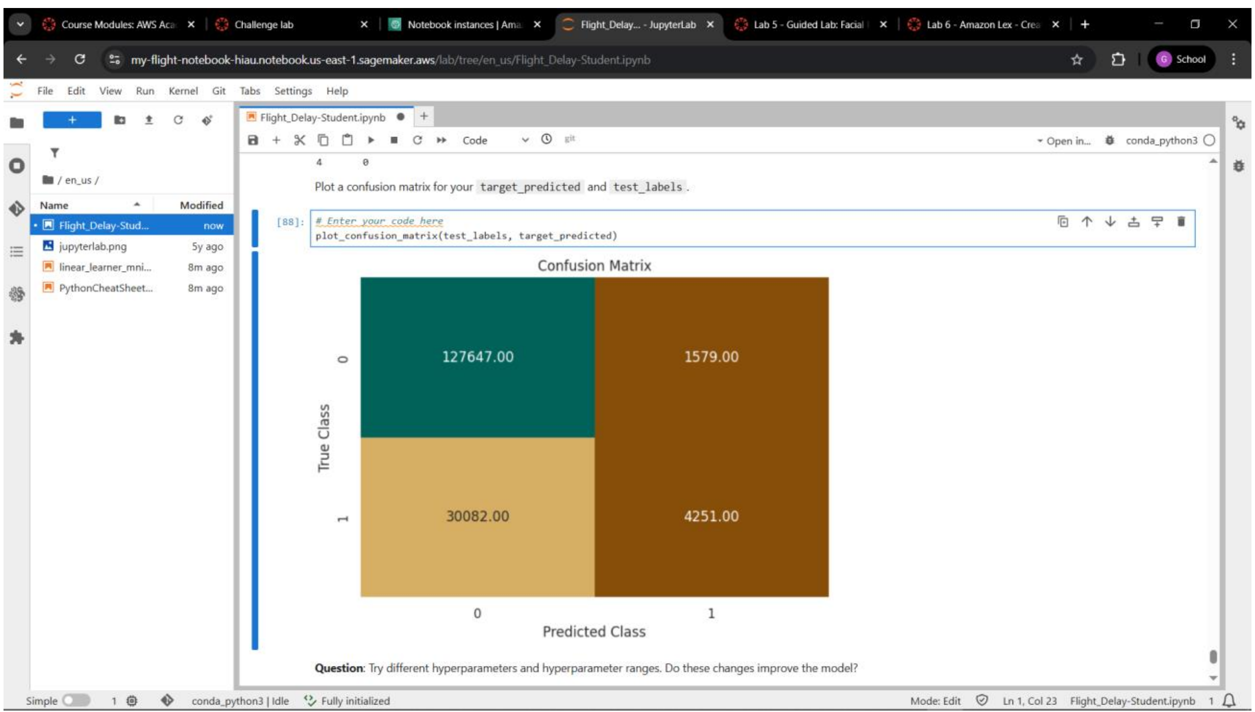

Course Modules: AWS Aca
Challenge lab
Notebook instances | Ama
Flight_Delay... - JupyterLab
Lab 5 - Guided Lab: Facial
Lab 6 - Amazon Lex - Crea
my-flight-notebook-hiau.notebook.us-east-1.sagemaker.aws/lab/tree/en_us/Flight_Delay-Student.ipynb
School
File Edit View Run Kernel Git Tabs Settings Help
/ en_us /
Name Modified
Flight_Delay-Stud... now
jupyterlab.png 5y ago
linear_learner_mni... 8m ago
PythonCheatSheet... 8m ago
Flight_Delay-Student.ipynb
Code
Open in...
conda_python3
"training"}, "Metrics": {"initialize.time": {"sum": 279.0954113006592, "count": 1, "min": 279.0954113006592, "max": 279.0954113006592}, "epochs": {"sum": 15.0, "count": 1, "min": 15, "max": 15}, "check_early_stopping.time": {"sum": 2.2656917572021484, "count": 6, "min": 0.1614093780517578, "max": 0.8072853088378906}, "update.time": {"sum": 176755.446434021, "count": 5, "min": 34637.386322021484, "max": 36447.35622406006}, "finalize.time": {"sum": 3536.5140438079834, "count": 1, "min": 3536.5140438079834, "max": 3536.5140438079834}, "setuptime": {"sum": 2.808809280395508, "count": 1, "min": 2.808809280395508, "max": 2.808809280395508}, "totaltime": {"sum": 181544.9457168579, "count": 1, "min": 181544.9457168579, "max": 181544.9457168579}}}
2026-04-29 11:39:46 Uploading - Uploading generated training model
2026-04-29 11:39:46 Completed - Training job completed
Training seconds: 341
Billable seconds: 341
Perform a batch prediction by using the newly trained model.
[69]: # Enter your code here
test_labels, target_predicted = batch_linear_predict(test, classifier_estimator2)
INFO:sagemaker.image_uris:Same images used for training and inference. Defaulting to image scope: inference.
INFO:sagemaker.image_uris:Ignoring unnecessary instance type: None.
INFO:sagemaker:Creating model with name: linear-learner-2026-04-29-11-40-13-535
INFO:sagemaker:Creating transform job with name: linear-learner-2026-04-29-11-40-14-138
....Docker entrypoint called with argument(s): serve
Running default environment configuration script
[04/29/2026 11:47:04 INFO 140622025713472] Memory profiler is not enabled by the environment variable ENABLE_PROFILER.
/opt/amazon/lib/python3.8/site-packages/mxnet/model.py:97: SyntaxWarning: "is" with a literal. Did you mean "=="?
if num_device is 1 and 'dist' not in kvstore:
/opt/amazon/lib/python3.8/site-packages/scipy/optimize/_shgo.py:495: SyntaxWarning: "is" with a literal. Did you mean "=="?
if cons['type'] is 'ineq':
/opt/amazon/lib/python3.8/site-packages/scipy/optimize/_shgo.py:743: SyntaxWarning: "is not" with a literal. Did you mean "!="?
if len(self.X_min) is not 0:
[04/29/2026 11:47:08 WARNING 140622025713472] Loggers have already been setup.
[04/29/2026 11:47:08 INFO 140622025713472] loaded entry point class algorithm.serve.server_config:config_api
[04/29/2026 11:47:08 INFO 140622025713472] loading entry points
[04/29/2026 11:47:08 INFO 140622025713472] loaded request iterator application/json
[04/29/2026 11:47:08 INFO 140622025713472] loaded request iterator application/jsonlines
[04/29/2026 11:47:08 INFO 140622025713472] loaded request iterator application/x-recordio-protobuf
[04/29/2026 11:47:08 INFO 140622025713472] loaded request iterator text/csv
[04/29/2026 11:47:08 INFO 140622025713472] loaded response encoder application/json
[04/29/2026 11:47:08 INFO 140622025713472] loaded response encoder application/jsonlines
Simple 1 conda_python3 | Idle Fully initialized
Mode: Edit Ln 2, Col 28 Flight_Delay-Student.ipynb 1

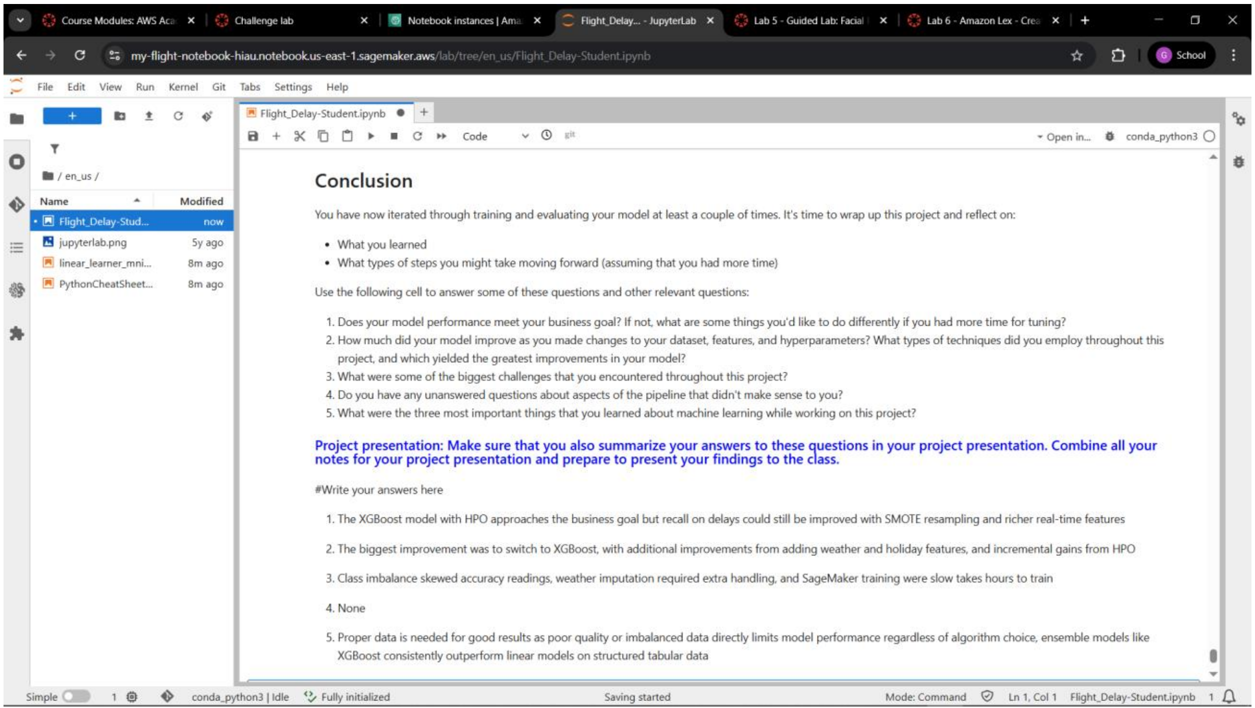

Course Modules: AWS Aca
Challenge lab
Notebook instances | Ama
Flight_Delay... - JupyterLab
Lab 5 - Guided Lab: Facial
Lab 6 - Amazon Lex - Crea
my-flight-notebook-hiau.notebook.us-east-1.sagemaker.aws/lab/tree/en_us/Flight_Delay-Student.ipynb
School
File Edit View Run Kernel Git Tabs Settings Help
/ en_us /
Name Modified
Flight_Delay-Stud... now
jupyterlab.png 5y ago
linear_learner_mni... 8m ago
PythonCheatSheet... 8m ago
Flight_Delay-Student.ipynb
Code
Open in...
conda_python3
2026-04-29T11:47:09.195:[sagemaker logs]: MaxConcurrentTransforms=4, MaxPayloadInMB=6, BatchStrategy=MULTI_RECORD
Plot a confusion matrix.
[70]: # Enter your code here
plot_confusion_matrix(test_labels, target_predicted)
Confusion Matrix
True Class
0
1
127657.00
1569.00
32302.00
2031.00
0
1
Predicted Class
The linear model shows only a little improvement in performance. Try a tree-based ensemble model, which is called XGBoost, with Amazon SageMaker.
Simple 1 conda_python3 | Idle Fully initialized
Mode: Edit Ln 2, Col 27 Flight_Delay-Student.ipynb 1


Course Modules: AWS Aca
Challenge lab
Notebook instances | Ama
Flight_Delay... - JupyterLab
Lab 5 - Guided Lab: Facial
Lab 6 - Amazon Lex - Crea
my-flight-notebook-hiau.notebook.us-east-1.sagemaker.aws/lab/tree/en_us/Flight_Delay-Student.ipynb
School
File Edit View Run Kernel Git Tabs Settings Help
/ en_us /
Name
Modified
Flight_Delay-Stud... now
jupyterlab.png 5y ago
linear_learner_mni... 8m ago
PythonCheatSheet... 8m ago
Flight_Delay-Student.ipynb
Code
Open in...
conda_python3
Plot a confusion matrix for your target_predicted and test_labels .
[79]: # Enter your code here
plot_confusion_matrix(test_labels, target_predicted)
Confusion Matrix
127836.00
1390.00
30901.00
3432.00
True Class
0
1
Predicted Class
0
1
Try different thresholds
Simple
conda_python3 | Idle
Fully initialized
Mode: Edit
Ln 2, Col 36
Flight_Delay-Student.ipynb

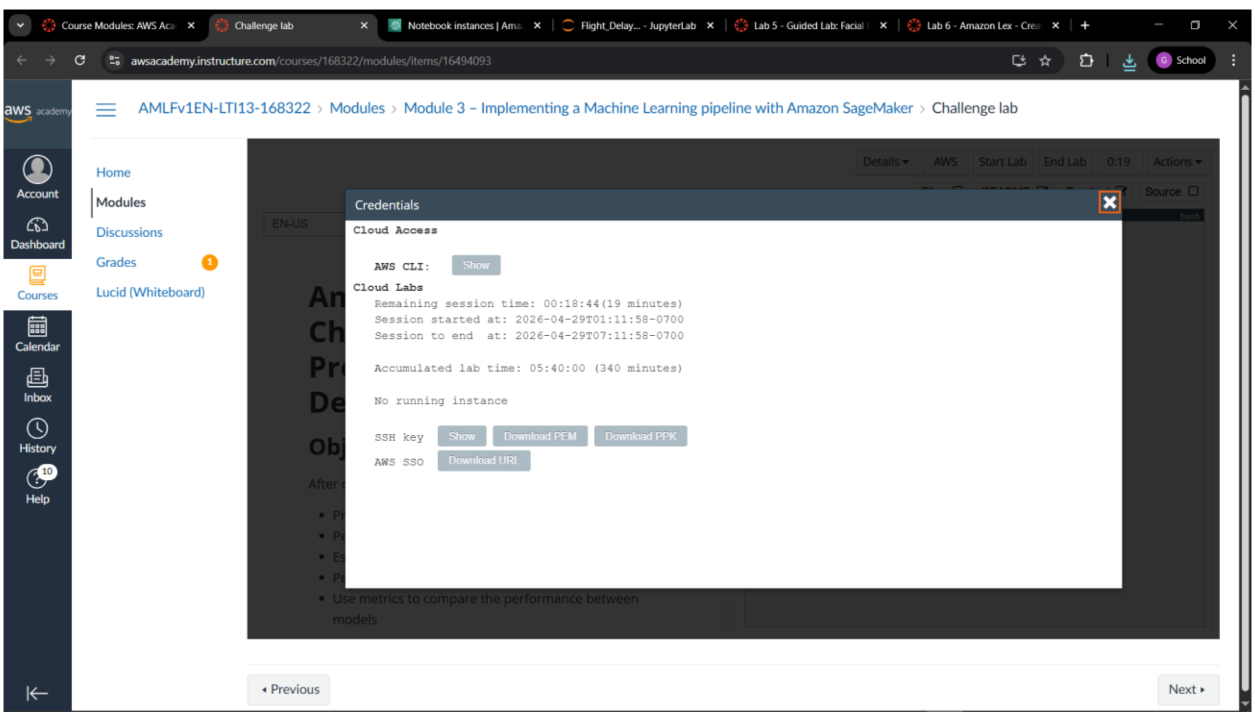

Course Modules: AWS Aca
Challenge lab
Notebook instances | Ama
Flight_Delay... - JupyterLab
Lab 5 - Guided Lab: Facial
Lab 6 - Amazon Lex - Crea
my-flight-notebook-hiau.notebook.us-east-1.sagemaker.aws/lab/tree/en_us/Flight_Delay-Student.ipynb
School
File Edit View Run Kernel Git Tabs Settings Help
/ en_us /
Name
Modified
Flight_Delay-Stud... now
jupyterlab.png 5y ago
linear_learner_mni... 8m ago
PythonCheatSheet... 8m ago
Flight_Delay-Student.ipynb
Markdown
Open in...
conda_python3
30901.00
3432.00
0
1
Predicted Class
Try different thresholds
Question: Based on how well the model handled the test set, what can you conclude?
#Enter your answer here
XGBoost outperforms the linear model in AUC and recall, and adjusting the threshold trades off precision against recall where the optimal value depends on whether missing a real delay or sending a false alarm is more costly to the business
Hyperparameter optimization (HPO)
[80]: from sagemaker.tuner import IntegerParameter, CategoricalParameter, ContinuousParameter, HyperparameterTuner
You can spin up multiple instances to do hyperparameter optimization in parallel
xgb = sagemaker.estimator.Estimator(container,
role=sagemaker.get_execution_role(),
instance_count= 1, # make sure you have a limit set for these instances
instance_type=instance_type,
output_path='s3://{}/{}/output'.format(bucket, prefix),
sagemaker_session=sess)
Simple
conda_python3 | Idle
Fully initialized
Mode: Command
Ln 3, Col 240
Flight_Delay-Student.ipynb


Course Modules: AWS Aca
Challenge lab
Notebook instances | Ama
Flight_Delay... - JupyterLab
Lab 5 - Guided Lab: Facial
Lab 6 - Amazon Lex - Crea
my-flight-notebook-hiau.notebook.us-east-1.sagemaker.aws/lab/tree/en_us/Flight_Delay-Student.ipynb
School
File Edit View Run Kernel Git Tabs Settings Help
Flight_Delay-Student.ipynb
Code
Open in...
conda_python3
/ en_us /
Name
Modified
Flight_Delay-Stud... now
jupyterlab.png 5y ago
linear_learner_mni... 8m ago
PythonCheatSheet... 8m ago
1202-007-cf9574ce
12:37:07+00:00
12:40:22+00:00
4 3.577422 0.158537 5.667987 86.0 0.877425 sagemaker-xgboost-260429-1202-006-d48266cb Completed 0.73124 2026-04-29 12:31:34+00:00 2026-04-29 12:36:03+00:00
Use the estimator best_estimator and train it by using the data.
Tip: See the previous XGBoost estimator fit function.
[84]:
# Enter your code here'
best_estimator.fit(inputs=data_channels)
INFO:sagemaker:Creating training-job with name: sagemaker-xgboost-2026-04-29-12-54-57-357
2026-04-29 12:54:57 Starting - Starting the training job...
2026-04-29 12:55:14 Starting - Preparing the instances for training...
2026-04-29 12:55:36 Downloading - Downloading input data...
2026-04-29 12:56:06 Downloading - Downloading the training image......
2026-04-29 12:57:02 Training - Training image download completed. Training in progress.[2026-04-29 12:57:15.841 ip-10-0-115-5.ec2.internal:7 INFO util
s.py:27] RULE_JOB_STOP_SIGNAL_FILENAME: None
INFO:sagemaker-containers:Imported framework sagemaker_xgboost_container.training
INFO:sagemaker-containers:Failed to parse hyperparameter _tuning_objective_metric value validation:auc to Json.
Returning the value itself
INFO:sagemaker-containers:Failed to parse hyperparameter eval_metric value auc to Json.
Returning the value itself
INFO:sagemaker-containers:Failed to parse hyperparameter objective value binary:logistic to Json.
Returning the value itself
INFO:sagemaker-containers:No GPUs detected (normal if no gpus installed)
INFO:sagemaker_xgboost_container.training:Running XGBoost Sagemaker in algorithm mode
INFO:root:Determined delimiter of CSV input is ','
INFO:root:Determined delimiter of CSV input is ','
INFO:root:Determined delimiter of CSV input is ','
INFO:root:Determined delimiter of CSV input is ','
INFO:root:Single node training.
Simple 1 conda_python3 | Idle Fully initialized
Mode: Edit Ln 1, Col 24 Flight_Delay-Student.ipynb 1


Course Modules: AWS Aca
Challenge lab
Notebook instances | Ama
Flight_Delay... - JupyterLab
Lab 5 - Guided Lab: Facial
Lab 6 - Amazon Lex - Crea
my-flight-notebook-hiau.notebook.us-east-1.sagemaker.aws/lab/tree/en_us/Flight_Delay-Student.ipynb
School
File Edit View Run Kernel Git Tabs Settings Help
Flight_Delay-Student.ipynb
Code
Open in...
conda_python3
/ en_us /
Name
Modified
Flight_Delay-Stud... now
jupyterlab.png 5y ago
linear_learner_mni... 8m ago
PythonCheatSheet... 8m ago
4 0
Plot a confusion matrix for your target_predicted and test_labels.
[88]:
# Enter your code here
plot_confusion_matrix(test_labels, target_predicted)
Confusion Matrix
127647.00
1579.00
30082.00
4251.00
True Class
0
1
Predicted Class
0
1
Question: Try different hyperparameters and hyperparameter ranges. Do these changes improve the model?
Simple 1 conda_python3 | Idle Fully initialized
Mode: Edit Ln 1, Col 23 Flight_Delay-Student.ipynb 1



Canvas Details



JOSE RIZAL UNIVERSITY

COLLEGE OF COMPUTER STUDIES AND ENGINEERING

COMPUTER ENGINEERING DEPARTMENT

CPE C313

ARTIFICIAL INTELLIGENCE

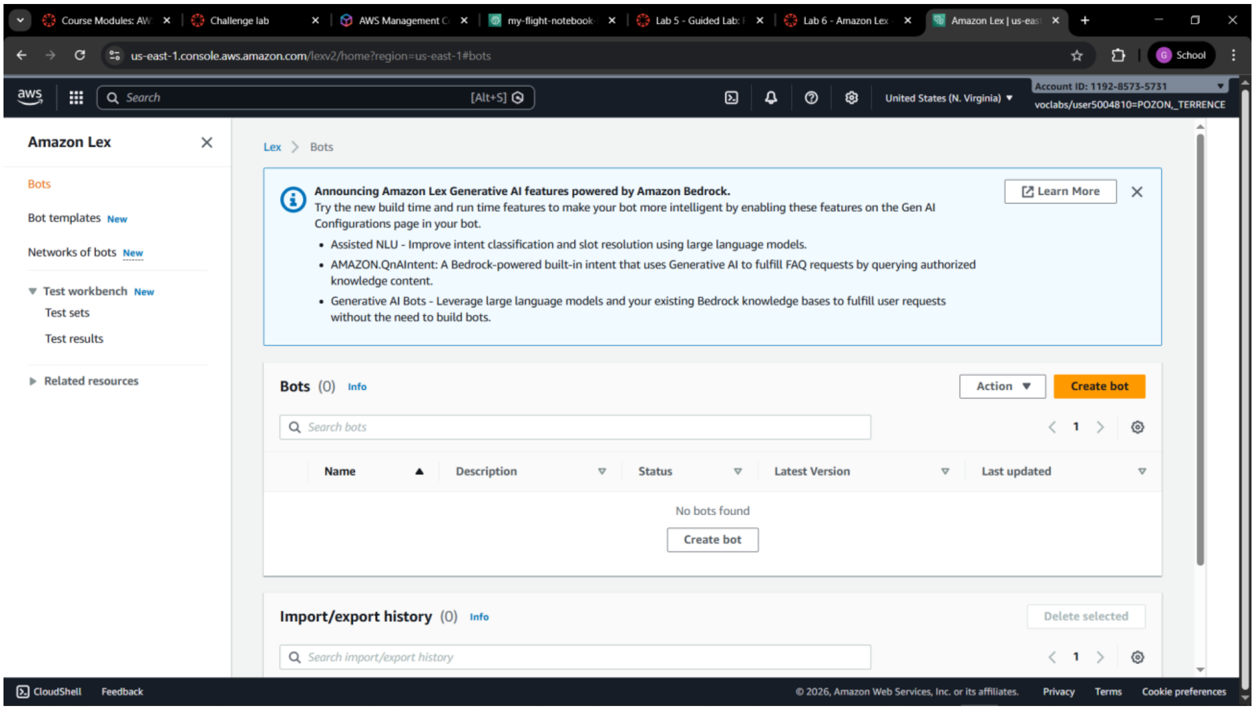
<u>LABORATORY</u>

6

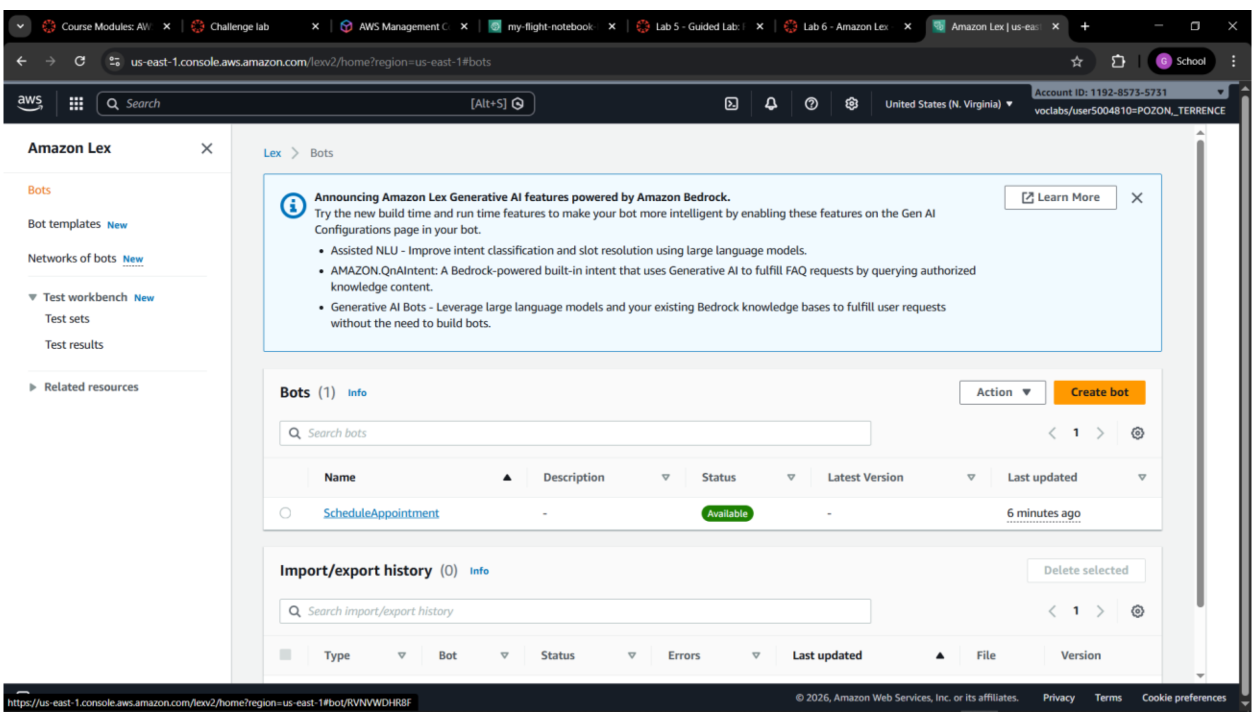
Pozon, Gion Terrence B.
301G
2026-04-29

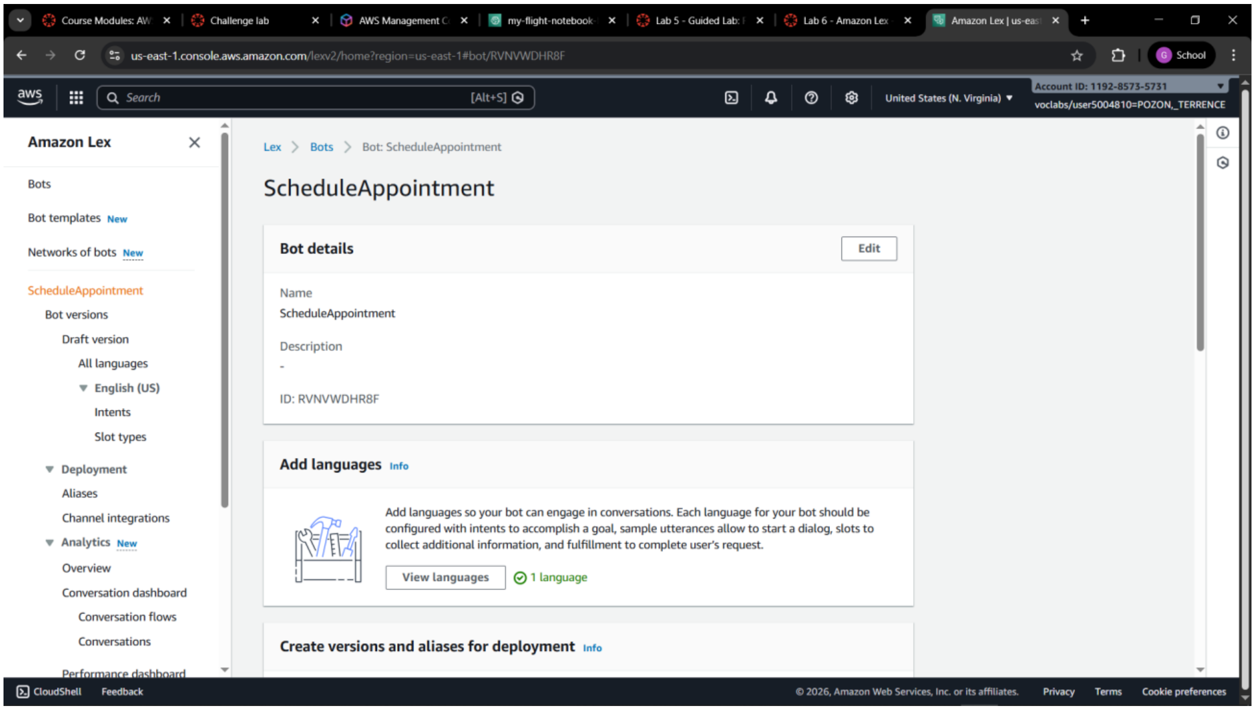
Lab 6: Amazon Lex

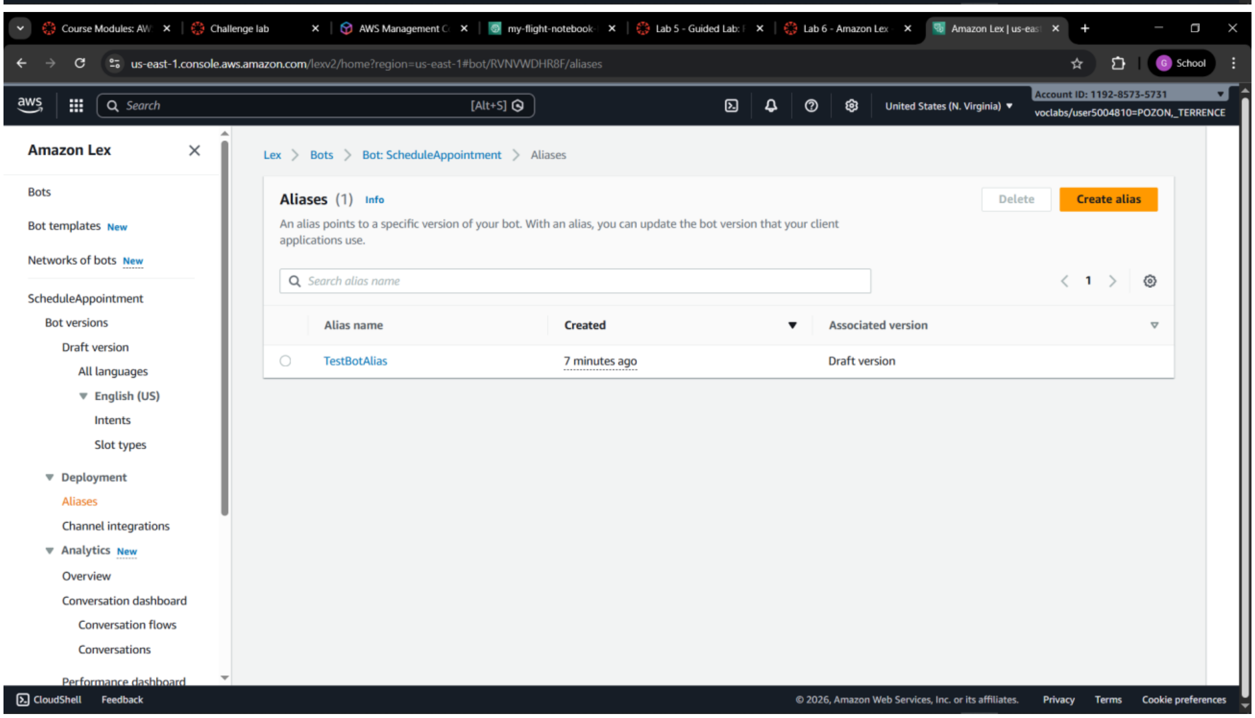
AWS Console account

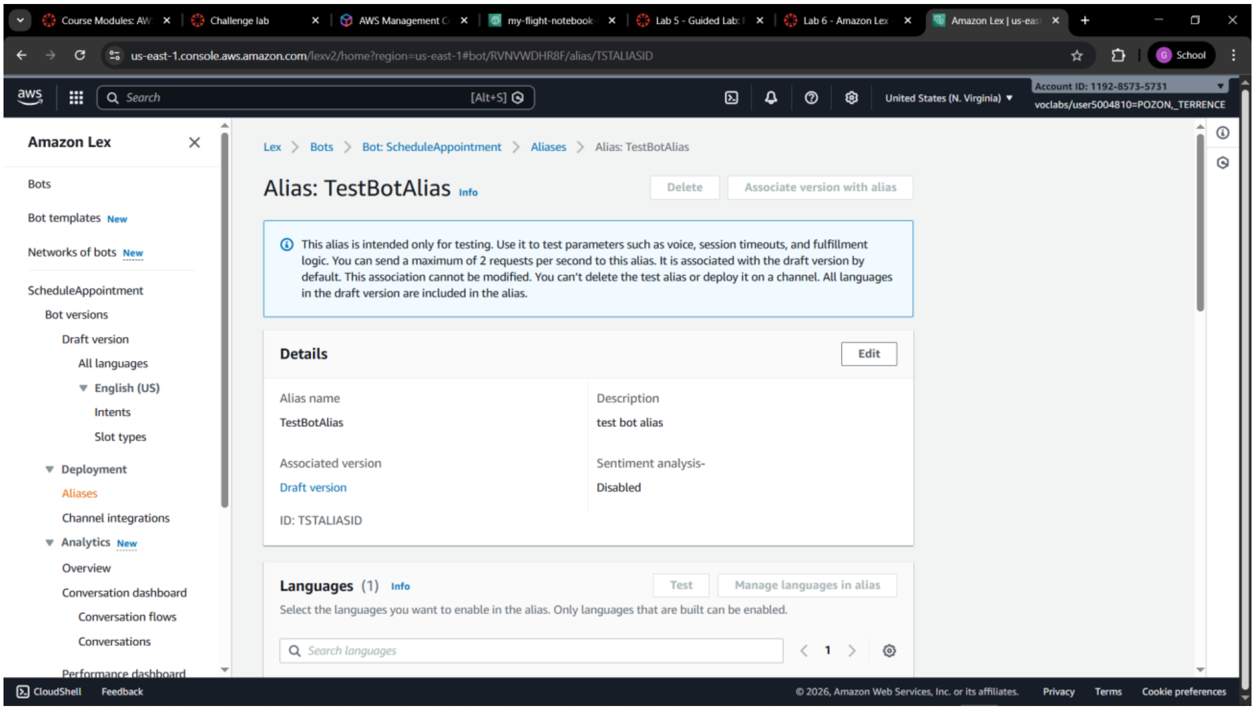


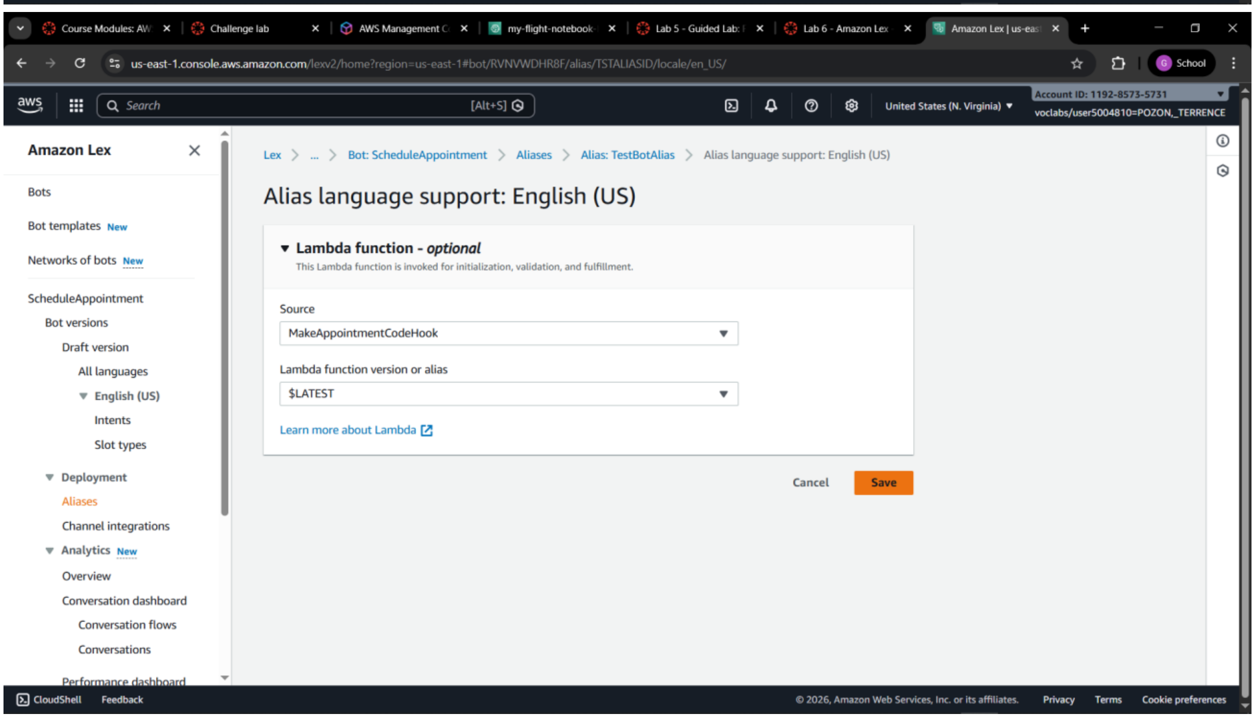
Task 3

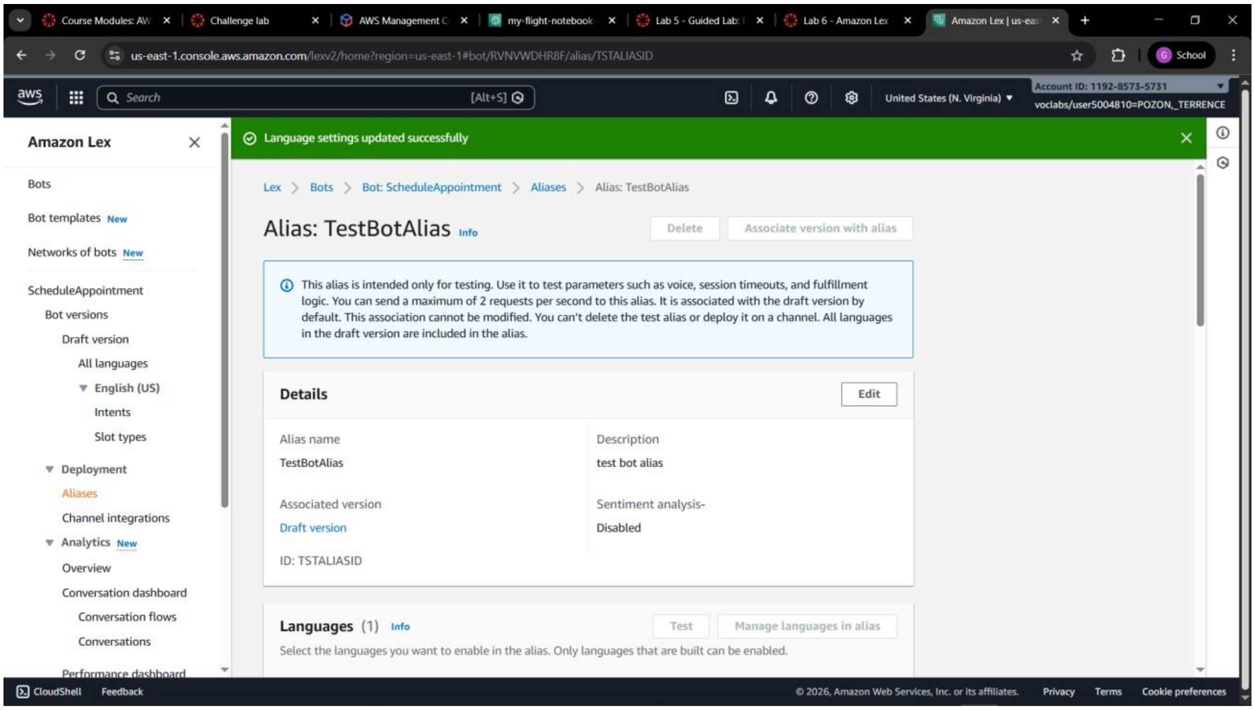



us-east-1.console.aws.amazon.com/lexv2/home?region=us-east-1#bot/RVNVWDHR8F
Amazon Lex
Bots
Bot templates New
Networks of bots New
ScheduleAppointment
Bot versions
Draft version
All languages
English (US)
Intents
Slot types
Deployment
Aliases
Channel integrations
Analytics New
Overview
Conversation dashboard
Conversation flows
Conversations
Performance dashboard
Lex > Bots > Bot: ScheduleAppointment
ScheduleAppointment
Bot details
Edit
Name
ScheduleAppointment
Description
-
ID: RVNVWDHR8F
Add languages Info
Add languages so your bot can engage in conversations. Each language for your bot should be configured with intents to accomplish a goal, sample utterances allow to start a dialog, slots to collect additional information, and fulfillment to complete user's request.
View languages
1 language
Create versions and aliases for deployment Info
CloudShell
Feedback
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences


us-east-1.console.aws.amazon.com/lexv2/home?region=us-east-1#bot/RVNVWDHR8F/aliases
Amazon Lex
Bots
Bot templates New
Networks of bots New
ScheduleAppointment
Bot versions
Draft version
All languages
English (US)
Intents
Slot types
Deployment
Aliases
Channel integrations
Analytics New
Overview
Conversation dashboard
Conversation flows
Conversations
Performance dashboard
Lex > Bots > Bot: ScheduleAppointment > Aliases
Aliases (1) Info
Delete
Create alias
An alias points to a specific version of your bot. With an alias, you can update the bot version that your client applications use.
Search alias name
Alias name
Created
Associated version
TestBotAlias
7 minutes ago
Draft version
CloudShell
Feedback
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences


us-east-1.console.aws.amazon.com/lexv2/home?region=us-east-1#bot/RVNVWDHR8F/alias/TSTALIASID
Amazon Lex
Bots
Bot templates New
Networks of bots New
ScheduleAppointment
Bot versions
Draft version
All languages
English (US)
Intents
Slot types
Deployment
Aliases
Channel integrations
Analytics New
Overview
Conversation dashboard
Conversation flows
Conversations
Performance dashboard
Lex > Bots > Bot: ScheduleAppointment > Aliases > Alias: TestBotAlias
Alias: TestBotAlias Info
Delete
Associate version with alias
This alias is intended only for testing. Use it to test parameters such as voice, session timeouts, and fulfillment logic. You can send a maximum of 2 requests per second to this alias. It is associated with the draft version by default. This association cannot be modified. You can't delete the test alias or deploy it on a channel. All languages in the draft version are included in the alias.
Details
Edit
Alias name
TestBotAlias
Description
test bot alias
Associated version
Draft version
Sentiment analysis-
Disabled
ID: TSTALIASID
Languages (1) Info
Test
Manage languages in alias
Select the languages you want to enable in the alias. Only languages that are built can be enabled.
Search languages
CloudShell
Feedback
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences


us-east-1.console.aws.amazon.com/lexv2/home?region=us-east-1#bot/RVNVWDHR8F/alias/TSTALIASID/locale/en_US/
Amazon Lex
Lex > ... > Bot: ScheduleAppointment > Aliases > Alias: TestBotAlias > Alias language support: English (US)
Alias language support: English (US)
Lambda function - optional
This Lambda function is invoked for initialization, validation, and fulfillment.
Source
MakeAppointmentCodeHook
Lambda function version or alias
$LATEST
Learn more about Lambda
Cancel
Save


Language settings updated successfully
Lex > Bots > Bot: ScheduleAppointment > Aliases > Alias: TestBotAlias
Alias: TestBotAlias Info
Delete
Associate version with alias
This alias is intended only for testing. Use it to test parameters such as voice, session timeouts, and fulfillment logic. You can send a maximum of 2 requests per second to this alias. It is associated with the draft version by default. This association cannot be modified. You can't delete the test alias or deploy it on a channel. All languages in the draft version are included in the alias.
Details
Edit
Alias name
TestBotAlias
Description
test bot alias
Associated version
Draft version
Sentiment analysis-
Disabled
ID: TSTALIASID
Languages (1) Info
Test
Manage languages in alias
Select the languages you want to enable in the alias. Only languages that are built can be enabled.
Amazon Lex
Bots
Bot templates New
Networks of bots New
ScheduleAppointment
Bot versions
Draft version
All languages
English (US)
Intents
Slot types
Deployment
Aliases
Channel integrations
Analytics New
Overview
Conversation dashboard
Conversation flows
Conversations
Performance dashboard
us-east-1.console.aws.amazon.com/lexv2/home?region=us-east-1#bot/RVNVWDHR8F/alias/TSTALIASID
United States (N. Virginia)
Account ID: 1192-8573-5731
voclabs/user5004810=POZON,_TERRENCE
CloudShell
Feedback
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Privacy
Terms
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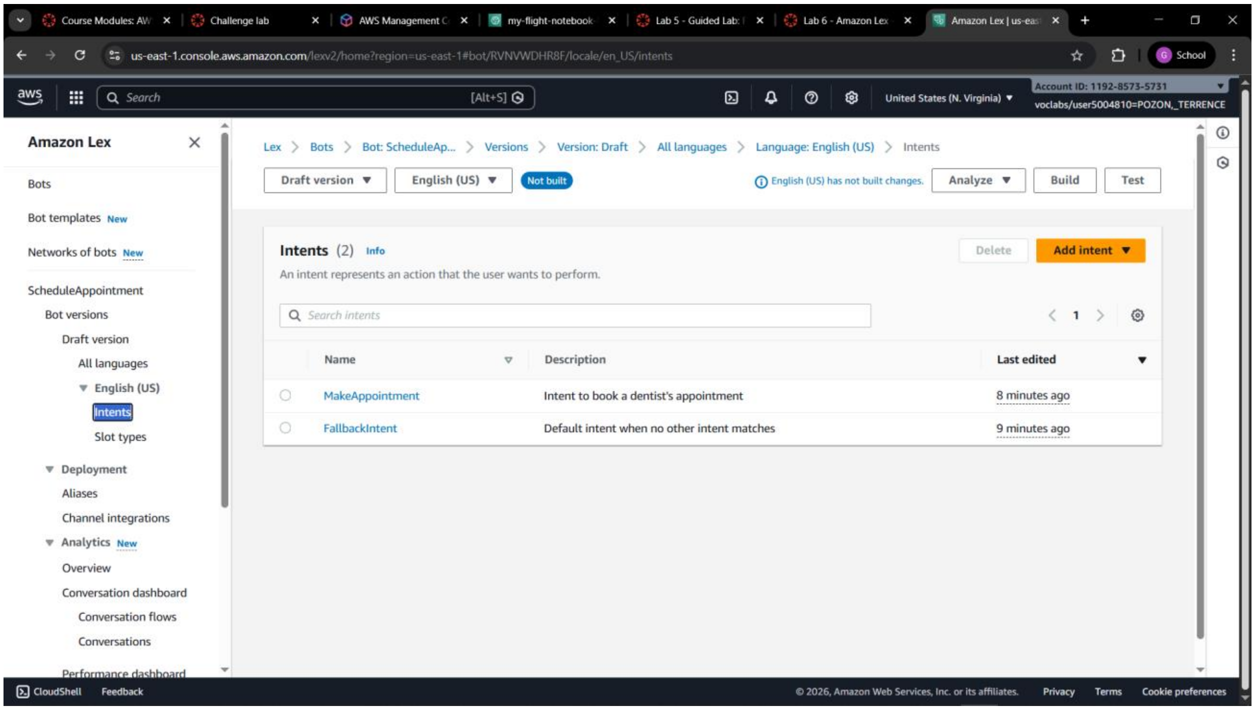
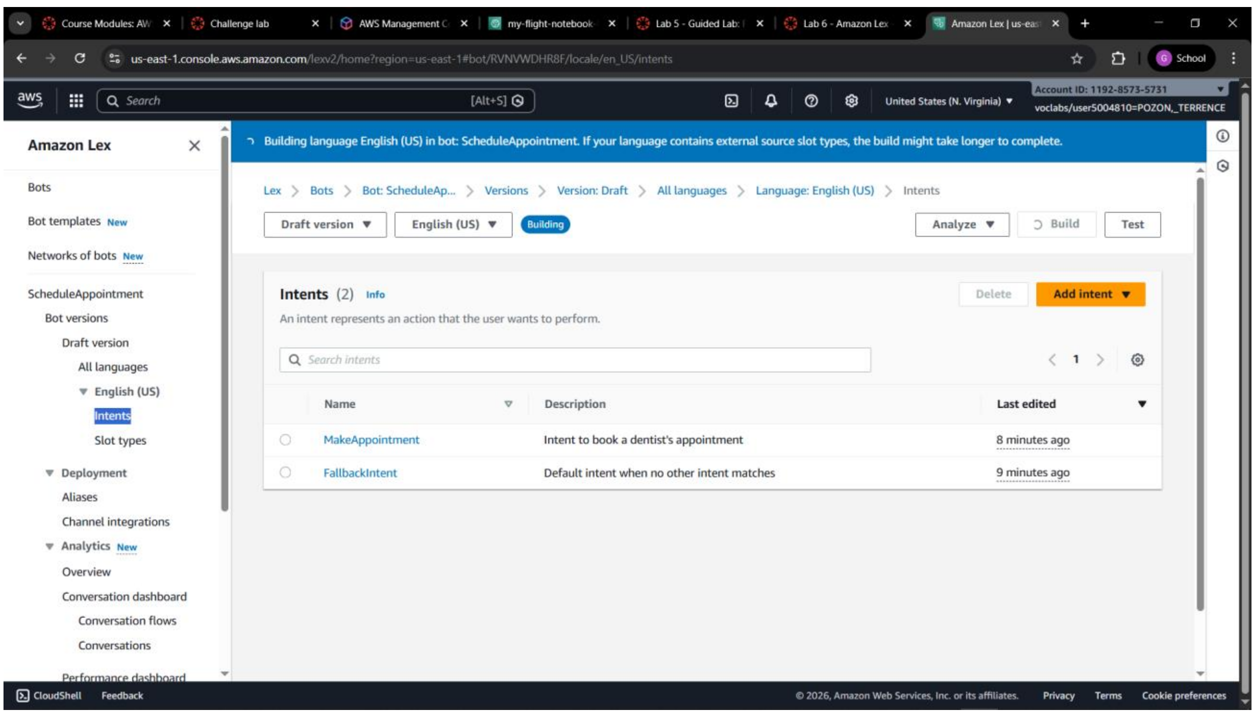
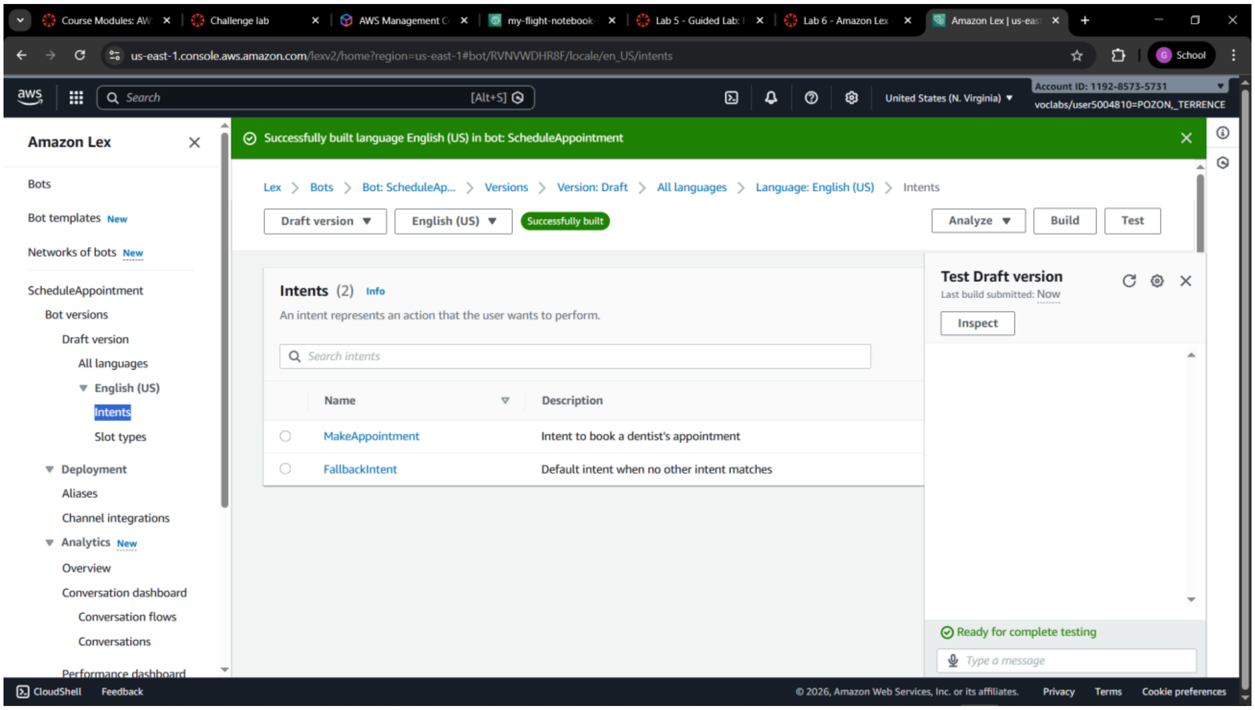
Task 4



Amazon Lex
Lex > Bots > Bot: ScheduleAp... > Versions > Version: Draft > All languages > Language: English (US) > Intents
Draft version
English (US)
Not built
English (US) has not built changes.
Analyze
Build
Test
Intents (2) Info
An intent represents an action that the user wants to perform.
Delete
Add intent
Search intents
Name
Description
Last edited
MakeAppointment
Intent to book a dentist's appointment
8 minutes ago
FallbackIntent
Default intent when no other intent matches
9 minutes ago



Building language English (US) in bot: ScheduleAppointment. If your language contains external source slot types, the build might take longer to complete.
Amazon Lex
Lex > Bots > Bot: ScheduleAp... > Versions > Version: Draft > All languages > Language: English (US) > Intents
Draft version
English (US)
Building
Analyze
Build
Test
Intents (2) Info
An intent represents an action that the user wants to perform.
Delete
Add intent
Search intents
Name
Description
Last edited
MakeAppointment
Intent to book a dentist's appointment
8 minutes ago
FallbackIntent
Default intent when no other intent matches
9 minutes ago


Amazon Lex
Successfully built language English (US) in bot: ScheduleAppointment
Lex > Bots > Bot: ScheduleAp... > Versions > Version: Draft > All languages > Language: English (US) > Intents
Draft version
English (US)
Successfully built
Analyze
Build
Test
Bots
Bot templates New
Networks of bots New
ScheduleAppointment
Bot versions
Draft version
All languages
English (US)
Intents
Slot types
Deployment
Aliases
Channel integrations
Analytics New
Overview
Conversation dashboard
Conversation flows
Conversations
Performance dashboard
Intents (2) Info
An intent represents an action that the user wants to perform.
Search intents
Name
Description
MakeAppointment
Intent to book a dentist's appointment
FallbackIntent
Default intent when no other intent matches
Test Draft version
Last build submitted: Now
Inspect
Ready for complete testing
Type a message
CloudShell
Feedback
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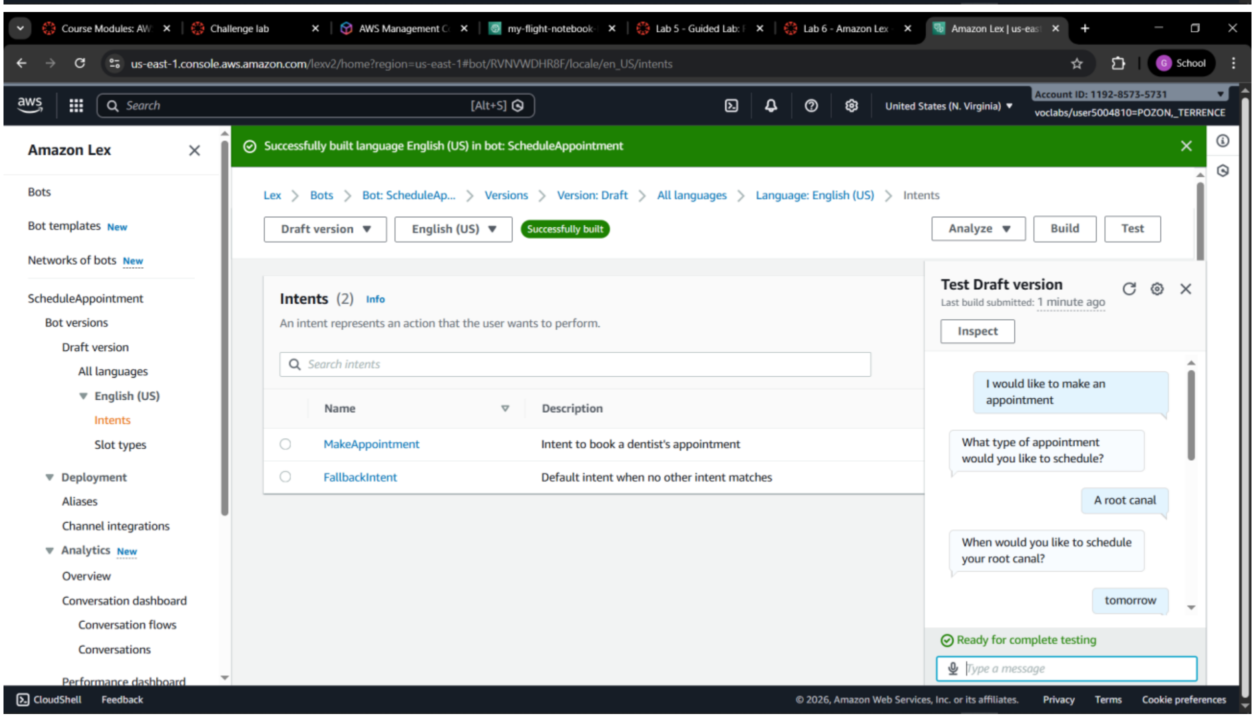
Privacy
Terms
Cookie preferences


Amazon Lex
Successfully built language English (US) in bot: ScheduleAppointment
Lex > Bots > Bot: ScheduleAp... > Versions > Version: Draft > All languages > Language: English (US) > Intents
Draft version
English (US)
Successfully built
Analyze
Build
Test
Intents (2) Info
An intent represents an action that the user wants to perform.
Search intents
Name
Description
MakeAppointment
Intent to book a dentist's appointment
FallbackIntent
Default intent when no other intent matches
Test Draft version
Last build submitted: 1 minute ago
Inspect
I would like to make an appointment
What type of appointment would you like to schedule?
A root canal
When would you like to schedule your root canal?
tomorrow
Ready for complete testing
Type a message
CloudShell
Feedback
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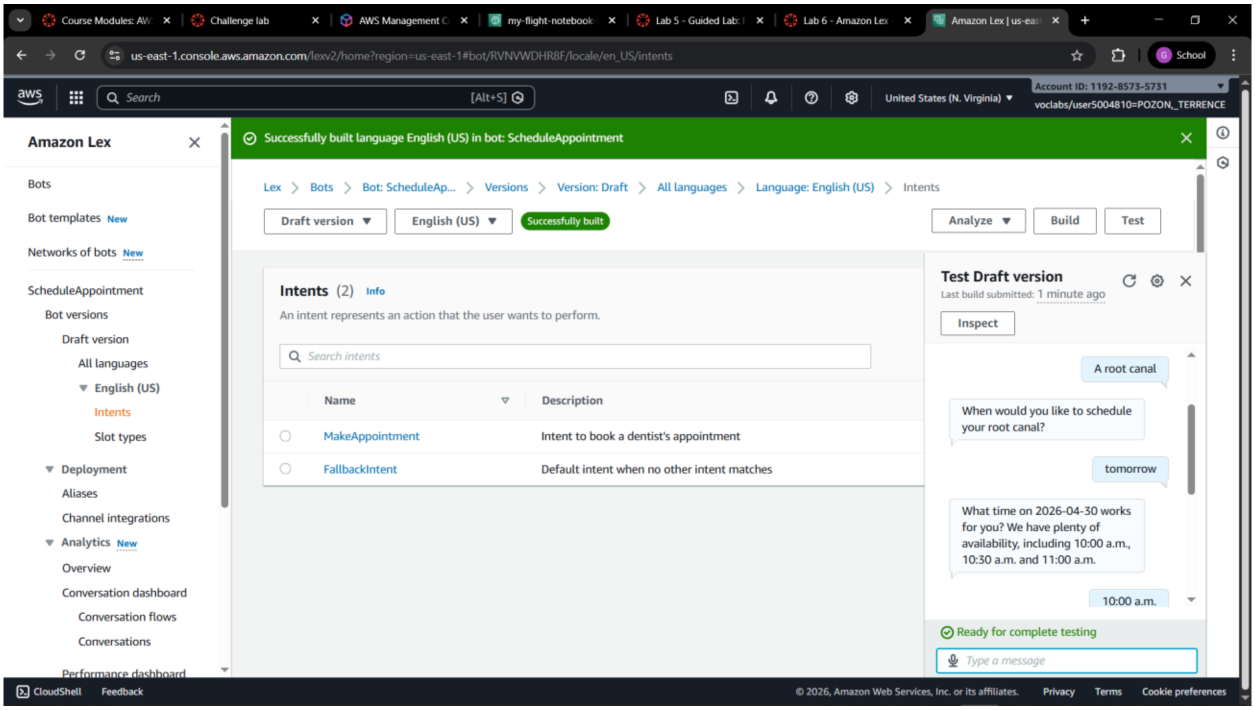
Cookie preferences


us-east-1.console.aws.amazon.com/lexv2/home?region=us-east-1#bot/RVNVWDHR8F/locale/en_US/intents
Account ID: 1192-8573-5731
voclabs/user5004810=POZON,_TERRENCE
Amazon Lex
Successfully built language English (US) in bot: ScheduleAppointment
Lex > Bots > Bot: ScheduleAp... > Versions > Version: Draft > All languages > Language: English (US) > Intents
Draft version
English (US)
Successfully built
Analyze
Build
Test
Bots
Bot templates New
Networks of bots New
ScheduleAppointment
Bot versions
Draft version
All languages
English (US)
Intents
Slot types
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Channel integrations
Analytics New
Overview
Conversation dashboard
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Conversations
Performance dashboard
Intents (2) Info
An intent represents an action that the user wants to perform.
Search intents
Name
Description
MakeAppointment
Intent to book a dentist's appointment
FallbackIntent
Default intent when no other intent matches
Test Draft version
Last build submitted: 1 minute ago
Inspect
A root canal
When would you like to schedule your root canal?
tomorrow
What time on 2026-04-30 works for you? We have plenty of availability, including 10:00 a.m., 10:30 a.m. and 11:00 a.m.
10:00 a.m.
Ready for complete testing
Type a message
CloudShell
Feedback
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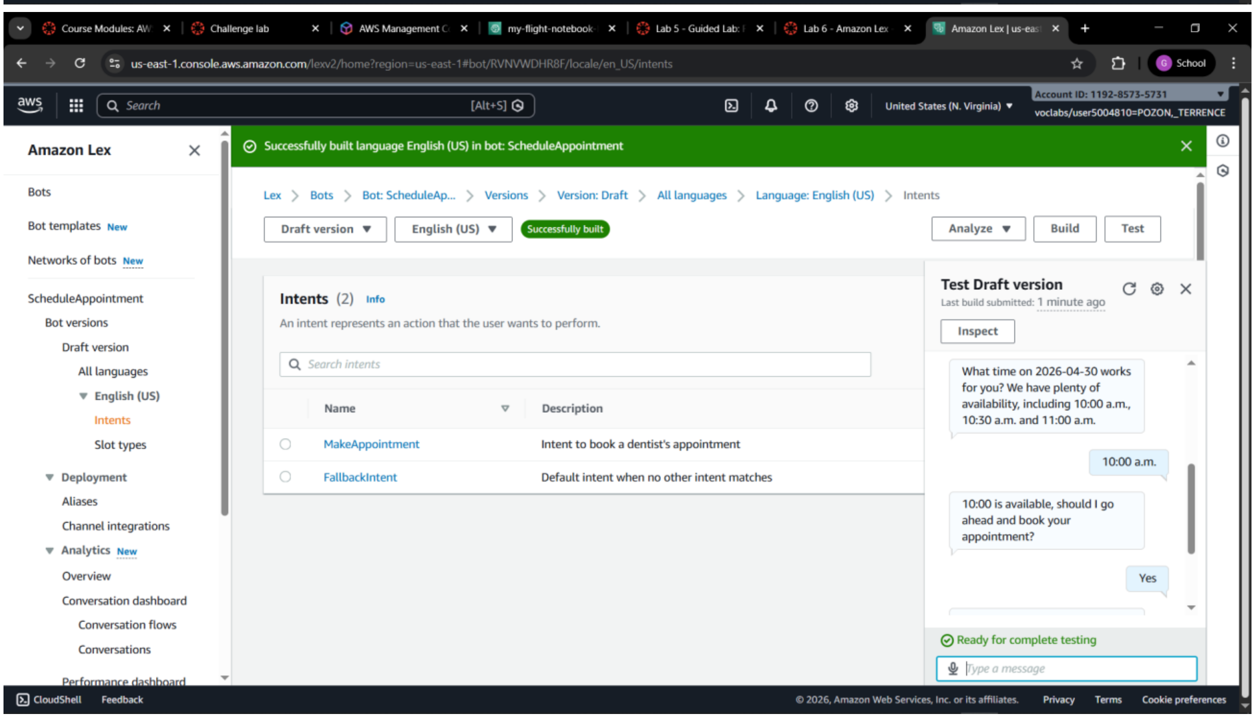
Terms
Cookie preferences


us-east-1.console.aws.amazon.com/lexv2/home?region=us-east-1#bot/RVNVWDHR8F/locale/en_US/intents
Account ID: 1192-8573-5731
voclabs/user5004810=POZON,_TERRENCE
Amazon Lex
Successfully built language English (US) in bot: ScheduleAppointment
Lex > Bots > Bot: ScheduleAp... > Versions > Version: Draft > All languages > Language: English (US) > Intents
Draft version
English (US)
Successfully built
Analyze
Build
Test
Intents (2) Info
An intent represents an action that the user wants to perform.
Search intents
Name
Description
MakeAppointment
Intent to book a dentist's appointment
FallbackIntent
Default intent when no other intent matches
Test Draft version
Last build submitted: 1 minute ago
Inspect
What time on 2026-04-30 works for you? We have plenty of availability, including 10:00 a.m., 10:30 a.m. and 11:00 a.m.
10:00 a.m.
10:00 is available, should I go ahead and book your appointment?
Yes
Ready for complete testing
Type a message
CloudShell
Feedback
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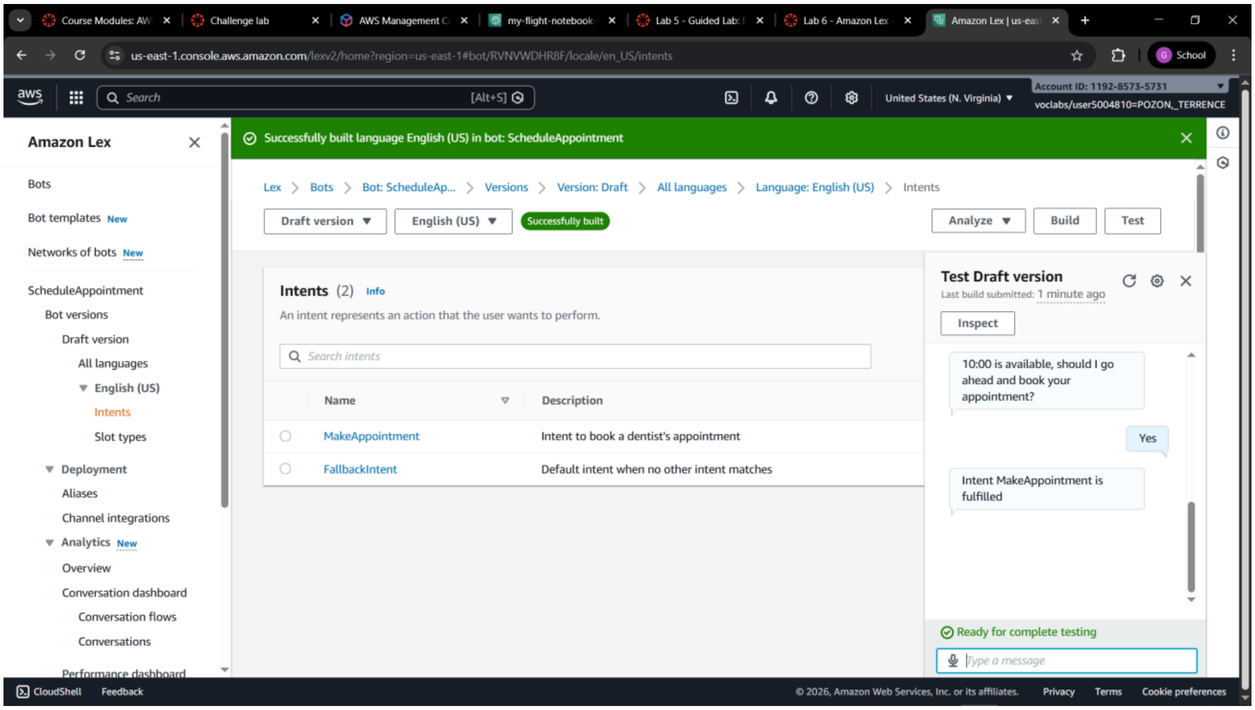
Course Modules: AW
Challenge lab
AWS Management C
my-flight-notebook
Lab 5 - Guided Lab
Lab 6 - Amazon Lex
Amazon Lex | us-eas
us-east-1.console.aws.amazon.com/lexv2/home?region=us-east-1#bot/RVNVWDHR8F/locale/en_US/intents
School
Search
[Alt+S]
United States (N. Virginia)
Account ID: 1192-8573-5731
voclabs/user5004810=POZON,_TERRENCE
Amazon Lex
Bots
Bot templates New
Networks of bots New
ScheduleAppointment
Bot versions
Draft version
All languages
English (US)
Intents
Slot types
Deployment
Aliases
Channel integrations
Analytics New
Overview
Conversation dashboard
Conversation flows
Conversations
Performance dashboard
Successfully built language English (US) in bot: ScheduleAppointment
Lex > Bots > Bot: ScheduleAp... > Versions > Version: Draft > All languages > Language: English (US) > Intents
Draft version
English (US)
Successfully built
Analyze
Build
Test
Intents (2) Info
An intent represents an action that the user wants to perform.
Search intents
Name
Description
MakeAppointment
Intent to book a dentist's appointment
FallbackIntent
Default intent when no other intent matches
Test Draft version
Last build submitted: 1 minute ago
Inspect
10:00 is available, should I go ahead and book your appointment?
Yes
Intent MakeAppointment is fulfilled
Ready for complete testing
Type a message
CloudShell
Feedback
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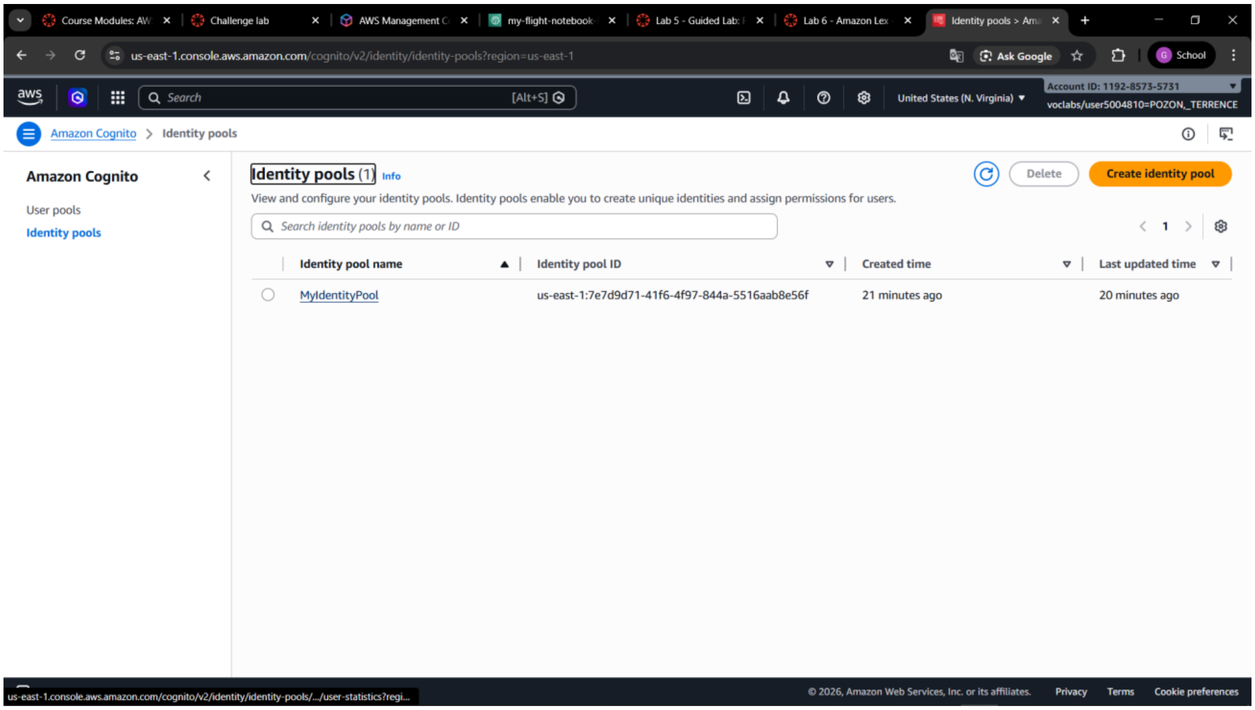
Task 5


us-east-1.console.aws.amazon.com/cognito/v2/identity/identity-pools?region=us-east-1
Account ID: 1192-8573-5731
voclabs/user5004810=POZON,_TERRENCE
United States (N. Virginia)
Amazon Cognito > Identity pools
Amazon Cognito
User pools
Identity pools
Identity pools (1) Info
Delete
Create identity pool
View and configure your identity pools. Identity pools enable you to create unique identities and assign permissions for users.
Search identity pools by name or ID
Identity pool name
Identity pool ID
Created time
Last updated time
MyIdentityPool
us-east-1:7e7d9d71-41f6-4f97-844a-5516aab8e56f
21 minutes ago
20 minutes ago
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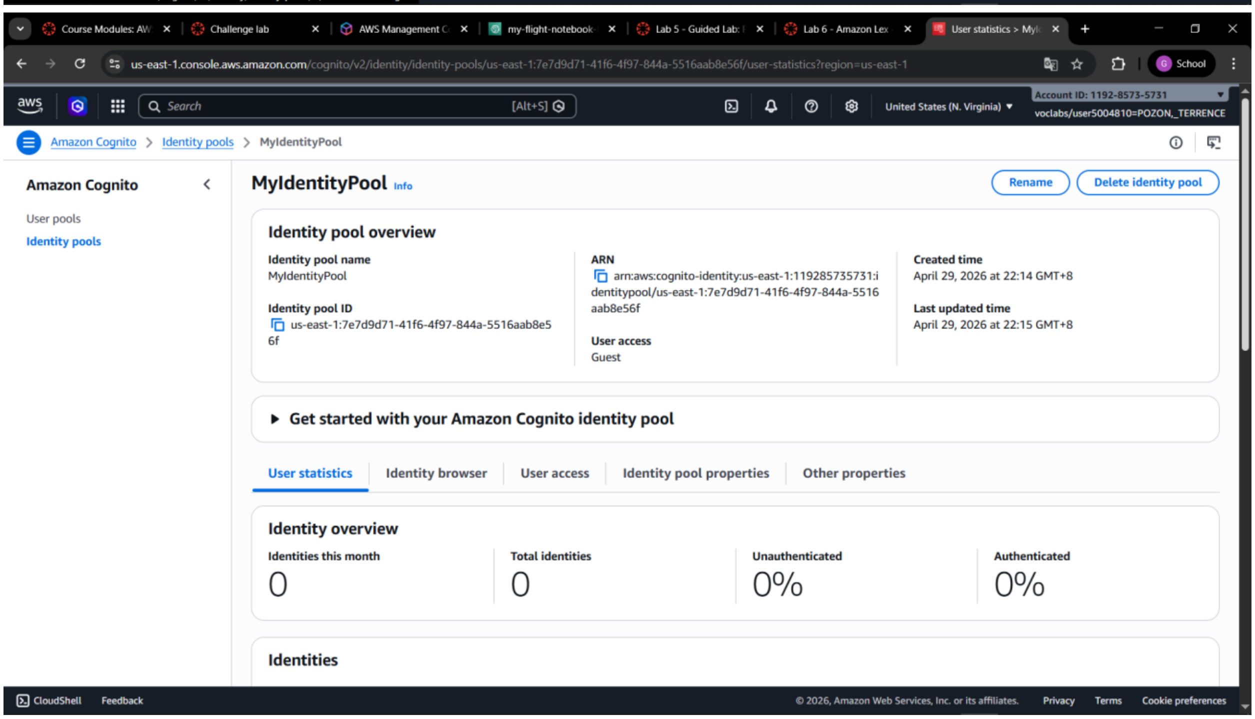
us-east-1.console.aws.amazon.com/cognito/v2/identity/identity-pools/us-east-1:7e7d9d71-41f6-4f97-844a-5516aab8e56f/user-statistics?region=us-east-1
Account ID: 1192-8573-5731
voclabs/user5004810=POZON,_TERRENCE
Amazon Cognito > Identity pools > MyIdentityPool
MyIdentityPool Info
Rename
Delete identity pool
Identity pool overview
Identity pool name
MyIdentityPool
Identity pool ID
us-east-1:7e7d9d71-41f6-4f97-844a-5516aab8e56f
ARN
arn:aws:cognito-identity:us-east-1:119285735731:identitypool/us-east-1:7e7d9d71-41f6-4f97-844a-5516aab8e56f
User access
Guest
Created time
April 29, 2026 at 22:14 GMT+8
Last updated time
April 29, 2026 at 22:15 GMT+8
Get started with your Amazon Cognito identity pool
User statistics
Identity browser
User access
Identity pool properties
Other properties
Identity overview
Identities this month
0
Total identities
0
Unauthenticated
0%
Authenticated
0%
Identities
CloudShell
Feedback
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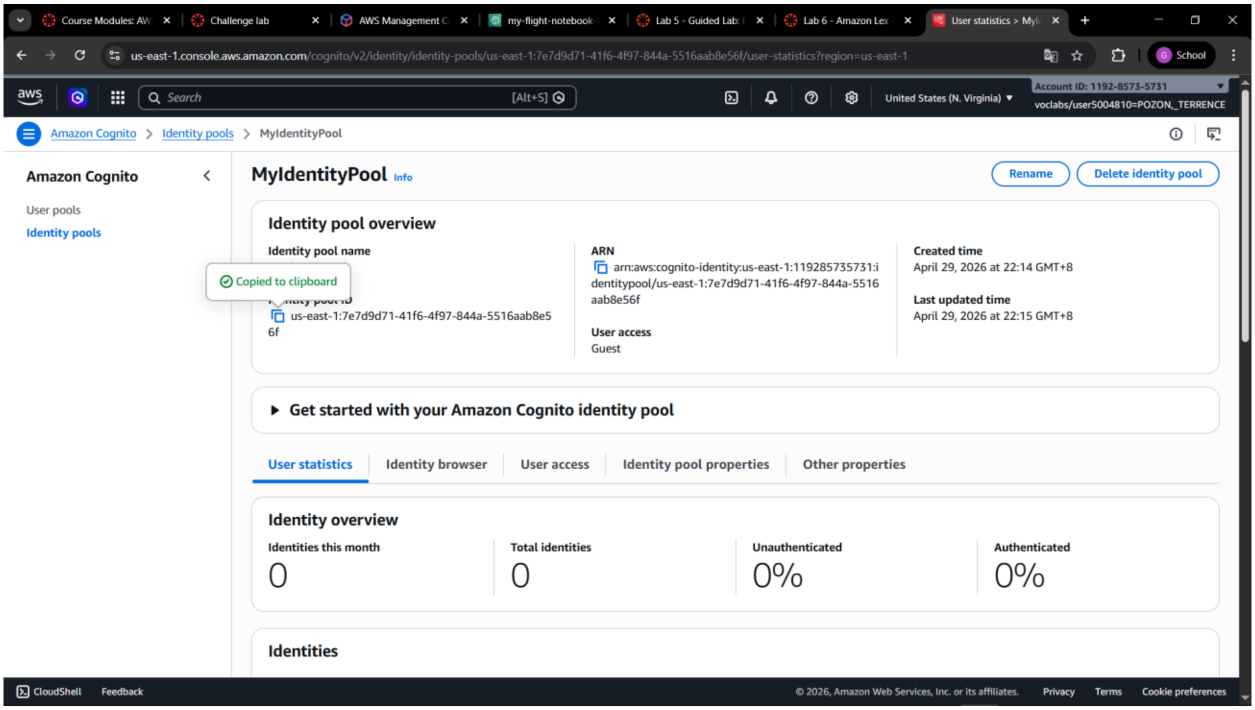
Terms
Cookie preferences


Course Modules: AW
Challenge lab
AWS Management C
my-flight-notebook
Lab 5 - Guided Lab
Lab 6 - Amazon Lex
User statistics > My
us-east-1.console.aws.amazon.com/cognito/v2/identity/identity-pools/us-east-1:7e7d9d71-41f6-4f97-844a-5516aab8e56f/user-statistics?region=us-east-1
School
Search
[Alt+S]
United States (N. Virginia)
Account ID: 1192-8573-5731
voclabs/user5004810=POZON,_TERRENCE
Amazon Cognito
Identity pools
MyIdentityPool
Amazon Cognito
User pools
Identity pools
MyIdentityPool
Info
Rename
Delete identity pool
Identity pool overview
Identity pool name
Copied to clipboard
us-east-1:7e7d9d71-41f6-4f97-844a-5516aab8e56f
ARN
arn:aws:cognito-identity:us-east-1:119285735731:identitypool/us-east-1:7e7d9d71-41f6-4f97-844a-5516aab8e56f
User access
Guest
Created time
April 29, 2026 at 22:14 GMT+8
Last updated time
April 29, 2026 at 22:15 GMT+8
Get started with your Amazon Cognito identity pool
User statistics
Identity browser
User access
Identity pool properties
Other properties
Identity overview
Identities this month
0
Total identities
0
Unauthenticated
0%
Authenticated
0%
Identities
CloudShell
Feedback
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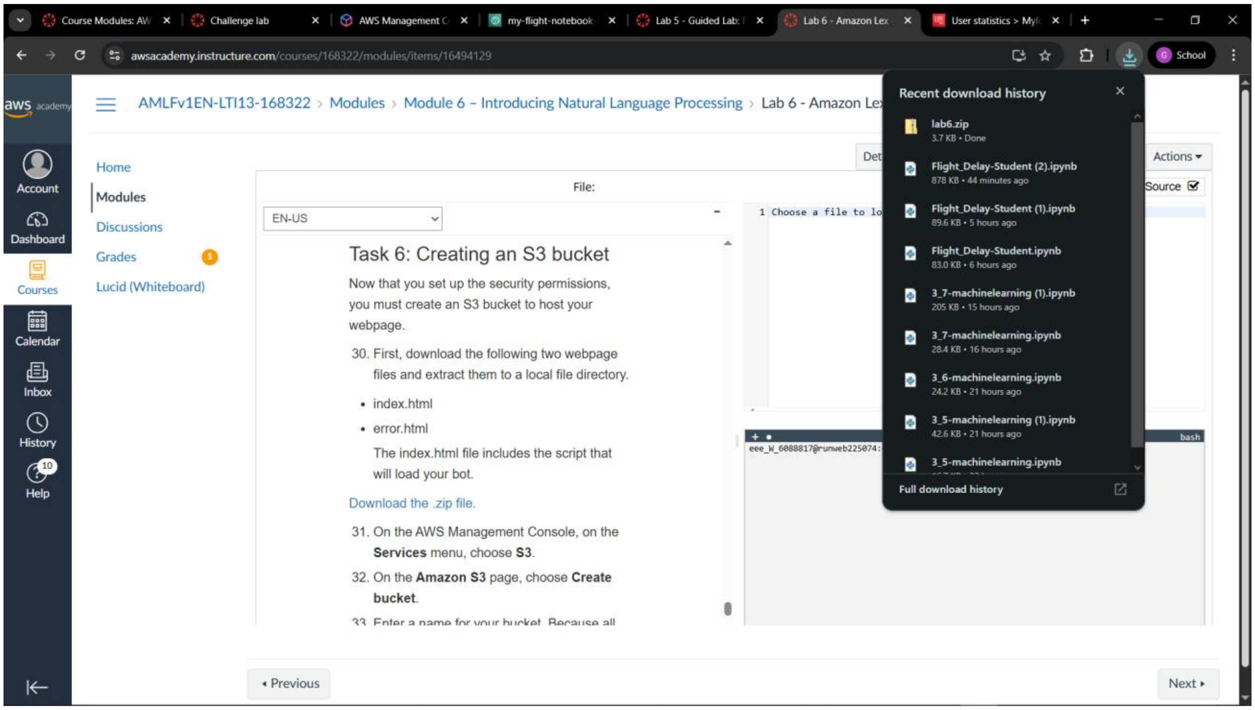
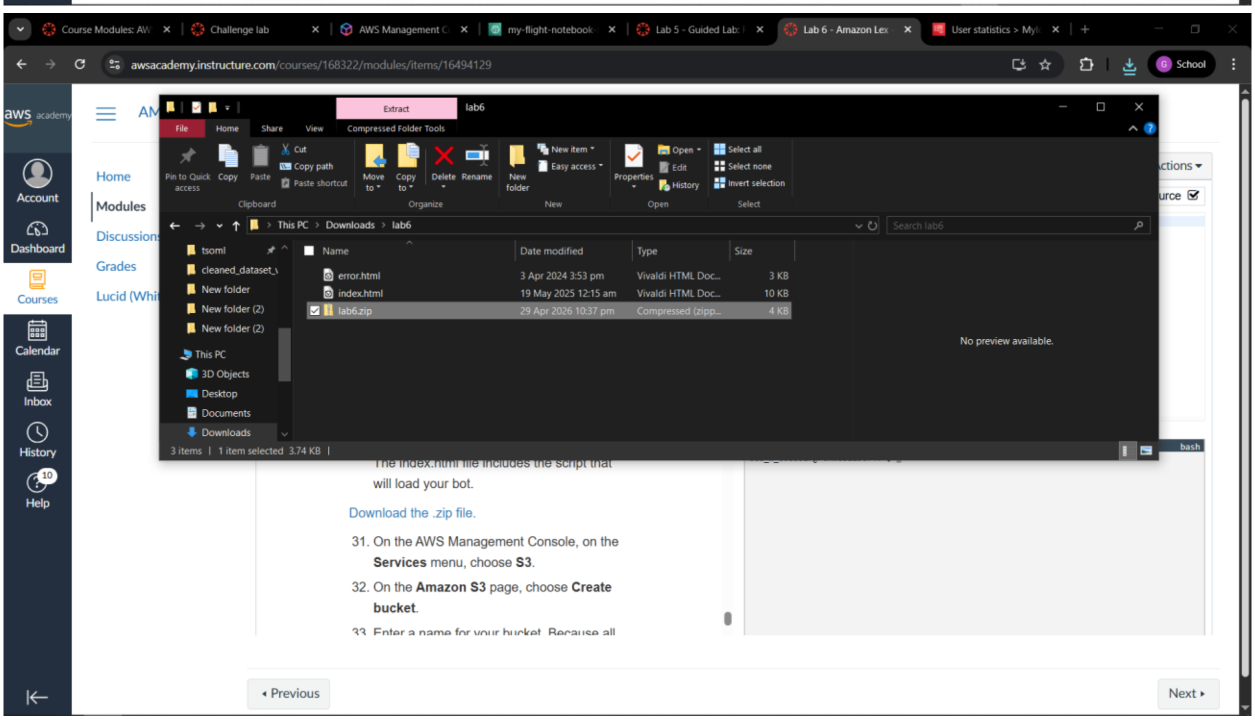
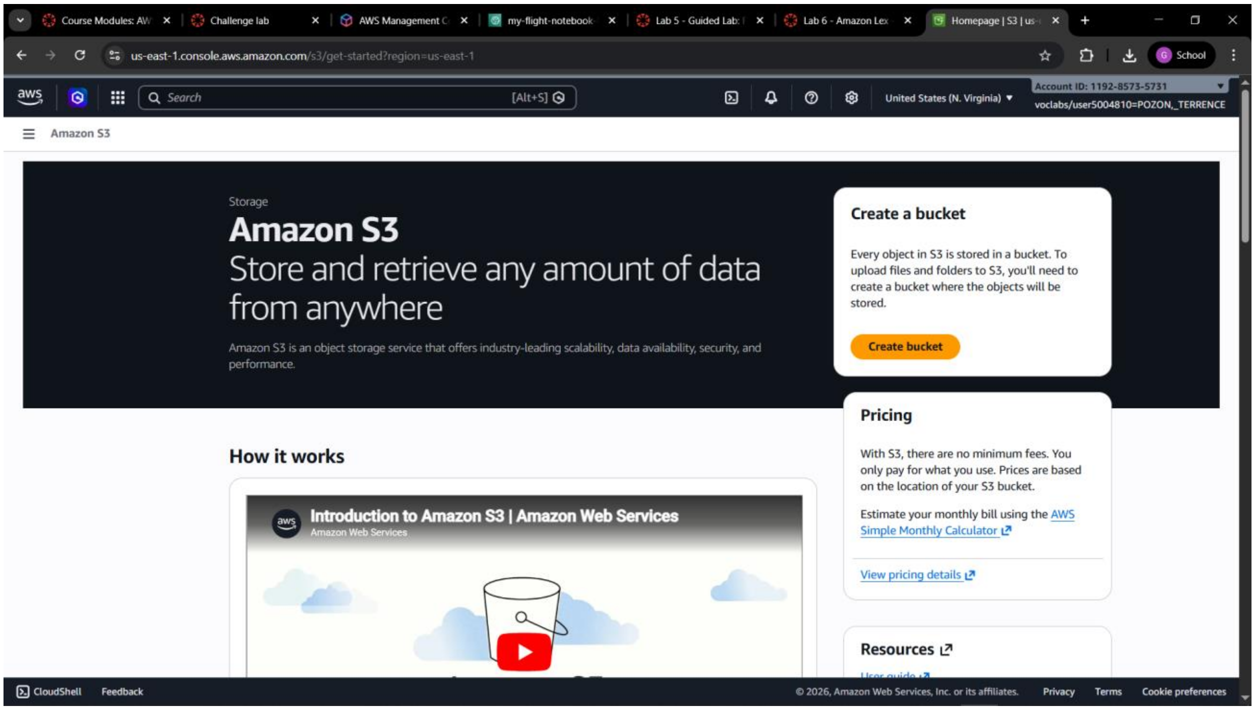
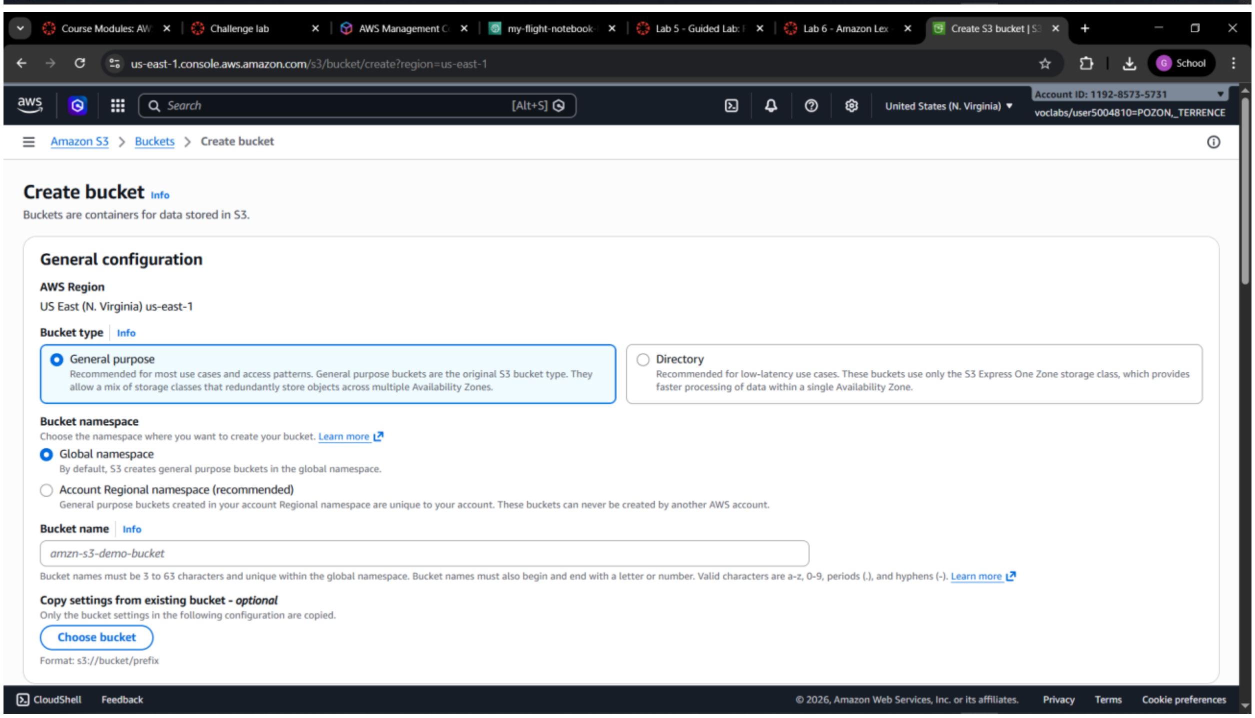
Task 6


awsacademy.instructure.com/courses/168322/modules/items/16494129
AMLFv1EN-LTI13-168322 › Modules › Module 6 – Introducing Natural Language Processing › Lab 6 - Amazon Lex
Home
Modules
Discussions
Grades
Lucid (Whiteboard)
Task 6: Creating an S3 bucket
Now that you set up the security permissions, you must create an S3 bucket to host your webpage.
30. First, download the following two webpage files and extract them to a local file directory.
index.html
error.html
The index.html file includes the script that will load your bot.
Download the .zip file.
31. On the AWS Management Console, on the Services menu, choose S3.
32. On the Amazon S3 page, choose Create bucket.
Recent download history
lab6.zip
3.7 KB • Done
Flight_Delay-Student (2).ipynb
Flight_Delay-Student (1).ipynb
Flight_Delay-Student.ipynb
3_7-machinelearning (1).ipynb
3_7-machinelearning.ipynb
3_6-machinelearning.ipynb
3_5-machinelearning (1).ipynb
3_5-machinelearning.ipynb
Full download history
Previous
Next

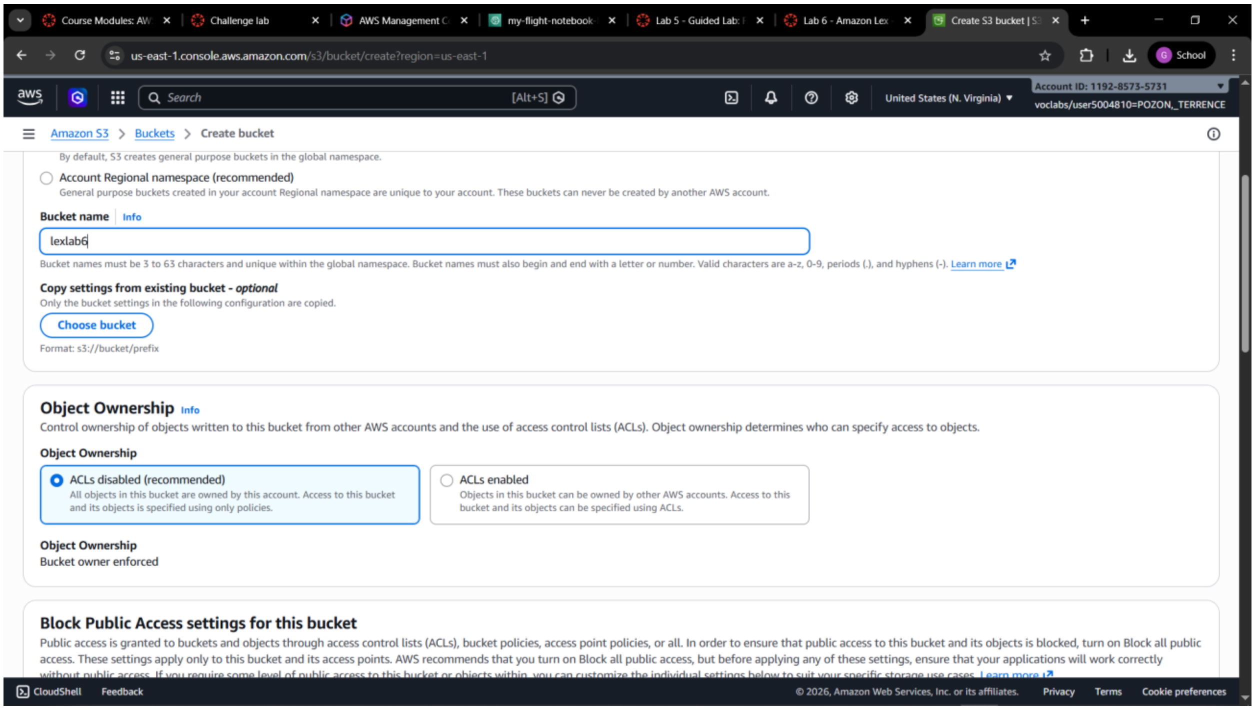

Extract
lab6
This PC › Downloads › lab6
Name
Date modified
Type
Size
error.html
3 Apr 2024 3:53 pm
Vivaldi HTML Doc...
3 KB
index.html
19 May 2025 12:15 am
Vivaldi HTML Doc...
10 KB
lab6.zip
29 Apr 2026 10:37 pm
Compressed (zipp...
4 KB
No preview available.
3 items | 1 item selected 3.74 KB
Download the .zip file.
31. On the AWS Management Console, on the Services menu, choose S3.
32. On the Amazon S3 page, choose Create bucket.
Previous
Next



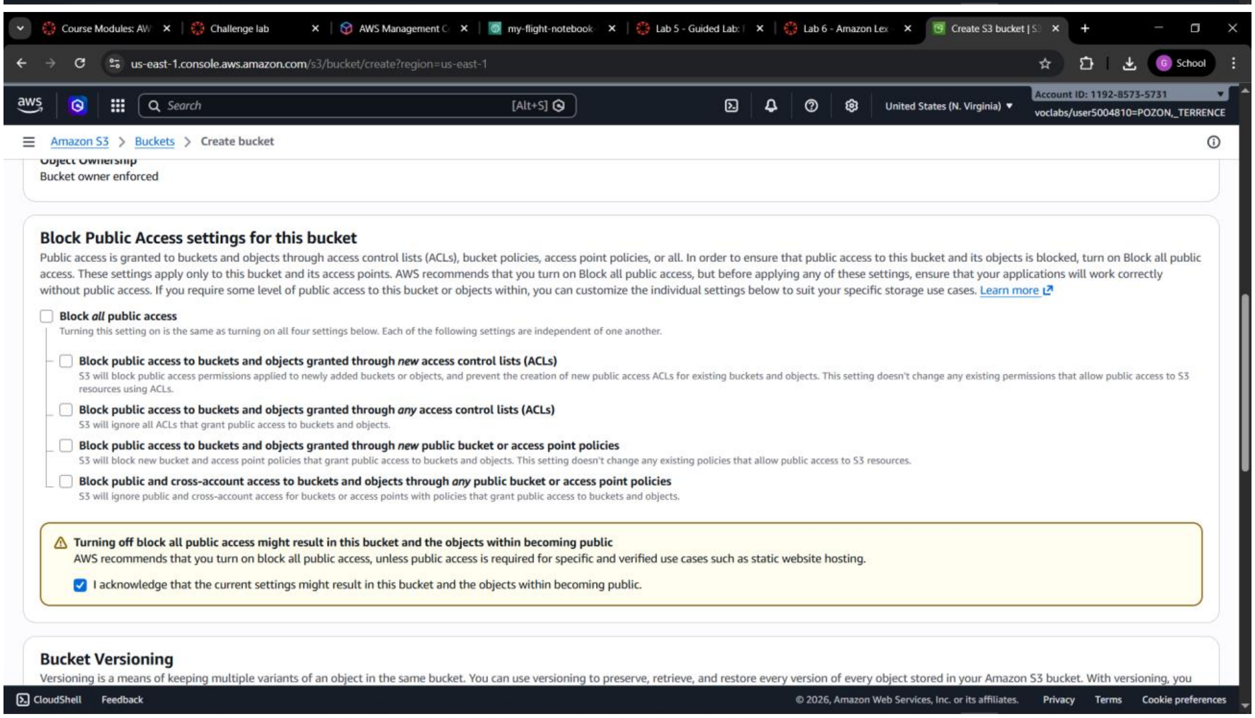
us-east-1.console.aws.amazon.com/s3/get-started?region=us-east-1
Amazon S3
Storage
Amazon S3
Store and retrieve any amount of data from anywhere
Amazon S3 is an object storage service that offers industry-leading scalability, data availability, security, and performance.
Create a bucket
Every object in S3 is stored in a bucket. To upload files and folders to S3, you'll need to create a bucket where the objects will be stored.
Create bucket
Pricing
With S3, there are no minimum fees. You only pay for what you use. Prices are based on the location of your S3 bucket.
Estimate your monthly bill using the AWS Simple Monthly Calculator
View pricing details
How it works
Introduction to Amazon S3 | Amazon Web Services
Resources



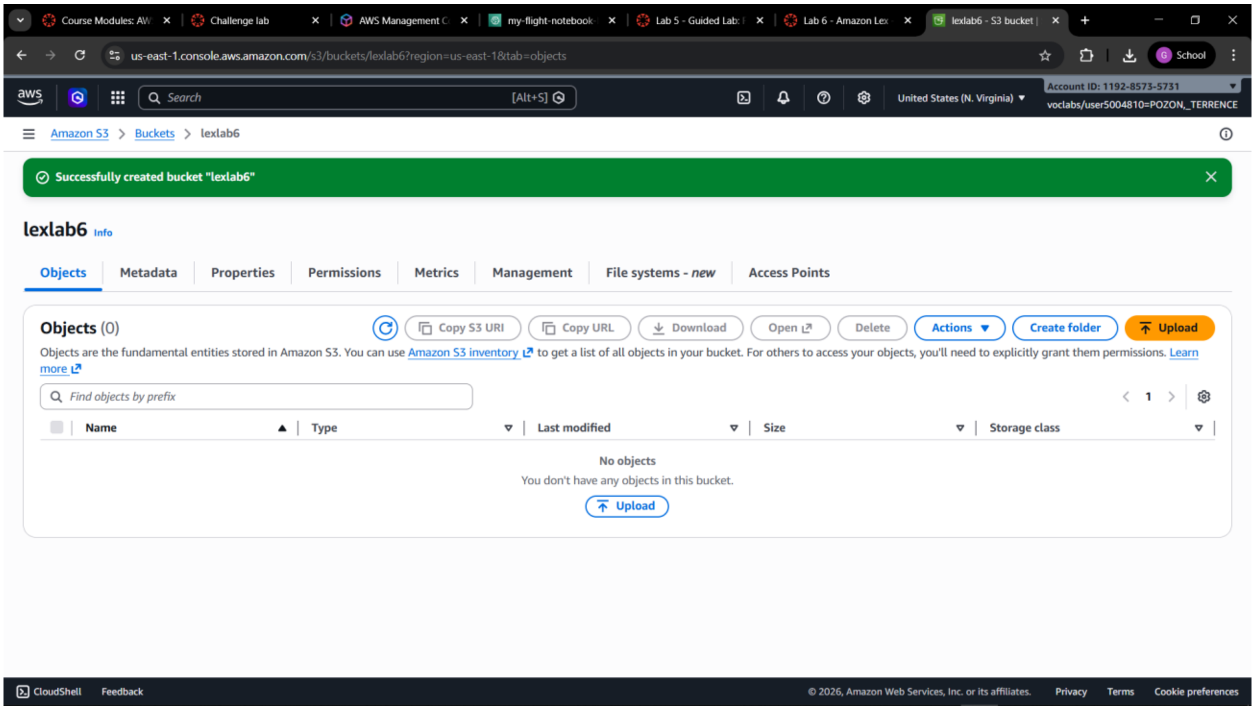
us-east-1.console.aws.amazon.com/s3/bucket/create?region=us-east-1
Amazon S3 > Buckets > Create bucket
Create bucket Info
Buckets are containers for data stored in S3.
General configuration
AWS Region
US East (N. Virginia) us-east-1
Bucket type Info
General purpose
Recommended for most use cases and access patterns. General purpose buckets are the original S3 bucket type. They allow a mix of storage classes that redundantly store objects across multiple Availability Zones.
Directory
Recommended for low-latency use cases. These buckets use only the S3 Express One Zone storage class, which provides faster processing of data within a single Availability Zone.
Bucket namespace
Choose the namespace where you want to create your bucket. Learn more
Global namespace
By default, S3 creates general purpose buckets in the global namespace.
Account Regional namespace (recommended)
General purpose buckets created in your account Regional namespace are unique to your account. These buckets can never be created by another AWS account.
Bucket name Info
amzn-s3-demo-bucket
Bucket names must be 3 to 63 characters and unique within the global namespace. Bucket names must also begin and end with a letter or number. Valid characters are a-z, 0-9, periods (.), and hyphens (-). Learn more
Copy settings from existing bucket - optional
Only the bucket settings in the following configuration are copied.
Choose bucket
Format: s3://bucket/prefix



us-east-1.console.aws.amazon.com/s3/bucket/create?region=us-east-1
Amazon S3 > Buckets > Create bucket
By default, S3 creates general purpose buckets in the global namespace.
Account Regional namespace (recommended)
General purpose buckets created in your account Regional namespace are unique to your account. These buckets can never be created by another AWS account.
Bucket name Info
lexlab6
Bucket names must be 3 to 63 characters and unique within the global namespace. Bucket names must also begin and end with a letter or number. Valid characters are a-z, 0-9, periods (.), and hyphens (-). Learn more
Copy settings from existing bucket - optional
Only the bucket settings in the following configuration are copied.
Choose bucket
Format: s3://bucket/prefix
Object Ownership Info
Control ownership of objects written to this bucket from other AWS accounts and the use of access control lists (ACLs). Object ownership determines who can specify access to objects.
Object Ownership
ACLs disabled (recommended)
All objects in this bucket are owned by this account. Access to this bucket and its objects is specified using only policies.
ACLs enabled
Objects in this bucket can be owned by other AWS accounts. Access to this bucket and its objects can be specified using ACLs.
Object Ownership
Bucket owner enforced
Block Public Access settings for this bucket
Public access is granted to buckets and objects through access control lists (ACLs), bucket policies, access point policies, or all. In order to ensure that public access to this bucket and its objects is blocked, turn on Block all public access. These settings apply only to this bucket and its access points. AWS recommends that you turn on Block all public access, but before applying any of these settings, ensure that your applications will work correctly



us-east-1.console.aws.amazon.com/s3/bucket/create?region=us-east-1
Amazon S3 > Buckets > Create bucket
Bucket owner enforced
Block Public Access settings for this bucket
Public access is granted to buckets and objects through access control lists (ACLs), bucket policies, access point policies, or all. In order to ensure that public access to this bucket and its objects is blocked, turn on Block all public access. These settings apply only to this bucket and its access points. AWS recommends that you turn on Block all public access, but before applying any of these settings, ensure that your applications will work correctly without public access. If you require some level of public access to this bucket or objects within, you can customize the individual settings below to suit your specific storage use cases. Learn more
Block all public access
Turning this setting on is the same as turning on all four settings below. Each of the following settings are independent of one another.
Block public access to buckets and objects granted through new access control lists (ACLs)
S3 will block public access permissions applied to newly added buckets or objects, and prevent the creation of new public access ACLs for existing buckets and objects. This setting doesn't change any existing permissions that allow public access to S3 resources using ACLs.
Block public access to buckets and objects granted through any access control lists (ACLs)
S3 will ignore all ACLs that grant public access to buckets and objects.
Block public access to buckets and objects granted through new public bucket or access point policies
S3 will block new bucket and access point policies that grant public access to buckets and objects. This setting doesn't change any existing policies that allow public access to S3 resources.
Block public and cross-account access to buckets and objects through any public bucket or access point policies
S3 will ignore public and cross-account access for buckets or access points with policies that grant public access to buckets and objects.
Turning off block all public access might result in this bucket and the objects within becoming public
AWS recommends that you turn on block all public access, unless public access is required for specific and verified use cases such as static website hosting.
I acknowledge that the current settings might result in this bucket and the objects within becoming public.
Bucket Versioning
Versioning is a means of keeping multiple variants of an object in the same bucket. You can use versioning to preserve, retrieve, and restore every version of every object stored in your Amazon S3 bucket. With versioning, you



Course Modules: AW
Challenge lab
AWS Management C
my-flight-notebook
Lab 5 - Guided Lab
Lab 6 - Amazon Lex
lexlab6 - S3 bucket
us-east-1.console.aws.amazon.com/s3/buckets/lexlab6?region=us-east-1&tab=objects
School
Search
[Alt+S]
United States (N. Virginia)
Account ID: 1192-8573-5731
voclabs/user5004810=POZON._TERRENCE
Amazon S3 > Buckets > lexlab6
Successfully created bucket "lexlab6"
lexlab6 Info
Objects
Metadata
Properties
Permissions
Metrics
Management
File systems - new
Access Points
Objects (0)
Copy S3 URI
Copy URL
Download
Open
Delete
Actions
Create folder
Upload
Objects are the fundamental entities stored in Amazon S3. You can use Amazon S3 inventory to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. Learn more
Find objects by prefix
Name
Type
Last modified
Size
Storage class
No objects
You don't have any objects in this bucket.
Upload
CloudShell
Feedback
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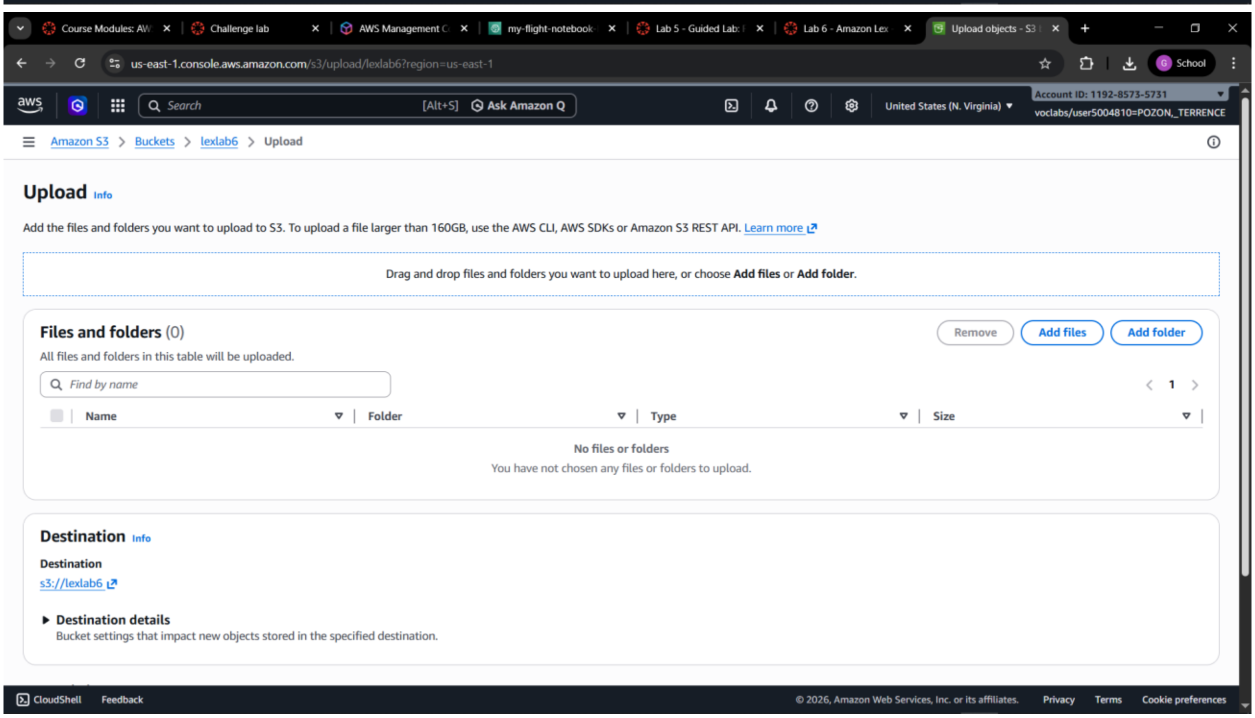

Course Modules: AW
Challenge lab
AWS Management C
my-flight-notebook
Lab 5 - Guided Lab
Lab 6 - Amazon Lex
Upload objects - S3
us-east-1.console.aws.amazon.com/s3/upload/lexlab6?region=us-east-1
School
Search
[Alt+S]
Ask Amazon Q
United States (N. Virginia)
Account ID: 1192-8573-5731
voclabs/user5004810=POZON._TERRENCE
Amazon S3 > Buckets > lexlab6 > Upload
Upload Info
Add the files and folders you want to upload to S3. To upload a file larger than 160GB, use the AWS CLI, AWS SDKs or Amazon S3 REST API. Learn more
Drag and drop files and folders you want to upload here, or choose Add files or Add folder.
Files and folders (0)
Remove
Add files
Add folder
All files and folders in this table will be uploaded.
Find by name
Name
Folder
Type
Size
No files or folders
You have not chosen any files or folders to upload.
Destination Info
Destination
s3://lexlab6
Destination details
Bucket settings that impact new objects stored in the specified destination.
CloudShell
Feedback
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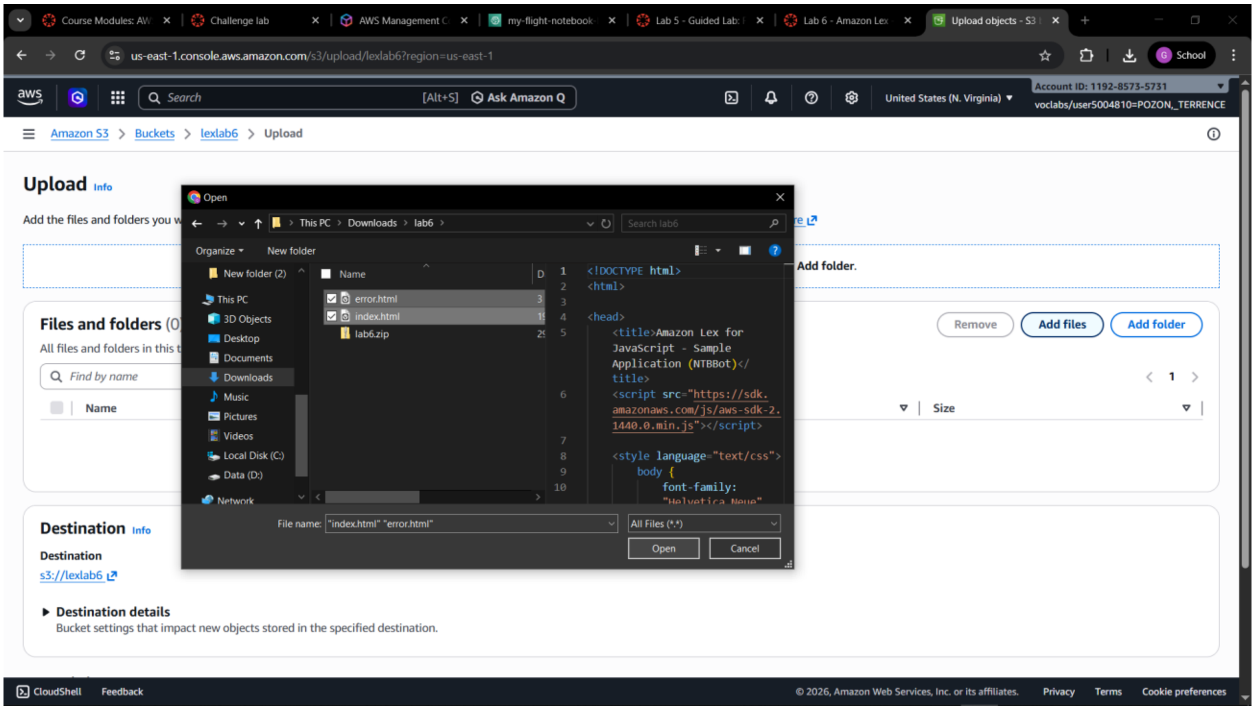
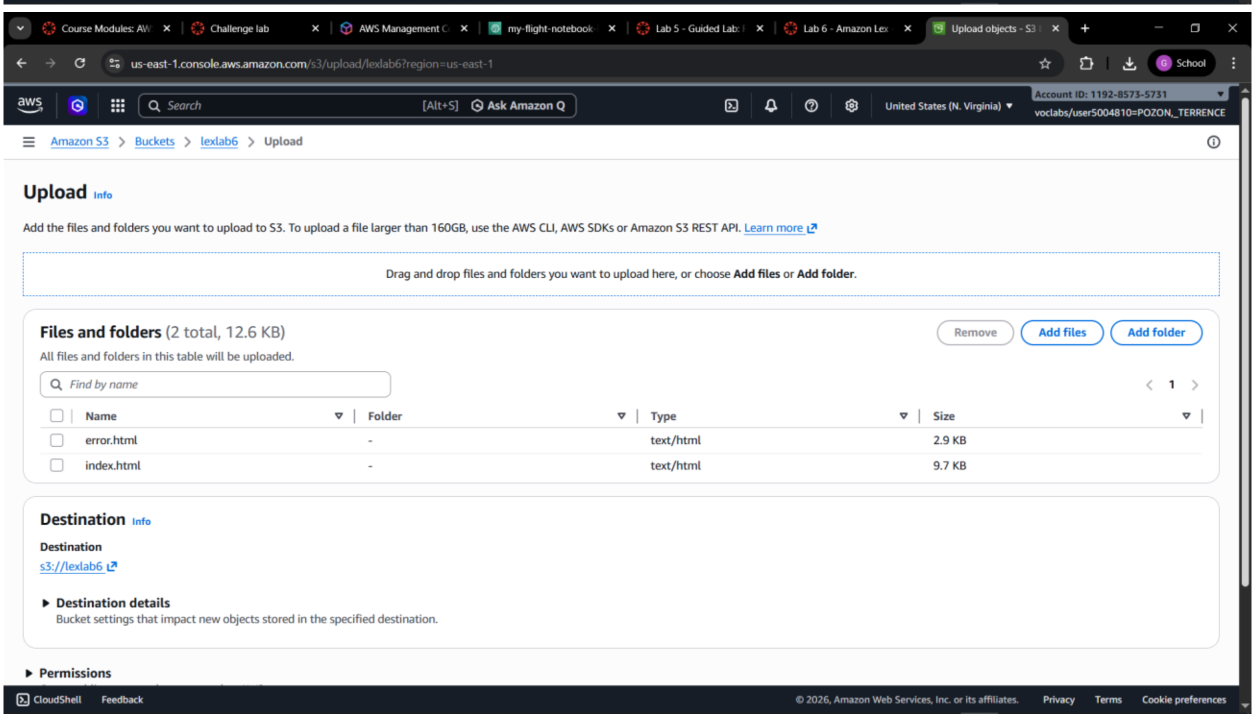
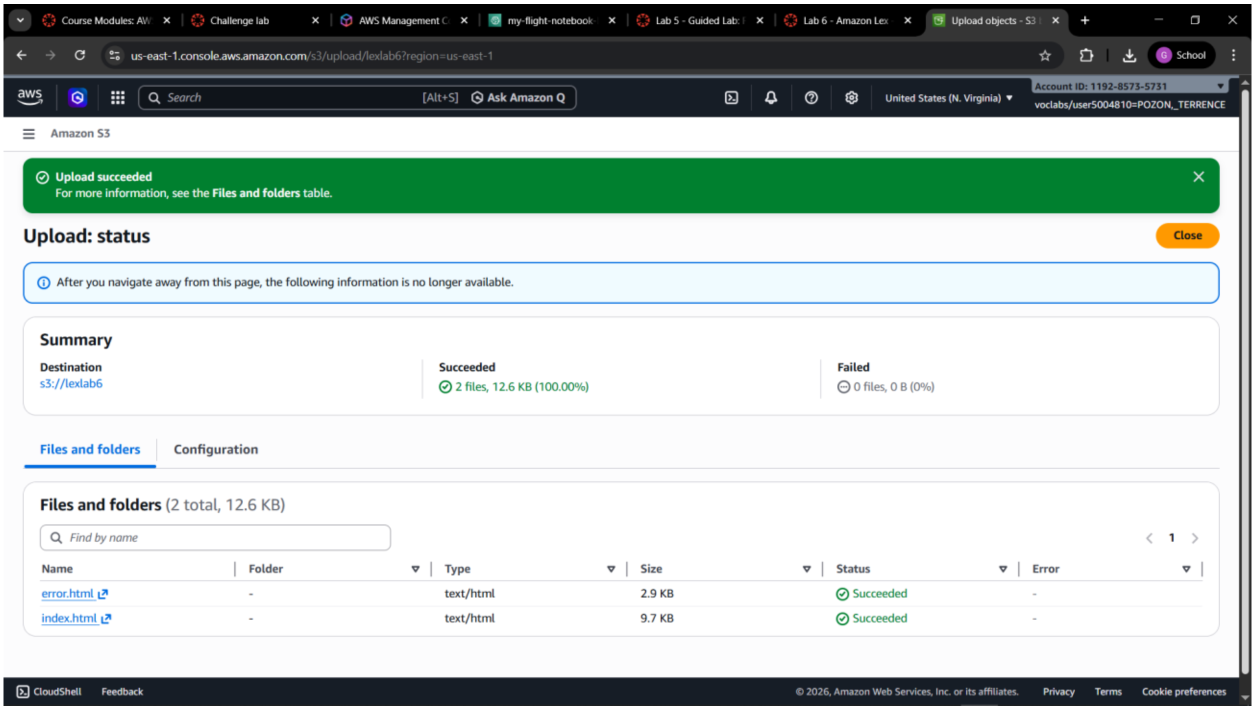
Privacy
Terms
Cookie preferences



us-east-1.console.aws.amazon.com/s3/upload/lexlab6?region=us-east-1
Amazon S3 > Buckets > lexlab6 > Upload
Upload Info
Open
This PC > Downloads > lab6
Organize New folder
error.html
index.html
lab6.zip
<!DOCTYPE html>
<html>
<head>
<title>Amazon Lex for JavaScript - Sample Application (NTBBot)</title>
<script src="https://sdk.amazonaws.com/js/aws-sdk-2.1440.0.min.js"></script>
<style language="text/css">
body {
font-family:
File name: "index.html" "error.html"
All Files (*.*)
Open
Cancel
Files and folders
Remove
Add files
Add folder
Destination Info
s3://lexlab6
Destination details
Bucket settings that impact new objects stored in the specified destination.



us-east-1.console.aws.amazon.com/s3/upload/lexlab6?region=us-east-1
Amazon S3 > Buckets > lexlab6 > Upload
Upload Info
Add the files and folders you want to upload to S3. To upload a file larger than 160GB, use the AWS CLI, AWS SDKs or Amazon S3 REST API. Learn more
Drag and drop files and folders you want to upload here, or choose Add files or Add folder.
Files and folders (2 total, 12.6 KB)
All files and folders in this table will be uploaded.
Remove
Add files
Add folder
Name | Folder | Type | Size
error.html | - | text/html | 2.9 KB
index.html | - | text/html | 9.7 KB
Destination Info
Destination
s3://lexlab6
Destination details
Bucket settings that impact new objects stored in the specified destination.
Permissions


us-east-1.console.aws.amazon.com/s3/upload/lexlab6?region=us-east-1
Amazon S3
Upload succeeded
For more information, see the Files and folders table.
Upload: status
Close
After you navigate away from this page, the following information is no longer available.
Summary
Destination
s3://lexlab6
Succeeded
2 files, 12.6 KB (100.00%)
Failed
0 files, 0 B (0%)
Files and folders
Configuration
Files and folders (2 total, 12.6 KB)
Name | Folder | Type | Size | Status | Error
error.html | - | text/html | 2.9 KB | Succeeded | -
index.html | - | text/html | 9.7 KB | Succeeded | -
CloudShell
Feedback
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Terms
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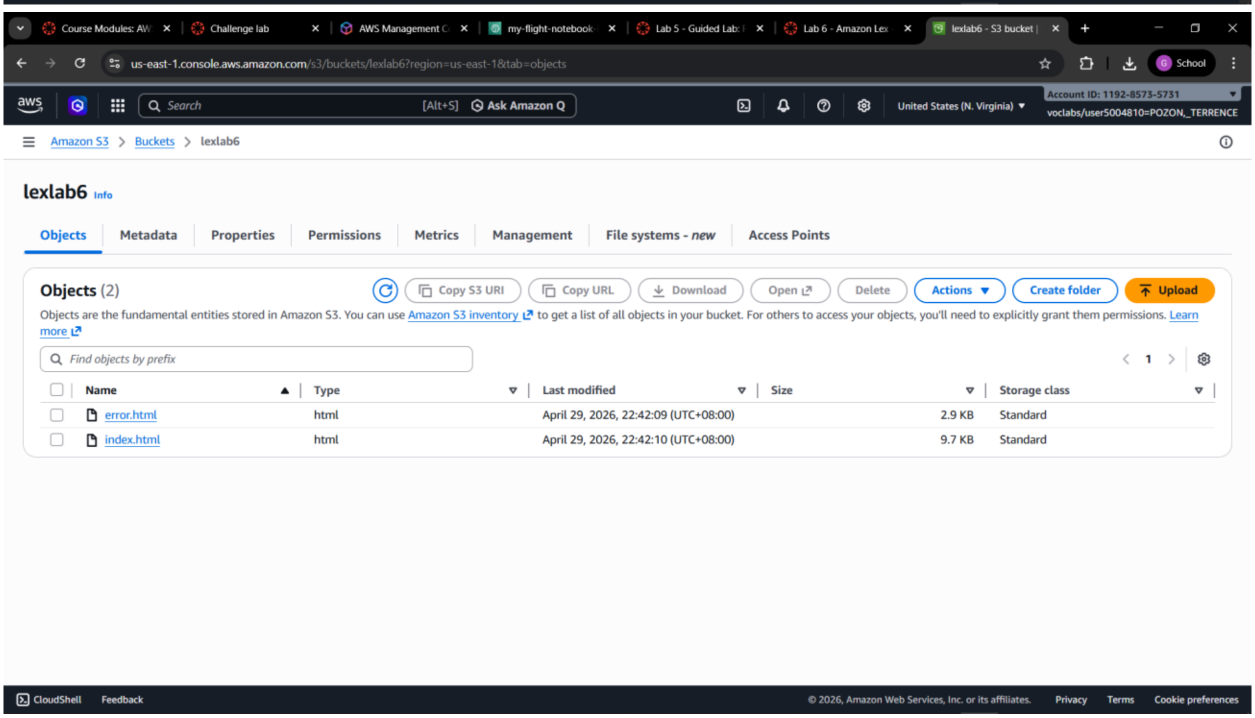
us-east-1.console.aws.amazon.com/s3/buckets/lexlab6?region=us-east-1&tab=objects
Amazon S3 > Buckets > lexlab6
lexlab6 Info
Objects
Metadata
Properties
Permissions
Metrics
Management
File systems - new
Access Points
Objects (2)
Copy S3 URI
Copy URL
Download
Open
Delete
Actions
Create folder
Upload
Objects are the fundamental entities stored in Amazon S3. You can use Amazon S3 inventory to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. Learn more
Name | Type | Last modified | Size | Storage class
error.html | html | April 29, 2026, 22:42:09 (UTC+08:00) | 2.9 KB | Standard
index.html | html | April 29, 2026, 22:42:10 (UTC+08:00) | 9.7 KB | Standard
CloudShell
Feedback
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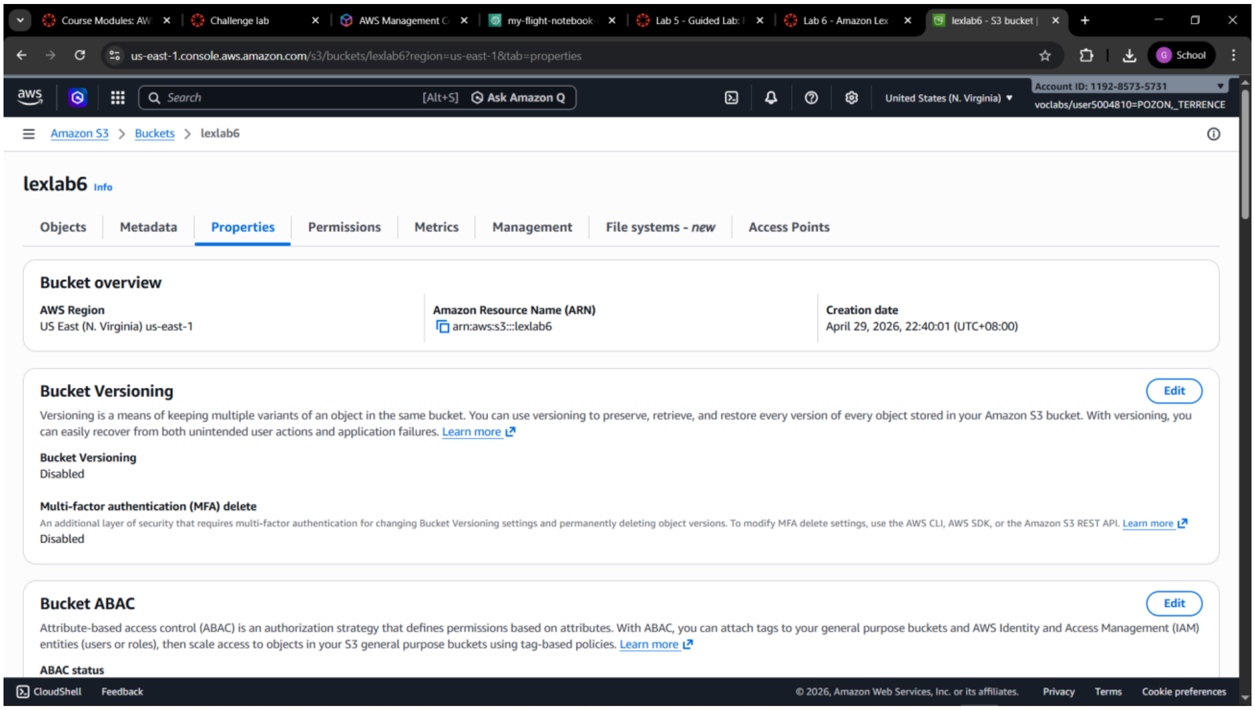
Privacy
Terms
Cookie preferences


Course Modules: AW
Challenge lab
AWS Management C
my-flight-notebook
Lab 5 - Guided Lab:
Lab 6 - Amazon Lex
lexlab6 - S3 bucket
us-east-1.console.aws.amazon.com/s3/buckets/lexlab6?region=us-east-1&tab=properties
School
Search
[Alt+S]
Ask Amazon Q
United States (N. Virginia)
Account ID: 1192-8573-5731
voclabs/user5004810=POZON_TERRENCE
Amazon S3 > Buckets > lexlab6
lexlab6 Info
Objects
Metadata
Properties
Permissions
Metrics
Management
File systems - new
Access Points
Bucket overview
AWS Region
US East (N. Virginia) us-east-1
Amazon Resource Name (ARN)
arn:aws:s3:::lexlab6
Creation date
April 29, 2026, 22:40:01 (UTC+08:00)
Bucket Versioning
Edit
Versioning is a means of keeping multiple variants of an object in the same bucket. You can use versioning to preserve, retrieve, and restore every version of every object stored in your Amazon S3 bucket. With versioning, you can easily recover from both unintended user actions and application failures. Learn more
Bucket Versioning
Disabled
Multi-factor authentication (MFA) delete
An additional layer of security that requires multi-factor authentication for changing Bucket Versioning settings and permanently deleting object versions. To modify MFA delete settings, use the AWS CLI, AWS SDK, or the Amazon S3 REST API. Learn more
Disabled
Bucket ABAC
Edit
Attribute-based access control (ABAC) is an authorization strategy that defines permissions based on attributes. With ABAC, you can attach tags to your general purpose buckets and AWS Identity and Access Management (IAM) entities (users or roles), then scale access to objects in your S3 general purpose buckets using tag-based policies. Learn more
ABAC status
CloudShell
Feedback
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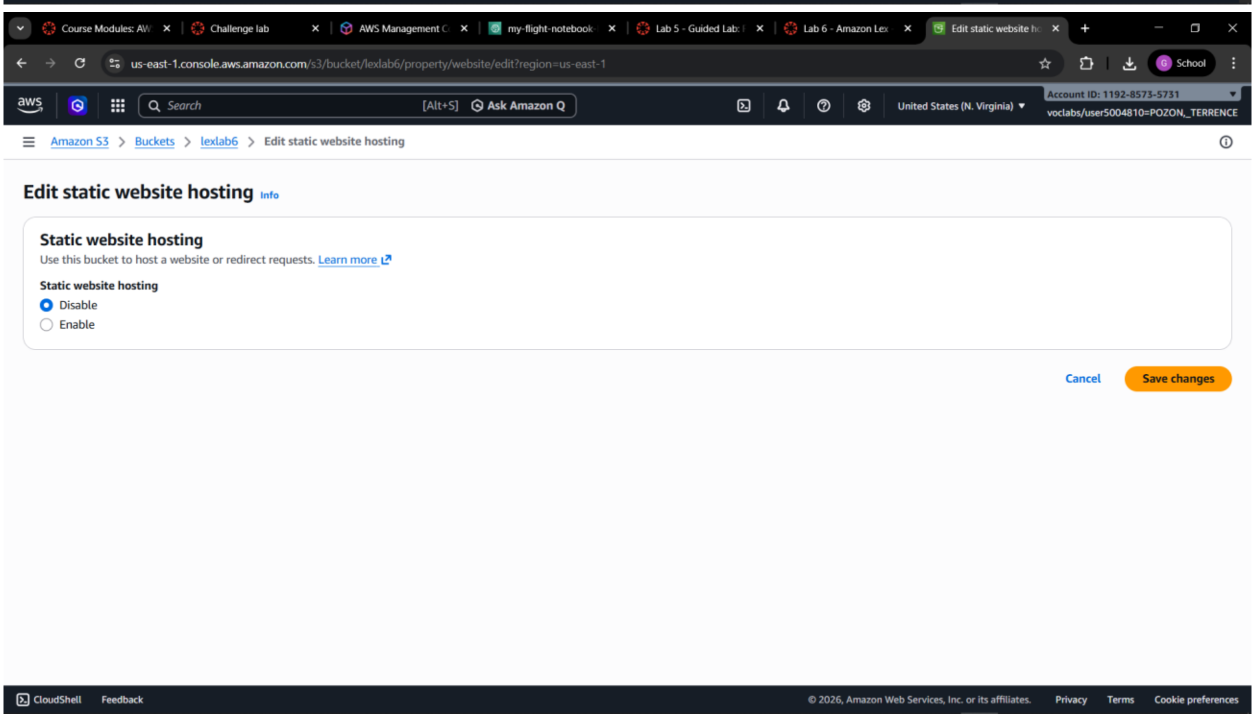
Course Modules: AW
Challenge lab
AWS Management C
my-flight-notebook
Lab 5 - Guided Lab:
Lab 6 - Amazon Lex
Edit static website h
us-east-1.console.aws.amazon.com/s3/bucket/lexlab6/property/website/edit?region=us-east-1
School
Search
[Alt+S]
Ask Amazon Q
United States (N. Virginia)
Account ID: 1192-8573-5731
voclabs/user5004810=POZON_TERRENCE
Amazon S3 > Buckets > lexlab6 > Edit static website hosting
Edit static website hosting Info
Static website hosting
Use this bucket to host a website or redirect requests. Learn more
Static website hosting
Disable
Enable
Cancel
Save changes
CloudShell
Feedback
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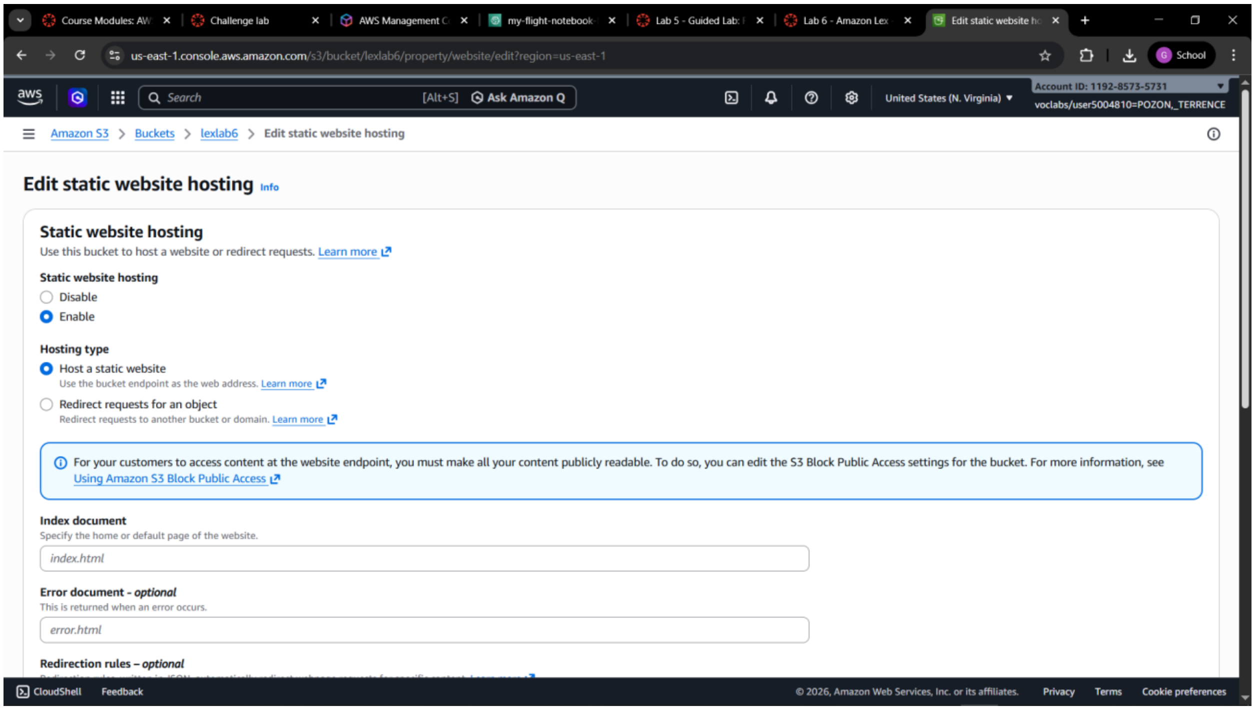
Cookie preferences


Course Modules: AW
Challenge lab
AWS Management C
my-flight-notebook
Lab 5 - Guided Lab
Lab 6 - Amazon Lex
Edit static website ho
us-east-1.console.aws.amazon.com/s3/bucket/lexlab6/property/website/edit?region=us-east-1
School
Search
[Alt+S]
Ask Amazon Q
United States (N. Virginia)
Account ID: 1192-8573-5731
voclabs/user5004810=POZON,_TERRENCE
Amazon S3 > Buckets > lexlab6 > Edit static website hosting
Edit static website hosting Info
Static website hosting
Use this bucket to host a website or redirect requests. Learn more
Static website hosting
Disable
Enable
Hosting type
Host a static website
Use the bucket endpoint as the web address. Learn more
Redirect requests for an object
Redirect requests to another bucket or domain. Learn more
For your customers to access content at the website endpoint, you must make all your content publicly readable. To do so, you can edit the S3 Block Public Access settings for the bucket. For more information, see Using Amazon S3 Block Public Access
Index document
Specify the home or default page of the website.
index.html
Error document - optional
This is returned when an error occurs.
error.html
Redirection rules – optional
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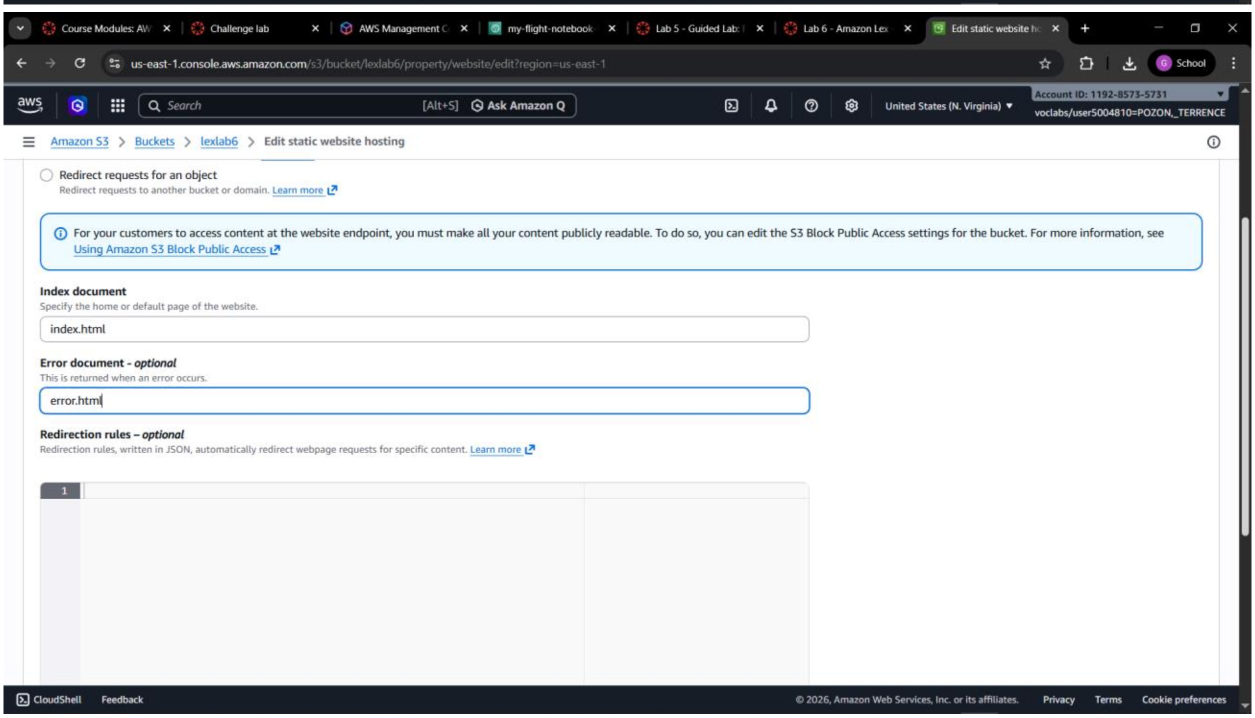
Course Modules: AW
Challenge lab
AWS Management C
my-flight-notebook
Lab 5 - Guided Lab
Lab 6 - Amazon Lex
Edit static website ho
us-east-1.console.aws.amazon.com/s3/bucket/lexlab6/property/website/edit?region=us-east-1
School
Search
[Alt+S]
Ask Amazon Q
United States (N. Virginia)
Account ID: 1192-8573-5731
voclabs/user5004810=POZON,_TERRENCE
Amazon S3 > Buckets > lexlab6 > Edit static website hosting
Redirect requests for an object
Redirect requests to another bucket or domain. Learn more
For your customers to access content at the website endpoint, you must make all your content publicly readable. To do so, you can edit the S3 Block Public Access settings for the bucket. For more information, see Using Amazon S3 Block Public Access
Index document
Specify the home or default page of the website.
index.html
Error document - optional
This is returned when an error occurs.
error.html
Redirection rules – optional
Redirection rules, written in JSON, automatically redirect webpage requests for specific content. Learn more
1
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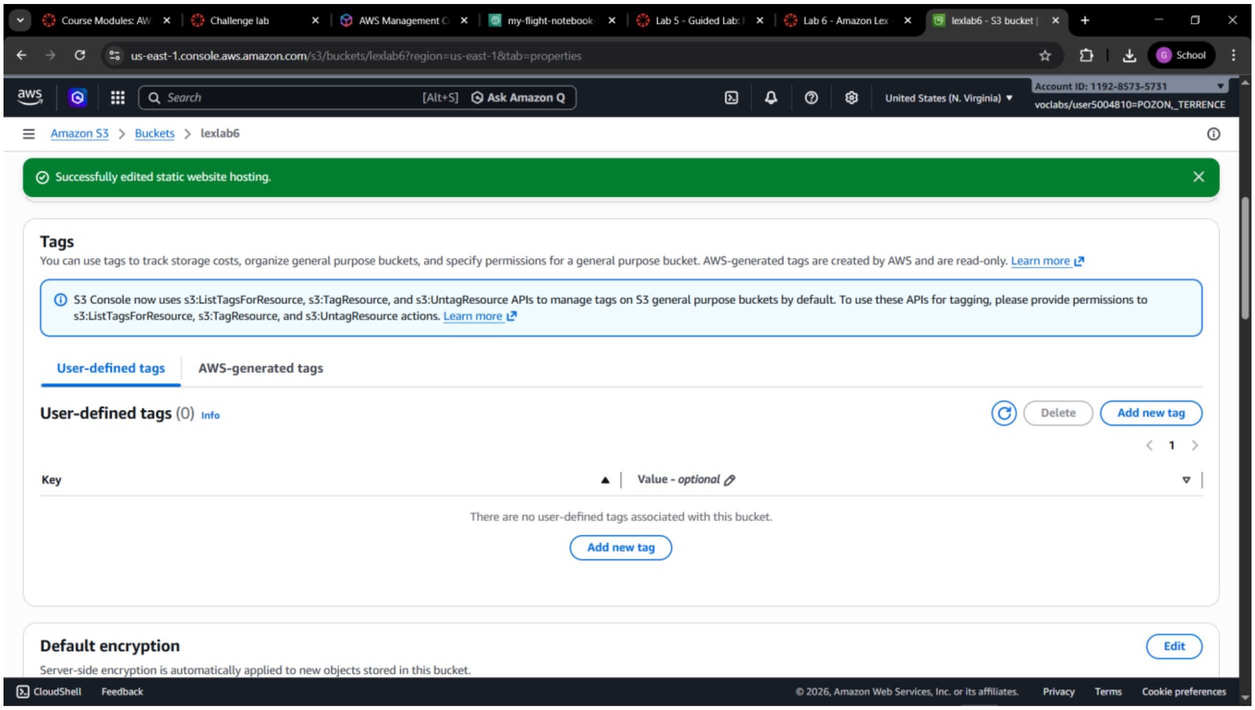
Course Modules: AW
Challenge lab
AWS Management C
my-flight-notebook
Lab 5 - Guided Lab:
Lab 6 - Amazon Lex
lexlab6 - S3 bucket |
us-east-1.console.aws.amazon.com/s3/buckets/lexlab6?region=us-east-1&tab=properties
School
Search
[Alt+S]
Ask Amazon Q
United States (N. Virginia)
Account ID: 1192-8573-5731
voclabs/user5004810=POZON,_TERRENCE
Amazon S3
Buckets
lexlab6
Successfully edited static website hosting.
Tags
You can use tags to track storage costs, organize general purpose buckets, and specify permissions for a general purpose bucket. AWS-generated tags are created by AWS and are read-only. Learn more
S3 Console now uses s3:ListTagsForResource, s3:TagResource, and s3:UntagResource APIs to manage tags on S3 general purpose buckets by default. To use these APIs for tagging, please provide permissions to s3:ListTagsForResource, s3:TagResource, and s3:UntagResource actions. Learn more
User-defined tags
AWS-generated tags
User-defined tags (0)
Info
Delete
Add new tag
1
Key
Value - optional
There are no user-defined tags associated with this bucket.
Add new tag
Default encryption
Edit
Server-side encryption is automatically applied to new objects stored in this bucket.
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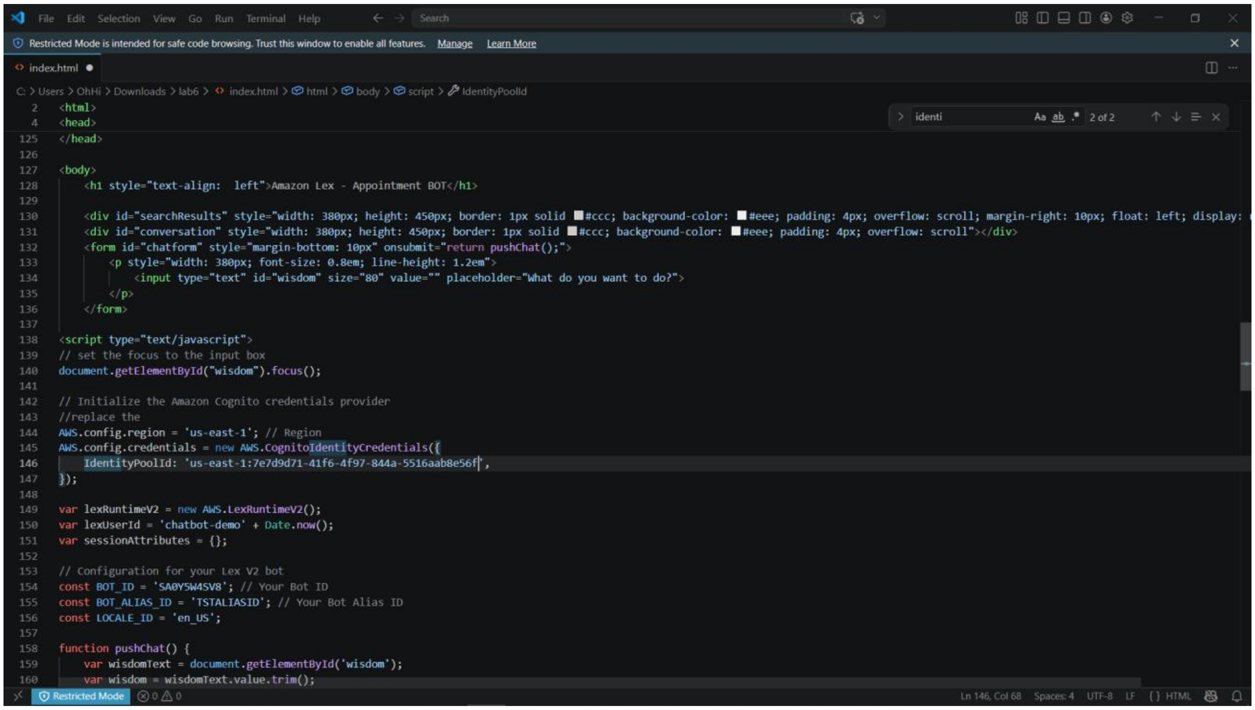
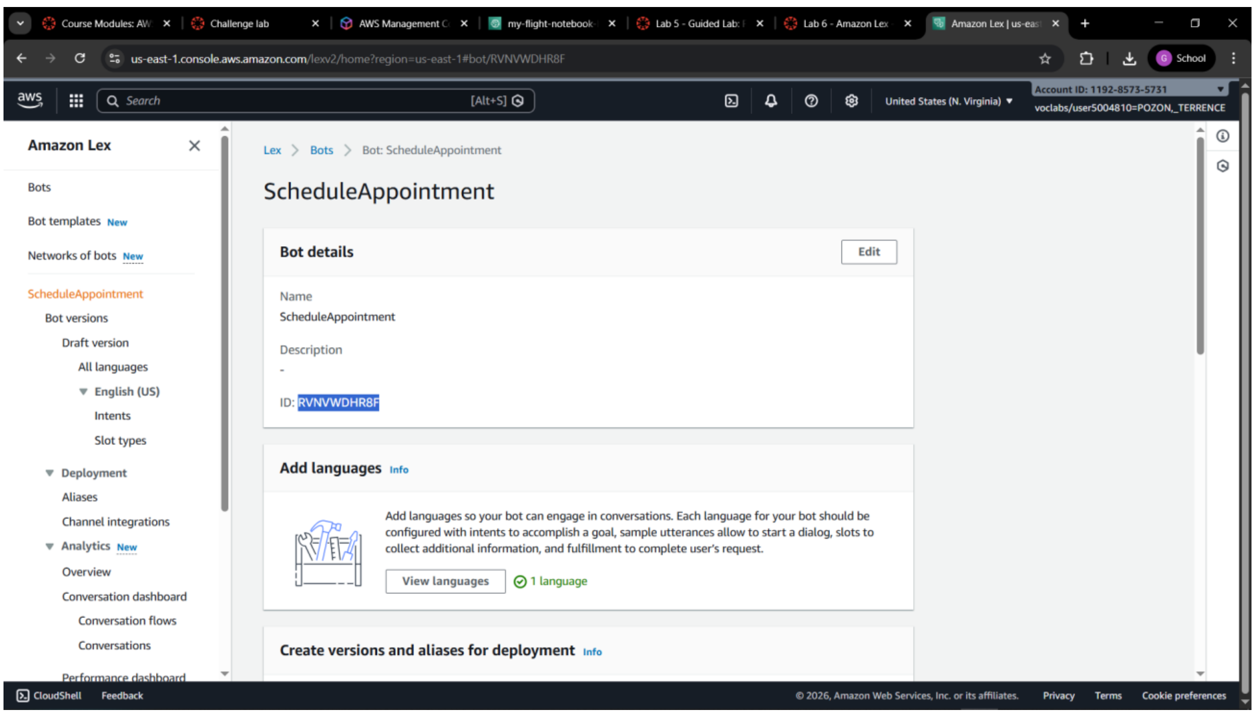
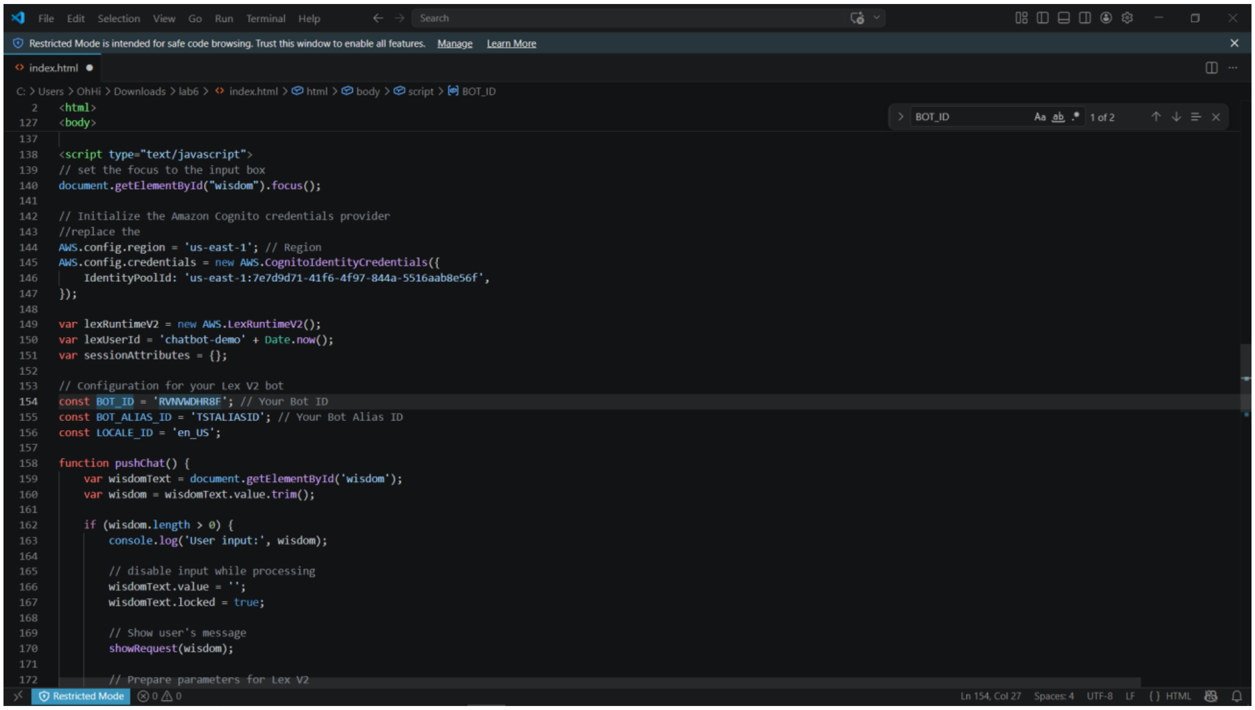
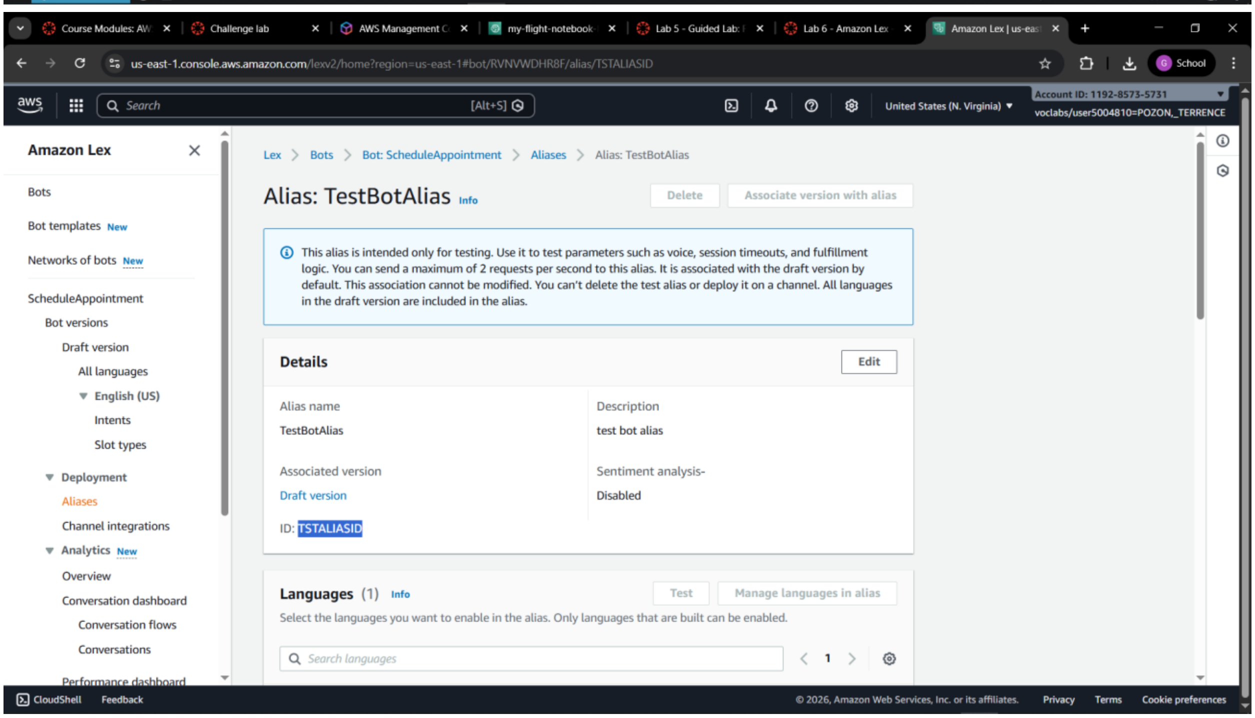
Task 7






File Edit Selection View Go Run Terminal Help
Restricted Mode is intended for safe code browsing. Trust this window to enable all features. Manage Learn More
index.html
C: > Users > OhHi > Downloads > lab6 > index.html > html > body > script > BOT_ID
BOT_ID
1 of 2
<html>
<body>
<script type="text/javascript">
// set the focus to the input box
document.getElementById("wisdom").focus();
// Initialize the Amazon Cognito credentials provider
//replace the
AWS.config.region = 'us-east-1'; // Region
AWS.config.credentials = new AWS.CognitoIdentityCredentials({
IdentityPoolId: 'us-east-1:7e7d9d71-41f6-4f97-844a-5516aab8e56f',
});
var lexRuntimeV2 = new AWS.LexRuntimeV2();
var lexUserId = 'chatbot-demo' + Date.now();
var sessionAttributes = {};
// Configuration for your Lex V2 bot
const BOT_ID = 'RVNVWDHR8F'; // Your Bot ID
const BOT_ALIAS_ID = 'TSTALIASID'; // Your Bot Alias ID
const LOCALE_ID = 'en_US';
function pushChat() {
var wisdomText = document.getElementById('wisdom');
var wisdom = wisdomText.value.trim();
if (wisdom.length > 0) {
console.log('User input:', wisdom);
// disable input while processing
wisdomText.value = '';
wisdomText.locked = true;
// Show user's message
showRequest(wisdom);
// Prepare parameters for Lex V2
Restricted Mode
Ln 154, Col 27 Spaces: 4 UTF-8 LF { } HTML


Course Modules: AW
Challenge lab
AWS Management C
my-flight-notebook
Lab 5 - Guided Lab
Lab 6 - Amazon Lex
Amazon Lex | us-eas
us-east-1.console.aws.amazon.com/lexv2/home?region=us-east-1#bot/RVNVWDHR8F/alias/TSTALIASID
School
Search [Alt+S]
United States (N. Virginia)
Account ID: 1192-8573-5731
voclabs/user5004810=POZON_TERRENCE
Amazon Lex
Bots
Bot templates New
Networks of bots New
ScheduleAppointment
Bot versions
Draft version
All languages
English (US)
Intents
Slot types
Deployment
Aliases
Channel integrations
Analytics New
Overview
Conversation dashboard
Conversation flows
Conversations
Performance dashboard
Lex > Bots > Bot: ScheduleAppointment > Aliases > Alias: TestBotAlias
Alias: TestBotAlias Info
Delete
Associate version with alias
This alias is intended only for testing. Use it to test parameters such as voice, session timeouts, and fulfillment logic. You can send a maximum of 2 requests per second to this alias. It is associated with the draft version by default. This association cannot be modified. You can't delete the test alias or deploy it on a channel. All languages in the draft version are included in the alias.
Details
Edit
Alias name
TestBotAlias
Description
test bot alias
Associated version
Draft version
Sentiment analysis-
Disabled
ID: TSTALIASID
Languages (1) Info
Test
Manage languages in alias
Select the languages you want to enable in the alias. Only languages that are built can be enabled.
Search languages
1
CloudShell Feedback
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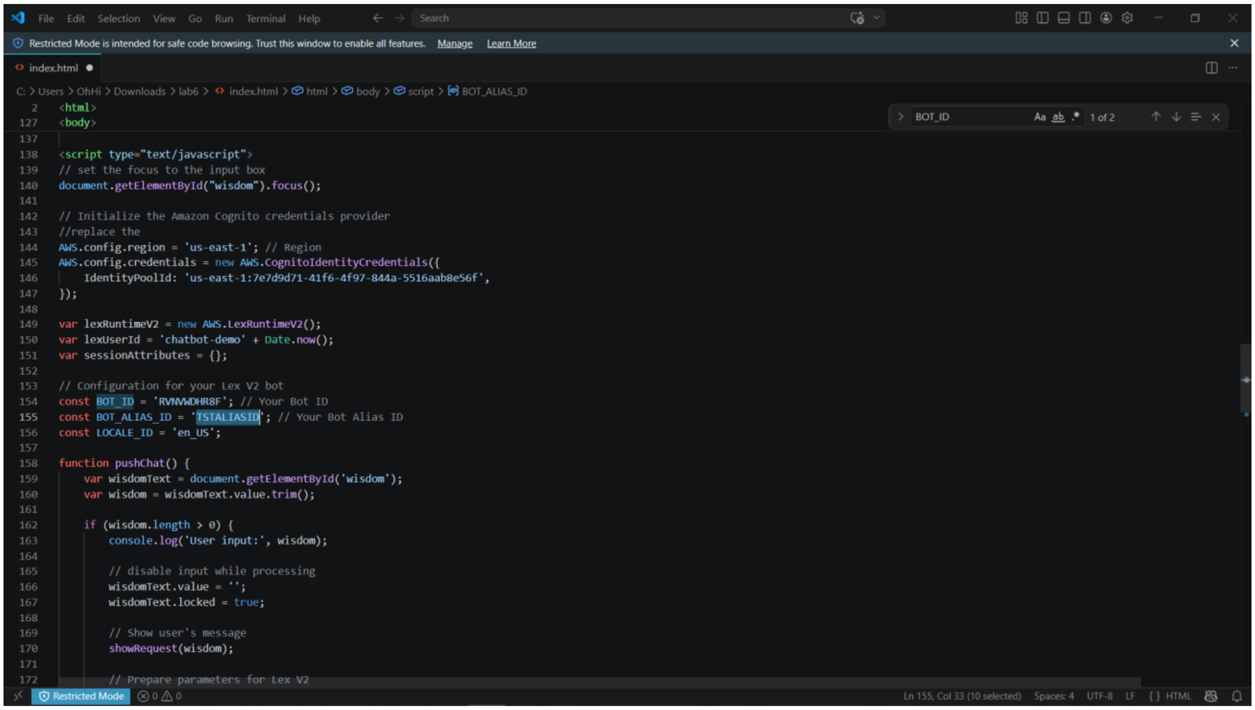

File Edit Selection View Go Run Terminal Help
Restricted Mode is intended for safe code browsing. Trust this window to enable all features. Manage Learn More
index.html
C: > Users > OhHi > Downloads > lab6 > index.html > html > body > script > BOT_ALIAS_ID
BOT_ID 1 of 2
<html>
<body>
<script type="text/javascript">
// set the focus to the input box
document.getElementById("wisdom").focus();
// Initialize the Amazon Cognito credentials provider
//replace the
AWS.config.region = 'us-east-1'; // Region
AWS.config.credentials = new AWS.CognitoIdentityCredentials({
IdentityPoolId: 'us-east-1:7e7d9d71-41f6-4f97-844a-5516aab8e56f',
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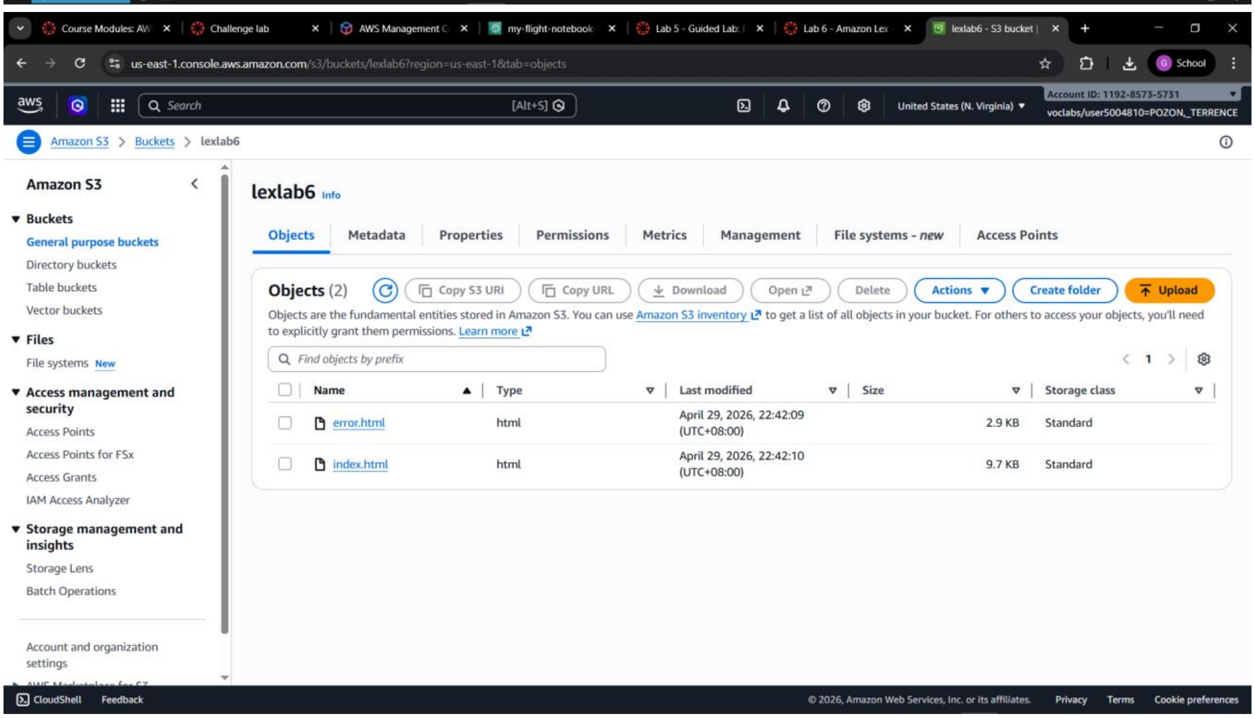
// disable input while processing
wisdomText.value = '';
wisdomText.locked = true;
// Show user's message
showRequest(wisdom);
// Prepare parameters for Lex V2
Restricted Mode
Ln 155, Col 33 (10 selected) Spaces: 4 UTF-8 LF HTML



Course Modules: AW
Challenge lab
AWS Management C
my-flight-notebook
Lab 5 - Guided Lab
Lab 6 - Amazon Lex
lexlab6 - S3 bucket
us-east-1.console.aws.amazon.com/s3/buckets/lexlab6?region=us-east-1&tab=objects
School
Search [Alt+S]
United States (N. Virginia)
Account ID: 1192-8573-5731
voclabs/user5004810=POZON_TERRENCE
Amazon S3 > Buckets > lexlab6
Amazon S3
Buckets
General purpose buckets
Directory buckets
Table buckets
Vector buckets
Files
File systems New
Access management and security
Access Points
Access Points for FSx
Access Grants
IAM Access Analyzer
Storage management and insights
Storage Lens
Batch Operations
Account and organization settings
lexlab6 Info
Objects Metadata Properties Permissions Metrics Management File systems - new Access Points
Objects (2)
Copy S3 URI Copy URL Download Open Delete Actions Create folder Upload
Objects are the fundamental entities stored in Amazon S3. You can use Amazon S3 inventory to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. Learn more
Find objects by prefix
Name Type Last modified Size Storage class
error.html html April 29, 2026, 22:42:09 (UTC+08:00) 2.9 KB Standard
index.html html April 29, 2026, 22:42:10 (UTC+08:00) 9.7 KB Standard
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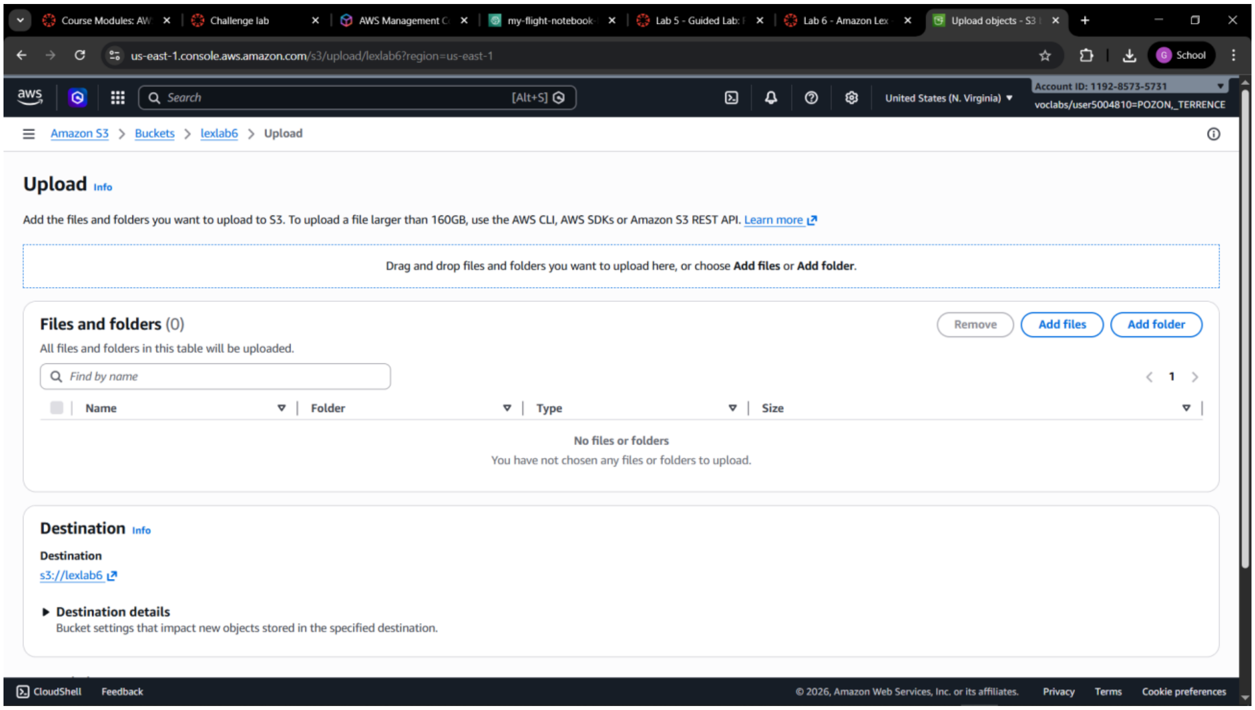

us-east-1.console.aws.amazon.com/s3/upload/lexlab6?region=us-east-1
Amazon S3 > Buckets > lexlab6 > Upload
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Drag and drop files and folders you want to upload here, or choose Add files or Add folder.
Files and folders (0)
All files and folders in this table will be uploaded.
Remove
Add files
Add folder
Find by name
Name
Folder
Type
Size
No files or folders
You have not chosen any files or folders to upload.
Destination Info
Destination
s3://lexlab6
Destination details
Bucket settings that impact new objects stored in the specified destination.
CloudShell
Feedback
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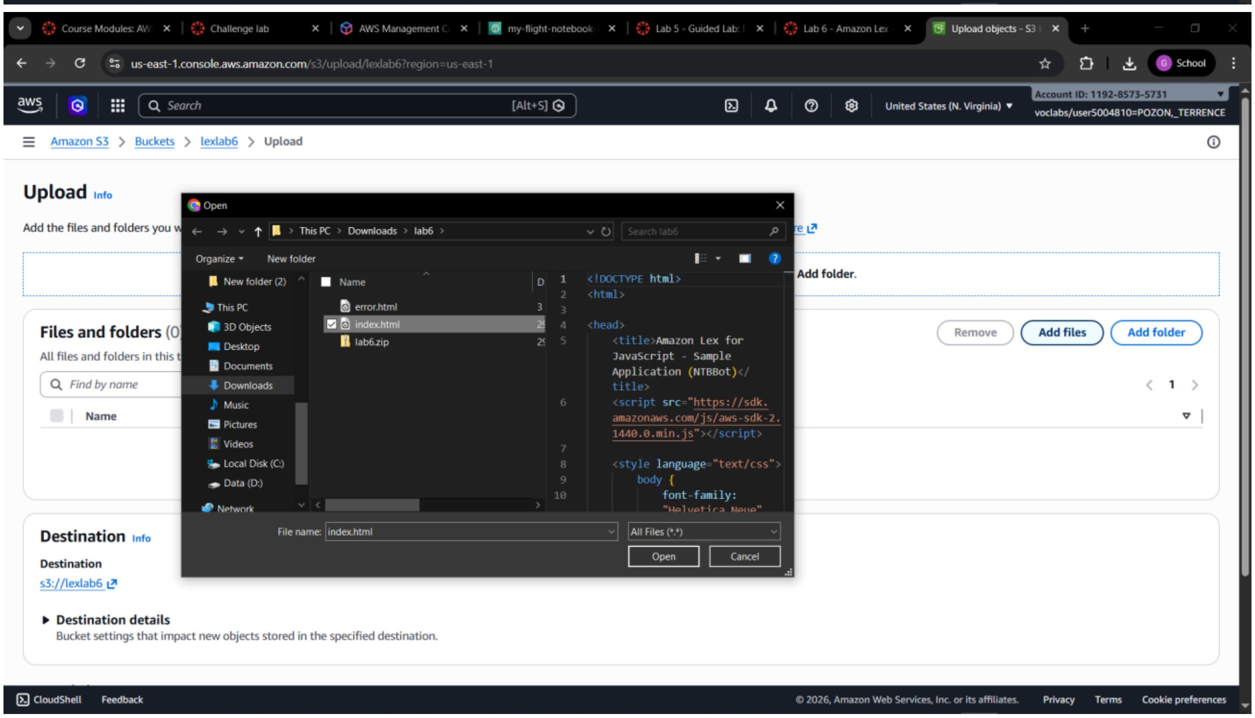
Terms
Cookie preferences



us-east-1.console.aws.amazon.com/s3/upload/lexlab6?region=us-east-1
Amazon S3 > Buckets > lexlab6 > Upload
Upload Info
Open
This PC > Downloads > lab6
Organize
New folder
error.html
index.html
lab6.zip
<!DOCTYPE html>
<html>
<head>
<title>Amazon Lex for JavaScript - Sample Application (NTBBot)</title>
File name: index.html
All Files (*.*)
Open
Cancel
Add folder.
Files and folders
Remove
Add files
Add folder
Destination Info
Destination
s3://lexlab6
Destination details
Bucket settings that impact new objects stored in the specified destination.
CloudShell
Feedback
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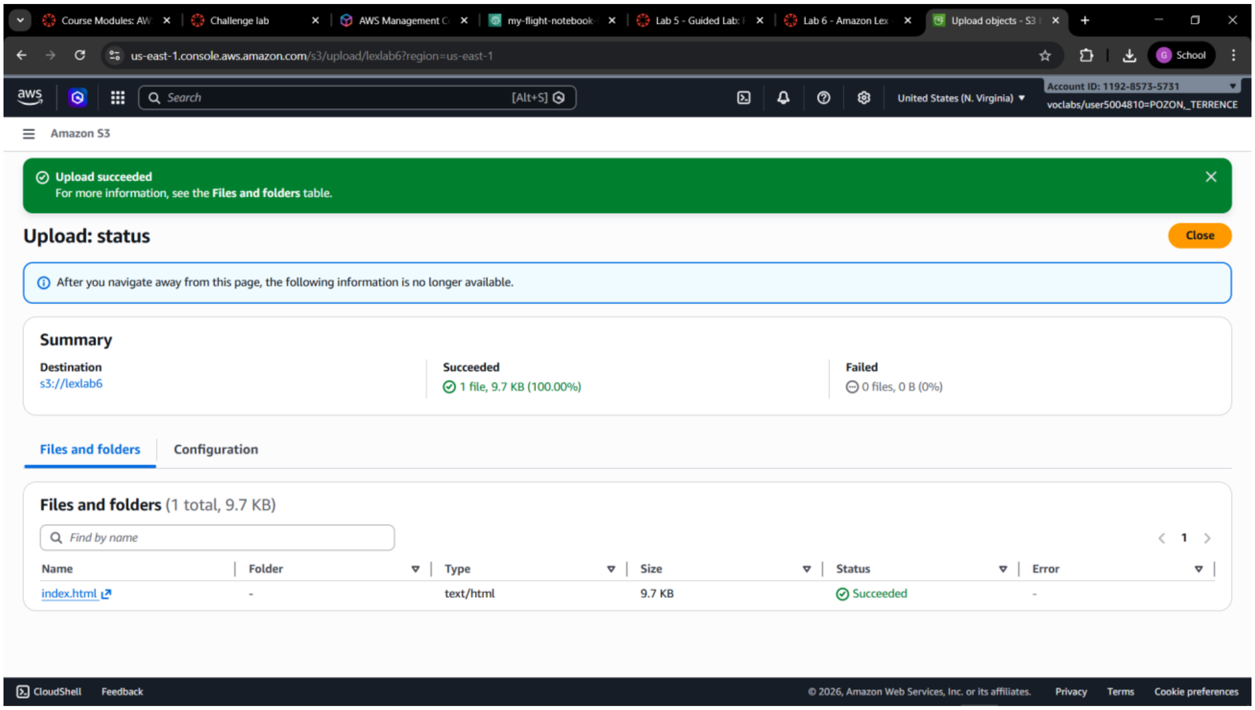
Terms
Cookie preferences


Course Modules: AW
Challenge lab
AWS Management C
my-flight-notebook
Lab 5 - Guided Lab
Lab 6 - Amazon Lex
Upload objects - S3
us-east-1.console.aws.amazon.com/s3/upload/lexlab6?region=us-east-1
School
Search
[Alt+S]
United States (N. Virginia)
Account ID: 1192-8573-5731
voclabs/user5004810=POZON_TERRENCE
Amazon S3
Upload succeeded
For more information, see the Files and folders table.
Upload: status
Close
After you navigate away from this page, the following information is no longer available.
Summary
Destination
s3://lexlab6
Succeeded
1 file, 9.7 KB (100.00%)
Failed
0 files, 0 B (0%)
Files and folders
Configuration
Files and folders (1 total, 9.7 KB)
Find by name
Name
Folder
Type
Size
Status
Error
index.html
-
text/html
9.7 KB
Succeeded
-
CloudShell
Feedback
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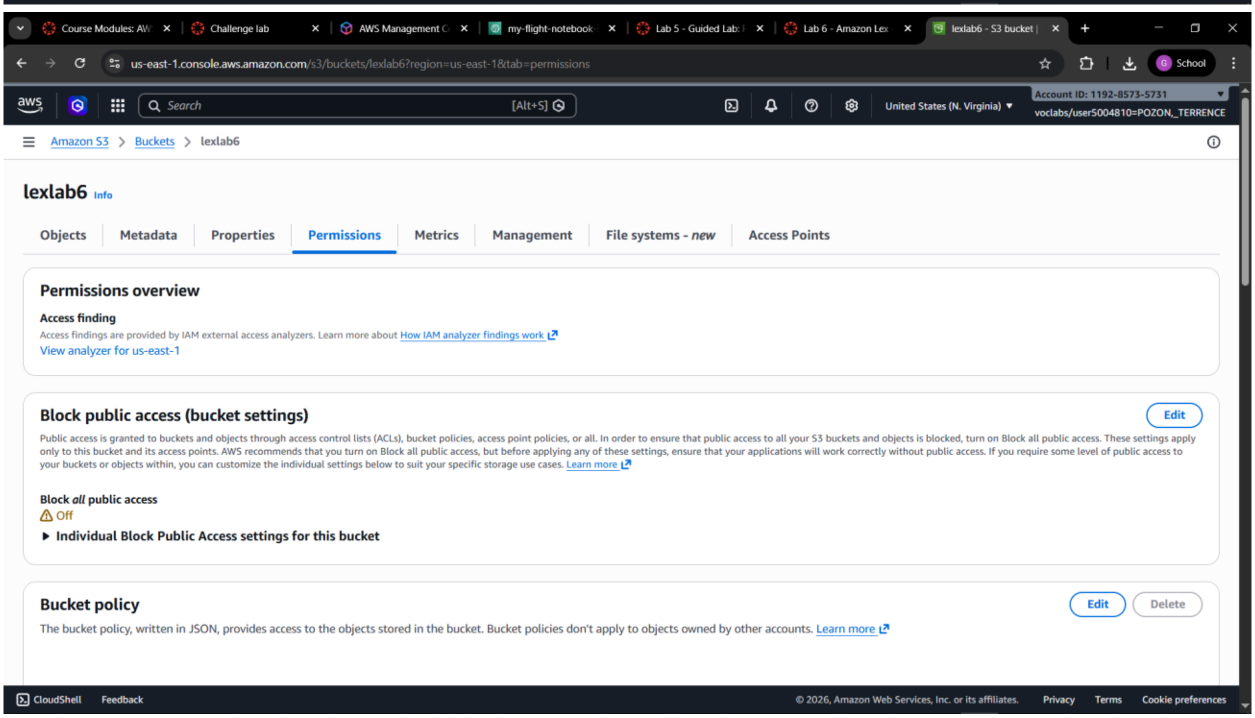
Terms
Cookie preferences


Course Modules: AW
Challenge lab
AWS Management C
my-flight-notebook
Lab 5 - Guided Lab
Lab 6 - Amazon Lex
lexlab6 - S3 bucket
us-east-1.console.aws.amazon.com/s3/buckets/lexlab6?region=us-east-1&tab=permissions
School
Search
[Alt+S]
United States (N. Virginia)
Account ID: 1192-8573-5731
voclabs/user5004810=POZON_TERRENCE
Amazon S3
Buckets
lexlab6
lexlab6
Info
Objects
Metadata
Properties
Permissions
Metrics
Management
File systems - new
Access Points
Permissions overview
Access finding
Access findings are provided by IAM external access analyzers. Learn more about How IAM analyzer findings work
View analyzer for us-east-1
Block public access (bucket settings)
Edit
Public access is granted to buckets and objects through access control lists (ACLs), bucket policies, access point policies, or all. In order to ensure that public access to all your S3 buckets and objects is blocked, turn on Block all public access. These settings apply only to this bucket and its access points. AWS recommends that you turn on Block all public access, but before applying any of these settings, ensure that your applications will work correctly without public access. If you require some level of public access to your buckets or objects within, you can customize the individual settings below to suit your specific storage use cases. Learn more
Block all public access
Off
Individual Block Public Access settings for this bucket
Bucket policy
Edit
Delete
The bucket policy, written in JSON, provides access to the objects stored in the bucket. Bucket policies don't apply to objects owned by other accounts. Learn more
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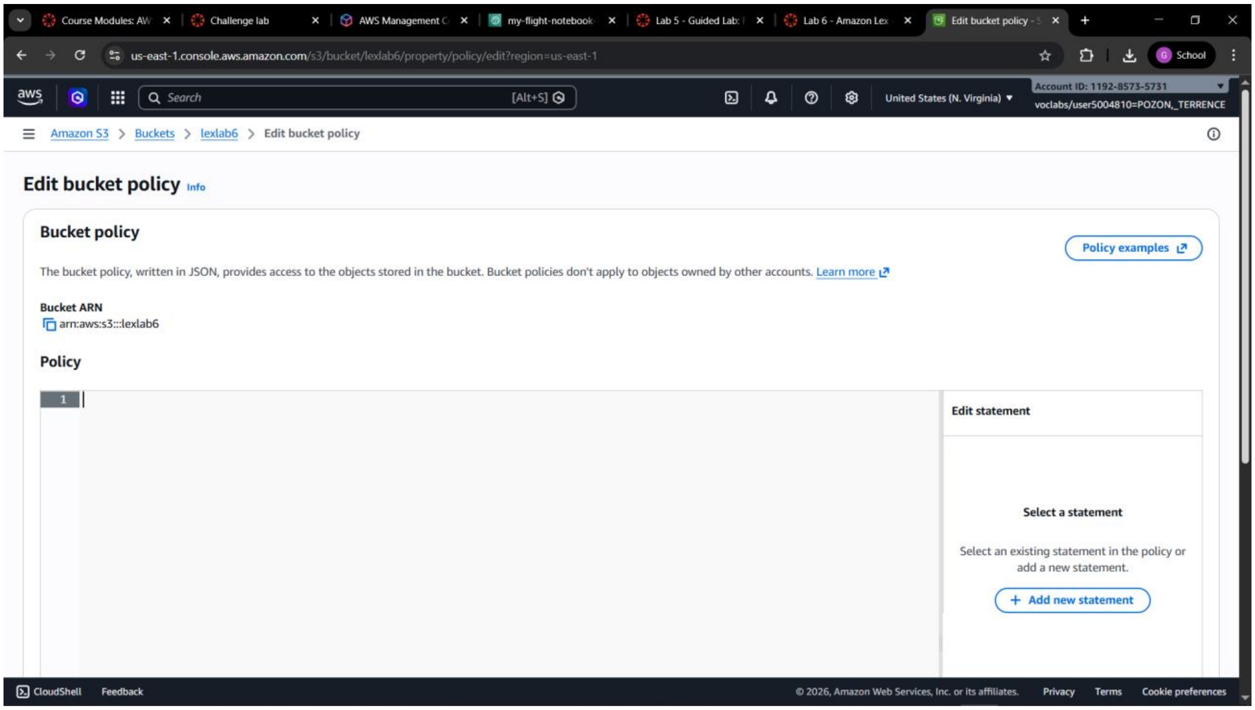
us-east-1.console.aws.amazon.com/s3/bucket/lexlab6/property/policy/edit?region=us-east-1
Amazon S3 > Buckets > lexlab6 > Edit bucket policy
Edit bucket policy Info
Bucket policy
Policy examples
The bucket policy, written in JSON, provides access to the objects stored in the bucket. Bucket policies don't apply to objects owned by other accounts. Learn more
Bucket ARN
arn:aws:s3:::lexlab6
Policy
Edit statement
Select a statement
Select an existing statement in the policy or add a new statement.
+ Add new statement
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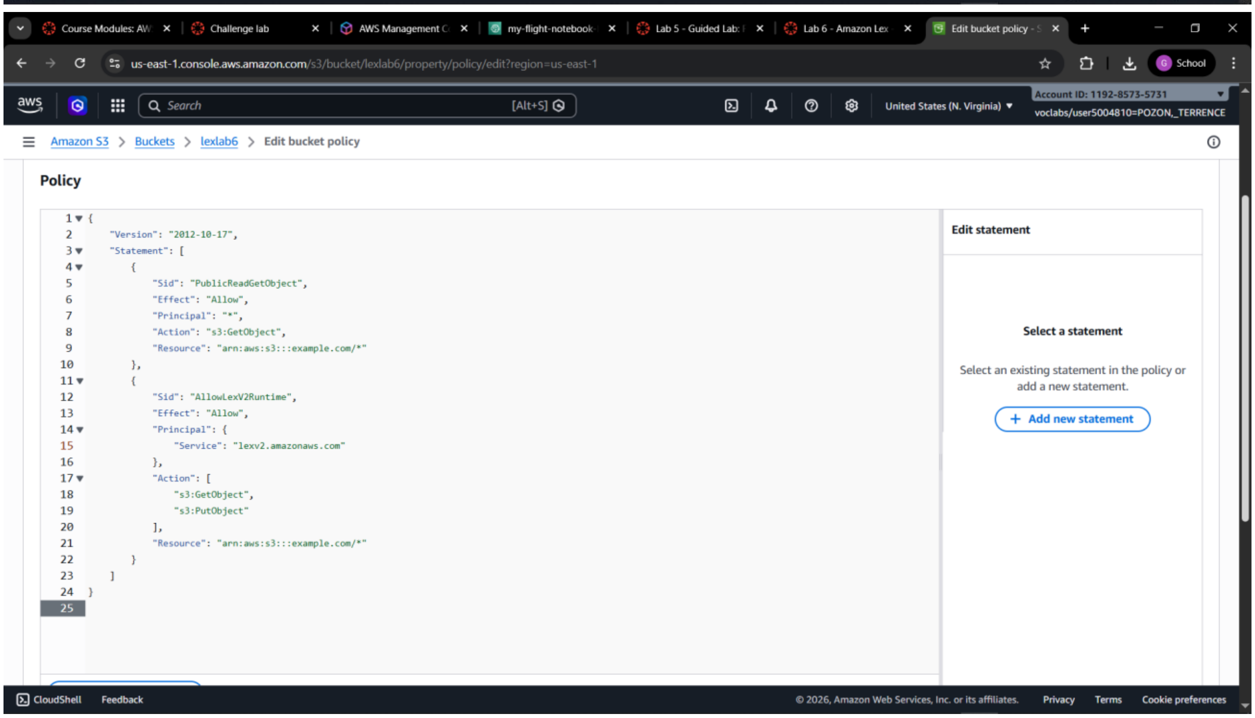
us-east-1.console.aws.amazon.com/s3/bucket/lexlab6/property/policy/edit?region=us-east-1
Amazon S3 > Buckets > lexlab6 > Edit bucket policy
Policy
{
"Version": "2012-10-17",
"Statement": [
{
"Sid": "PublicReadGetObject",
"Effect": "Allow",
"Principal": "*",
"Action": "s3:GetObject",
"Resource": "arn:aws:s3:::example.com/*"
},
{
"Sid": "AllowLexV2Runtime",
"Effect": "Allow",
"Principal": {
"Service": "lexv2.amazonaws.com"
},
"Action": [
"s3:GetObject",
"s3:PutObject"
],
"Resource": "arn:aws:s3:::example.com/*"
}
]
}
Edit statement
Select a statement
Select an existing statement in the policy or add a new statement.
+ Add new statement
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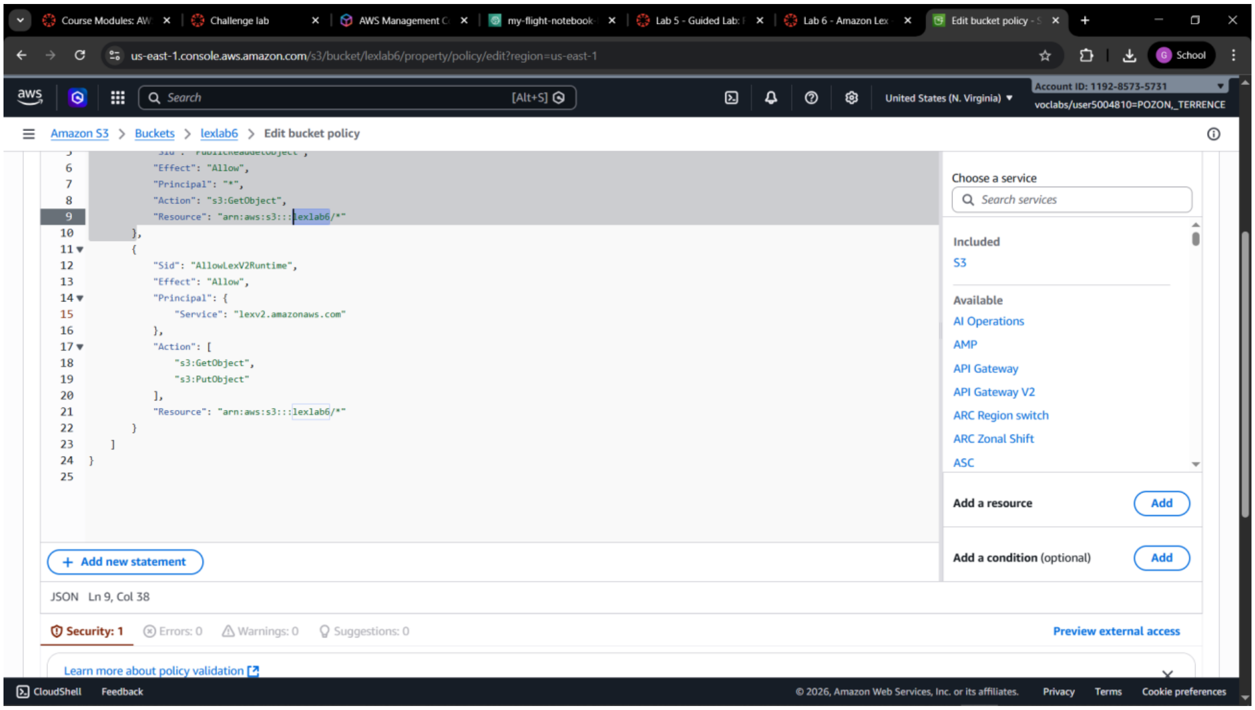
Course Modules: AW
Challenge lab
AWS Management C
my-flight-notebook
Lab 5 - Guided Lab:
Lab 6 - Amazon Lex
Edit bucket policy
us-east-1.console.aws.amazon.com/s3/bucket/lexlab6/property/policy/edit?region=us-east-1
School
Search
[Alt+S]
United States (N. Virginia)
Account ID: 1192-8573-5731
voclabs/user5004810=POZON_TERRENCE
Amazon S3 > Buckets > lexlab6 > Edit bucket policy
"Effect": "Allow",
"Principal": "*",
"Action": "s3:GetObject",
"Resource": "arn:aws:s3:::lexlab6/*"
},
{
"Sid": "AllowLexV2Runtime",
"Effect": "Allow",
"Principal": {
"Service": "lexv2.amazonaws.com"
},
"Action": [
"s3:GetObject",
"s3:PutObject"
],
"Resource": "arn:aws:s3:::lexlab6/*"
}
]
}
Choose a service
Search services
Included
S3
Available
AI Operations
AMP
API Gateway
API Gateway V2
ARC Region switch
ARC Zonal Shift
ASC
Add a resource
Add
Add a condition (optional)
Add
Add new statement
JSON Ln 9, Col 38
Security: 1
Errors: 0
Warnings: 0
Suggestions: 0
Preview external access
Learn more about policy validation
CloudShell
Feedback
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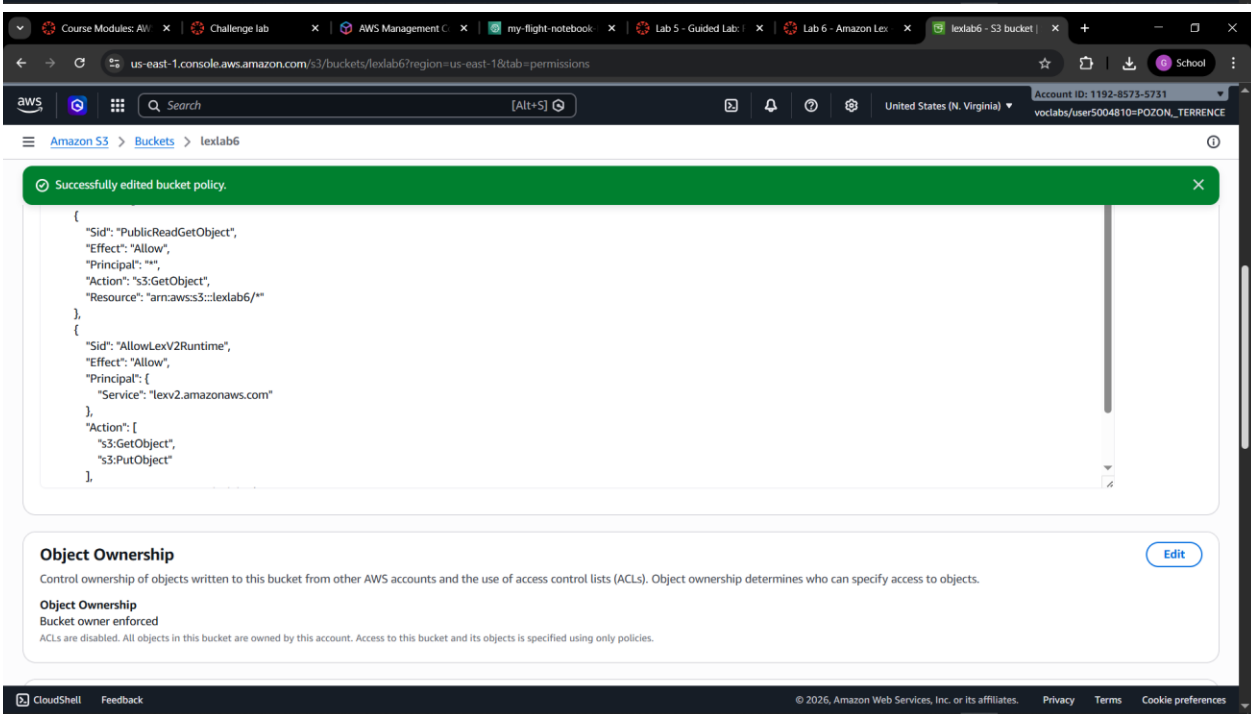
Course Modules: AW
Challenge lab
AWS Management C
my-flight-notebook
Lab 5 - Guided Lab:
Lab 6 - Amazon Lex
lexlab6 - S3 bucket
us-east-1.console.aws.amazon.com/s3/buckets/lexlab6?region=us-east-1&tab=permissions
School
Search
[Alt+S]
United States (N. Virginia)
Account ID: 1192-8573-5731
voclabs/user5004810=POZON_TERRENCE
Amazon S3 > Buckets > lexlab6
Successfully edited bucket policy.
{
"Sid": "PublicReadGetObject",
"Effect": "Allow",
"Principal": "*",
"Action": "s3:GetObject",
"Resource": "arn:aws:s3:::lexlab6/*"
},
{
"Sid": "AllowLexV2Runtime",
"Effect": "Allow",
"Principal": {
"Service": "lexv2.amazonaws.com"
},
"Action": [
"s3:GetObject",
"s3:PutObject"
],
Object Ownership
Edit
Control ownership of objects written to this bucket from other AWS accounts and the use of access control lists (ACLs). Object ownership determines who can specify access to objects.
Object Ownership
Bucket owner enforced
ACLs are disabled. All objects in this bucket are owned by this account. Access to this bucket and its objects is specified using only policies.
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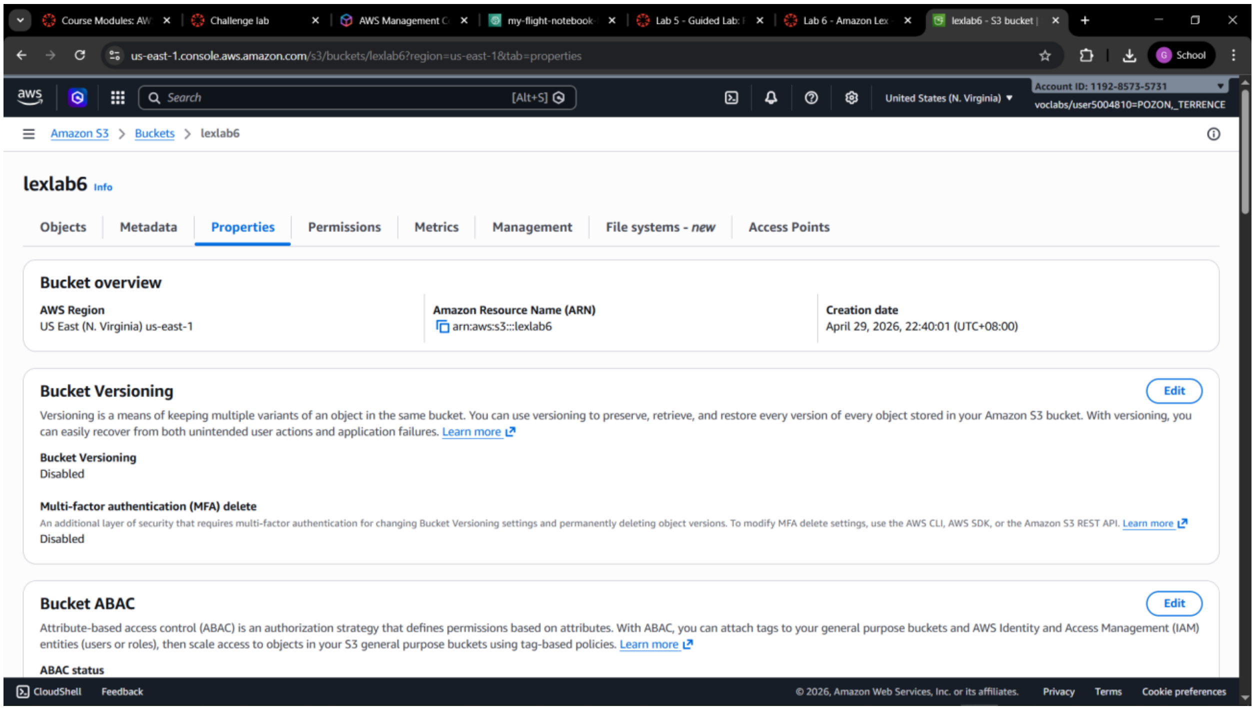

us-east-1.console.aws.amazon.com/s3/buckets/lexlab6?region=us-east-1&tab=properties
Amazon S3 > Buckets > lexlab6
lexlab6 Info
Objects
Metadata
Properties
Permissions
Metrics
Management
File systems - new
Access Points
Bucket overview
AWS Region
US East (N. Virginia) us-east-1
Amazon Resource Name (ARN)
arn:aws:s3:::lexlab6
Creation date
April 29, 2026, 22:40:01 (UTC+08:00)
Bucket Versioning
Edit
Versioning is a means of keeping multiple variants of an object in the same bucket. You can use versioning to preserve, retrieve, and restore every version of every object stored in your Amazon S3 bucket. With versioning, you can easily recover from both unintended user actions and application failures. Learn more
Bucket Versioning
Disabled
Multi-factor authentication (MFA) delete
An additional layer of security that requires multi-factor authentication for changing Bucket Versioning settings and permanently deleting object versions. To modify MFA delete settings, use the AWS CLI, AWS SDK, or the Amazon S3 REST API. Learn more
Disabled
Bucket ABAC
Edit
Attribute-based access control (ABAC) is an authorization strategy that defines permissions based on attributes. With ABAC, you can attach tags to your general purpose buckets and AWS Identity and Access Management (IAM) entities (users or roles), then scale access to objects in your S3 general purpose buckets using tag-based policies. Learn more
ABAC status
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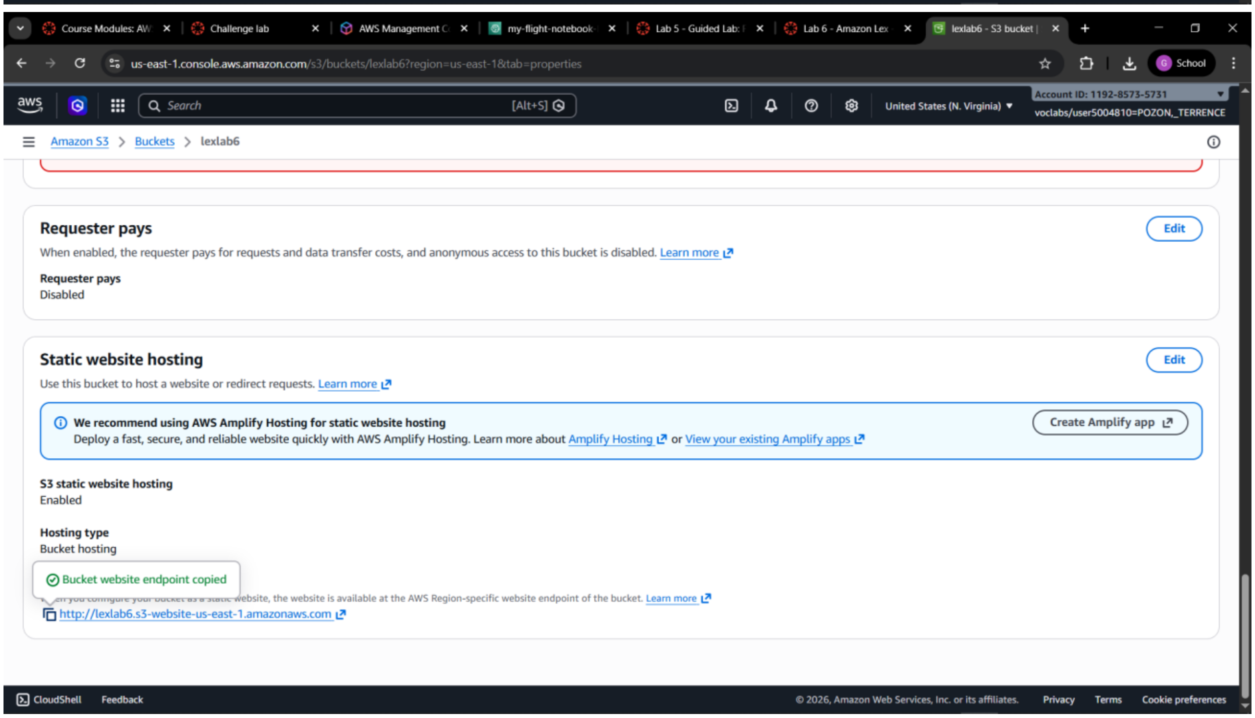

us-east-1.console.aws.amazon.com/s3/buckets/lexlab6?region=us-east-1&tab=properties
Amazon S3 > Buckets > lexlab6
Requester pays
Edit
When enabled, the requester pays for requests and data transfer costs, and anonymous access to this bucket is disabled. Learn more
Requester pays
Disabled
Static website hosting
Edit
Use this bucket to host a website or redirect requests. Learn more
We recommend using AWS Amplify Hosting for static website hosting
Deploy a fast, secure, and reliable website quickly with AWS Amplify Hosting. Learn more about Amplify Hosting or View your existing Amplify apps
Create Amplify app
S3 static website hosting
Enabled
Hosting type
Bucket hosting
Bucket website endpoint copied
website, the website is available at the AWS Region-specific website endpoint of the bucket. Learn more
http://lexlab6.s3-website-us-east-1.amazonaws.com
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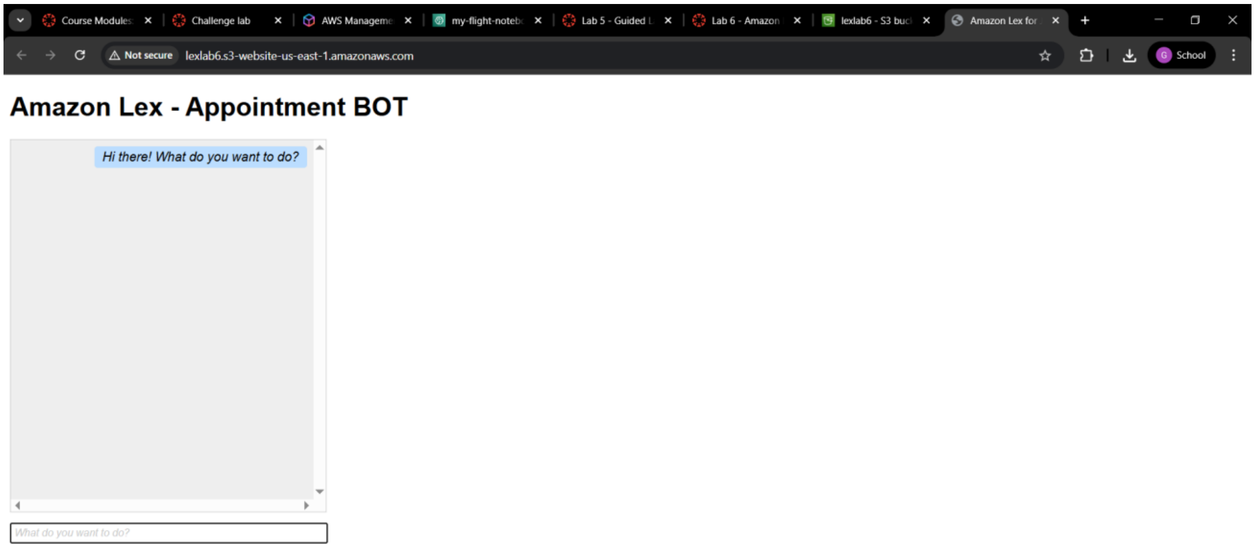
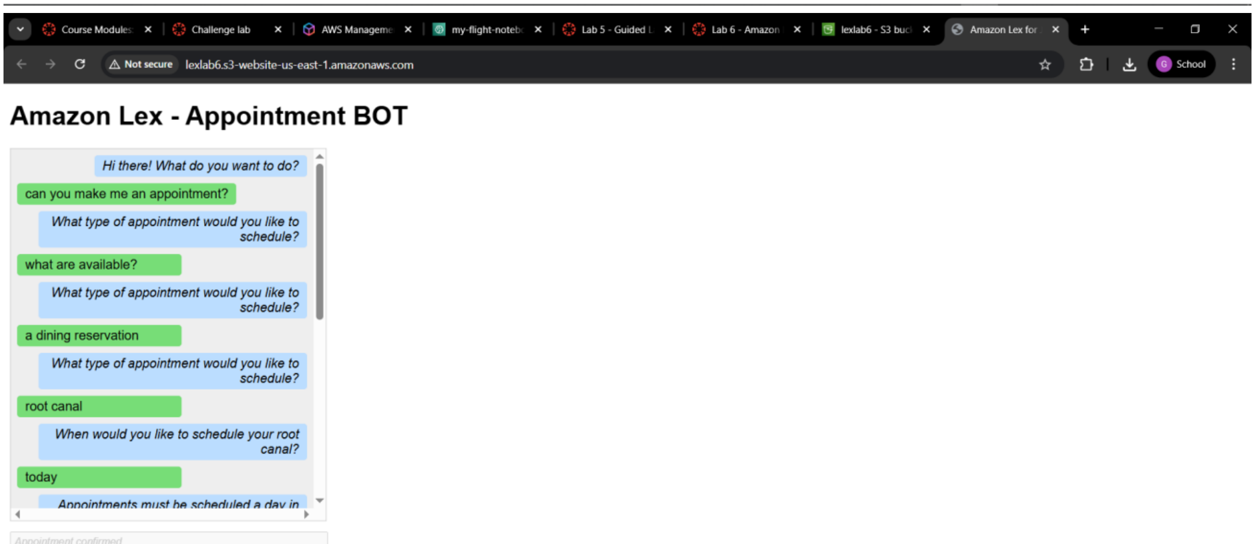

Course Modules
Challenge lab
AWS Manageme
my-flight-noteb
Lab 5 - Guided L
Lab 6 - Amazon
lexlab6 - S3 buc
Amazon Lex for
Not secure
lexlab6.s3-website-us-east-1.amazonaws.com
School
Amazon Lex - Appointment BOT
Hi there! What do you want to do?
What do you want to do?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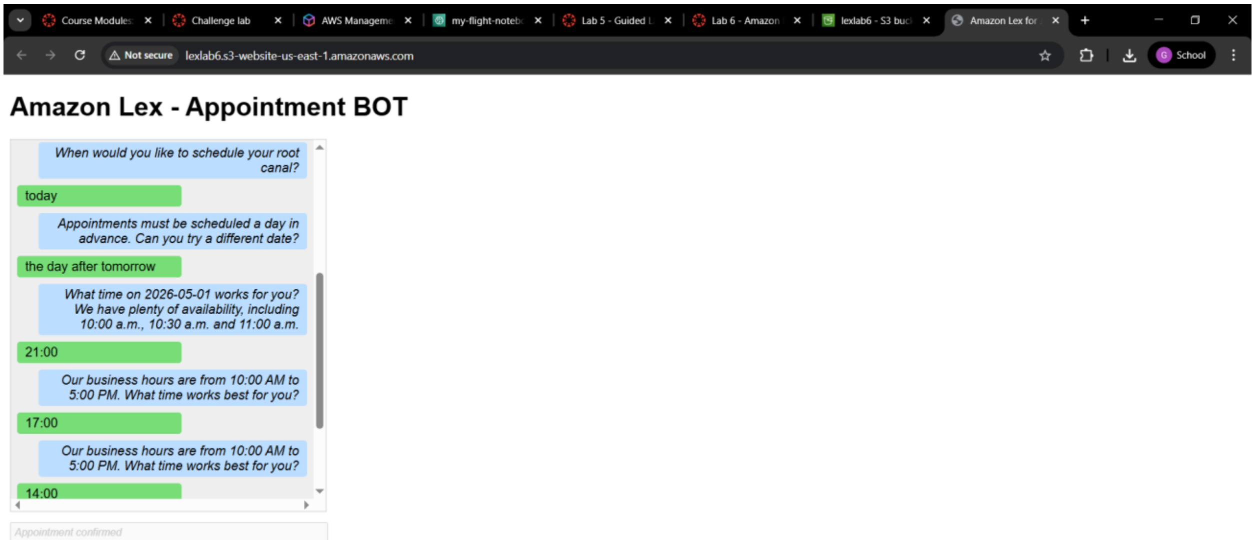
lexlab6 - S3 buc
Amazon Lex for
Not secure
lexlab6.s3-website-us-east-1.amazonaws.com
School
Amazon Lex - Appointment BOT
Hi there! What do you want to do?
can you make me an appointment?
What type of appointment would you like to schedule?
what are available?
What type of appointment would you like to schedule?
a dining reservation
What type of appointment would you like to schedule?
root canal
When would you like to schedule your root canal?
today
Appointment confirmed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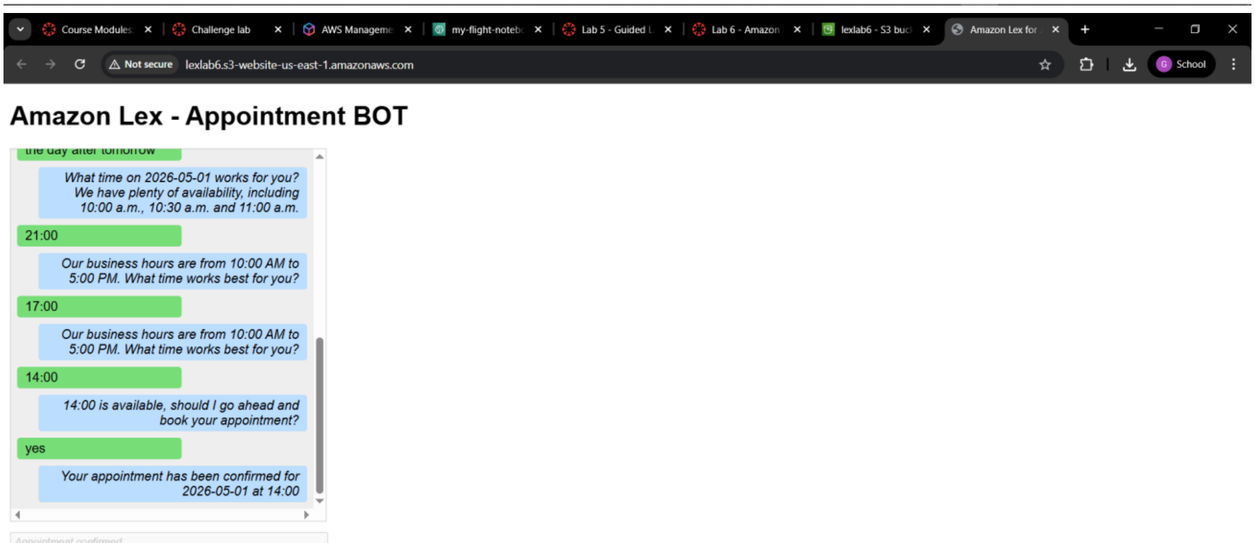
my-flight-notebo
Lab 5 - Guided L
Lab 6 - Amazon
lexlab6 - S3 buck
Amazon Lex for
Not secure
lexlab6.s3-website-us-east-1.amazonaws.com
School
Amazon Lex - Appointment BOT
When would you like to schedule your root canal?
today
Appointments must be scheduled a day in advance. Can you try a different date?
the day after tomorrow
What time on 2026-05-01 works for you? We have plenty of availability, including 10:00 a.m., 10:30 a.m. and 11:00 a.m.
21:00
Our business hours are from 10:00 AM to 5:00 PM. What time works best for you?
17:00
Our business hours are from 10:00 AM to 5:00 PM. What time works best for you?
14:00
Appointment confirmed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Lab 6 - Amazon
lexlab6 - S3 buck
Amazon Lex for
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lexlab6.s3-website-us-east-1.amazonaws.com
School
Amazon Lex - Appointment BOT
What time on 2026-05-01 works for you? We have plenty of availability, including 10:00 a.m., 10:30 a.m. and 11:00 a.m.
21:00
Our business hours are from 10:00 AM to 5:00 PM. What time works best for you?
17:00
Our business hours are from 10:00 AM to 5:00 PM. What time works best for you?
14:00
14:00 is available, should I go ahead and book your appointment?
yes
Your appointment has been confirmed for 2026-05-01 at 14:00
Appointment confirmed

Canvas Details